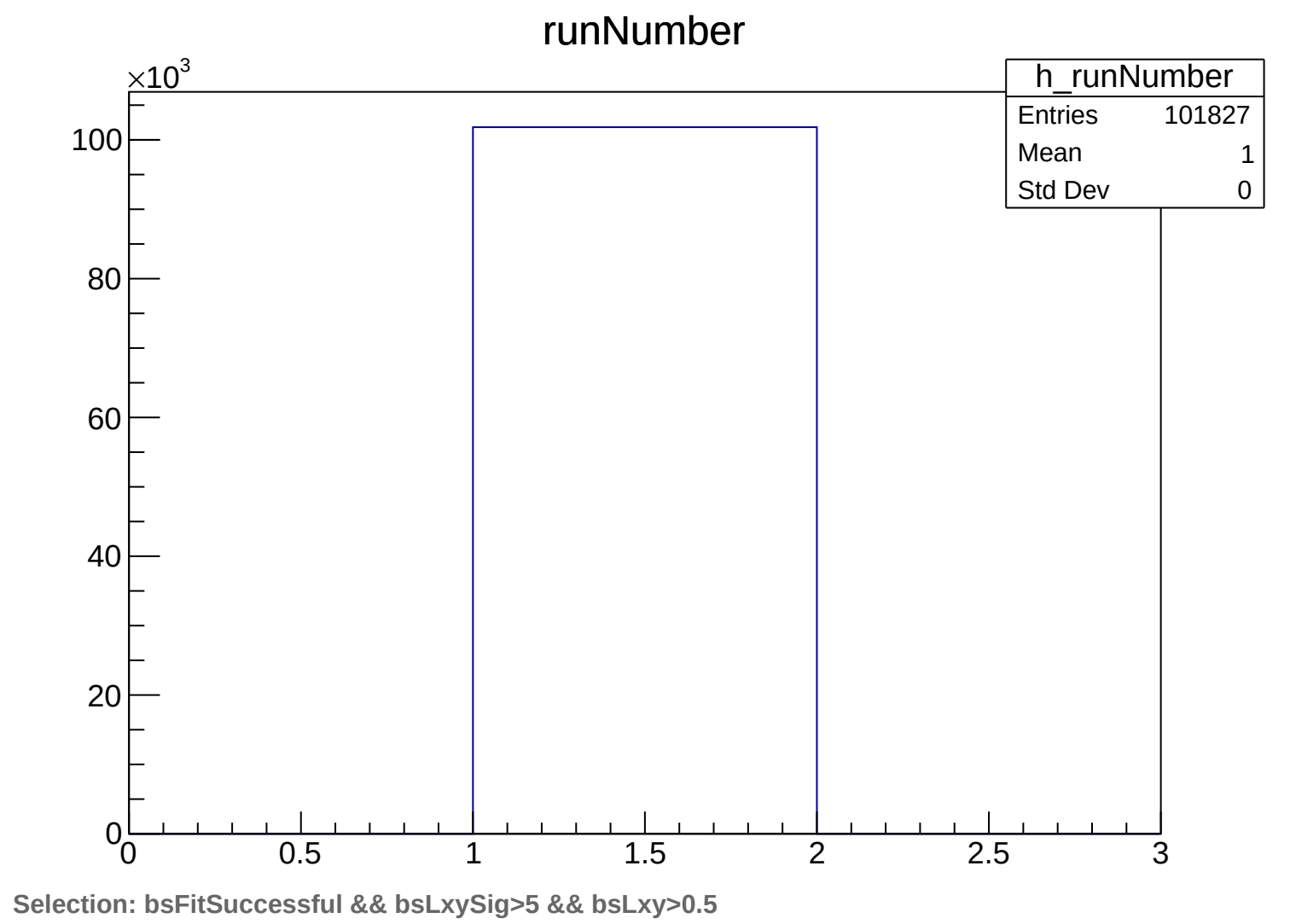
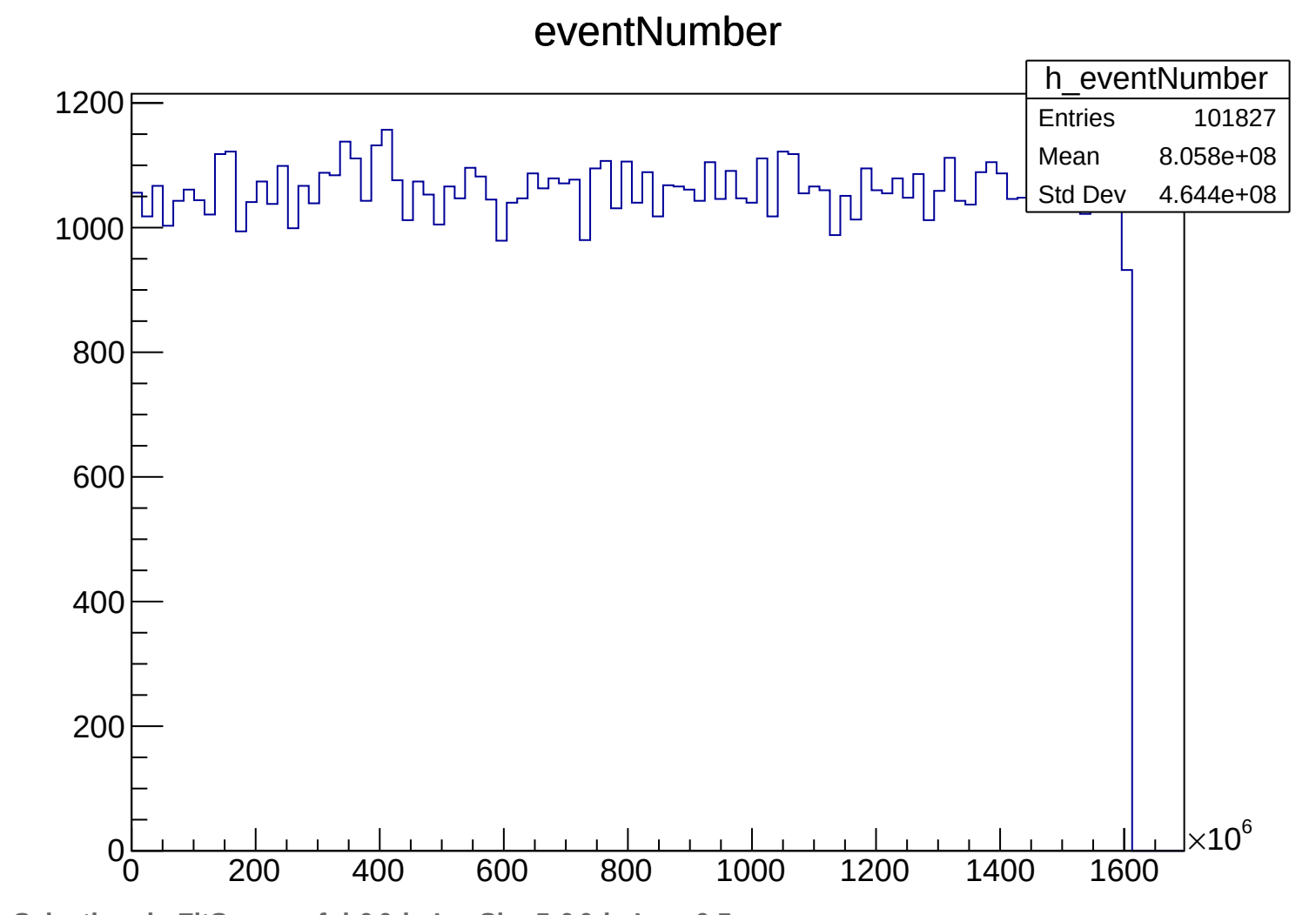
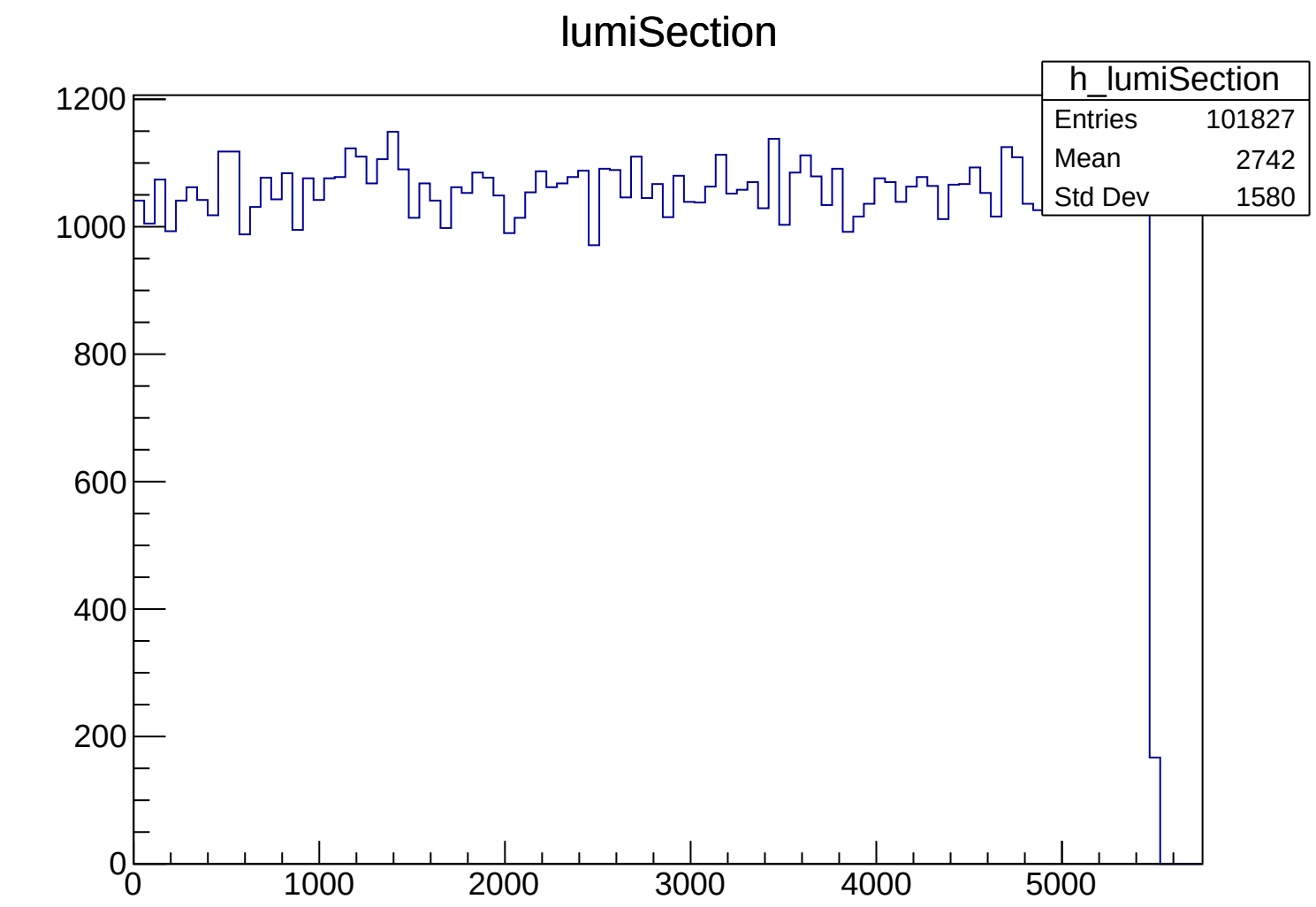


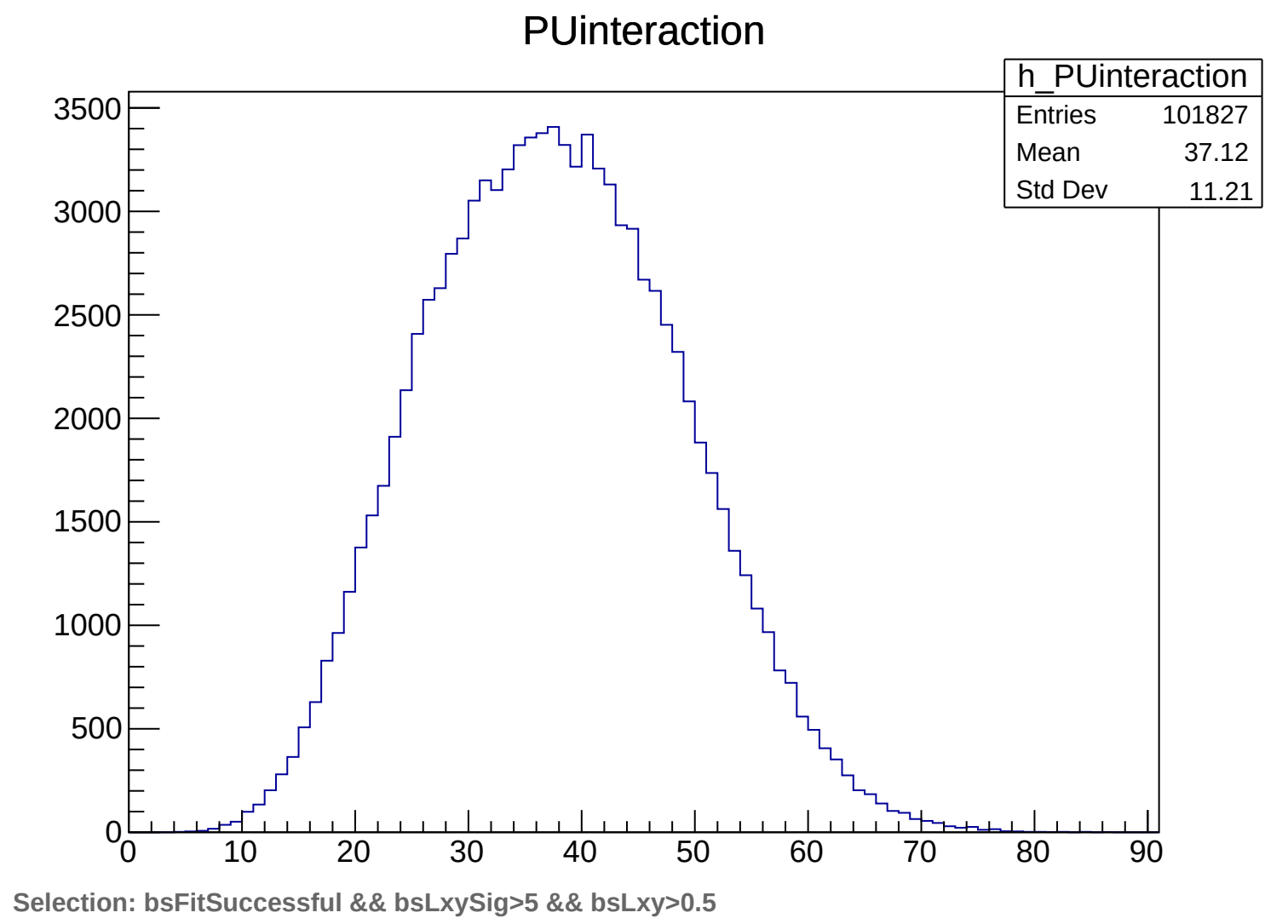
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

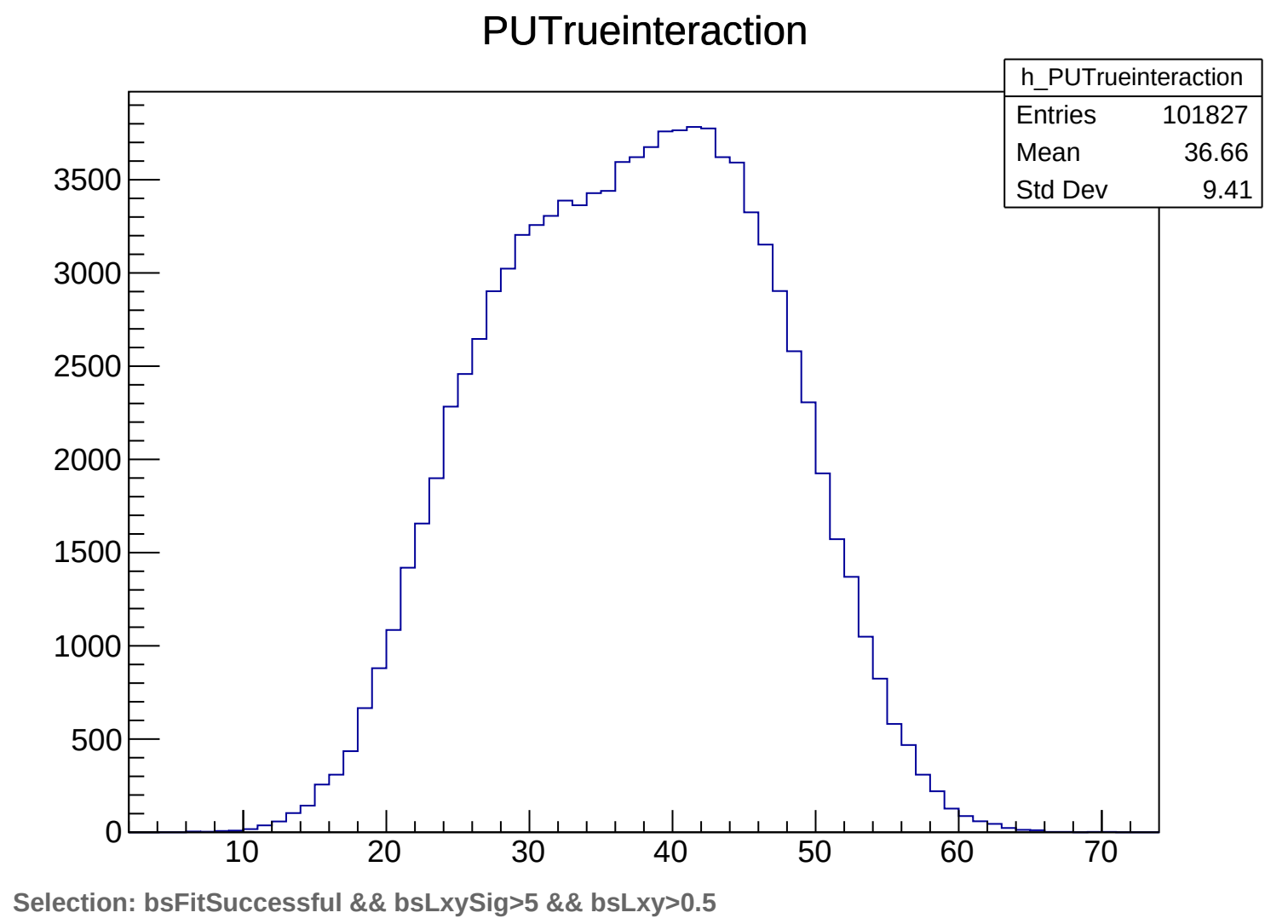


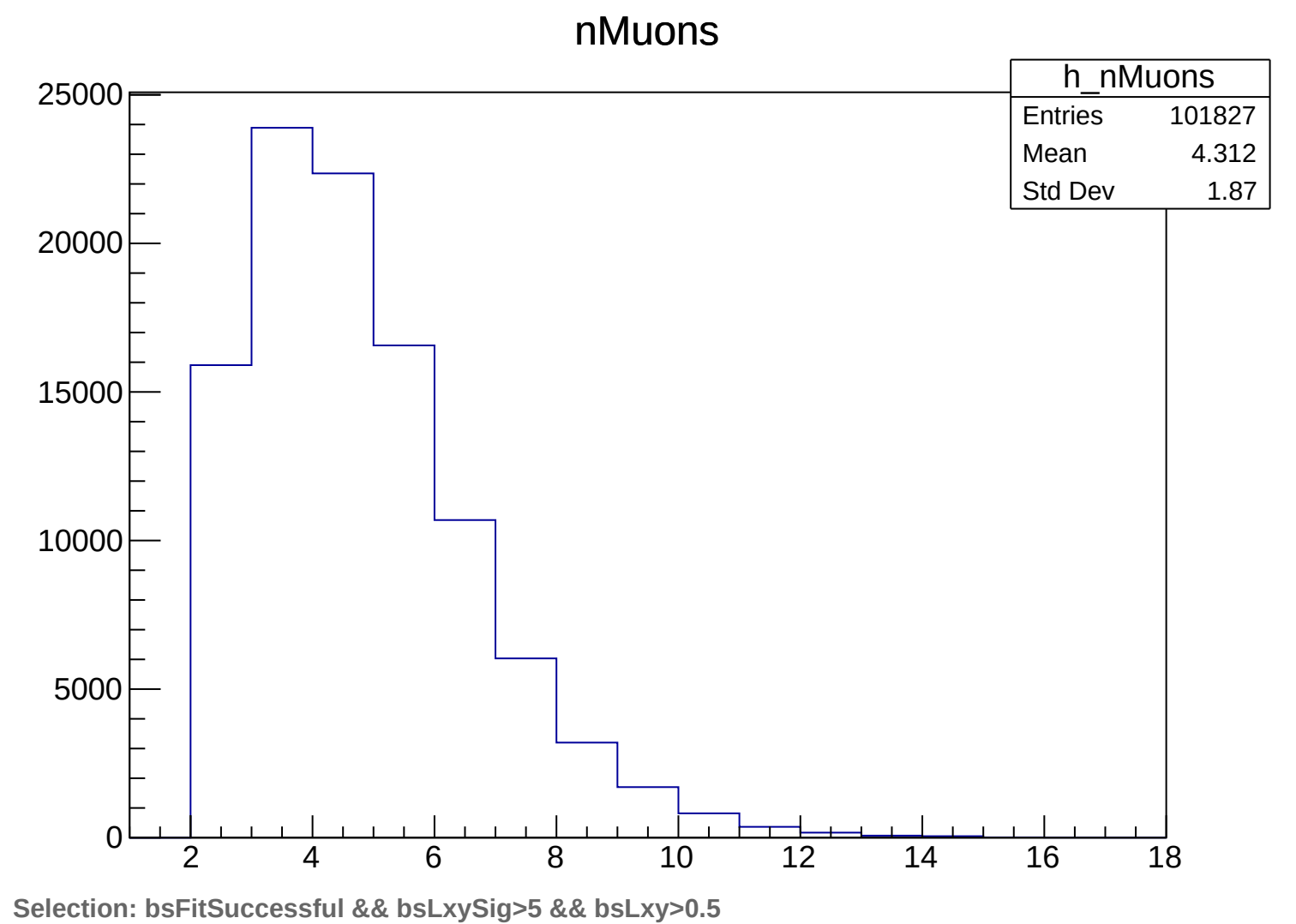
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

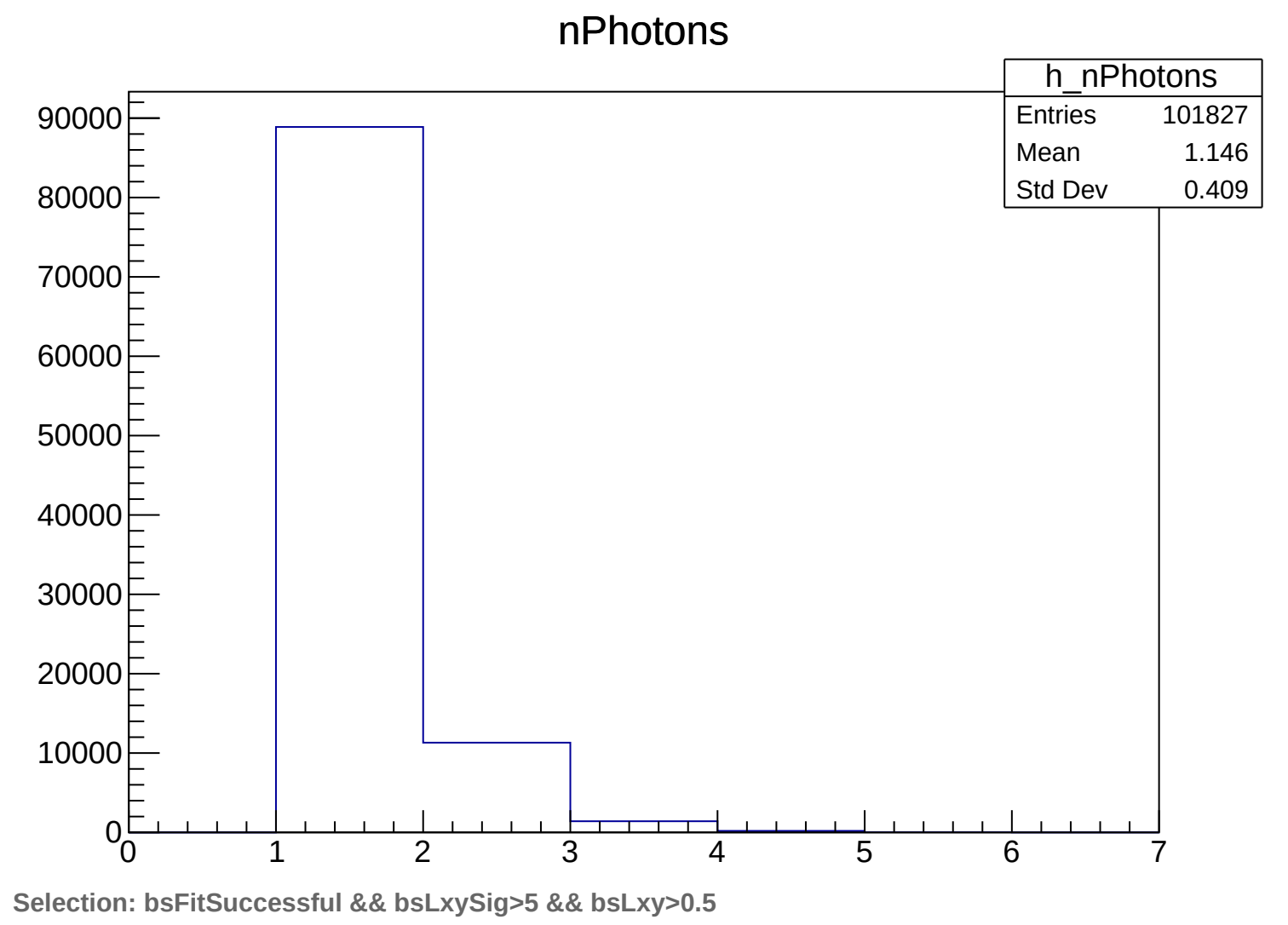


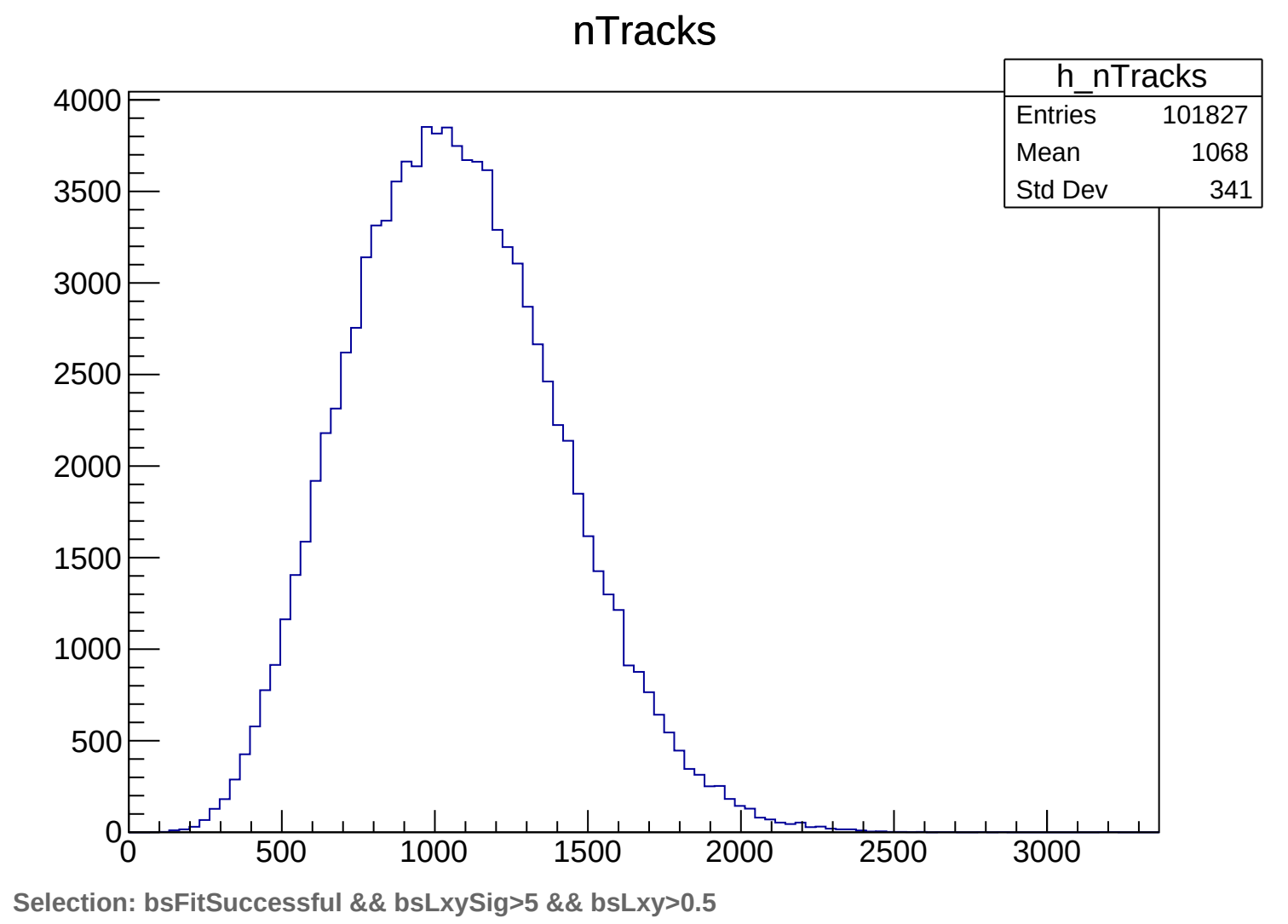
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

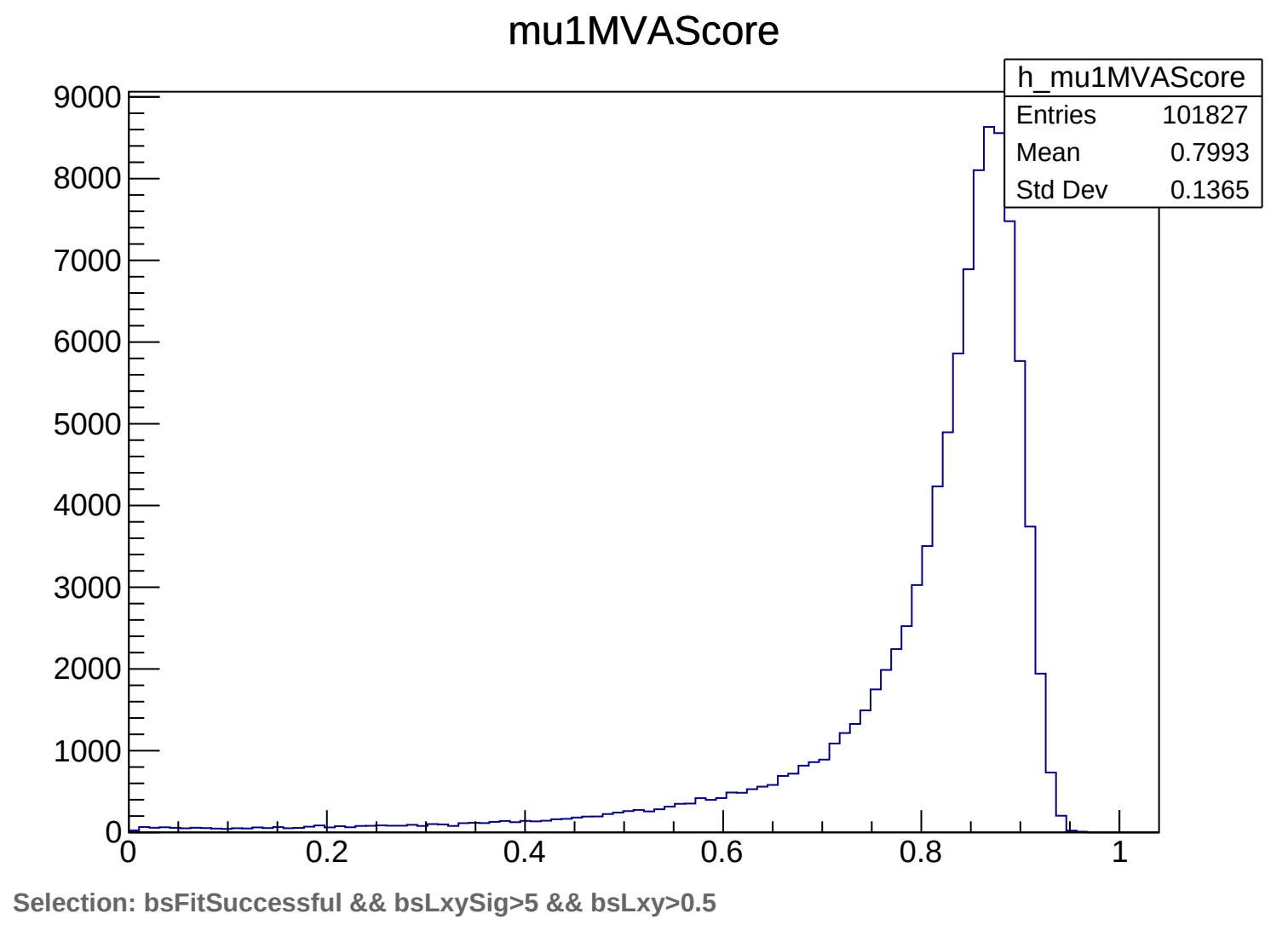

PUinteraction
h_PUinteraction
Entries 101827
Mean 37.12
Std Dev 11.21
3500
3000
2500
2000
1500
1000
500
0
0
10
20
30
40
50
60
70
80
90
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

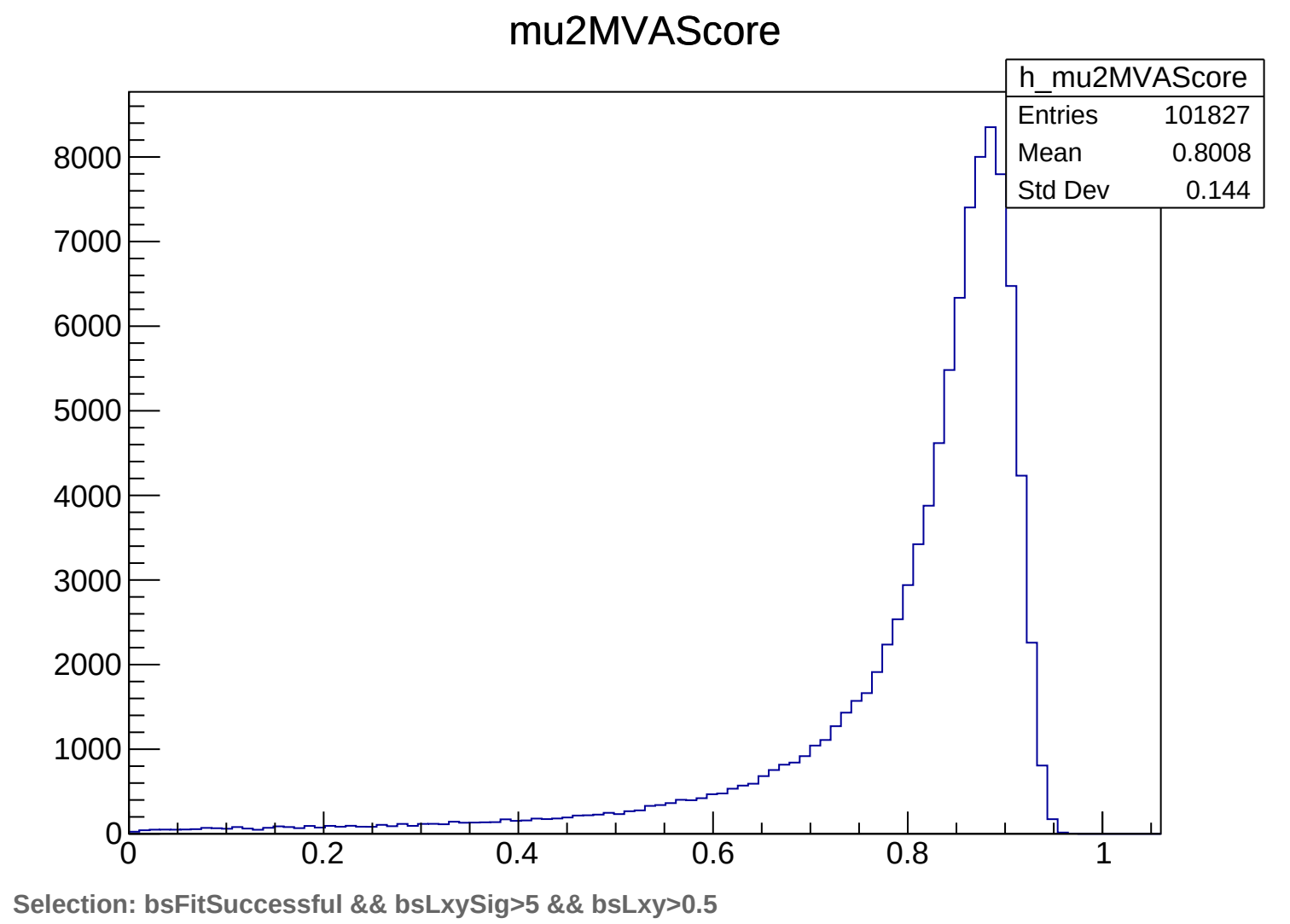

PUTrueinteraction
h_PUTrueinteraction
Entries 101827
Mean 36.66
Std Dev 9.41
3500
3000
2500
2000
1500
1000
500
0
10
20
30
40
50
60
70
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

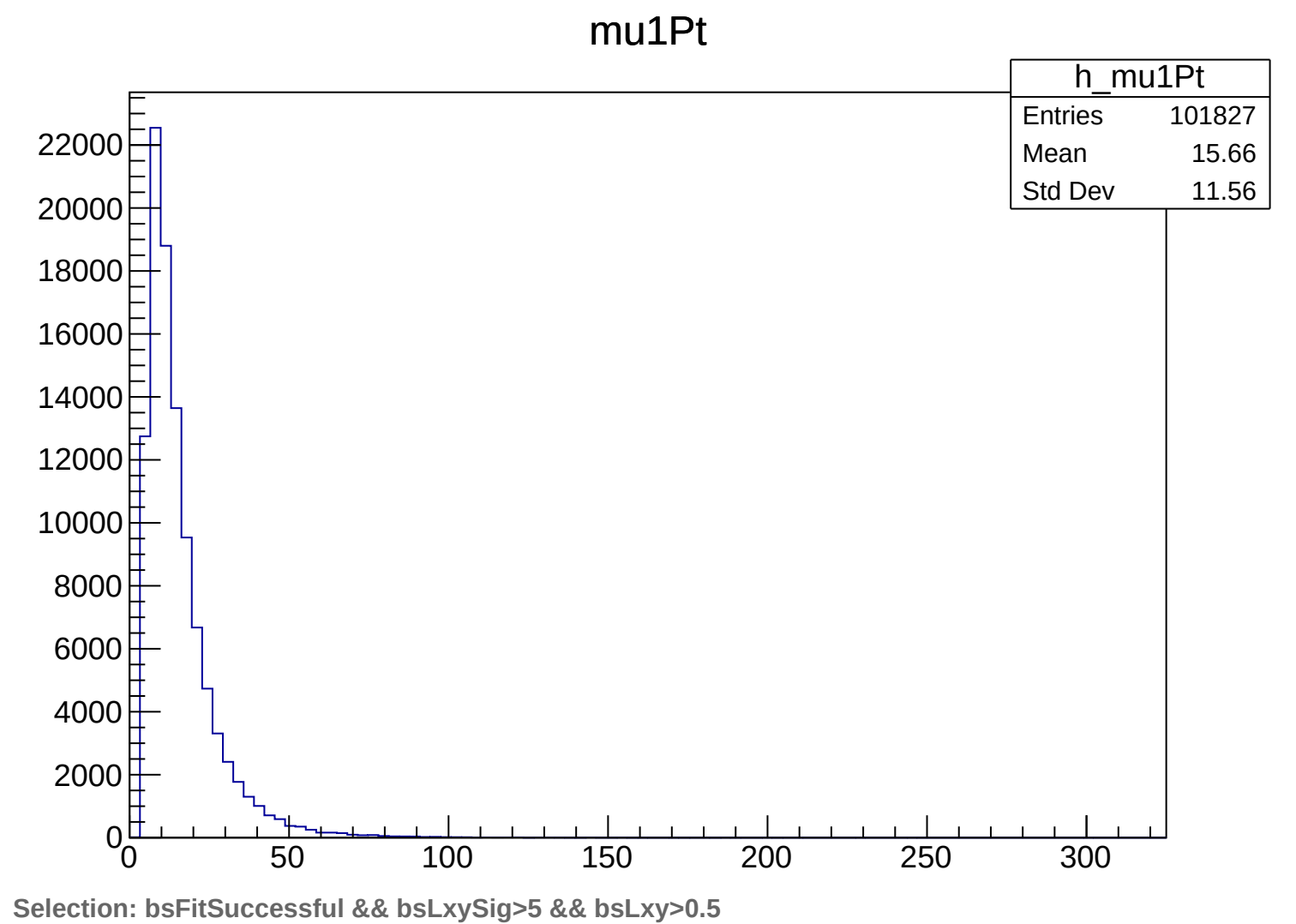

nMuons
h_nMuons
Entries 101827
Mean 4.312
Std Dev 1.87
25000
20000
15000
10000
5000
0
2
4
6
8
10
12
14
16
18
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


nPhotons
h_nPhotons
Entries 101827
Mean 1.146
Std Dev 0.409
90000
80000
70000
60000
50000
40000
30000
20000
10000
0
0
1
2
3
4
5
6
7
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

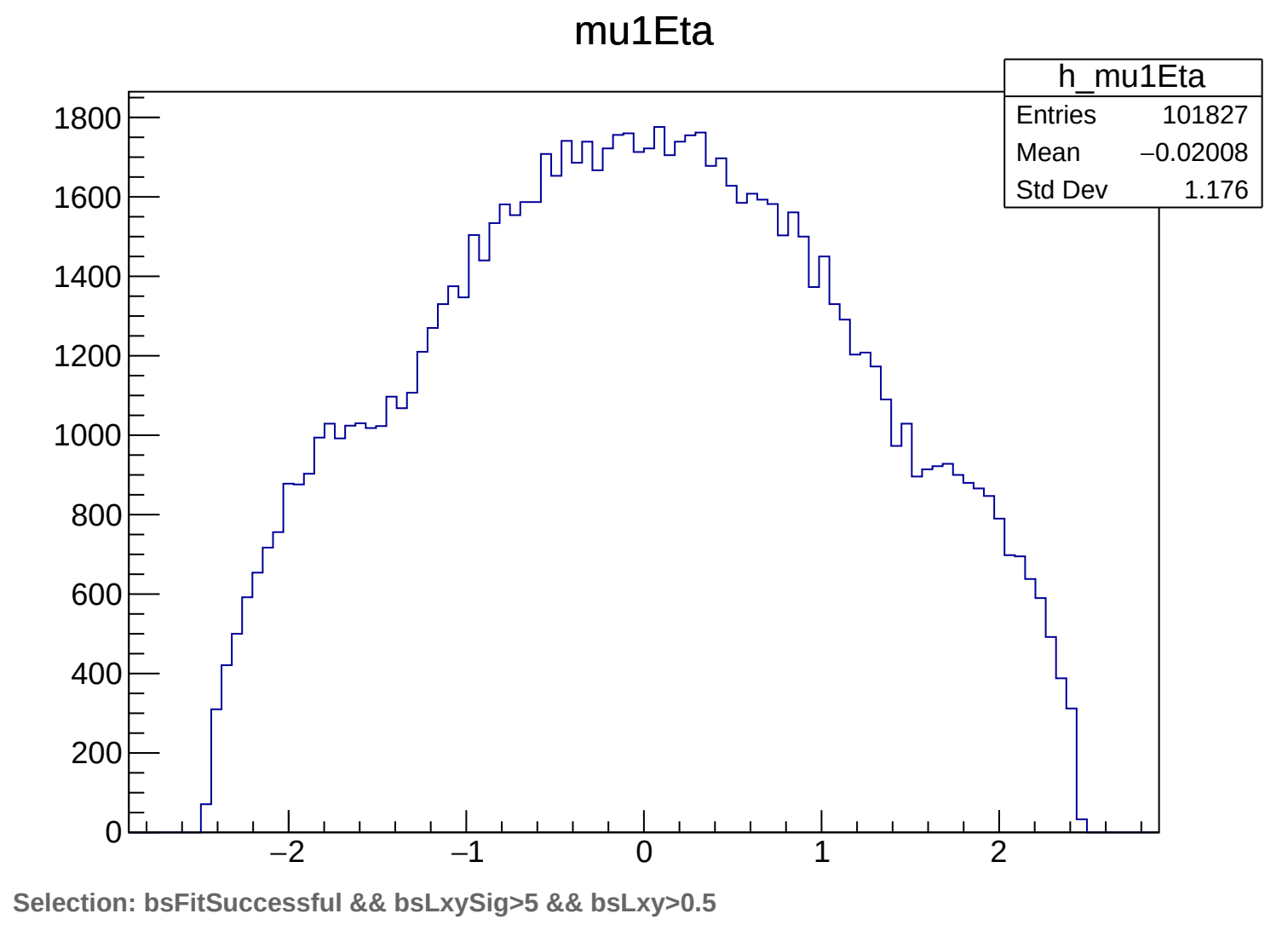

nTracks
h_nTracks
Entries 101827
Mean 1068
Std Dev 341
4000
3500
3000
2500
2000
1500
1000
500
0
0
500
1000
1500
2000
2500
3000
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


mu1MVAScore
h_mu1MVAScore
Entries 101827
Mean 0.7993
Std Dev 0.1365
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

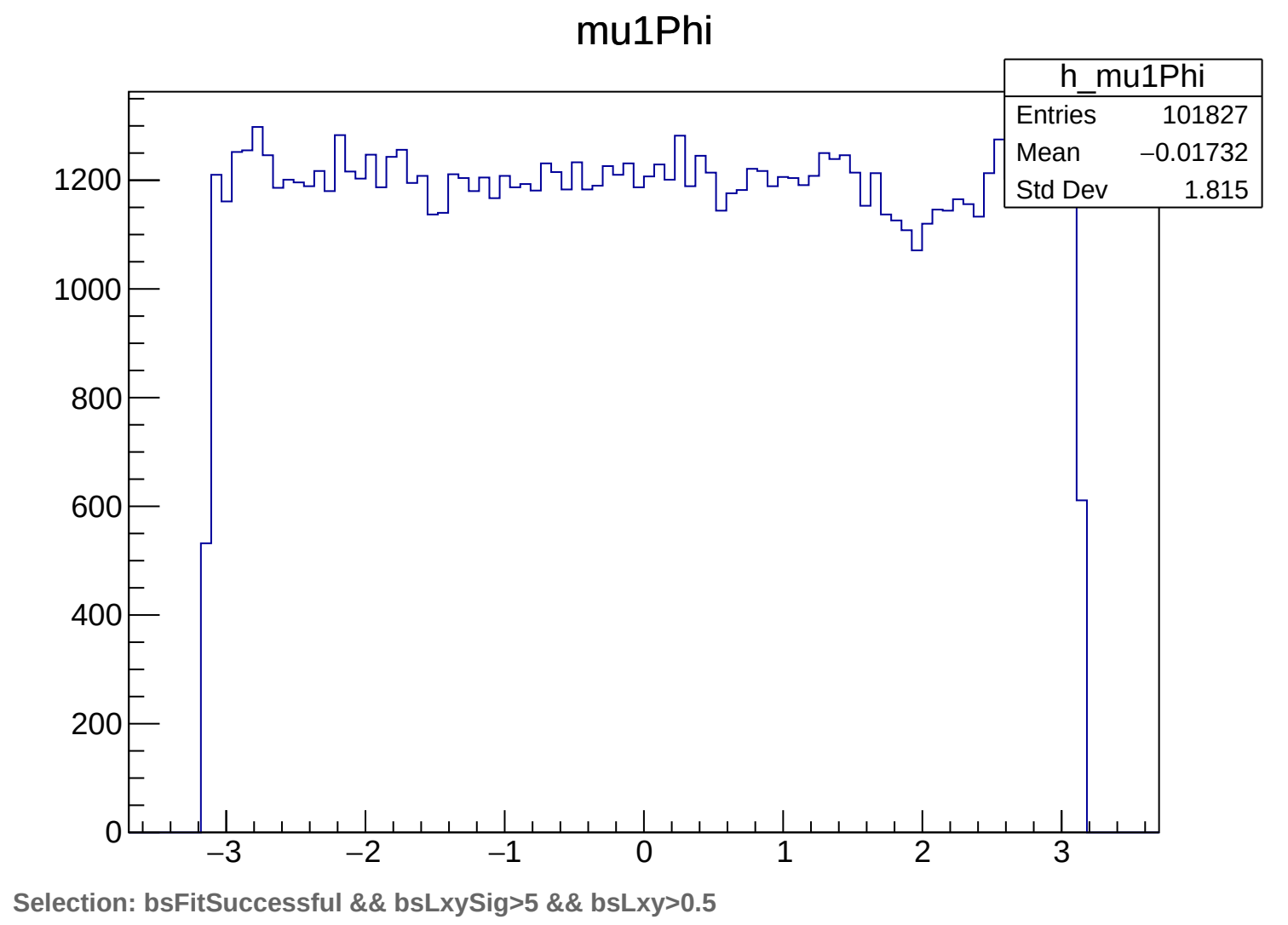

mu2MVAScore
h_mu2MVAScore
Entries 101827
Mean 0.8008
Std Dev 0.144
8000
7000
6000
5000
4000
3000
2000
1000
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5



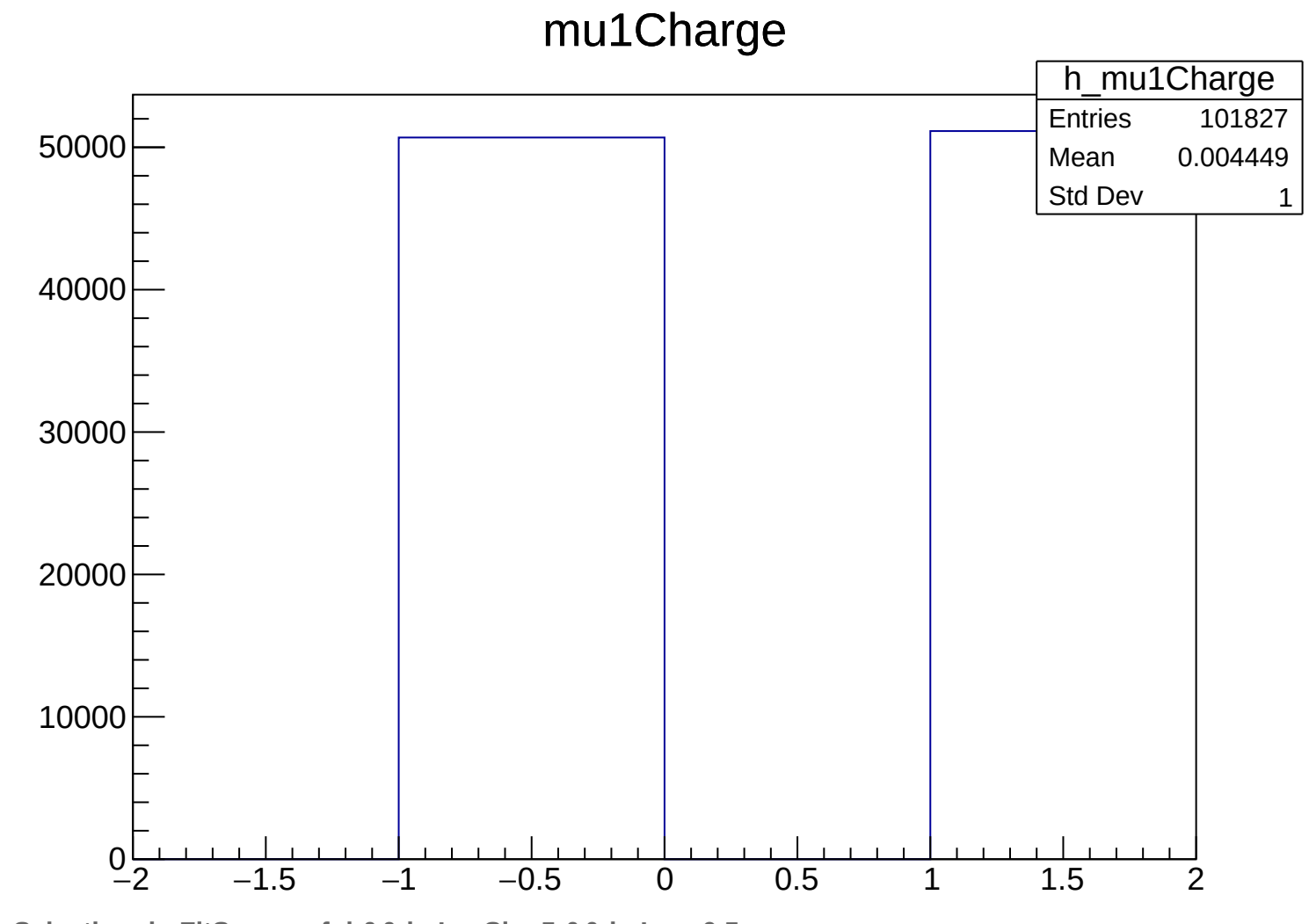
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5



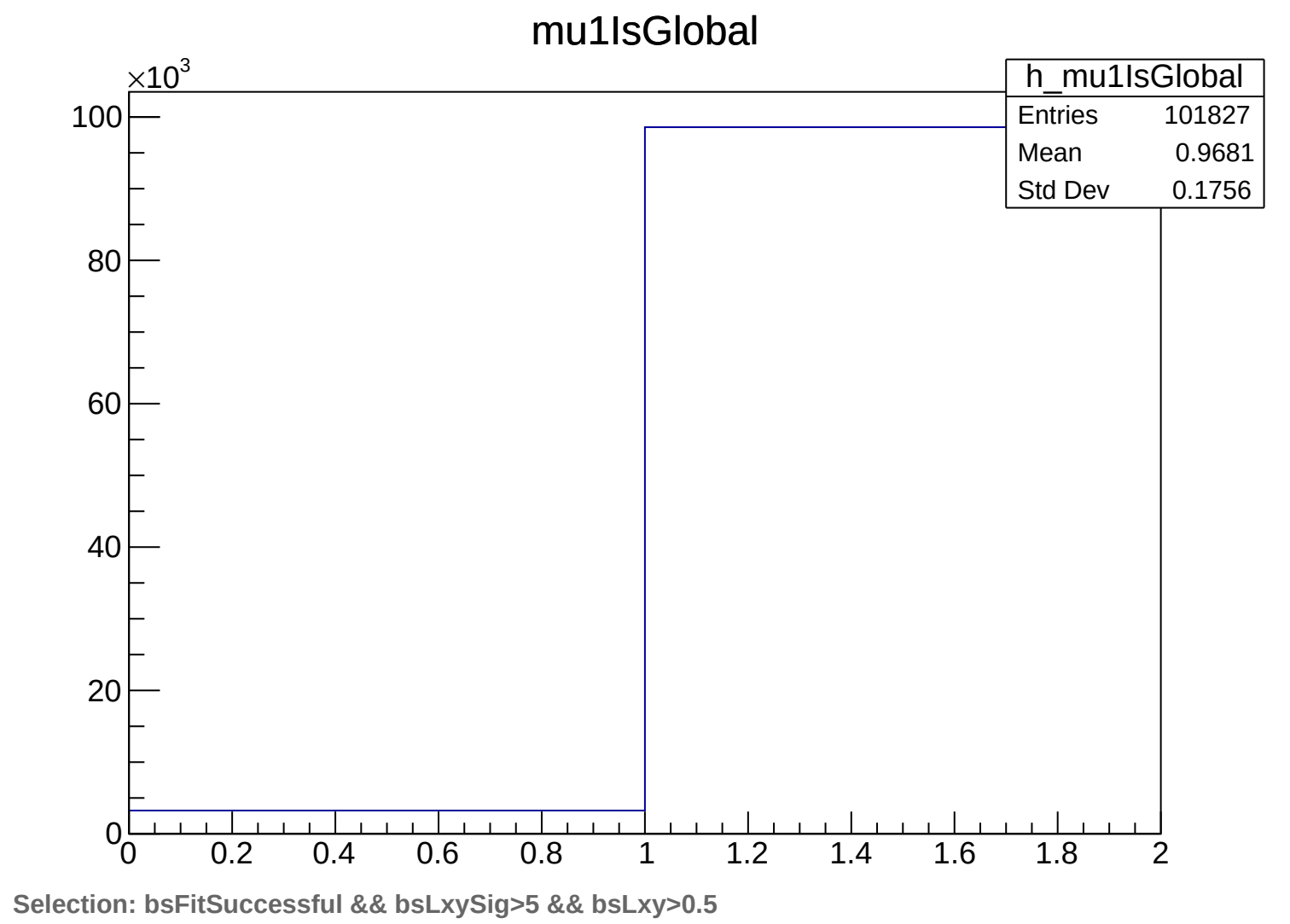
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

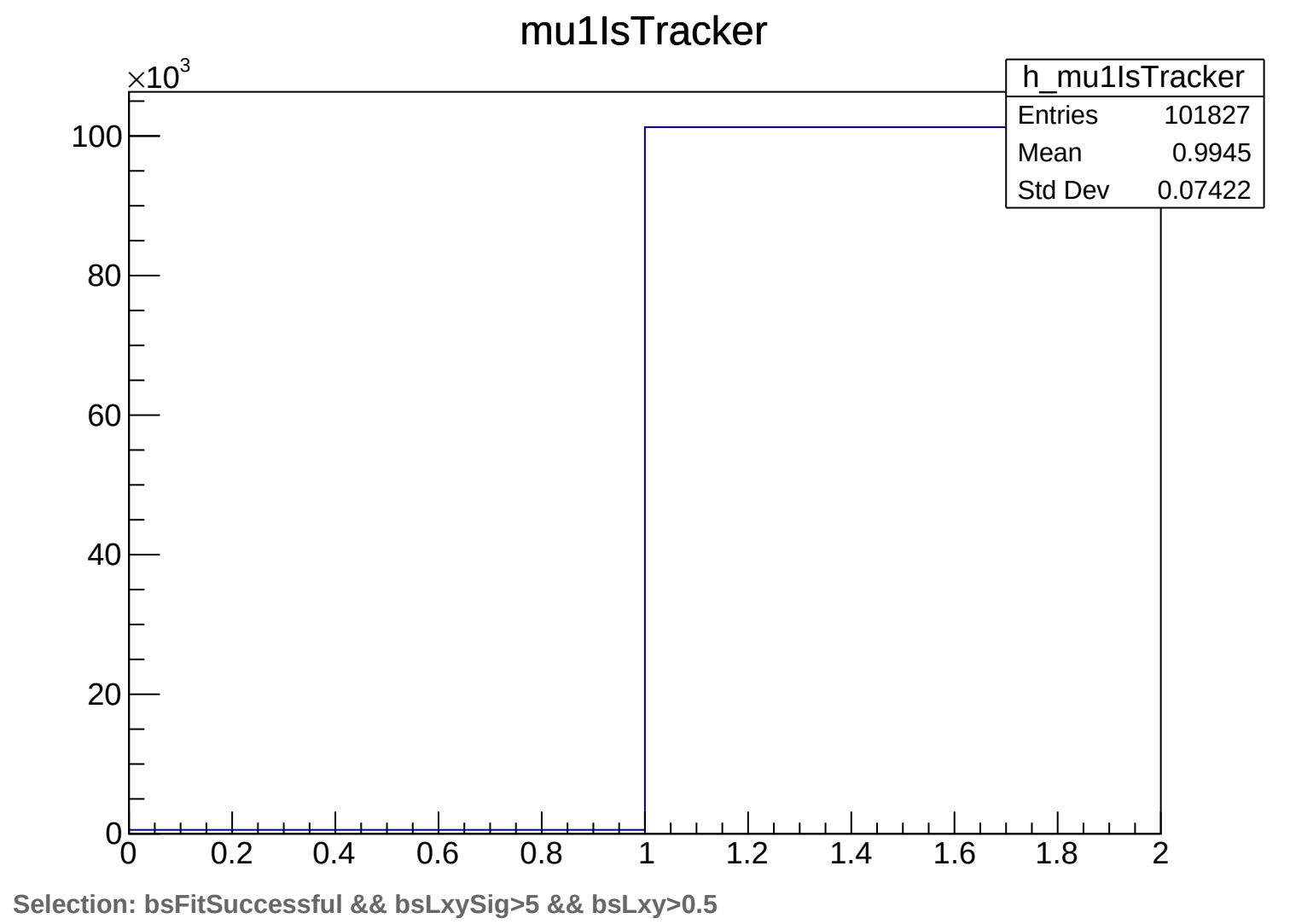


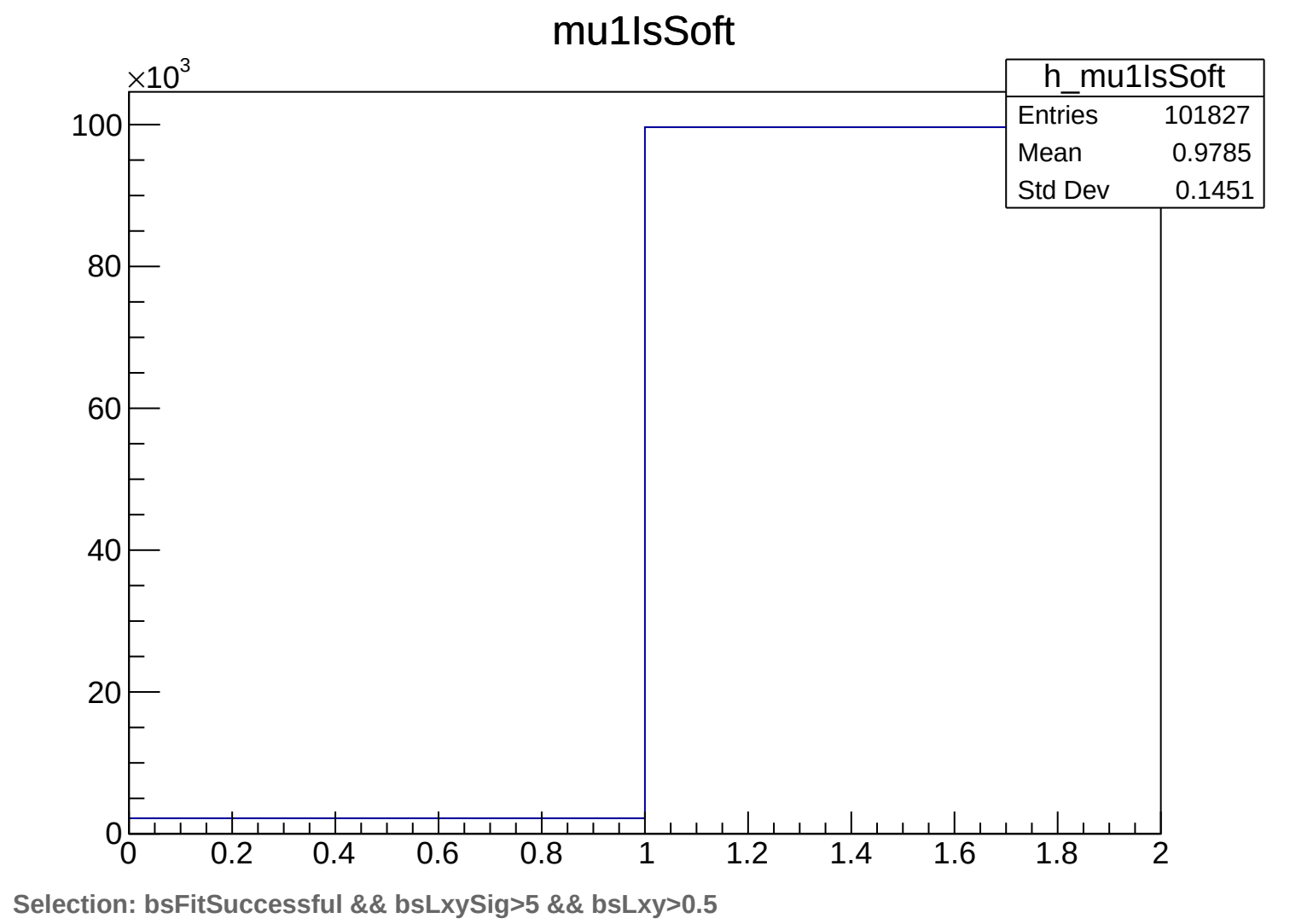
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

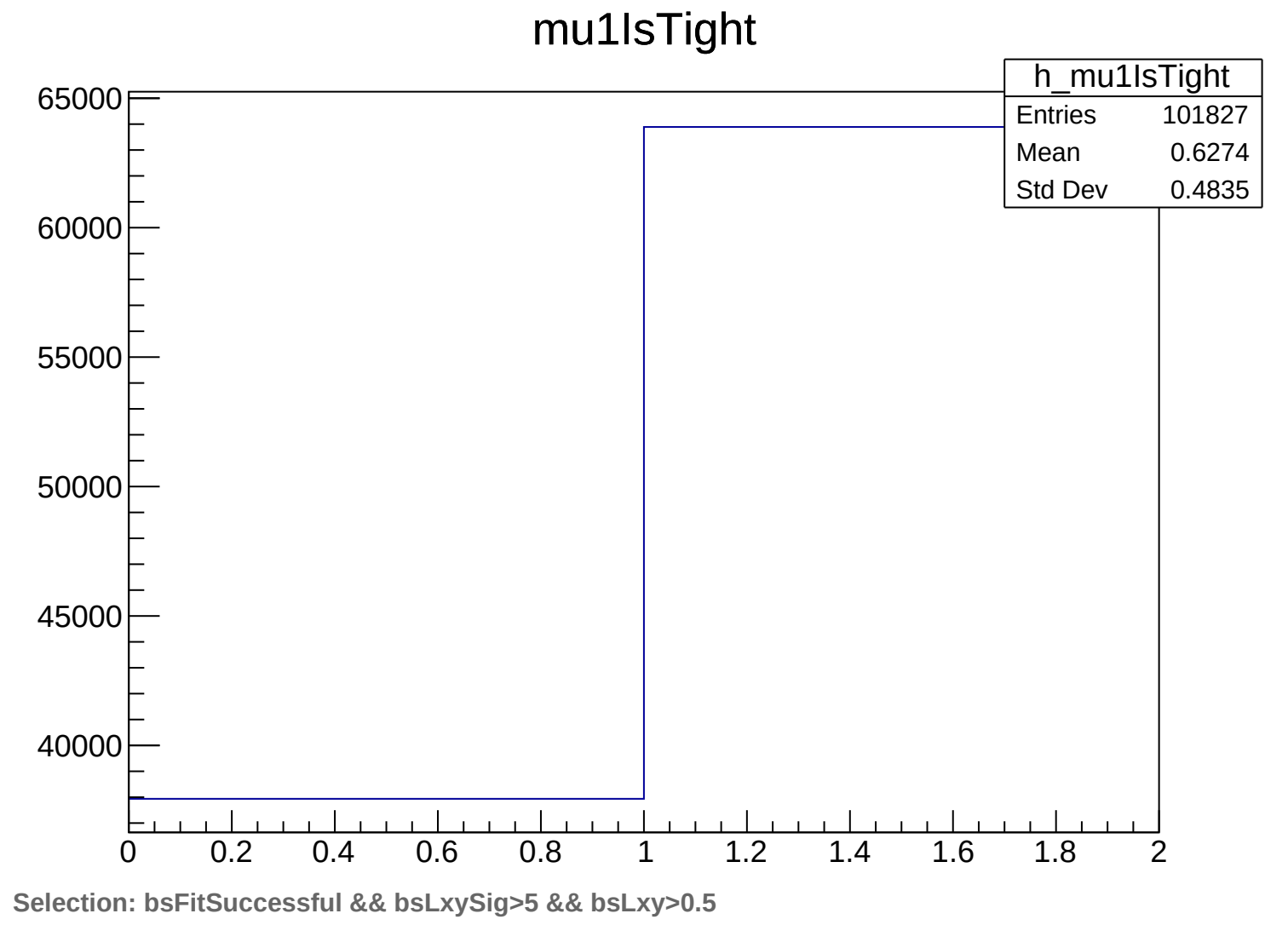


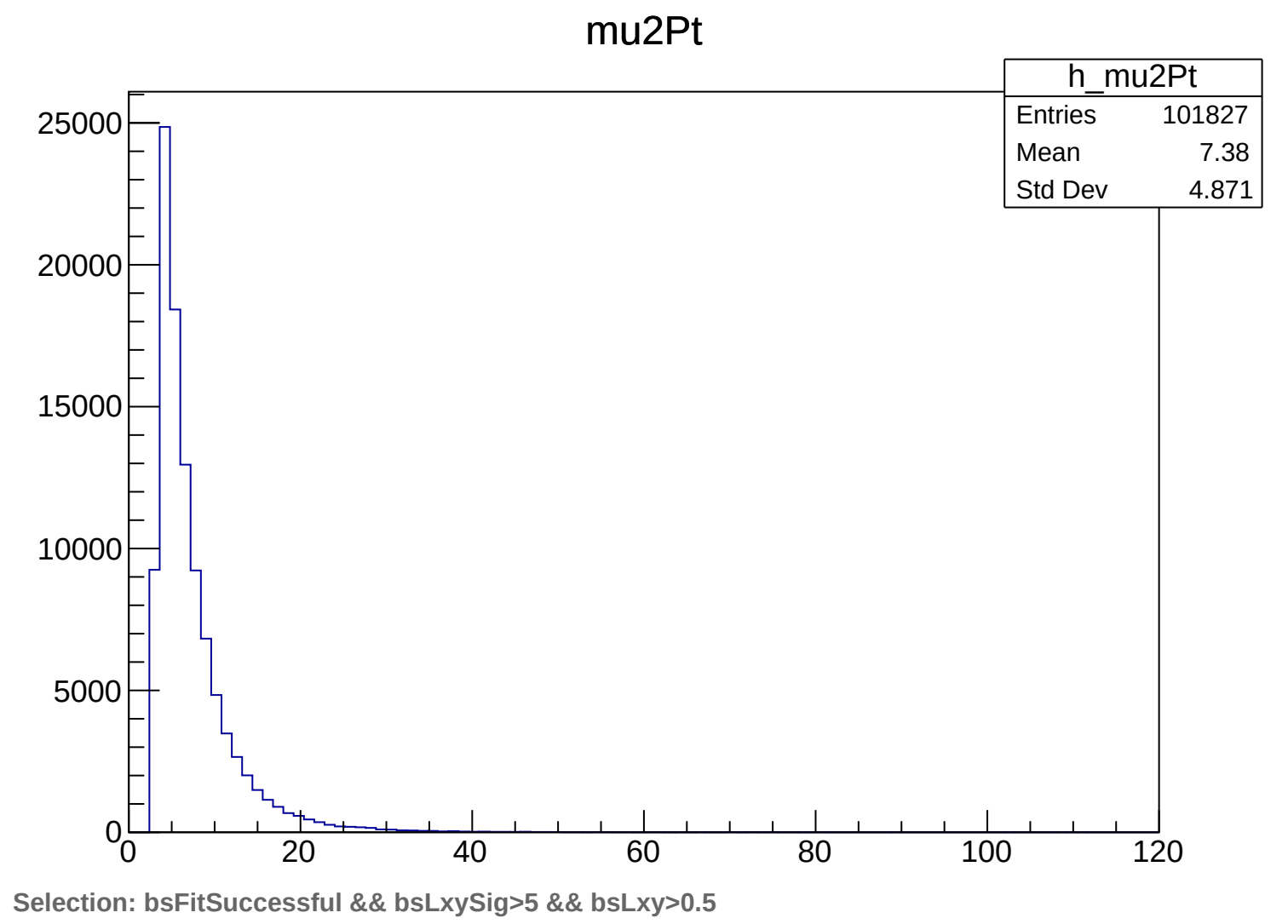
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

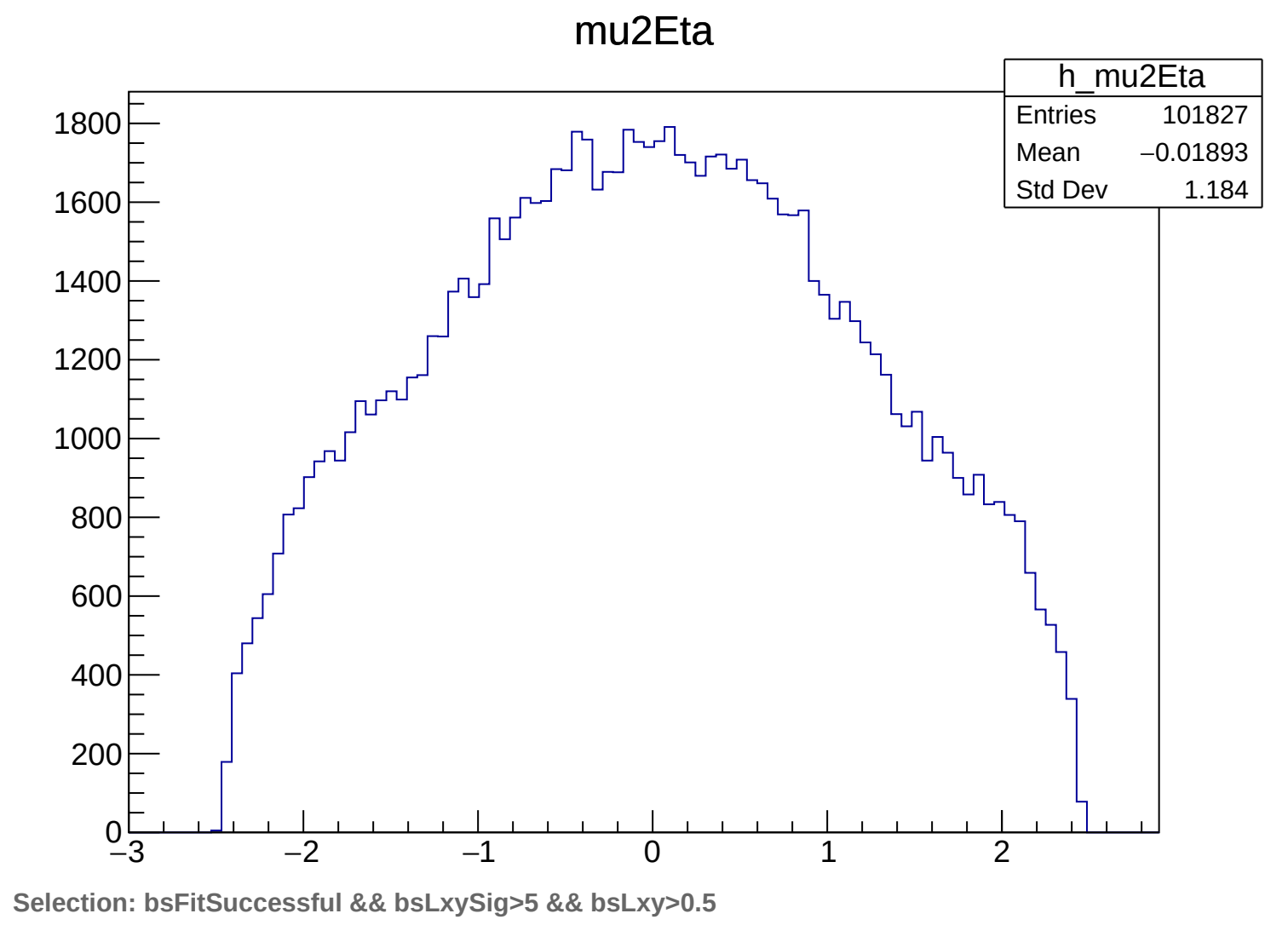

mu1IsGlobal
h_mu1IsGlobal
Entries 101827
Mean 0.9681
Std Dev 0.1756
×10^3
100
80
60
40
20
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

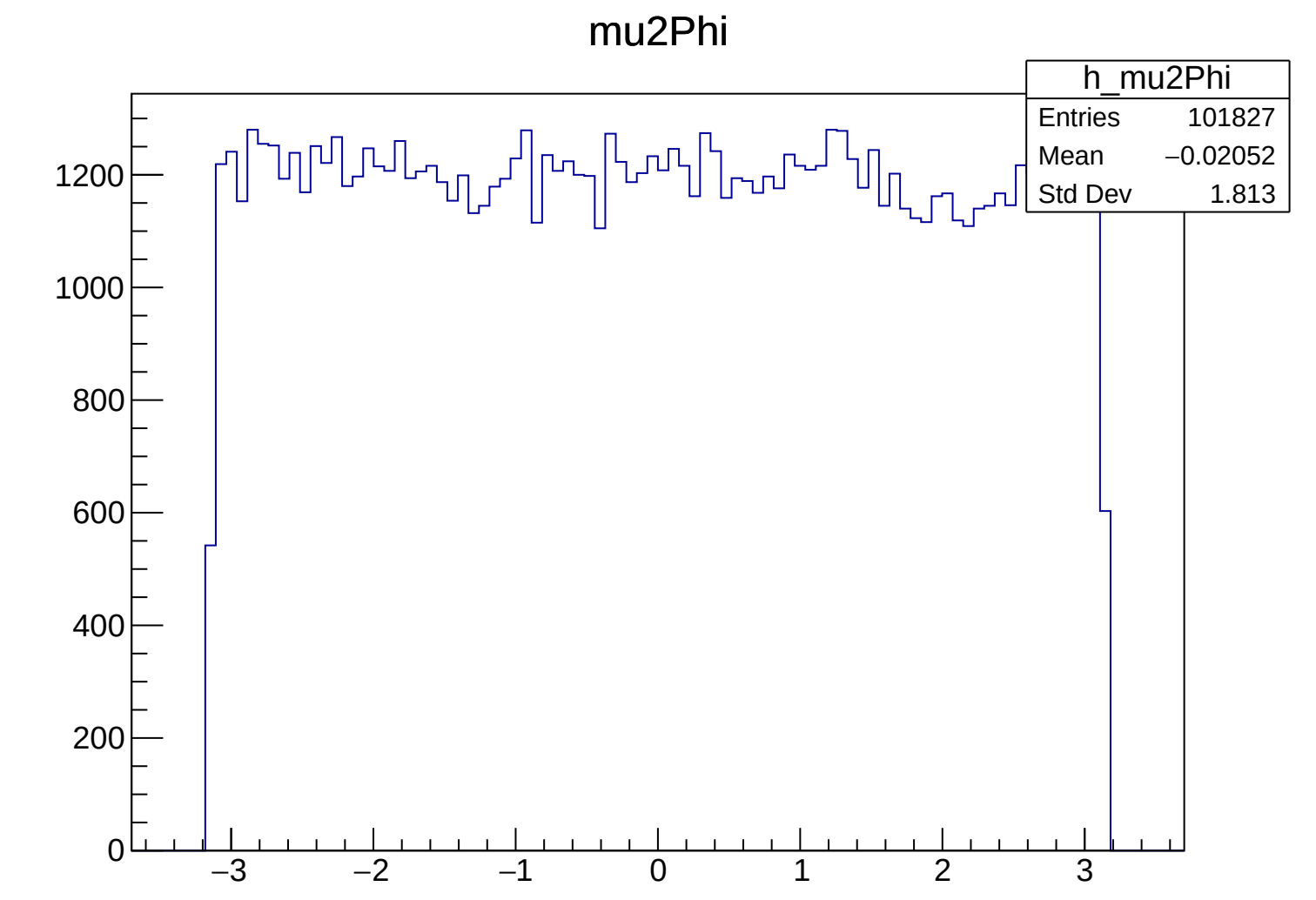

mu1IsTracker
h_mu1IsTracker
Entries 101827
Mean 0.9945
Std Dev 0.07422
×10^3
100
80
60
40
20
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


mu1IsSoft
h_mu1IsSoft
Entries 101827
Mean 0.9785
Std Dev 0.1451
×10^3
100
80
60
40
20
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

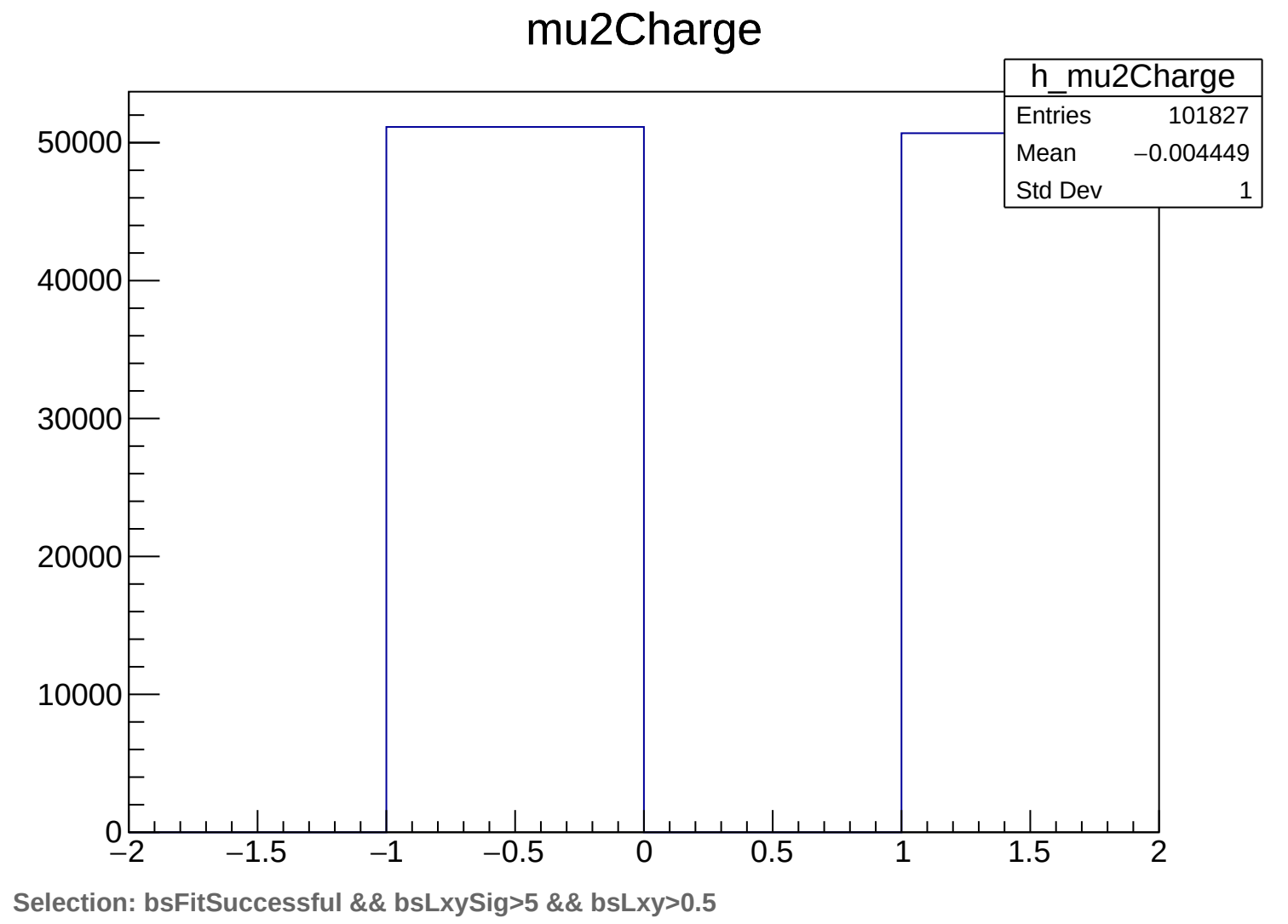

mu1IsTight
h_mu1IsTight
Entries 101827
Mean 0.6274
Std Dev 0.4835
65000
60000
55000
50000
45000
40000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


mu2Pt
h_mu2Pt
Entries 101827
Mean 7.38
Std Dev 4.871
25000
20000
15000
10000
5000
0
0
20
40
60
80
100
120
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

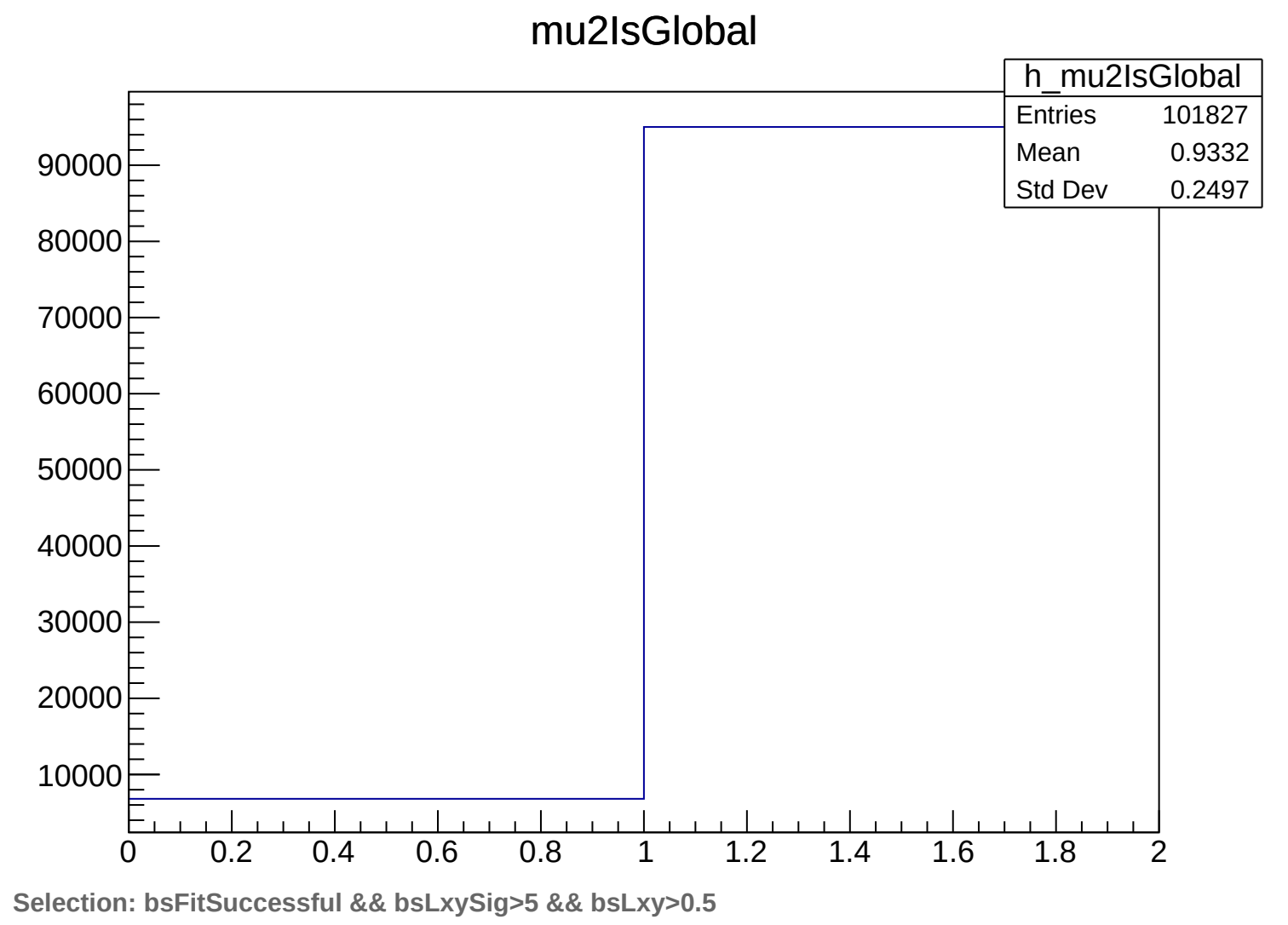

mu2Eta
h_mu2Eta
Entries 101827
Mean −0.01893
Std Dev 1.184
1800
1600
1400
1200
1000
800
600
400
200
0
−3
−2
−1
0
1
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

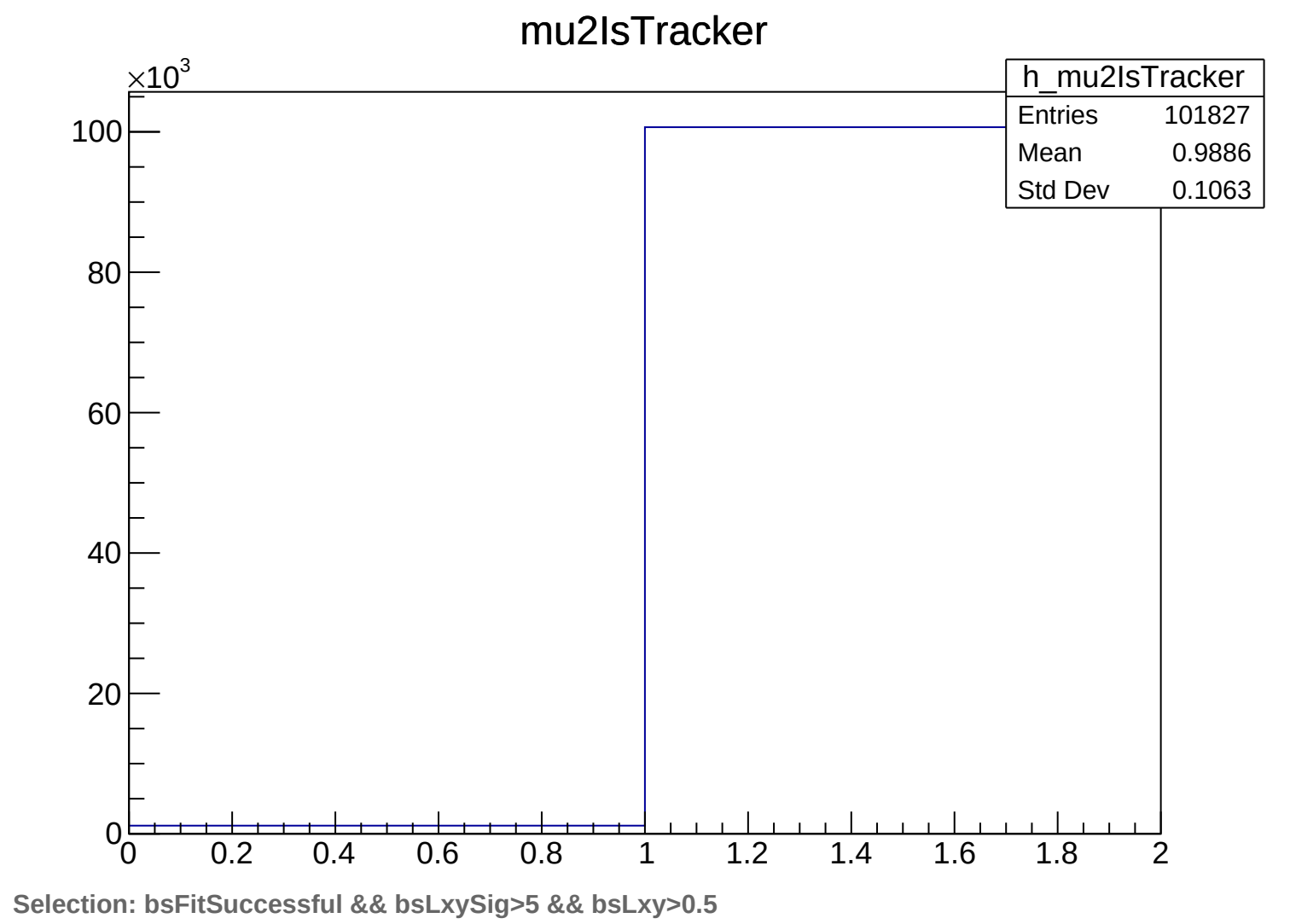


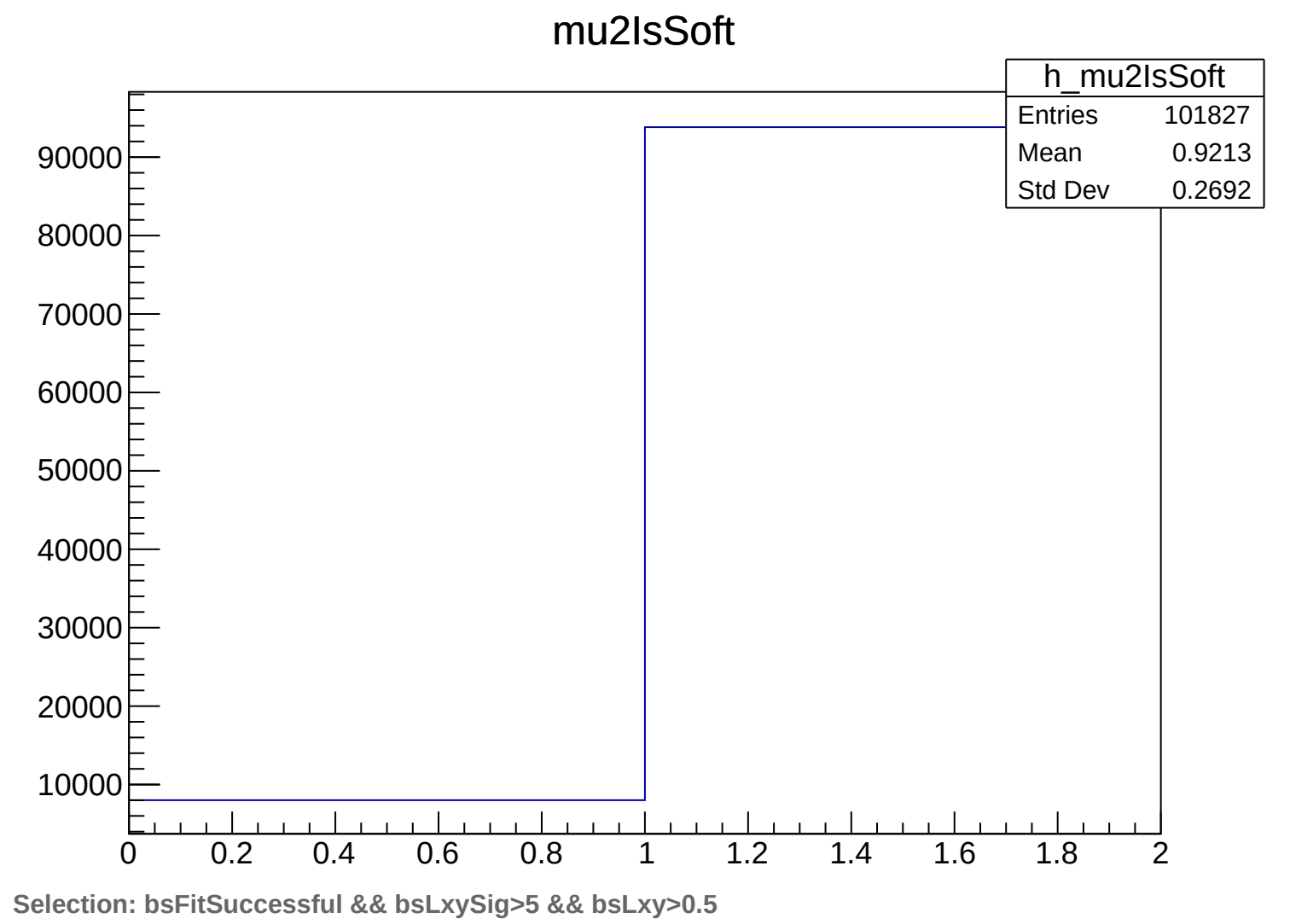
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

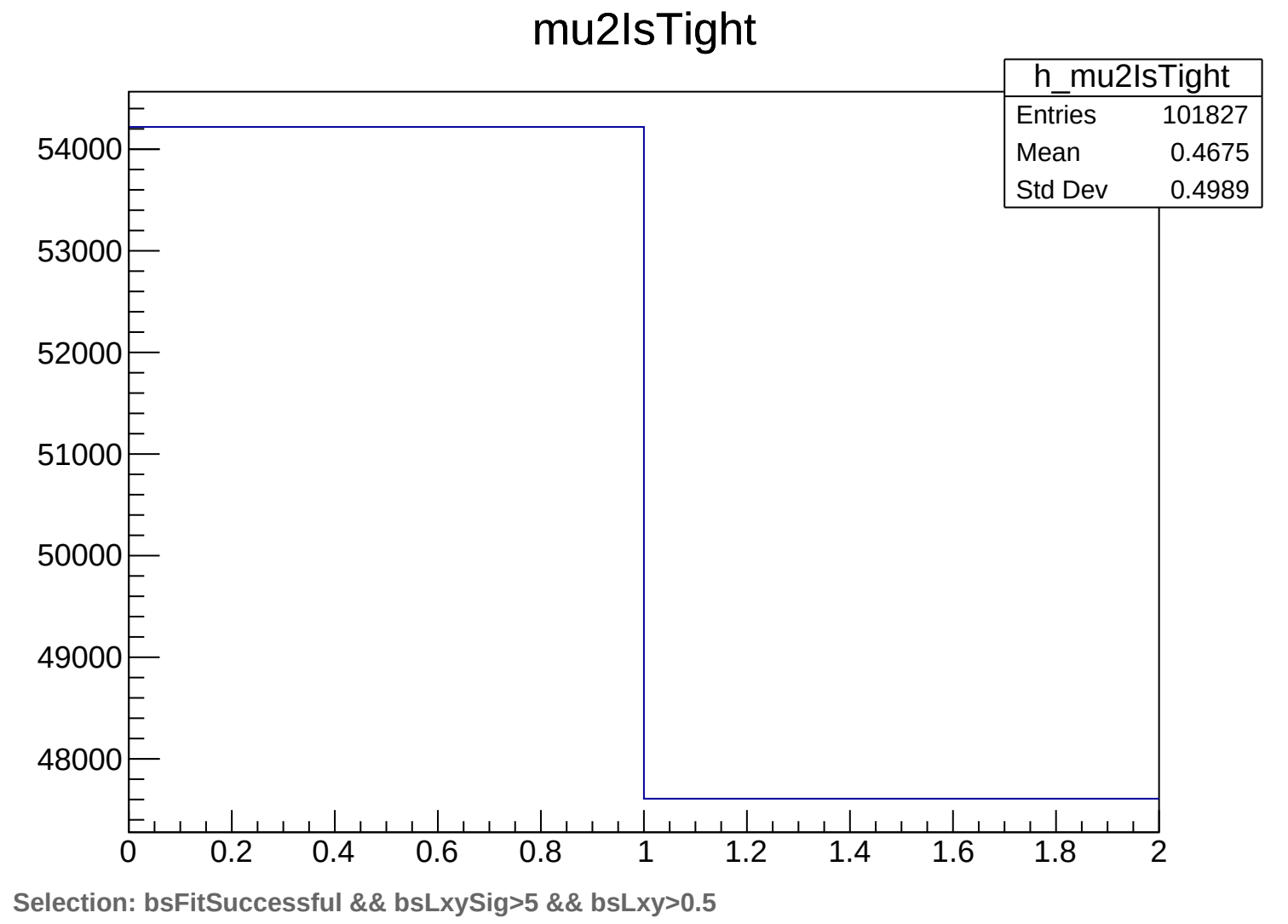


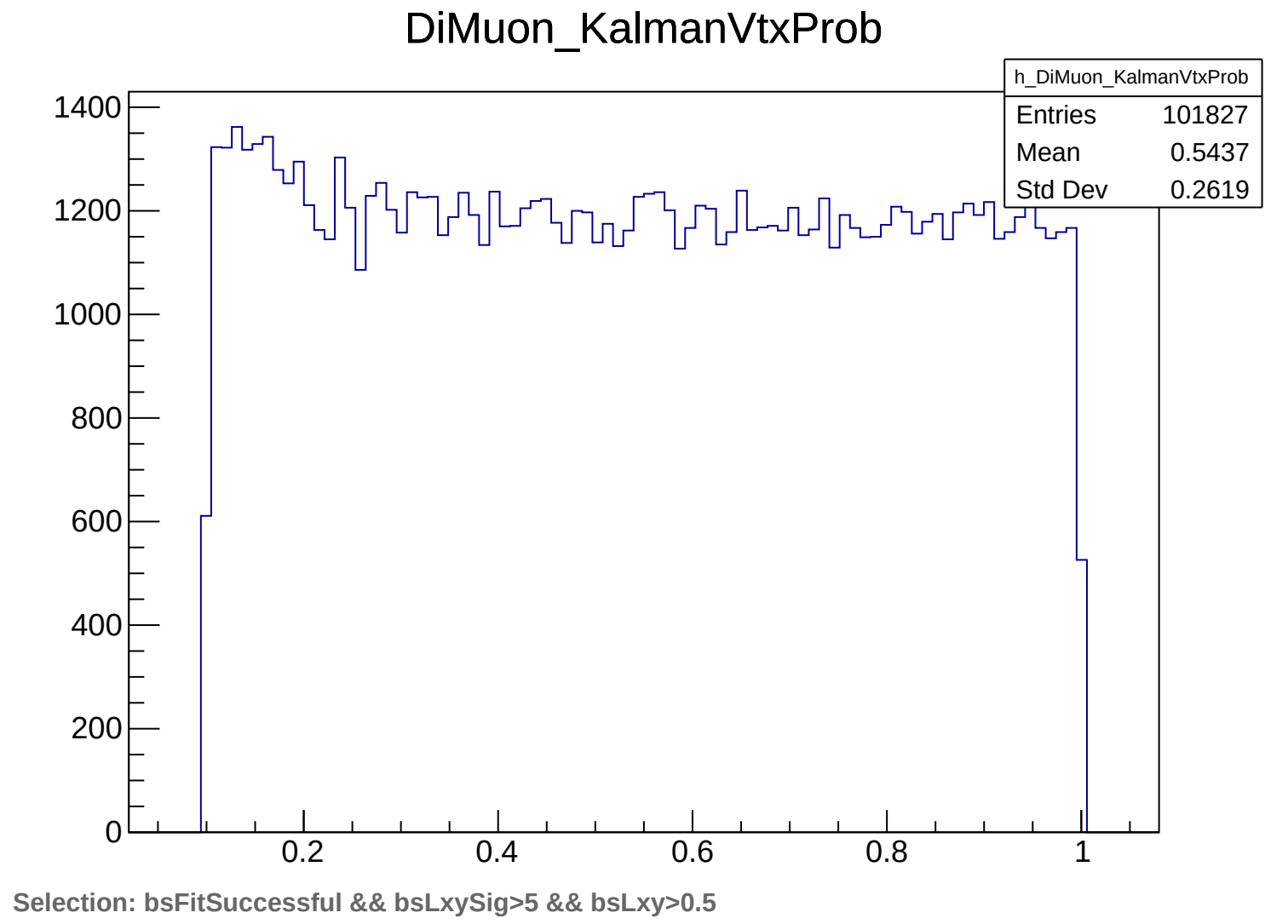
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

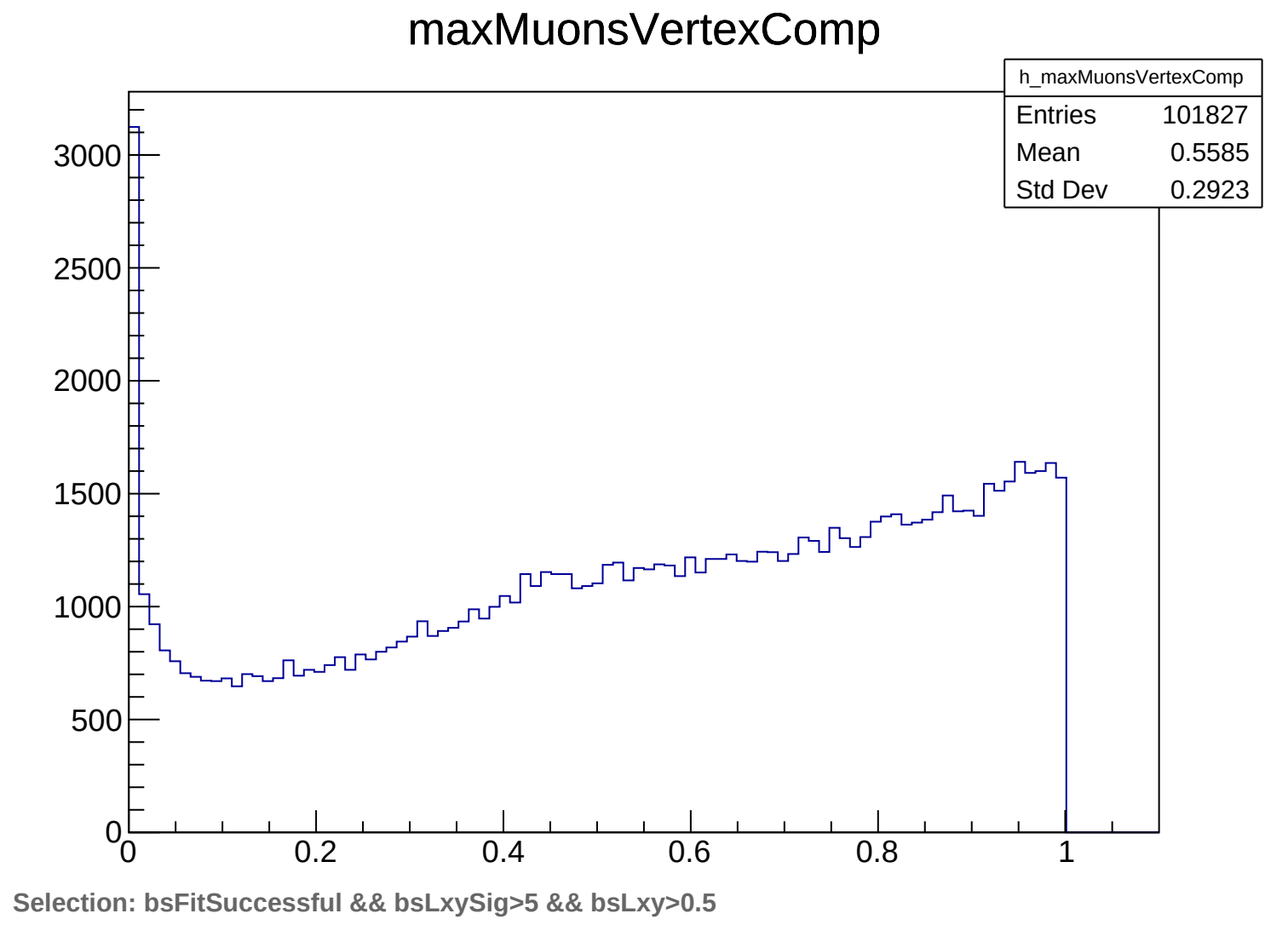

mu2IsGlobal
h_mu2IsGlobal
Entries 101827
Mean 0.9332
Std Dev 0.2497
90000
80000
70000
60000
50000
40000
30000
20000
10000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

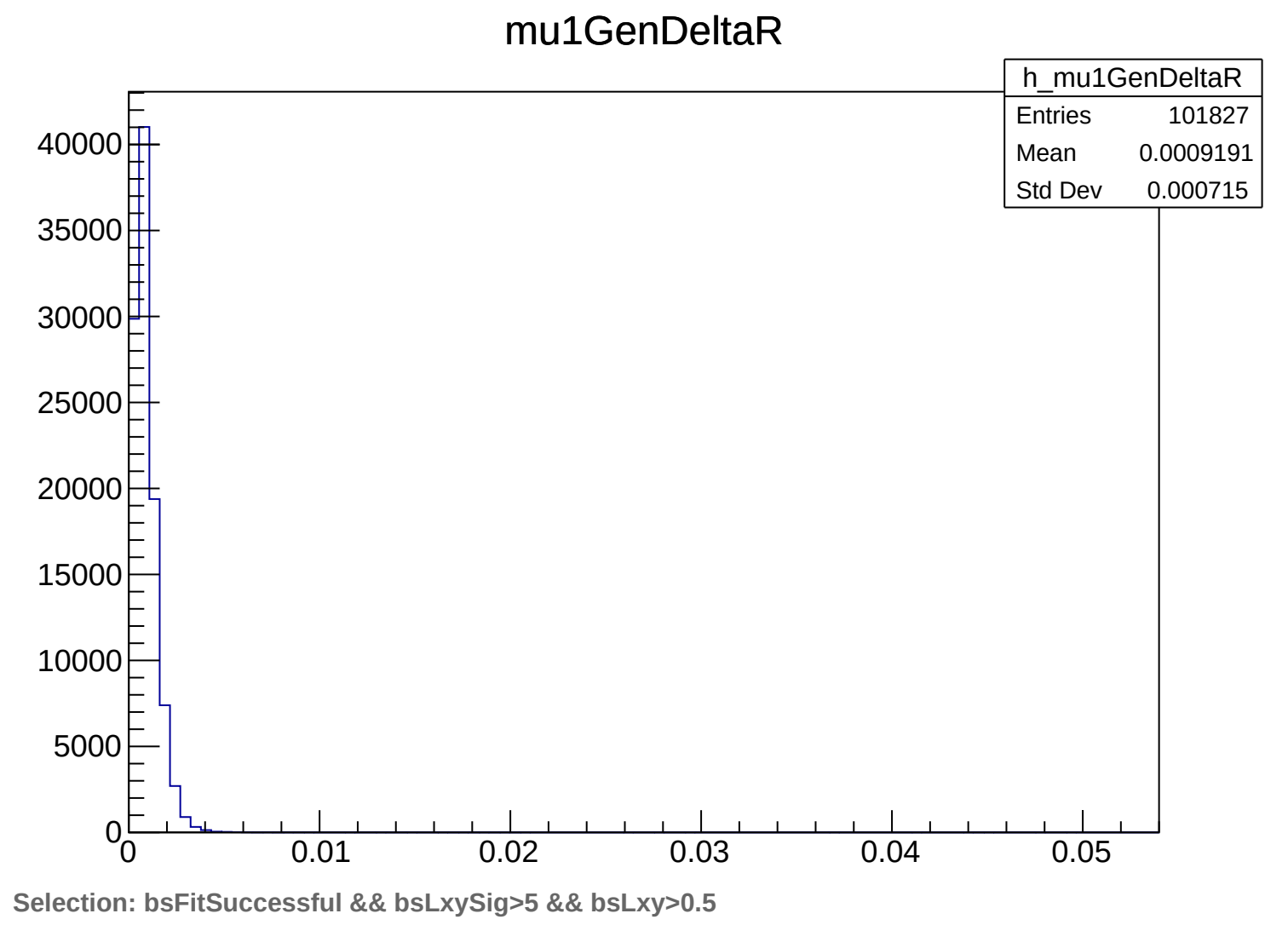

mu2IsTracker
h_mu2IsTracker
Entries 101827
Mean 0.9886
Std Dev 0.1063
×10^3
100
80
60
40
20
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

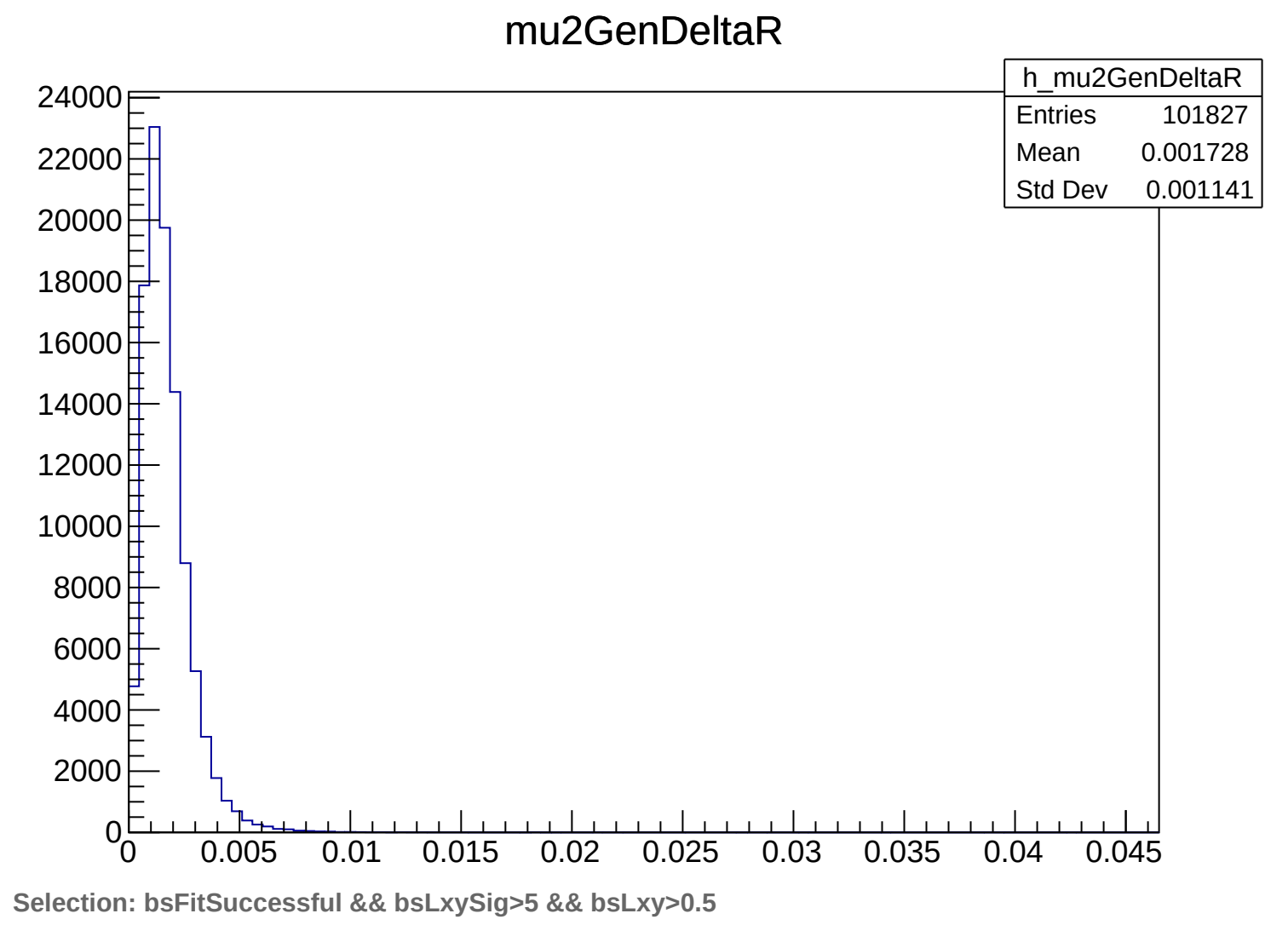

mu2IsSoft
h_mu2IsSoft
Entries 101827
Mean 0.9213
Std Dev 0.2692
90000
80000
70000
60000
50000
40000
30000
20000
10000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

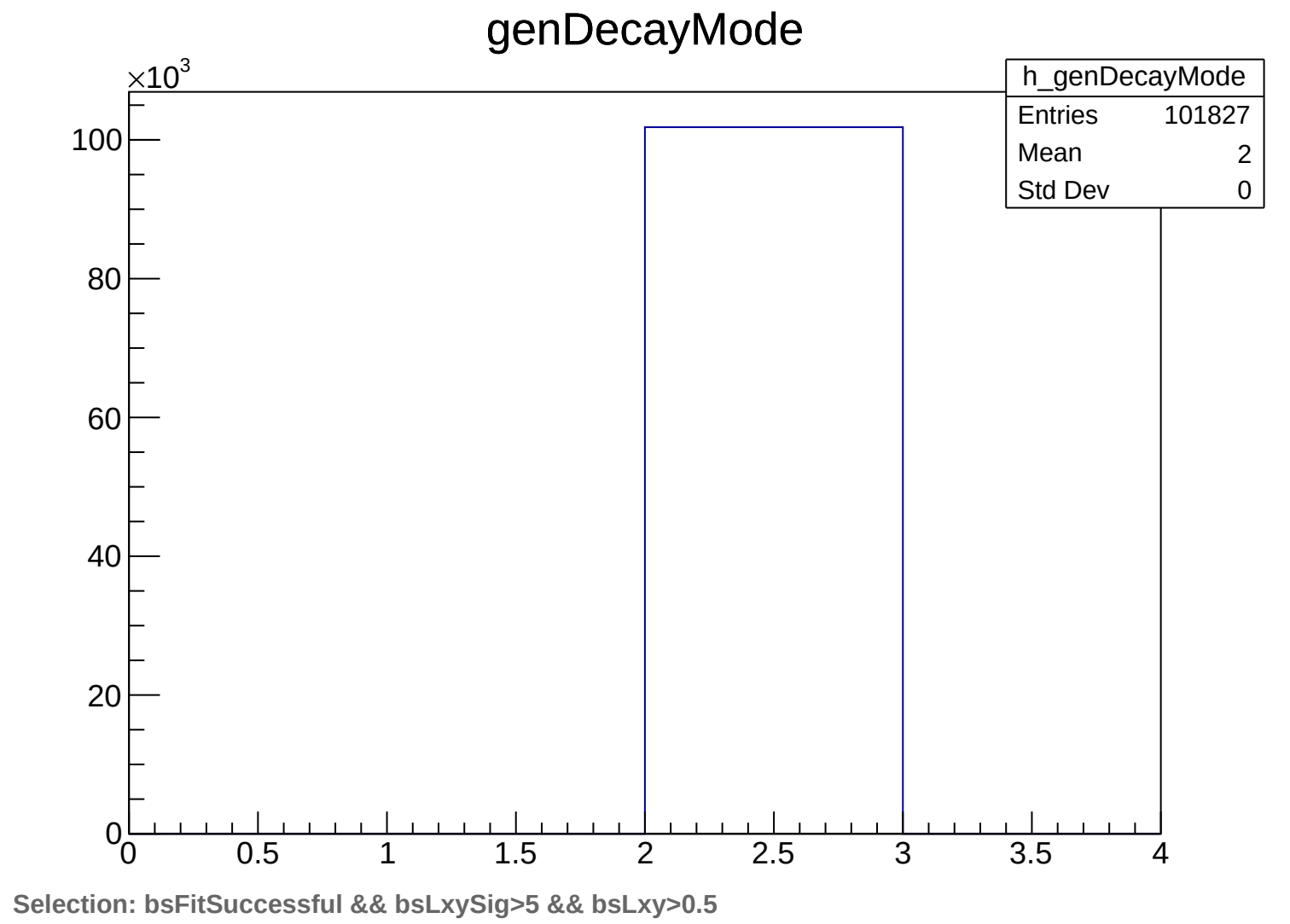

mu2IsTight
h_mu2IsTight
Entries 101827
Mean 0.4675
Std Dev 0.4989
54000
53000
52000
51000
50000
49000
48000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


DiMuon_KalmanVtxProb
h_DiMuon_KalmanVtxProb
Entries 101827
Mean 0.5437
Std Dev 0.2619
1400
1200
1000
800
600
400
200
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

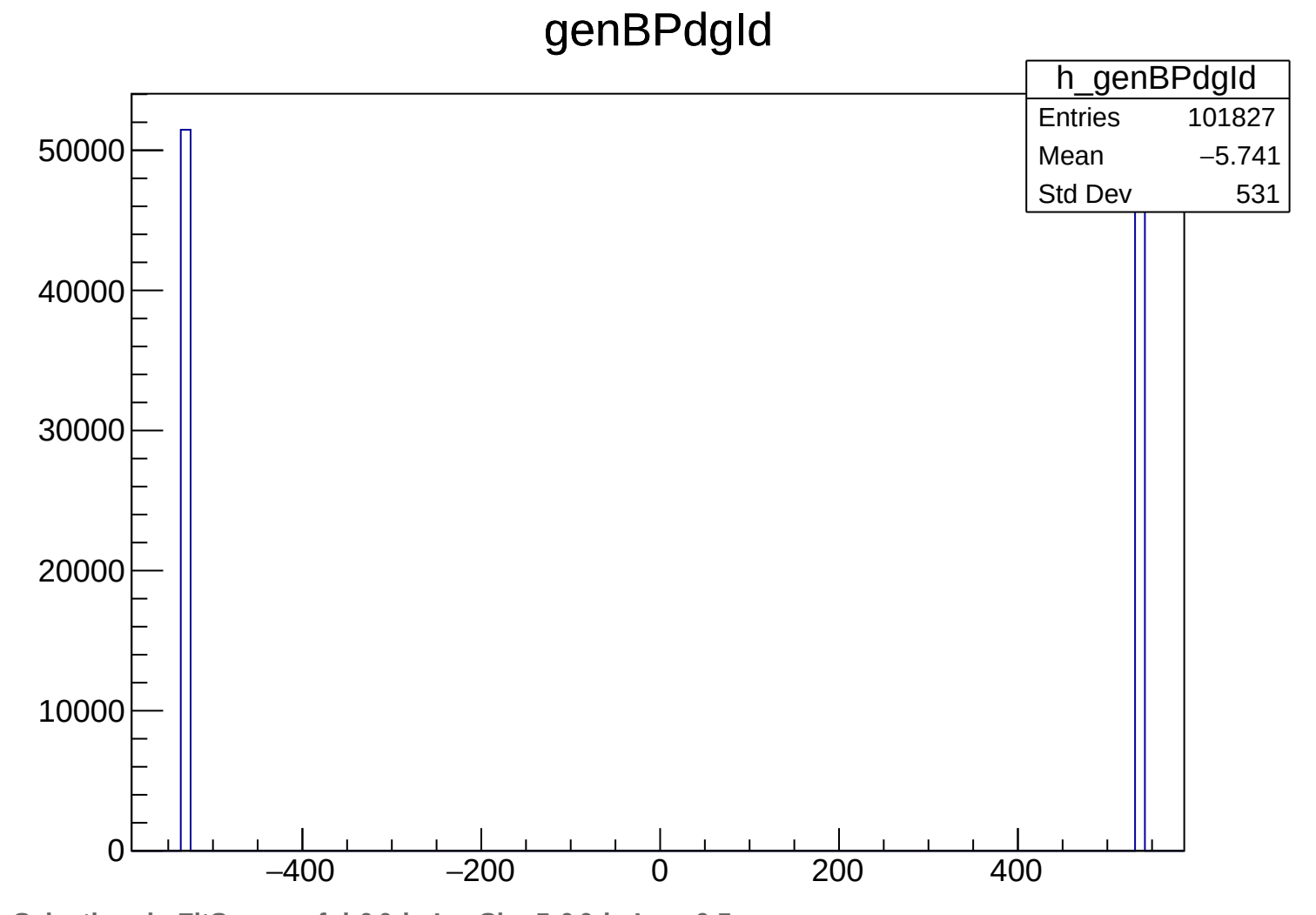

maxMuonsVertexComp
h_maxMuonsVertexComp
Entries 101827
Mean 0.5585
Std Dev 0.2923
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


mu1GenDeltaR
h_mu1GenDeltaR
Entries 101827
Mean 0.0009191
Std Dev 0.000715
40000
35000
30000
25000
20000
15000
10000
5000
0
0
0.01
0.02
0.03
0.04
0.05
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

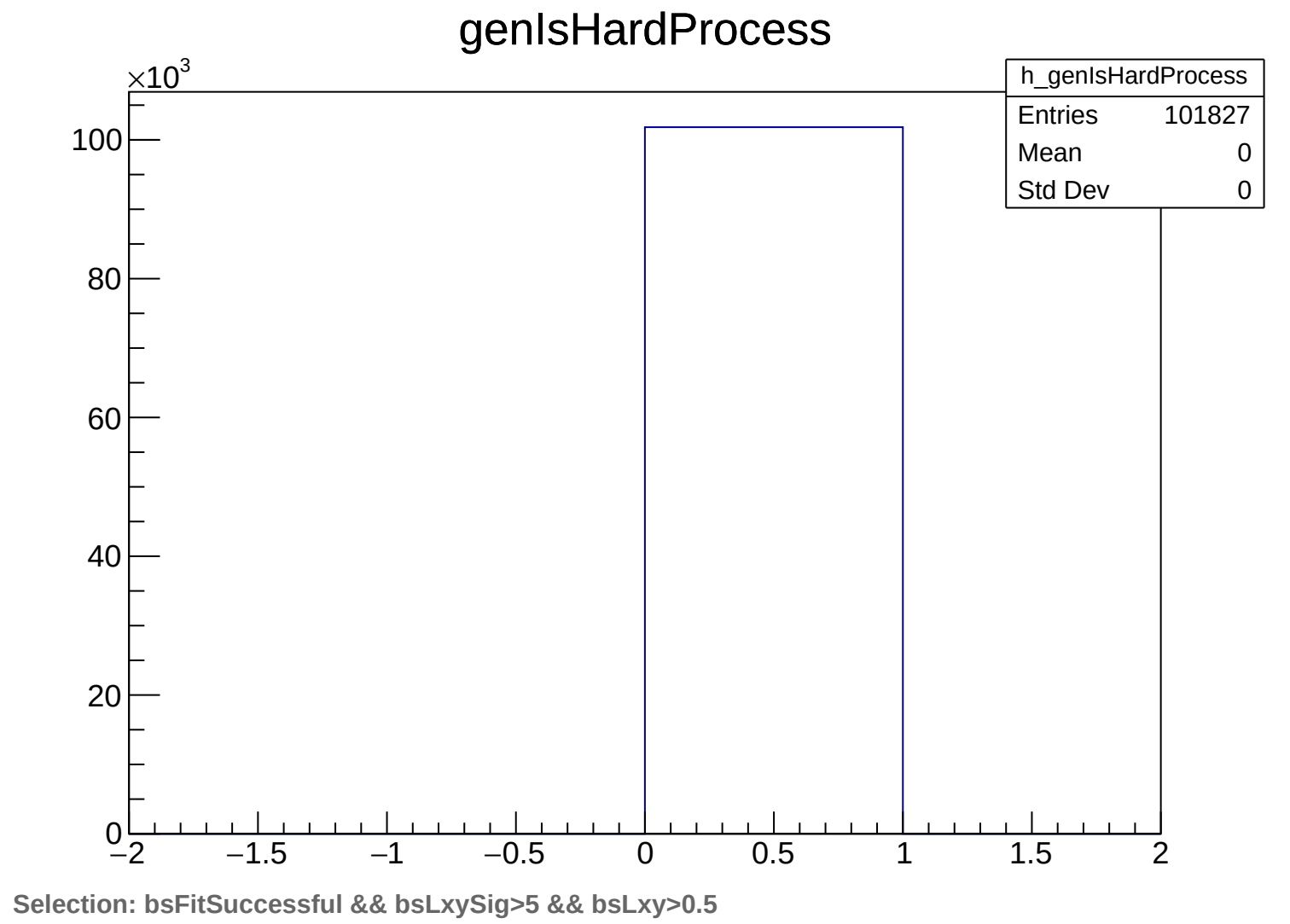

mu2GenDeltaR
h_mu2GenDeltaR
Entries 101827
Mean 0.001728
Std Dev 0.001141
24000
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
0.005
0.01
0.015
0.02
0.025
0.03
0.035
0.04
0.045
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5



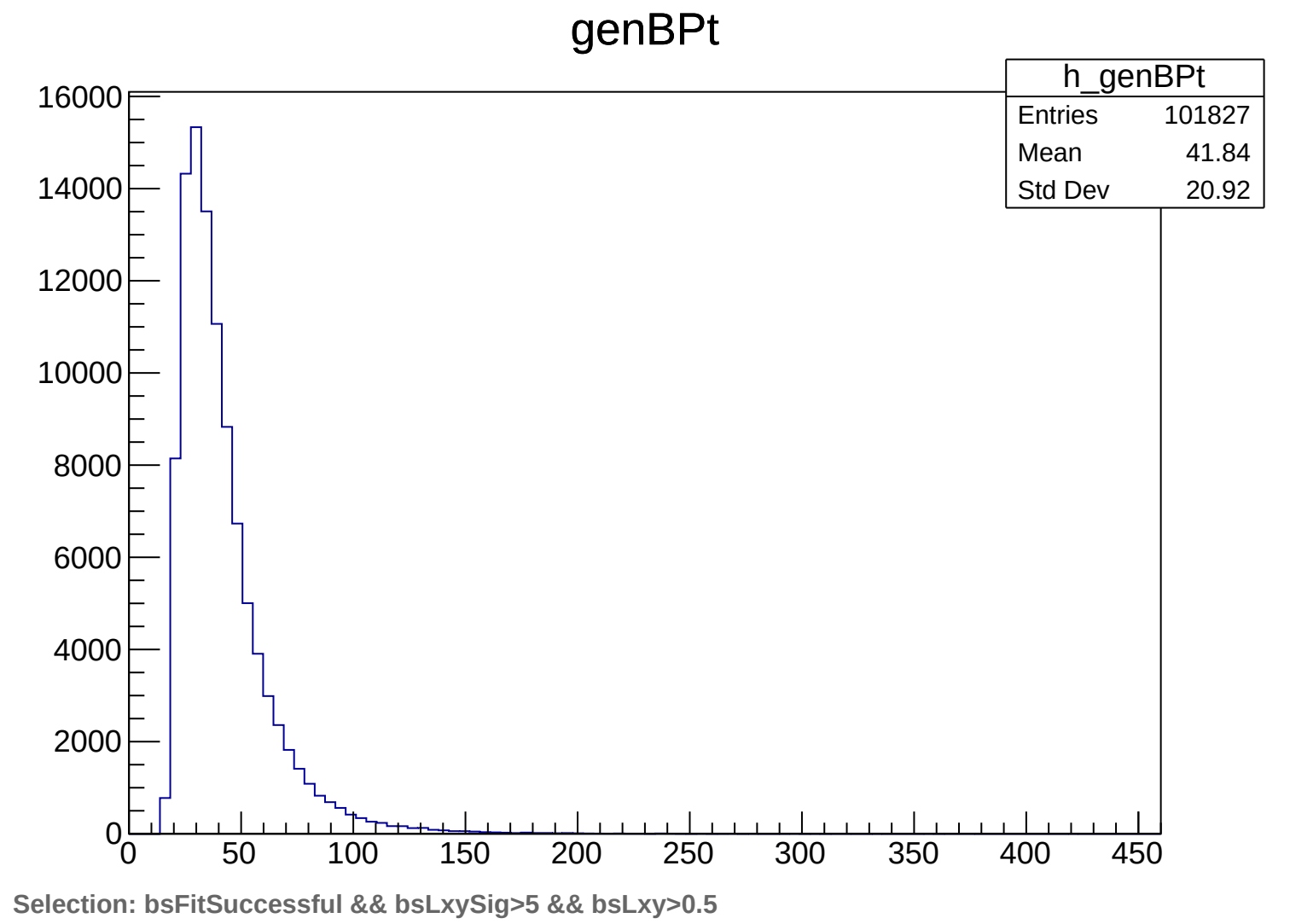
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

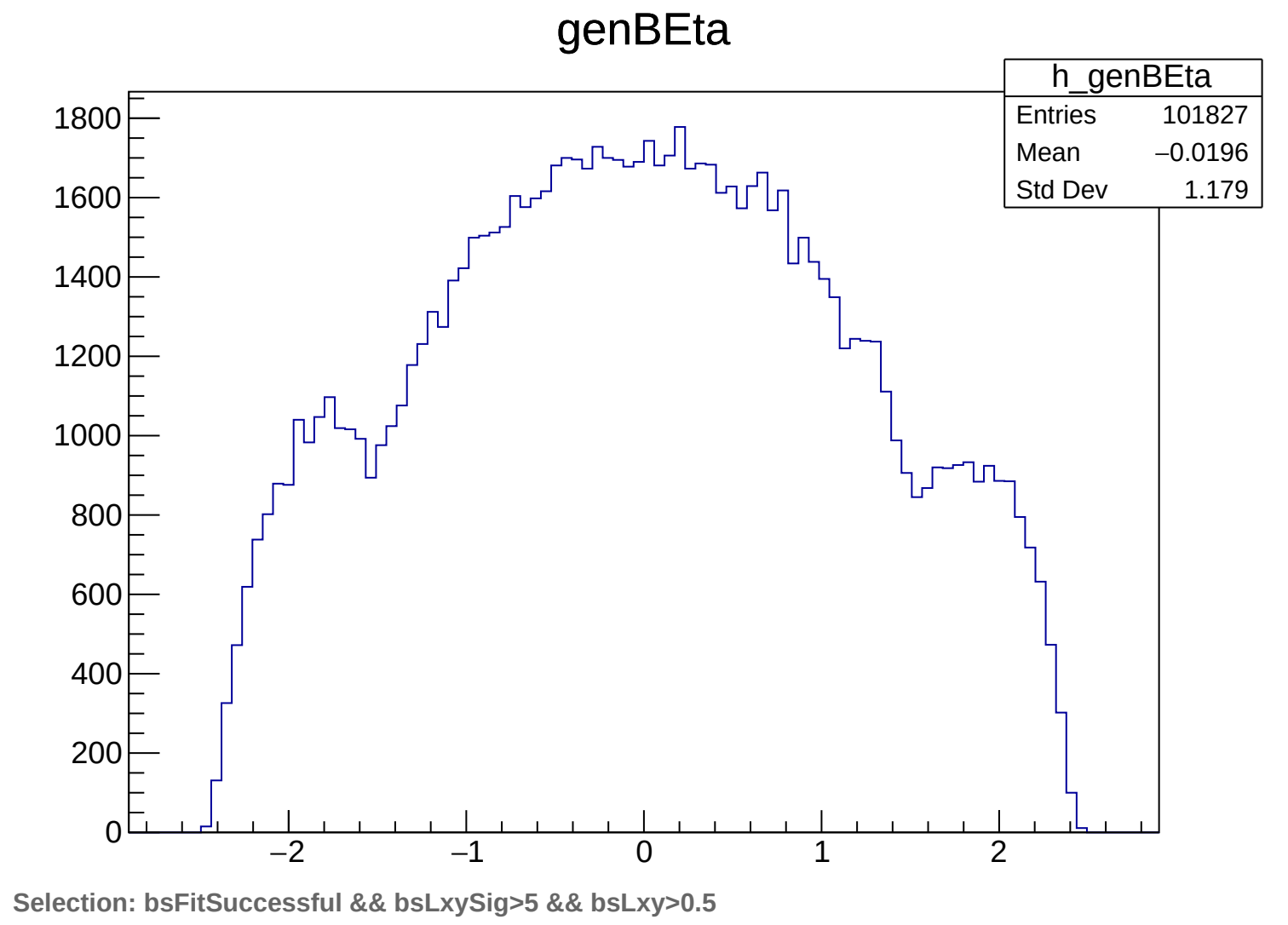


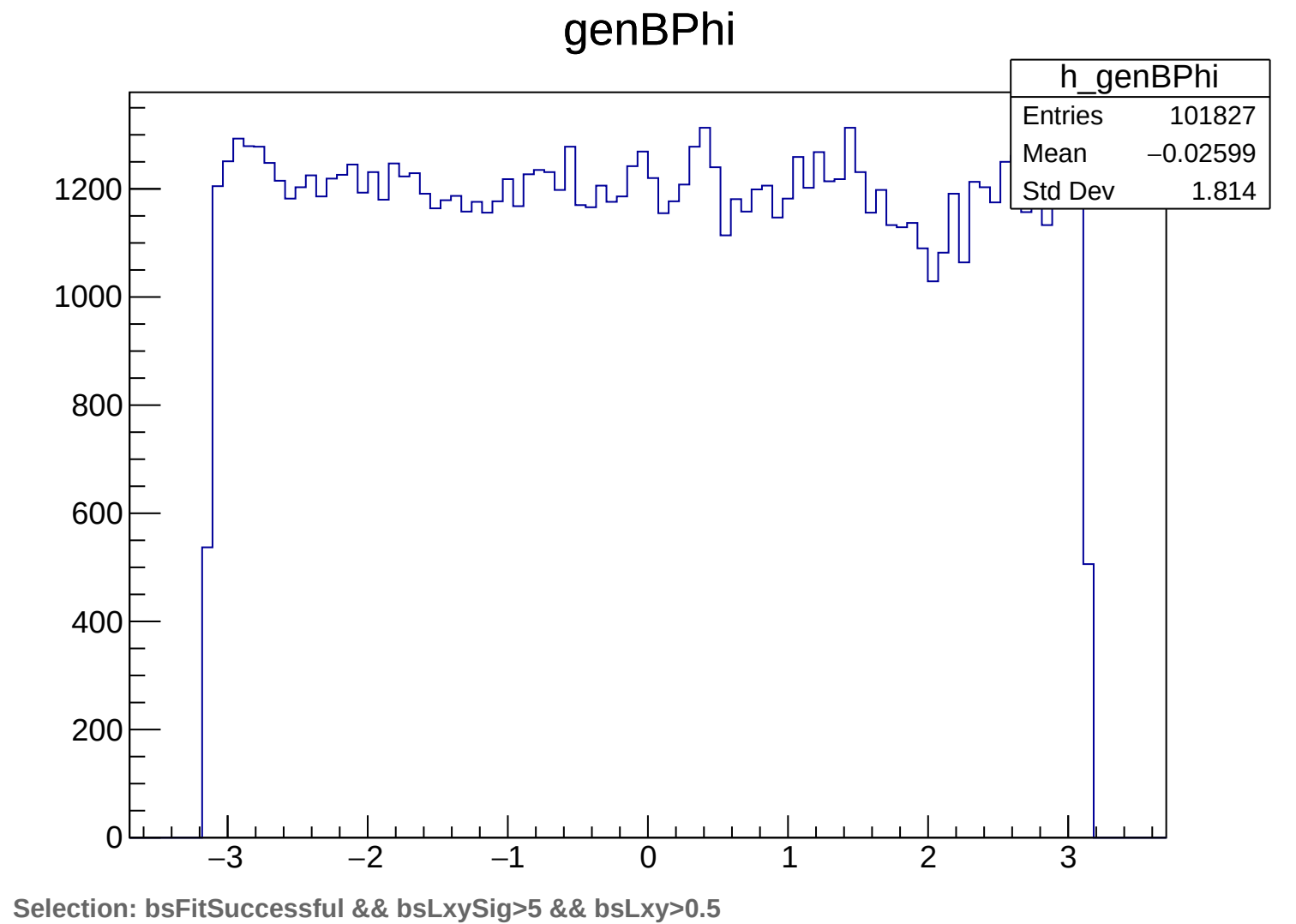
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

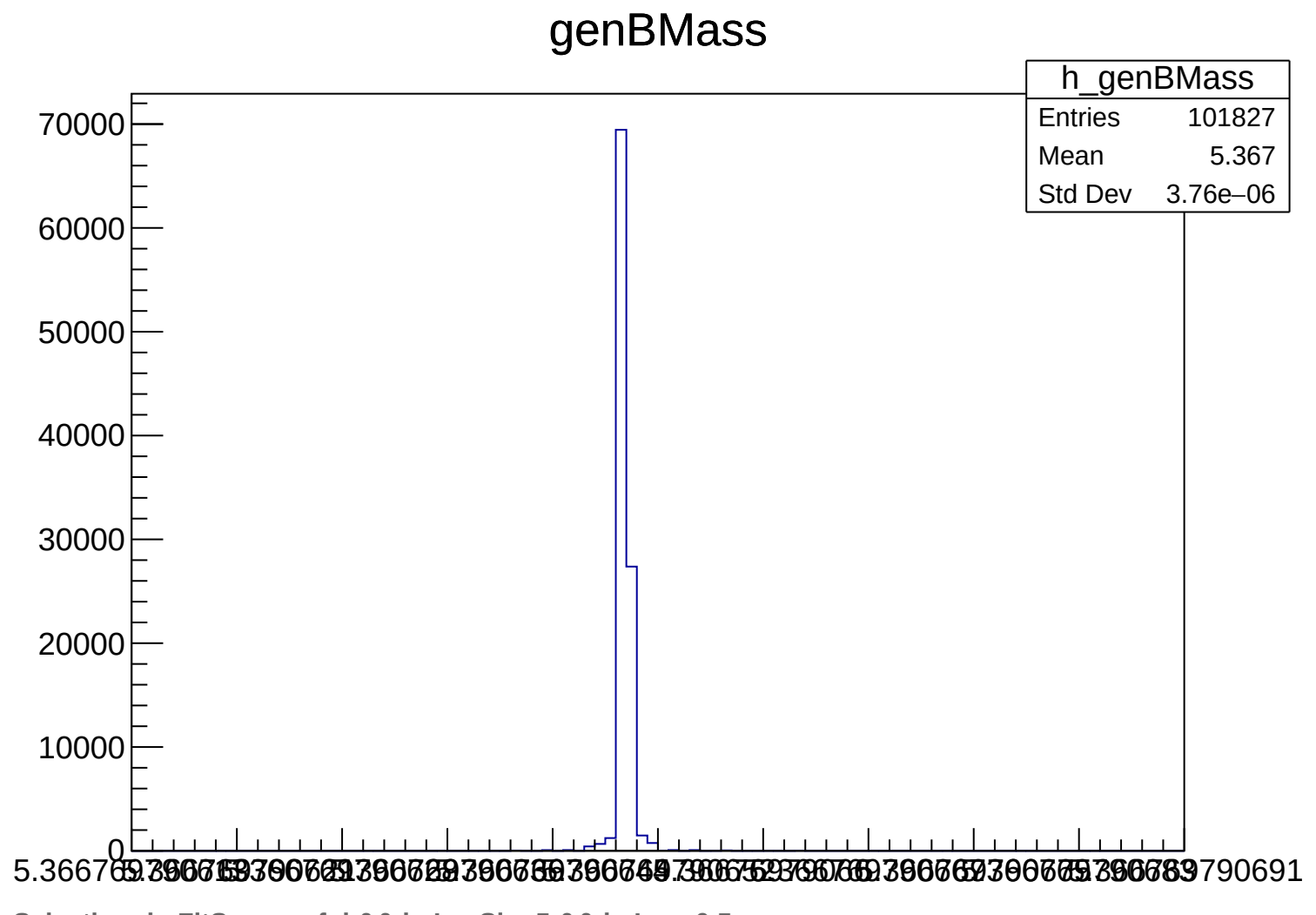


Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

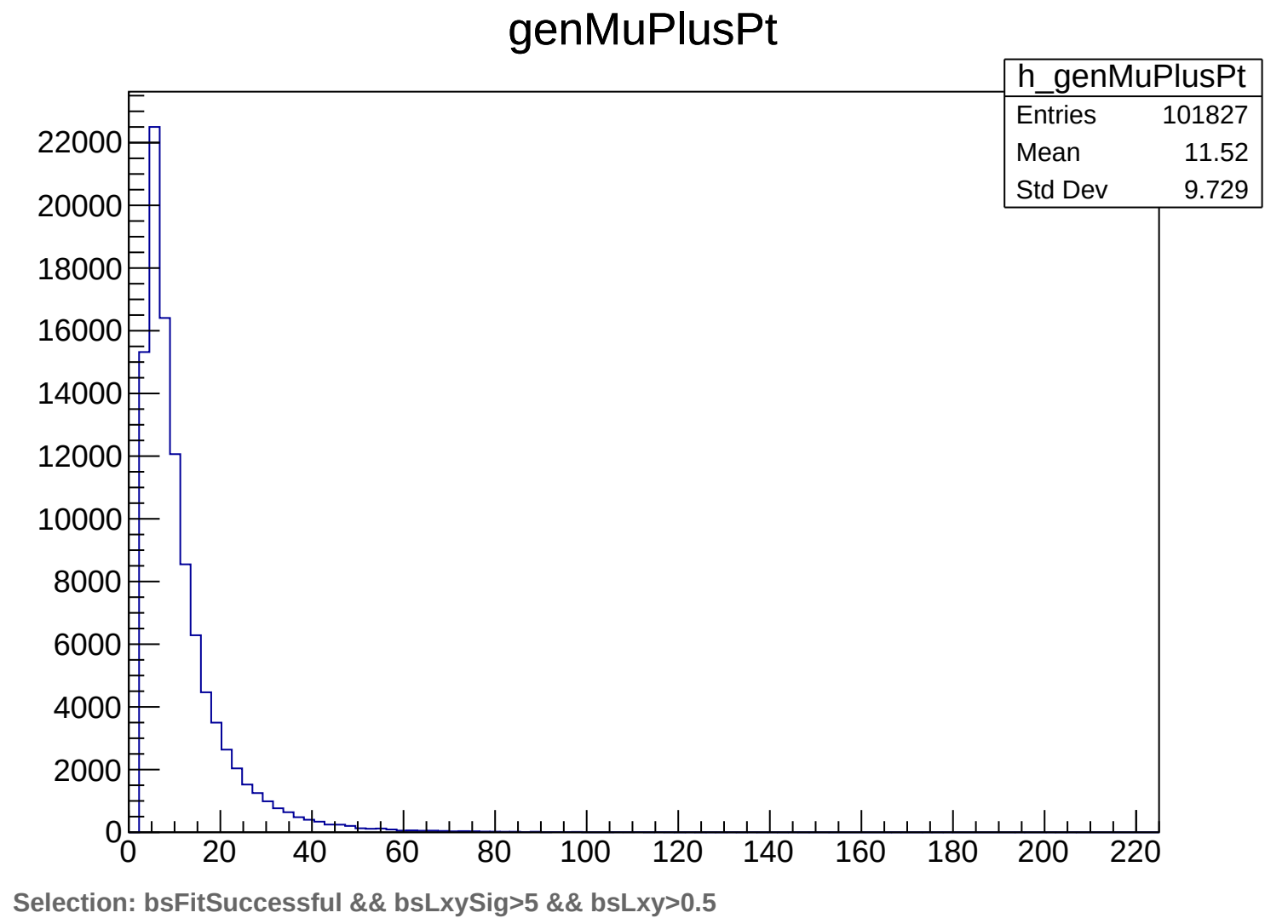

genBPt
h_genBPt
Entries 101827
Mean 41.84
Std Dev 20.92
16000
14000
12000
10000
8000
6000
4000
2000
0
0
50
100
150
200
250
300
350
400
450
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

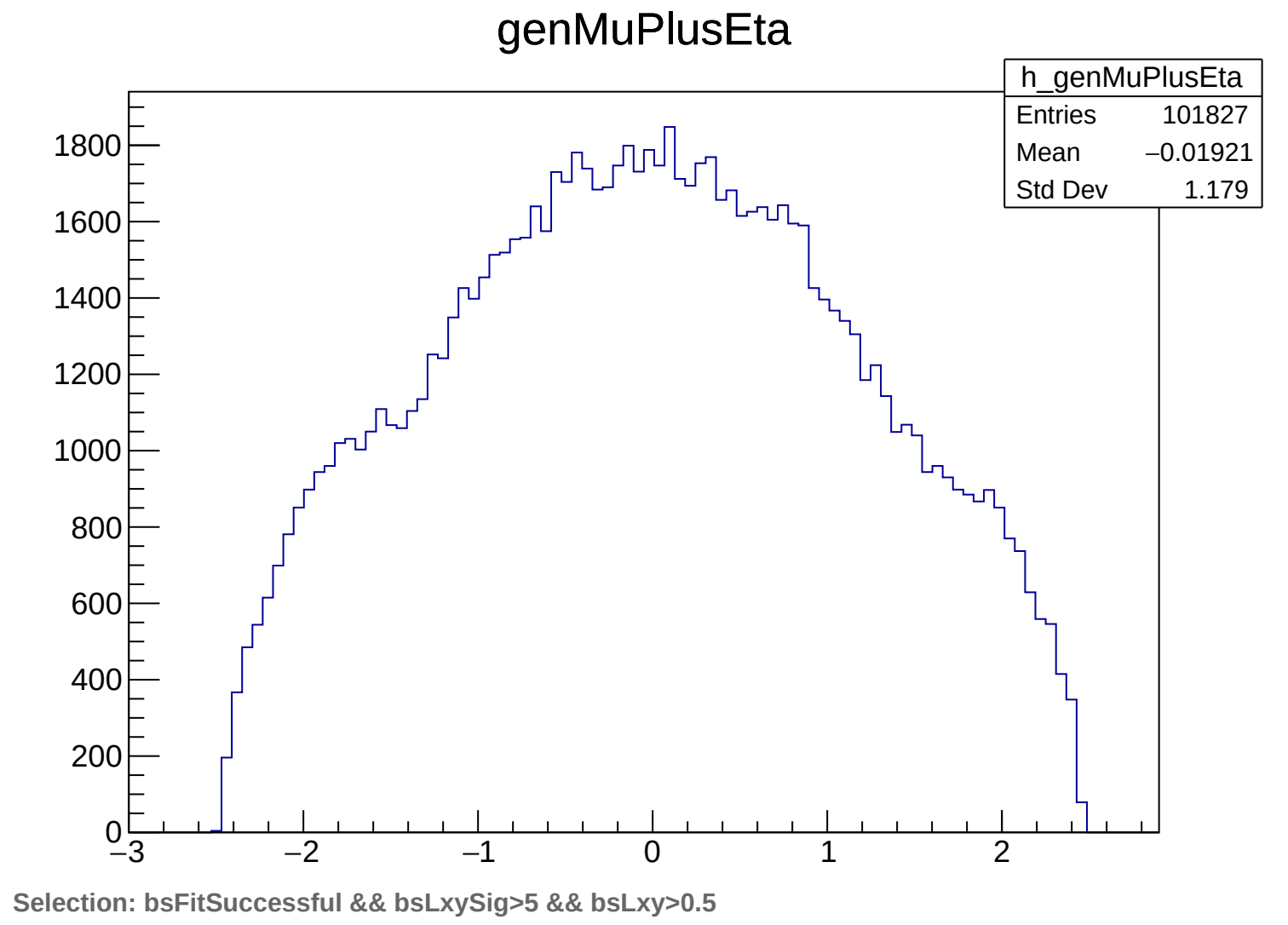

genBEta
h_genBEta
Entries 101827
Mean −0.0196
Std Dev 1.179
1800
1600
1400
1200
1000
800
600
400
200
0
−2
−1
0
1
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

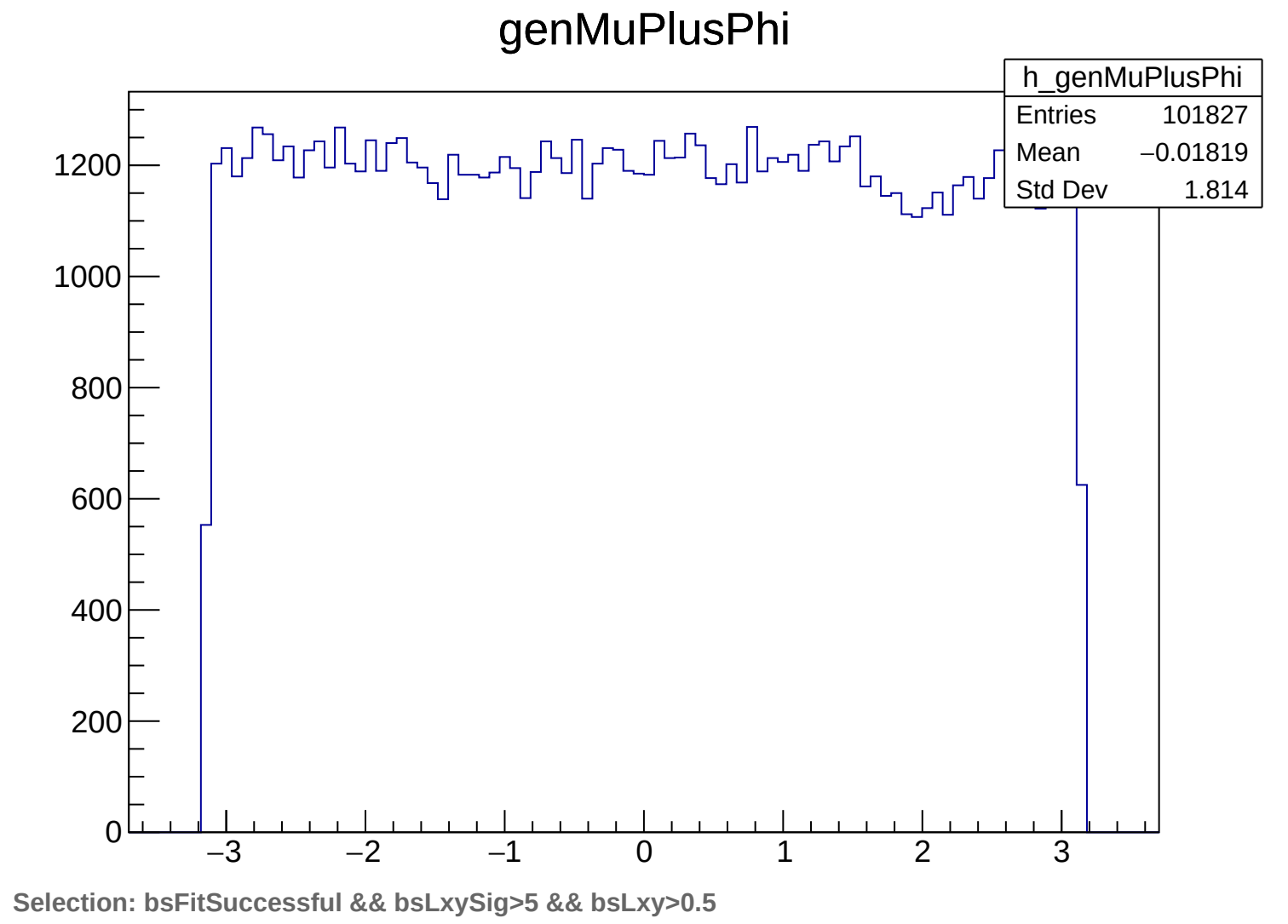

genBPhi
h_genBPhi
Entries 101827
Mean −0.02599
Std Dev 1.814
1200
1000
800
600
400
200
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

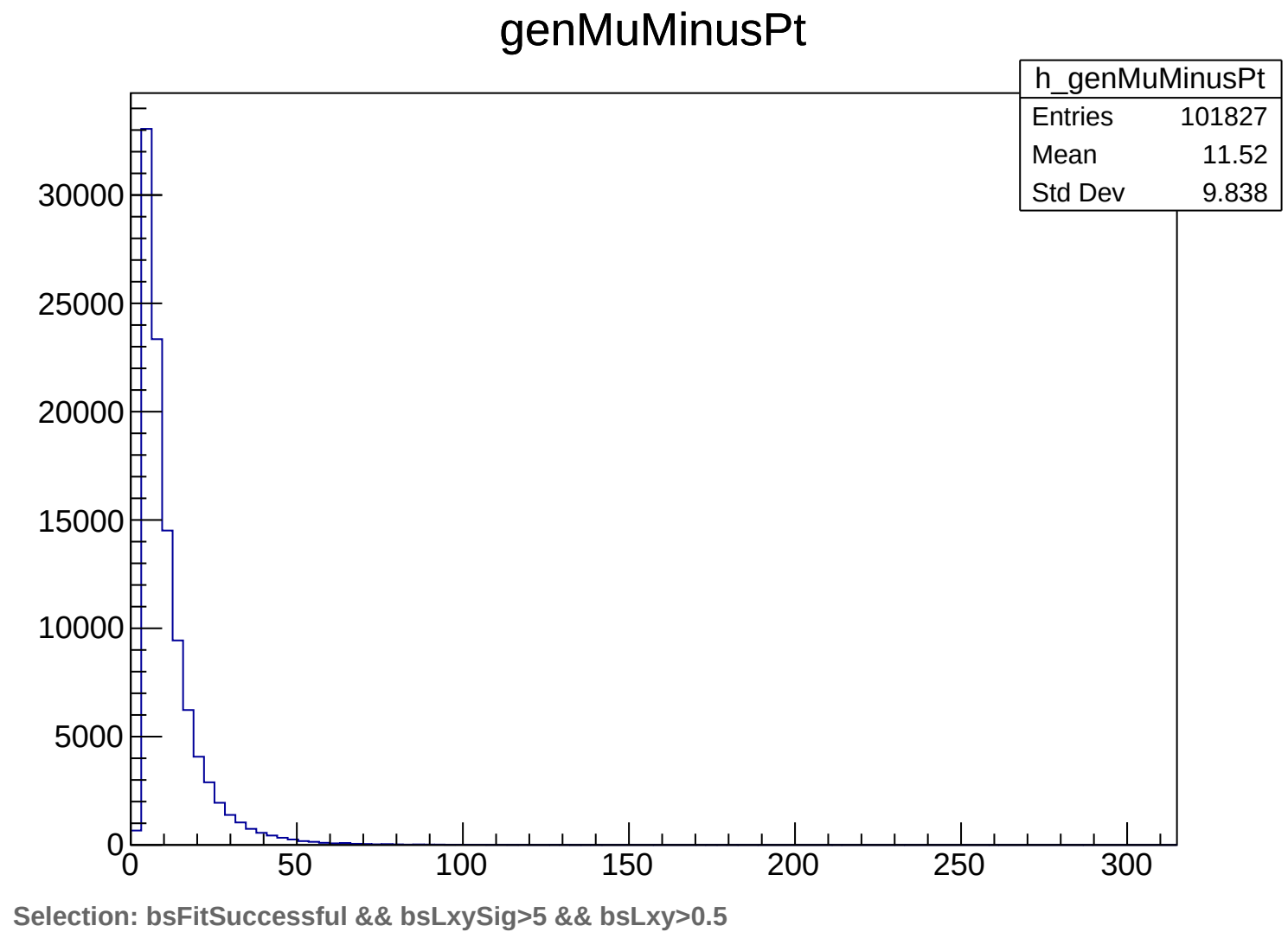


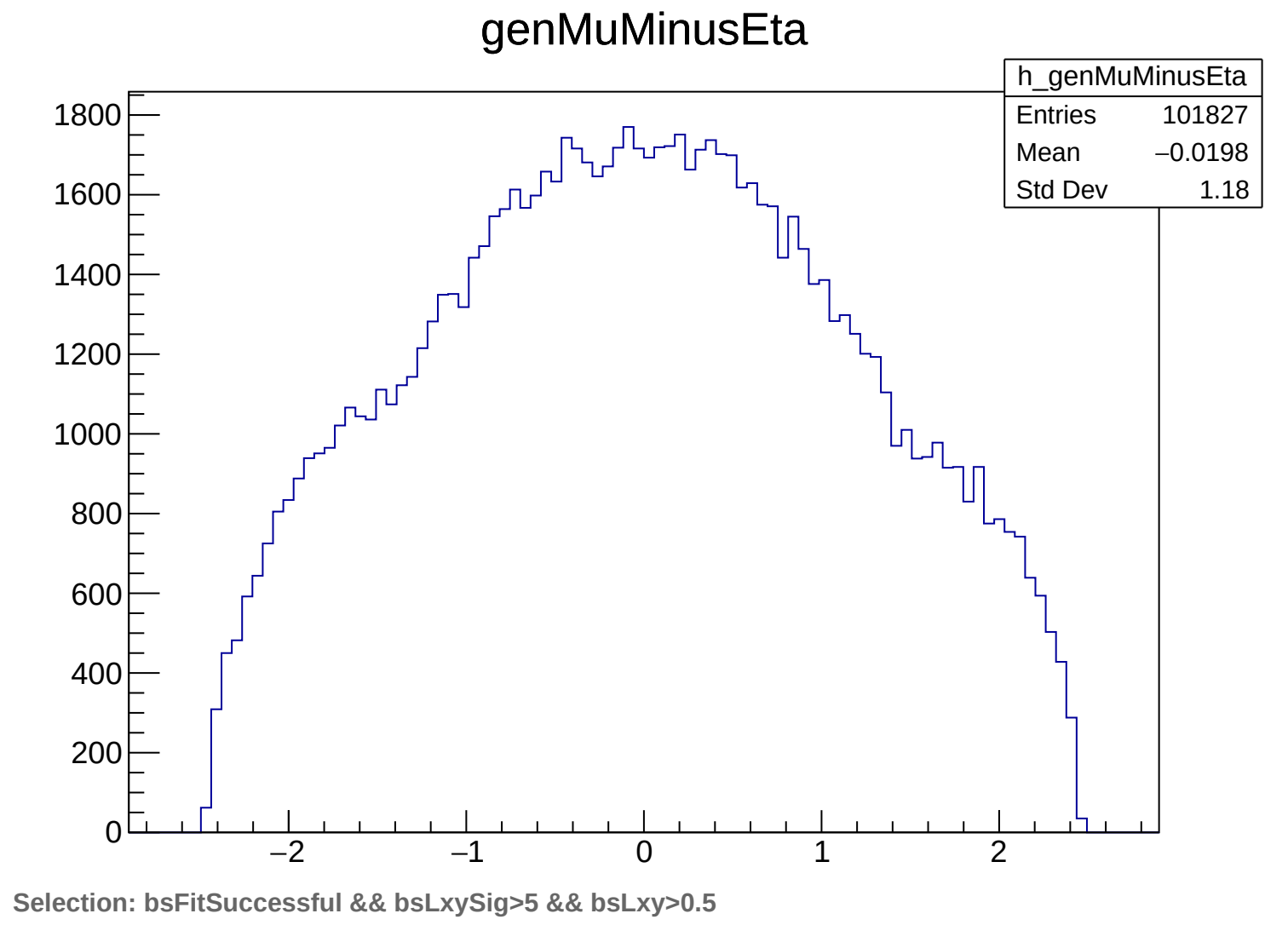
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

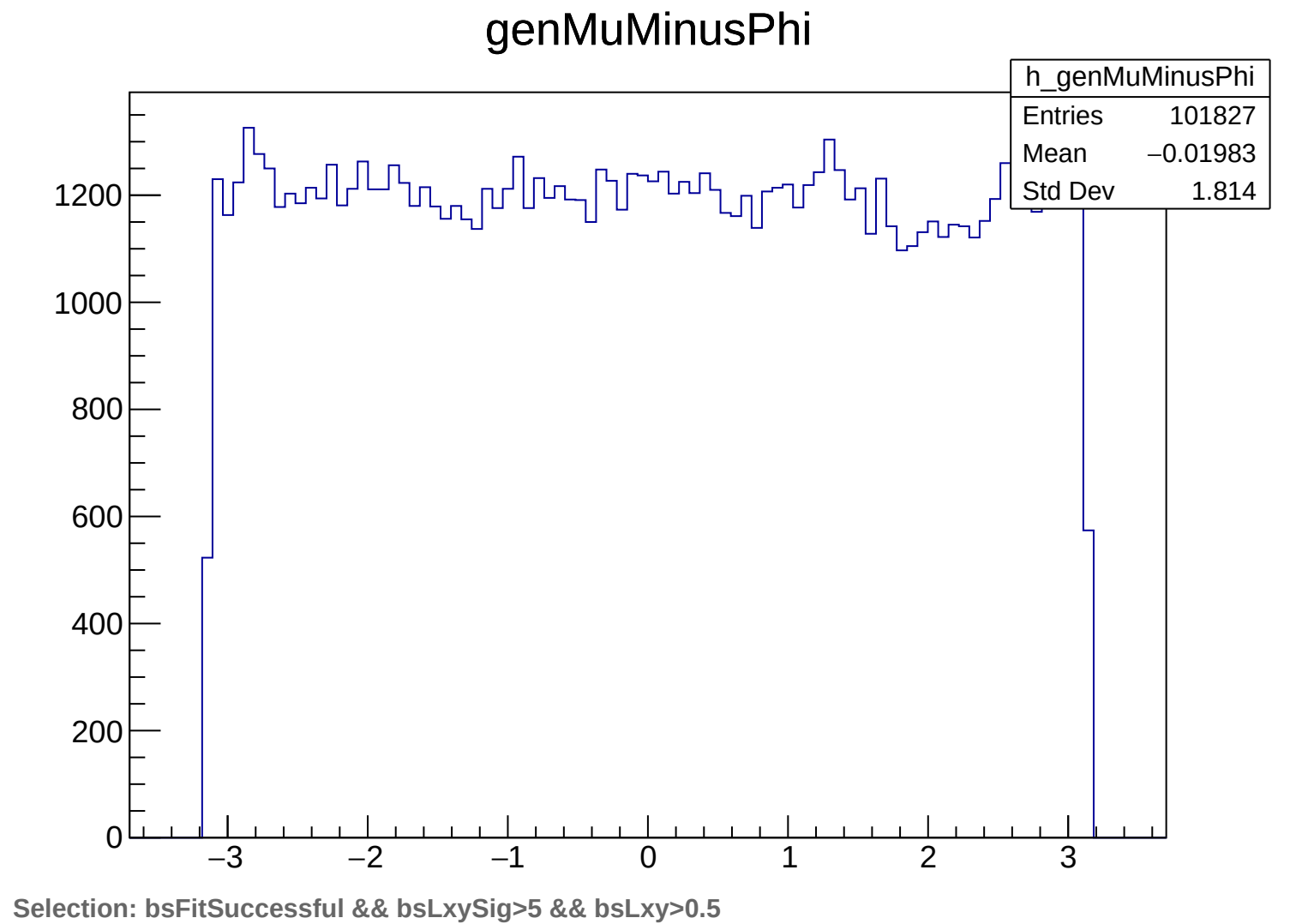

genMuPlusPt
h_genMuPlusPt
Entries 101827
Mean 11.52
Std Dev 9.729
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
20
40
60
80
100
120
140
160
180
200
220
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


genMuPlusEta
h_genMuPlusEta
Entries 101827
Mean −0.01921
Std Dev 1.179
1800
1600
1400
1200
1000
800
600
400
200
0
−3
−2
−1
0
1
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

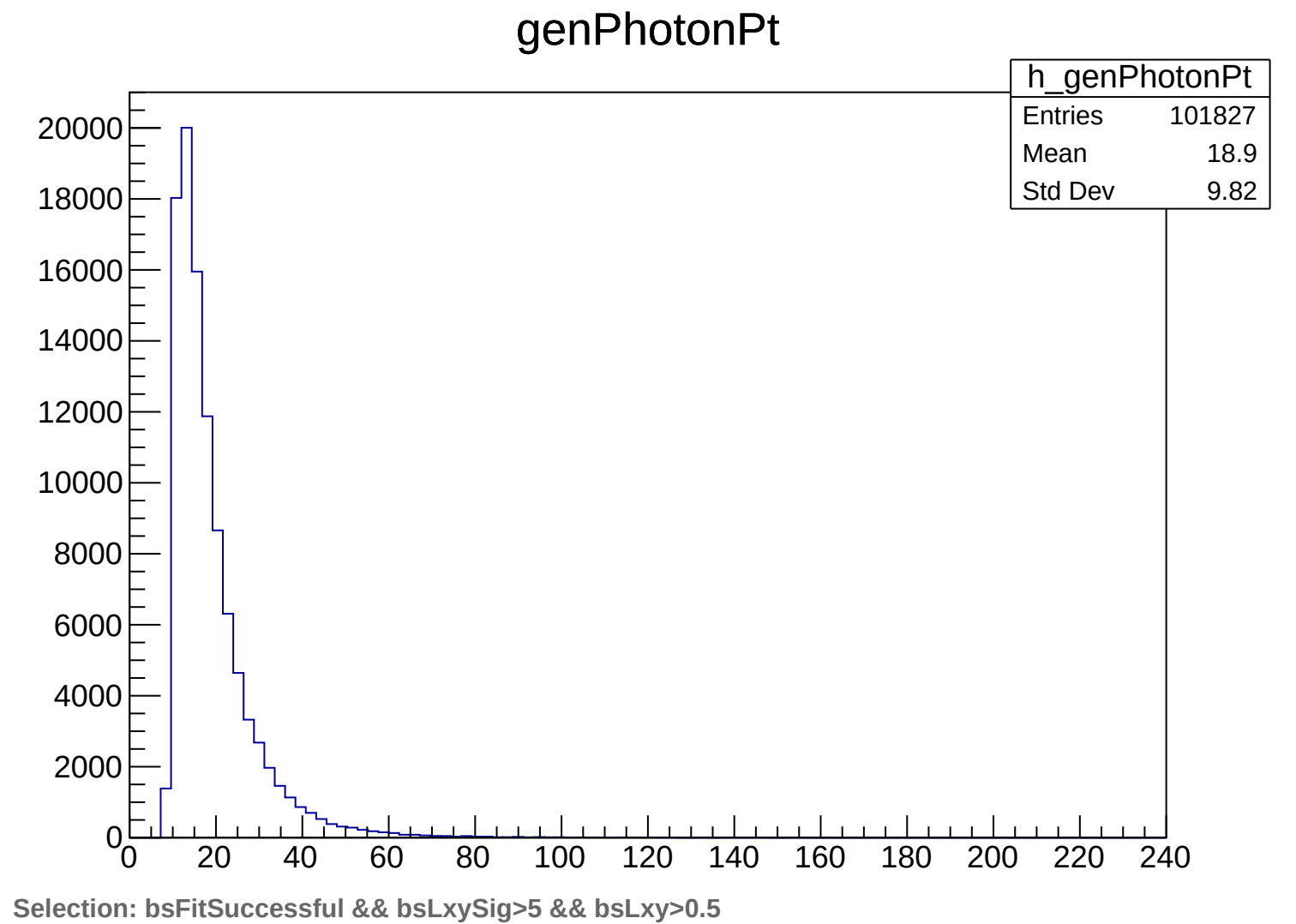

genMuPlusPhi
h_genMuPlusPhi
Entries 101827
Mean −0.01819
Std Dev 1.814
1200
1000
800
600
400
200
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

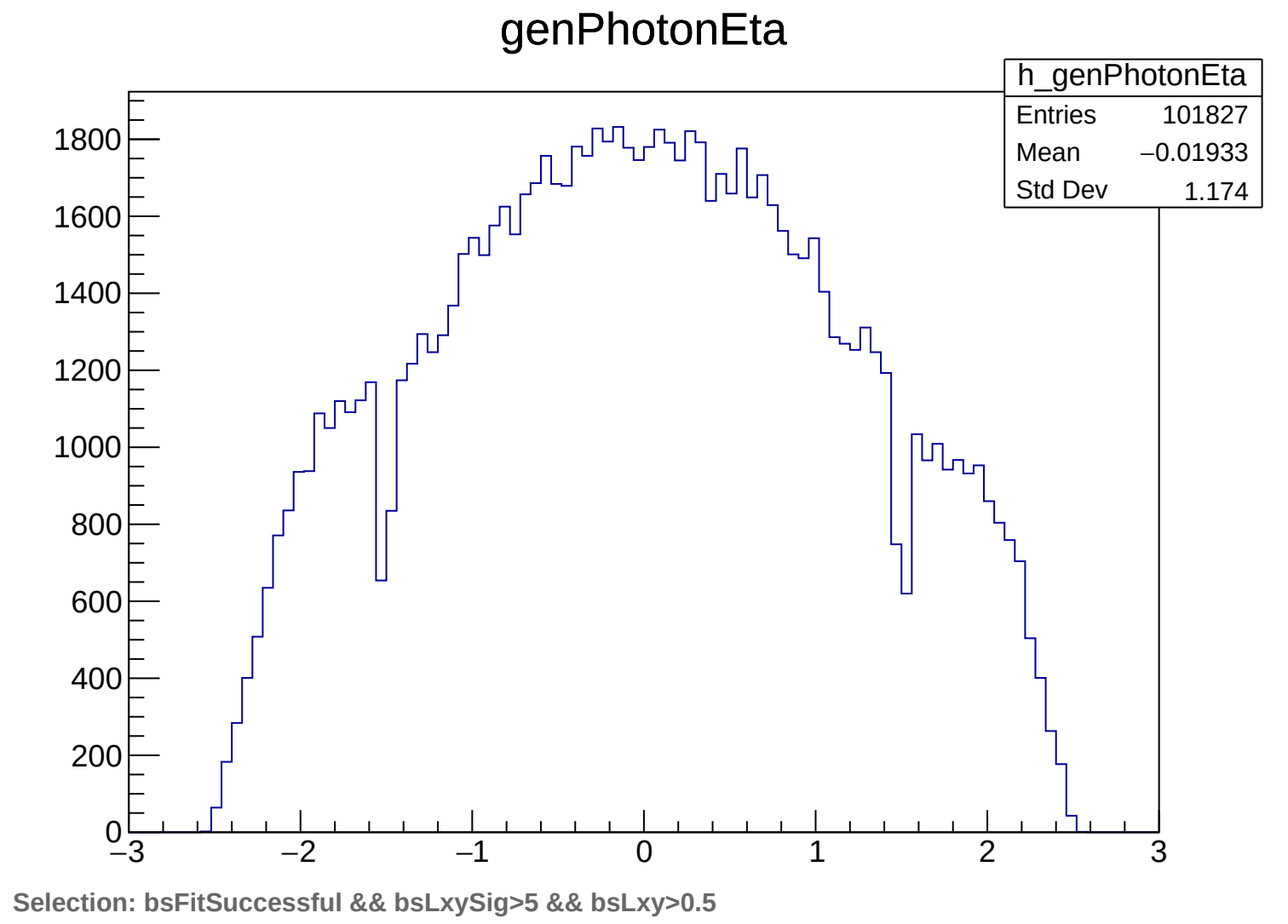

genMuMinusPt
h_genMuMinusPt
Entries 101827
Mean 11.52
Std Dev 9.838
30000
25000
20000
15000
10000
5000
0
0
50
100
150
200
250
300
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

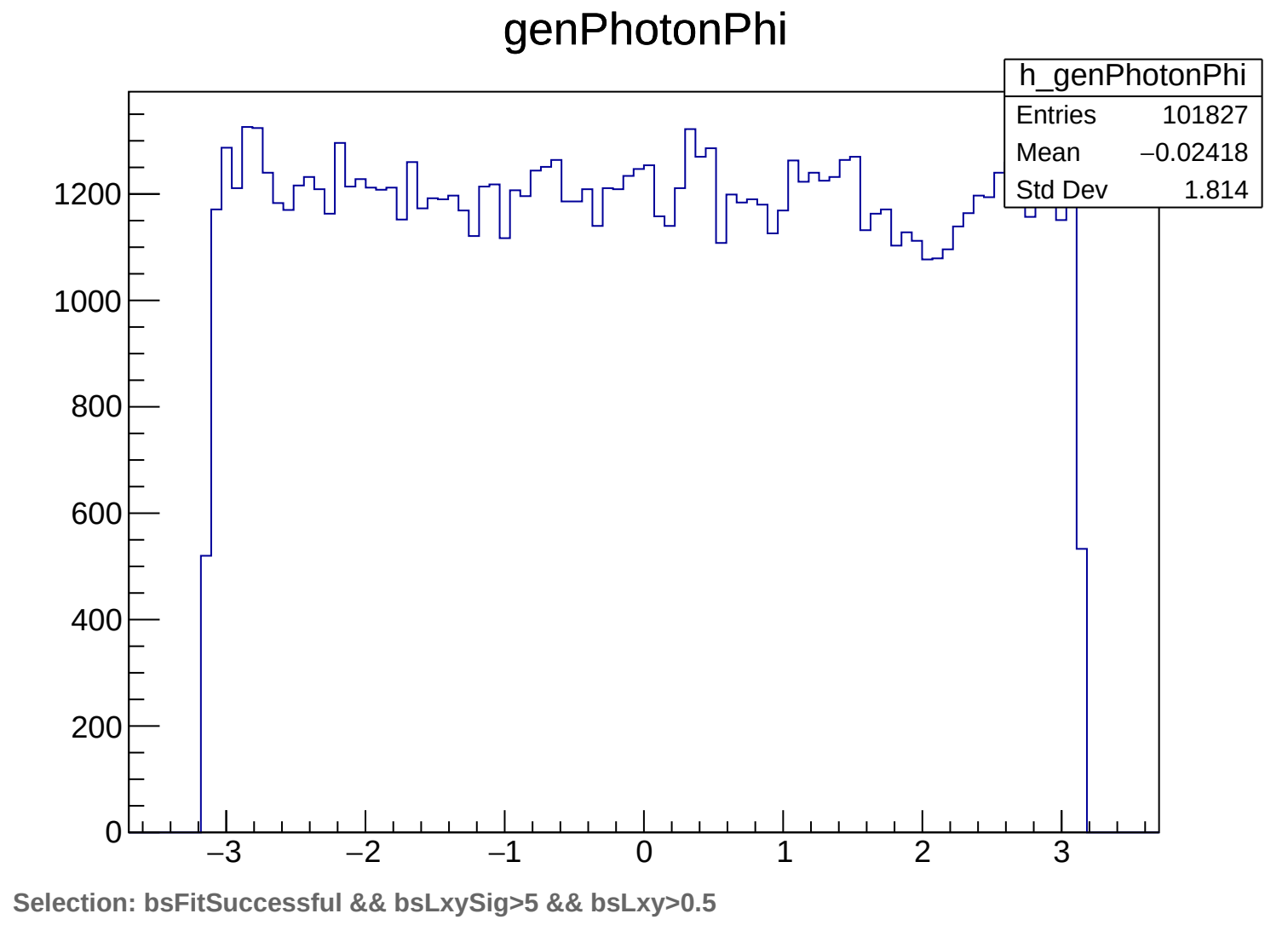

genMuMinusEta
h_genMuMinusEta
Entries 101827
Mean −0.0198
Std Dev 1.18
1800
1600
1400
1200
1000
800
600
400
200
0
−2
−1
0
1
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

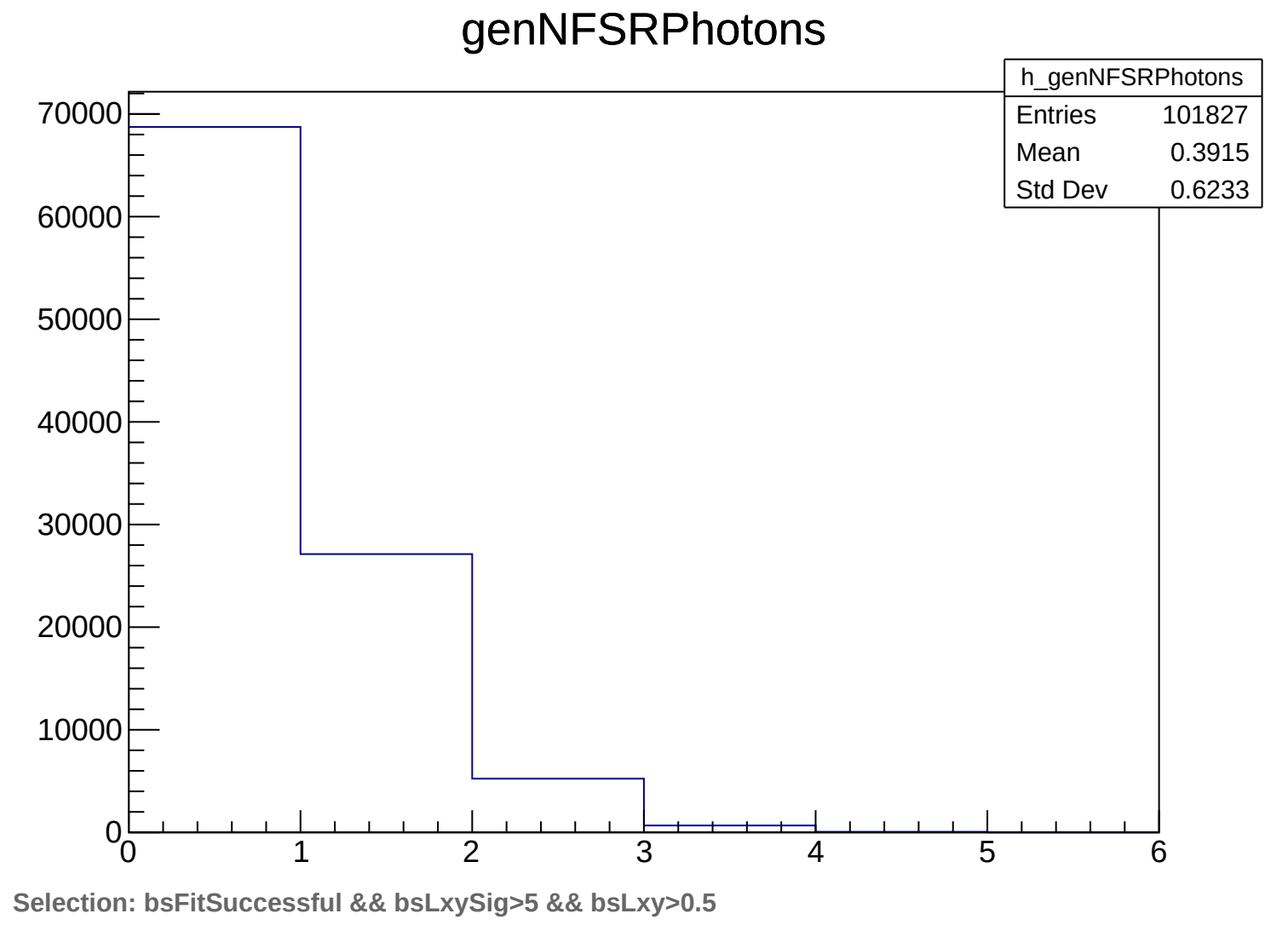


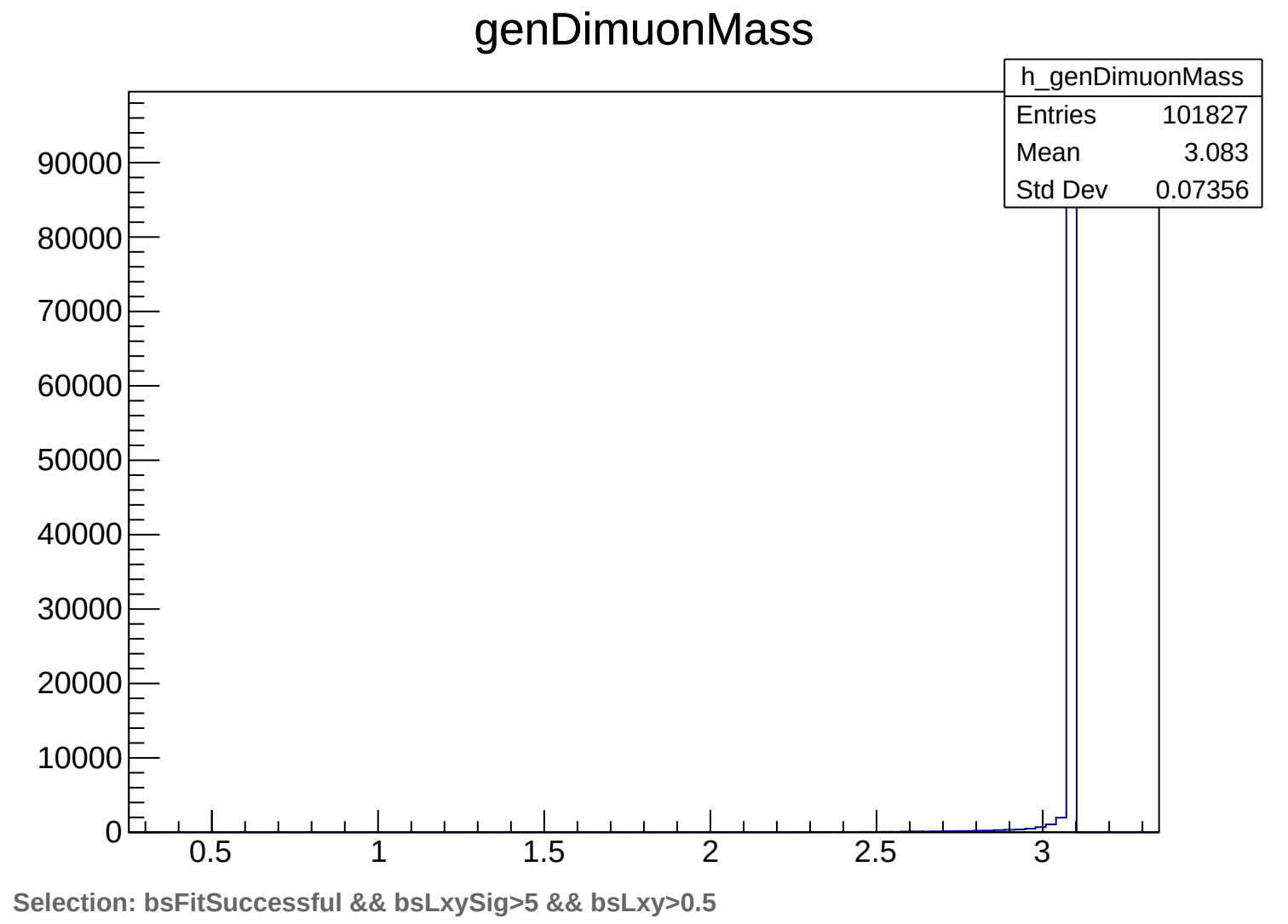
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

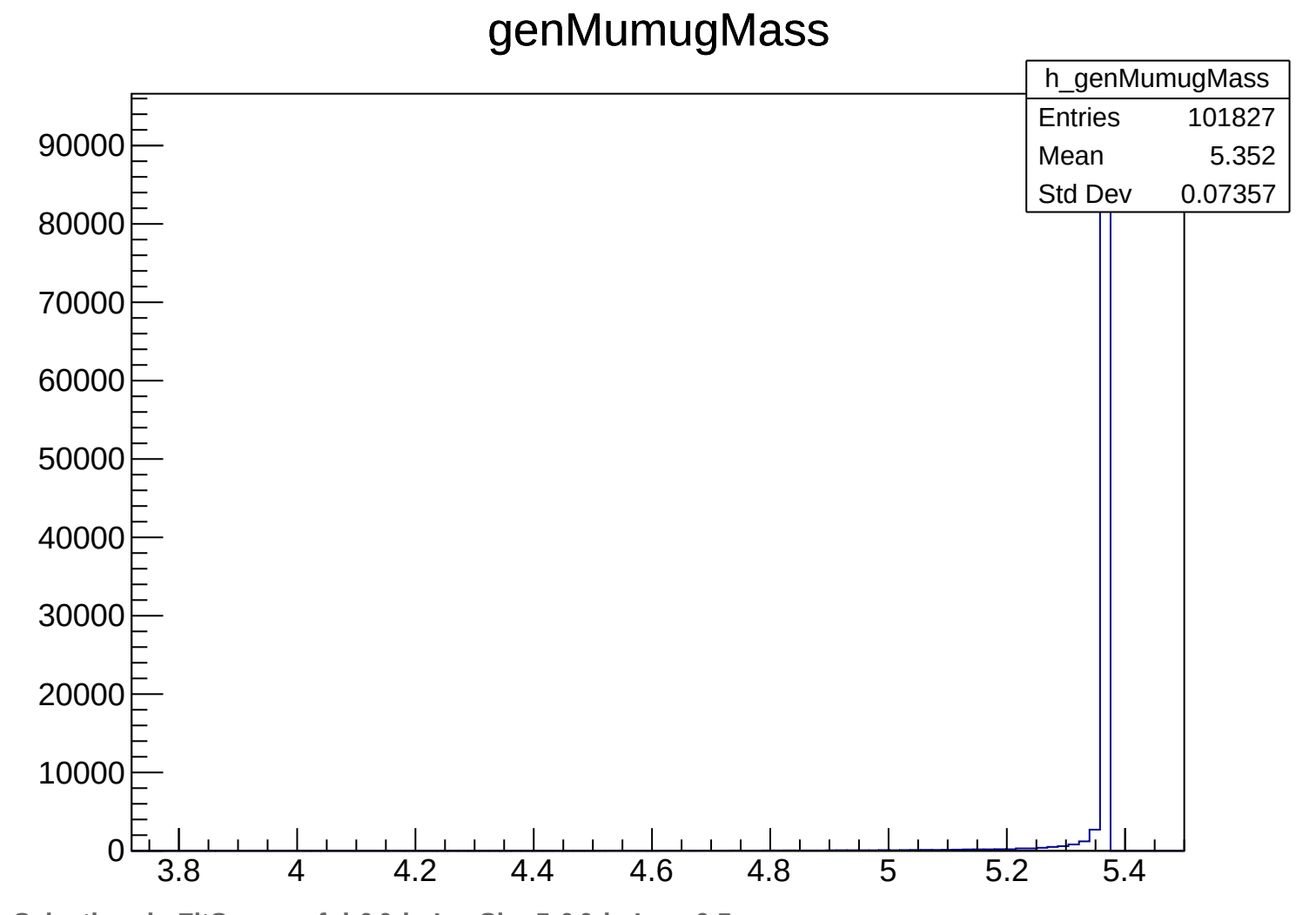

genPhotonPt
h_genPhotonPt
Entries 101827
Mean 18.9
Std Dev 9.82
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
20
40
60
80
100
120
140
160
180
200
220
240
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

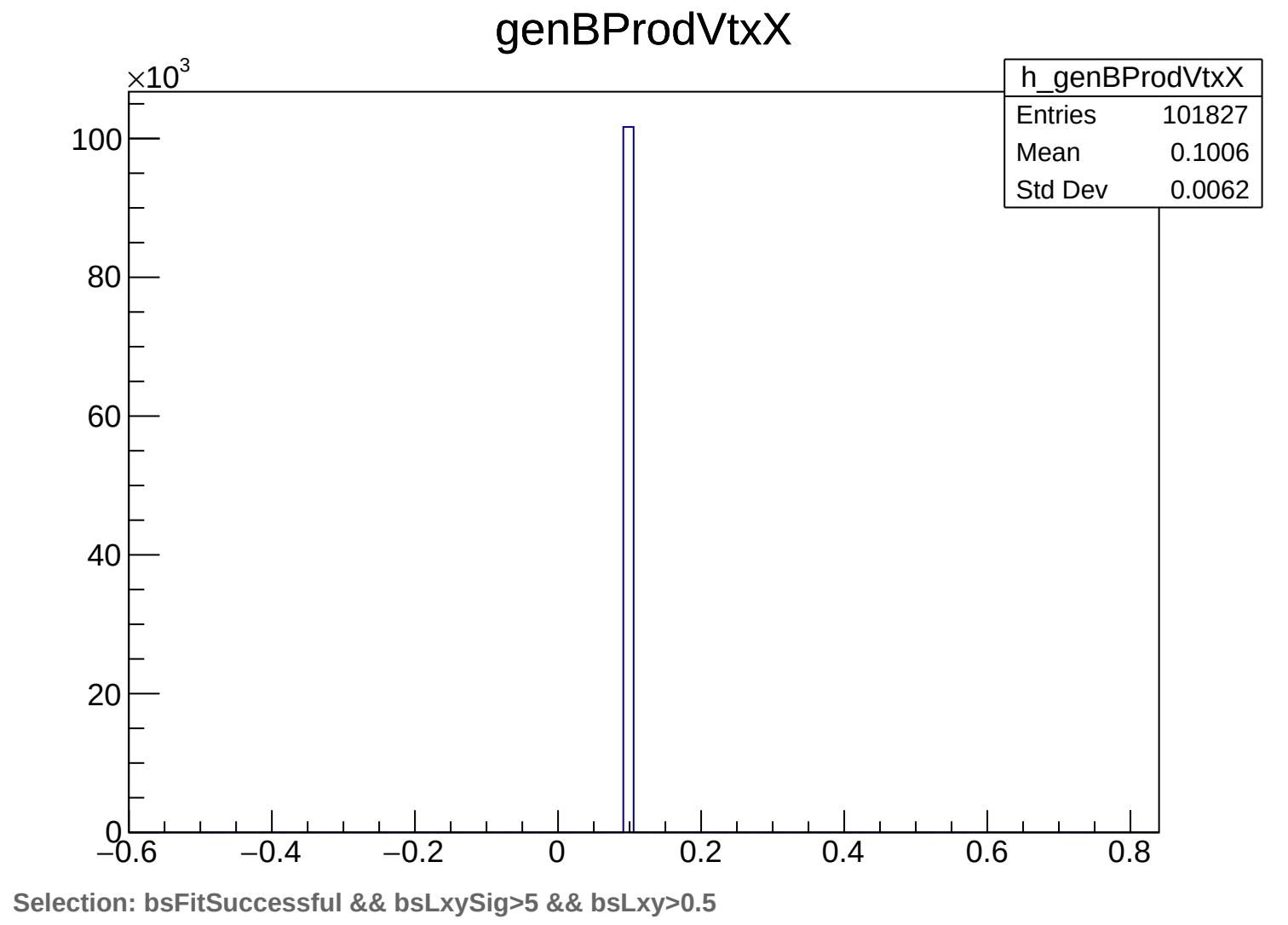

genPhotonEta
h_genPhotonEta
Entries 101827
Mean −0.01933
Std Dev 1.174
1800
1600
1400
1200
1000
800
600
400
200
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

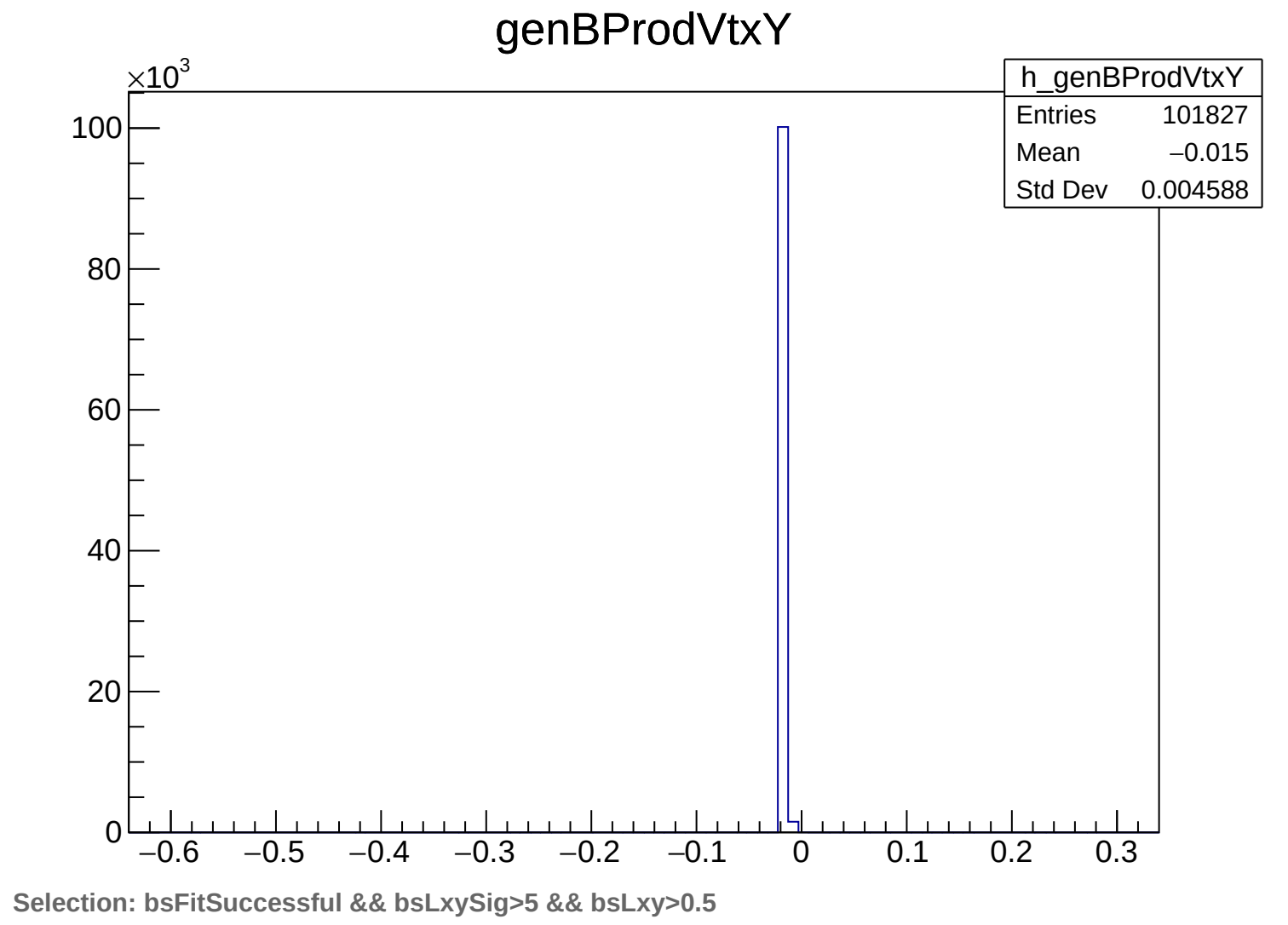

genPhotonPhi
h_genPhotonPhi
Entries 101827
Mean −0.02418
Std Dev 1.814
1200
1000
800
600
400
200
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

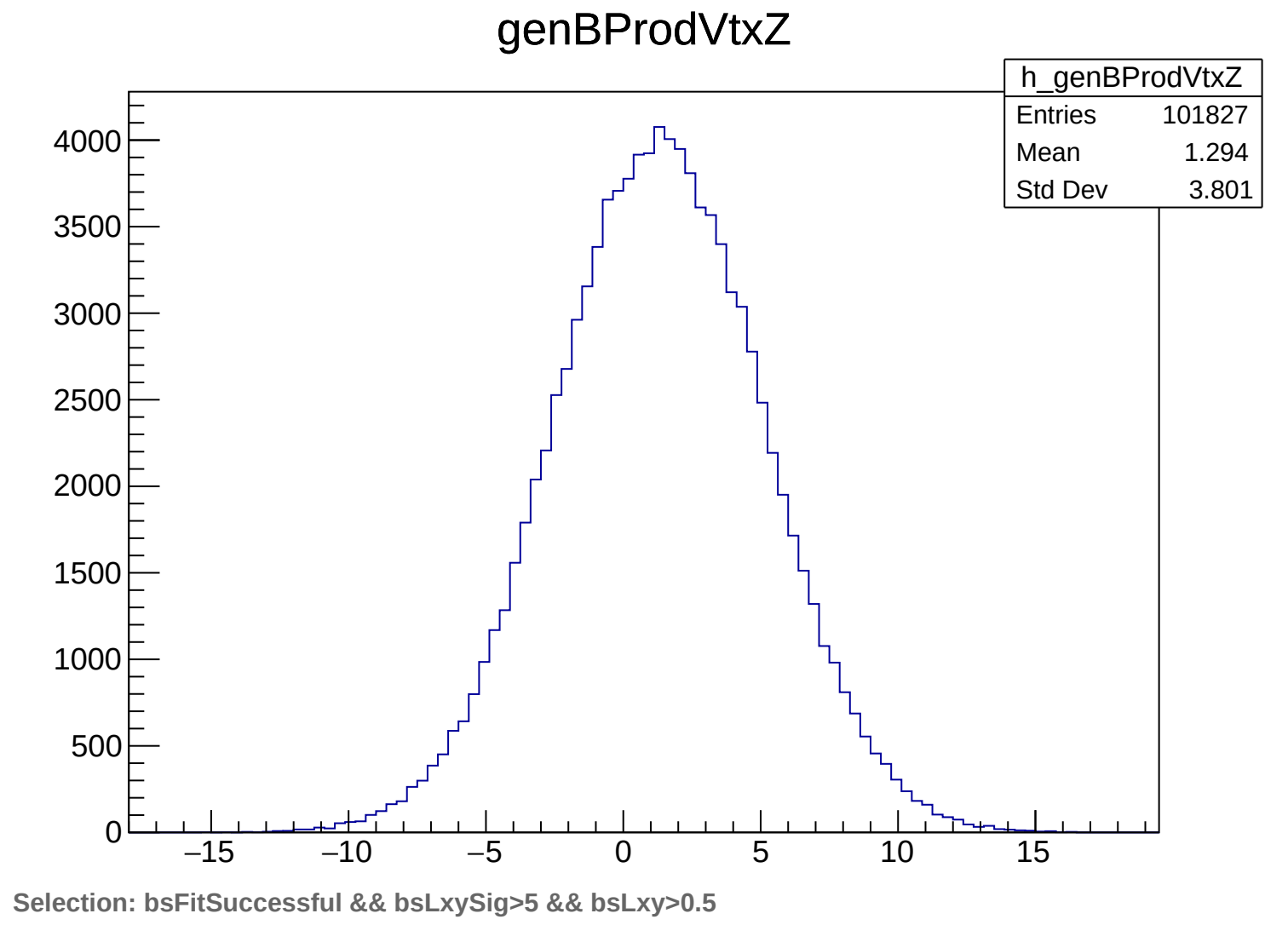

genNFSRPhotons
h_genNFSRPhotons
Entries 101827
Mean 0.3915
Std Dev 0.6233
70000
60000
50000
40000
30000
20000
10000
0
0
1
2
3
4
5
6
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

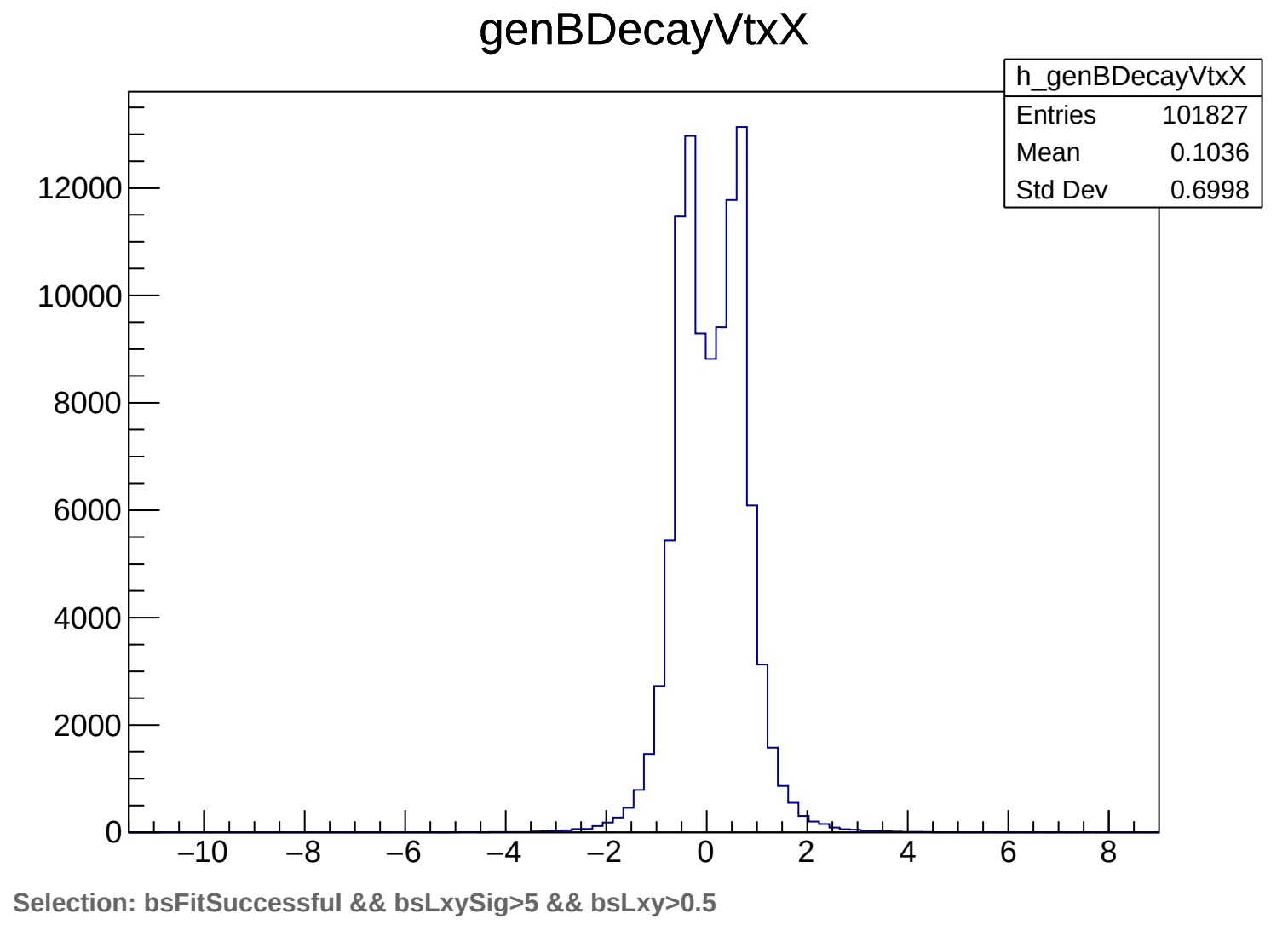

genDimuonMass
h_genDimuonMass
Entries 101827
Mean 3.083
Std Dev 0.07356
90000
80000
70000
60000
50000
40000
30000
20000
10000
0
0.5
1
1.5
2
2.5
3
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

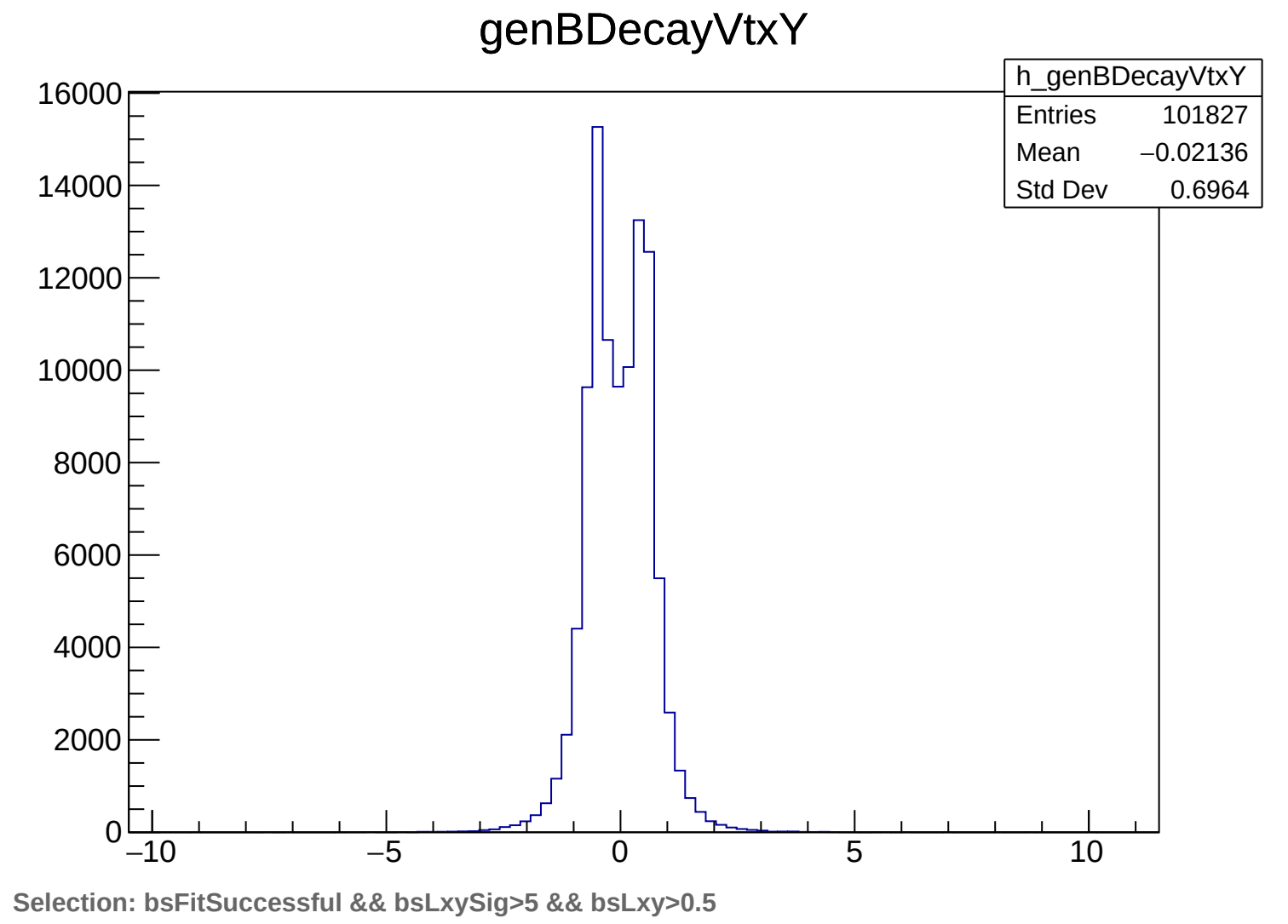

genMumugMass
h_genMumugMass
Entries 101827
Mean 5.352
Std Dev 0.07357
90000
80000
70000
60000
50000
40000
30000
20000
10000
0
3.8
4
4.2
4.4
4.6
4.8
5
5.2
5.4
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

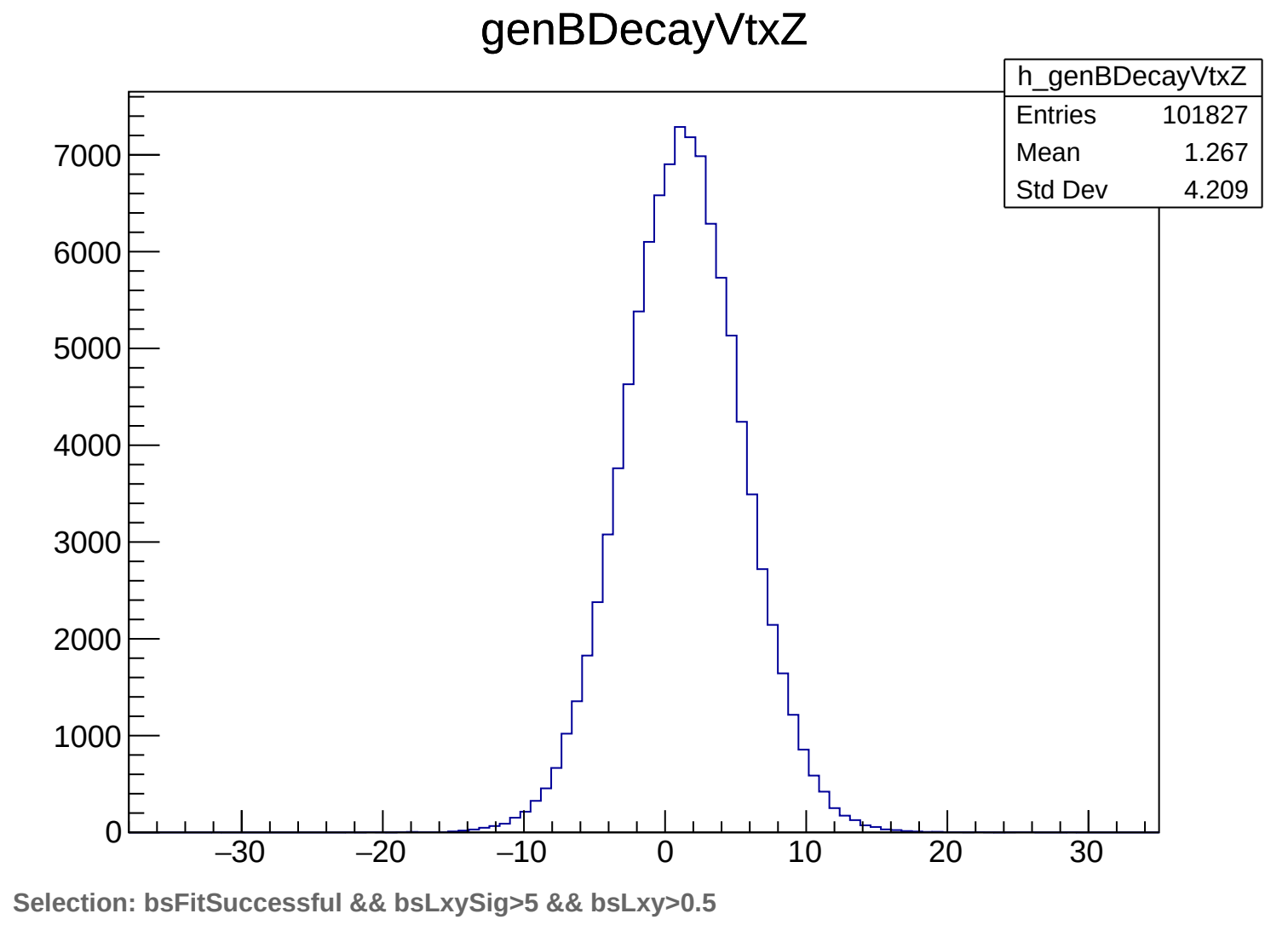

genBProdVtxX
h_genBProdVtxX
Entries 101827
Mean 0.1006
Std Dev 0.0062
×10^3
100
80
60
40
20
0
−0.6
−0.4
−0.2
0
0.2
0.4
0.6
0.8
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

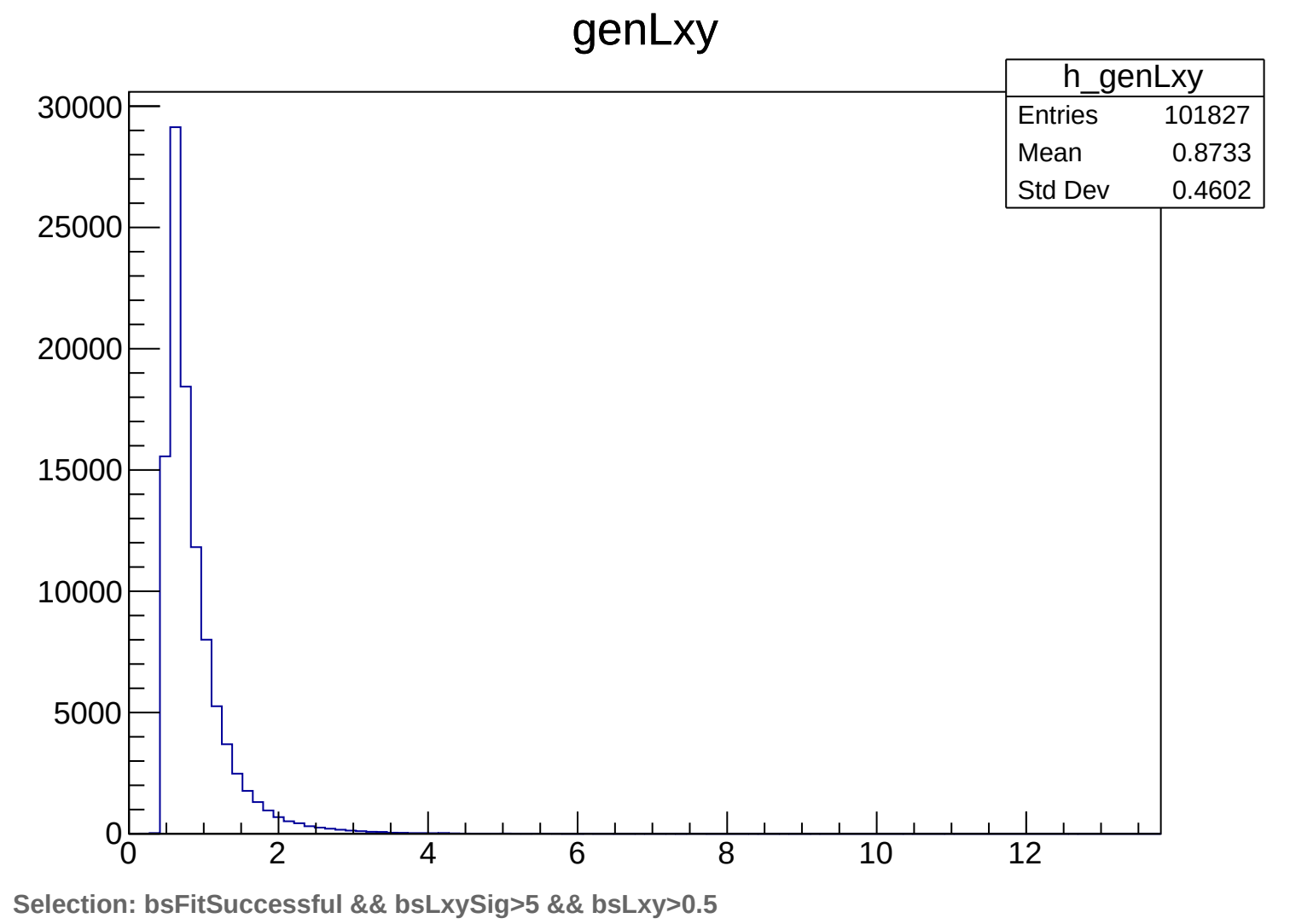

genBProdVtxY
×10^3
100
80
60
40
20
0
−0.6
−0.5
−0.4
−0.3
−0.2
−0.1
0
0.1
0.2
0.3
h_genBProdVtxY
Entries 101827
Mean −0.015
Std Dev 0.004588
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

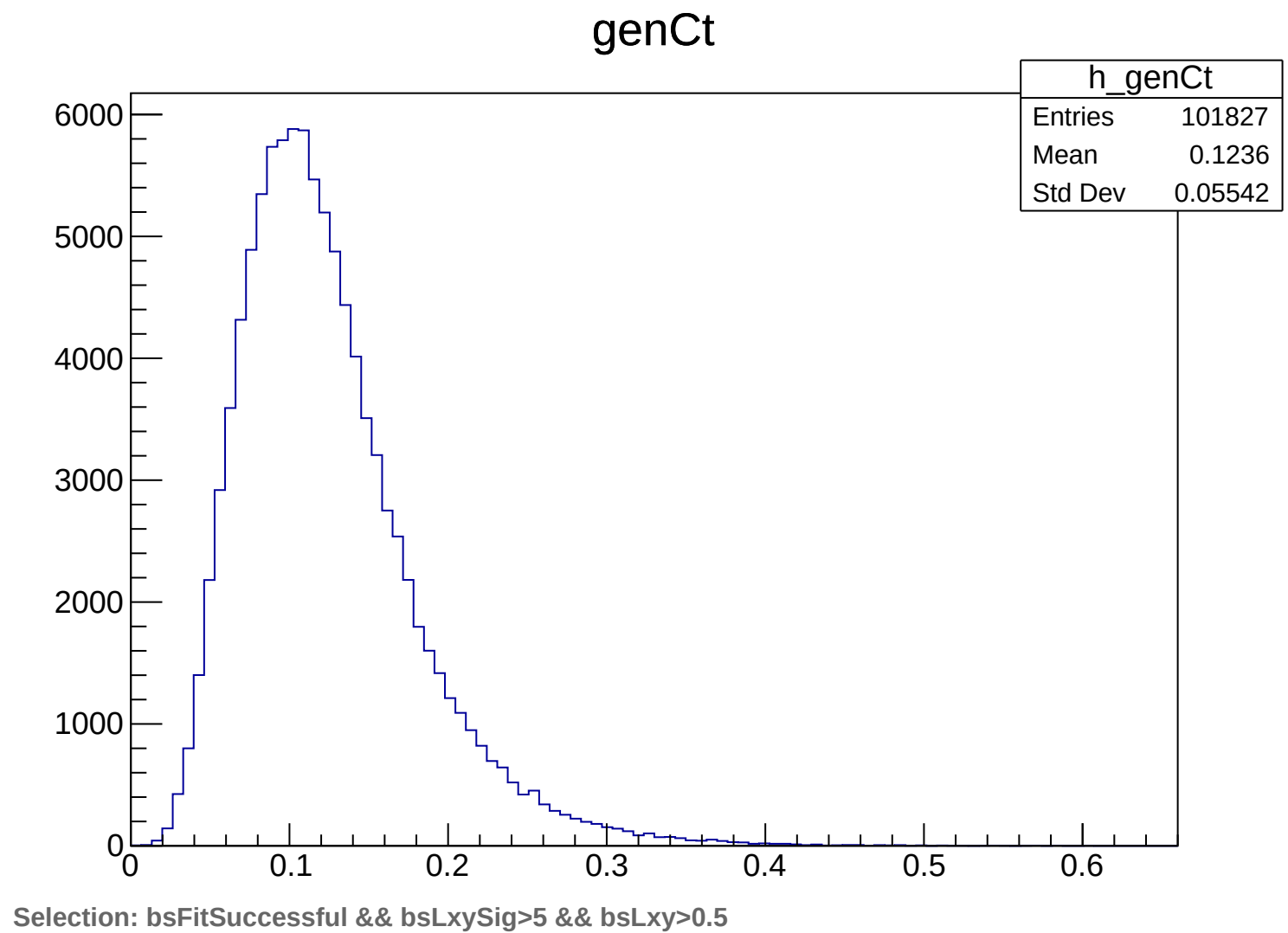

genBProdVtxZ
h_genBProdVtxZ
Entries 101827
Mean 1.294
Std Dev 3.801
4000
3500
3000
2500
2000
1500
1000
500
0
−15
−10
−5
0
5
10
15
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

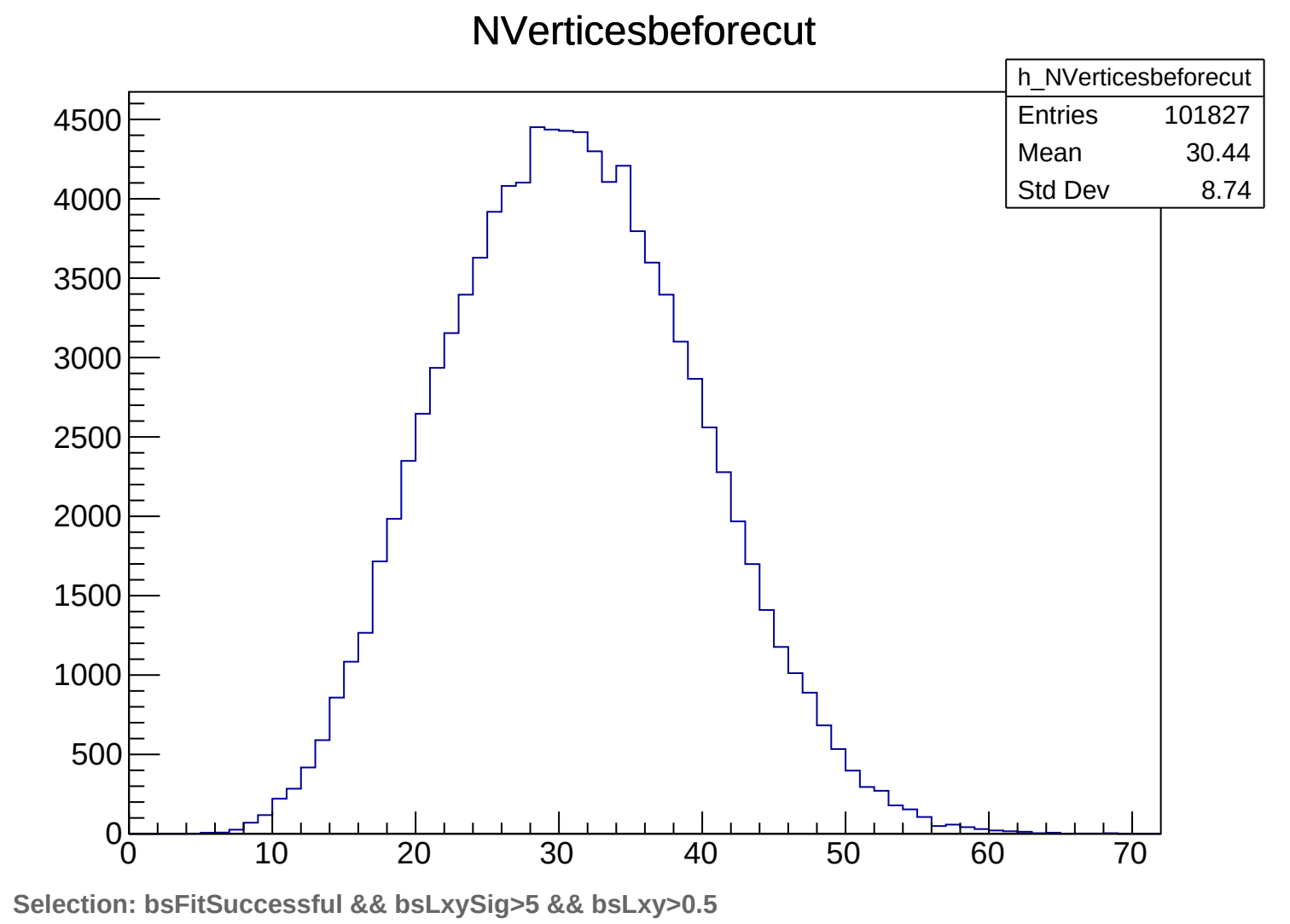

genBDecayVtxX
h_genBDecayVtxX
Entries 101827
Mean 0.1036
Std Dev 0.6998
12000
10000
8000
6000
4000
2000
0
−10
−8
−6
−4
−2
0
2
4
6
8
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

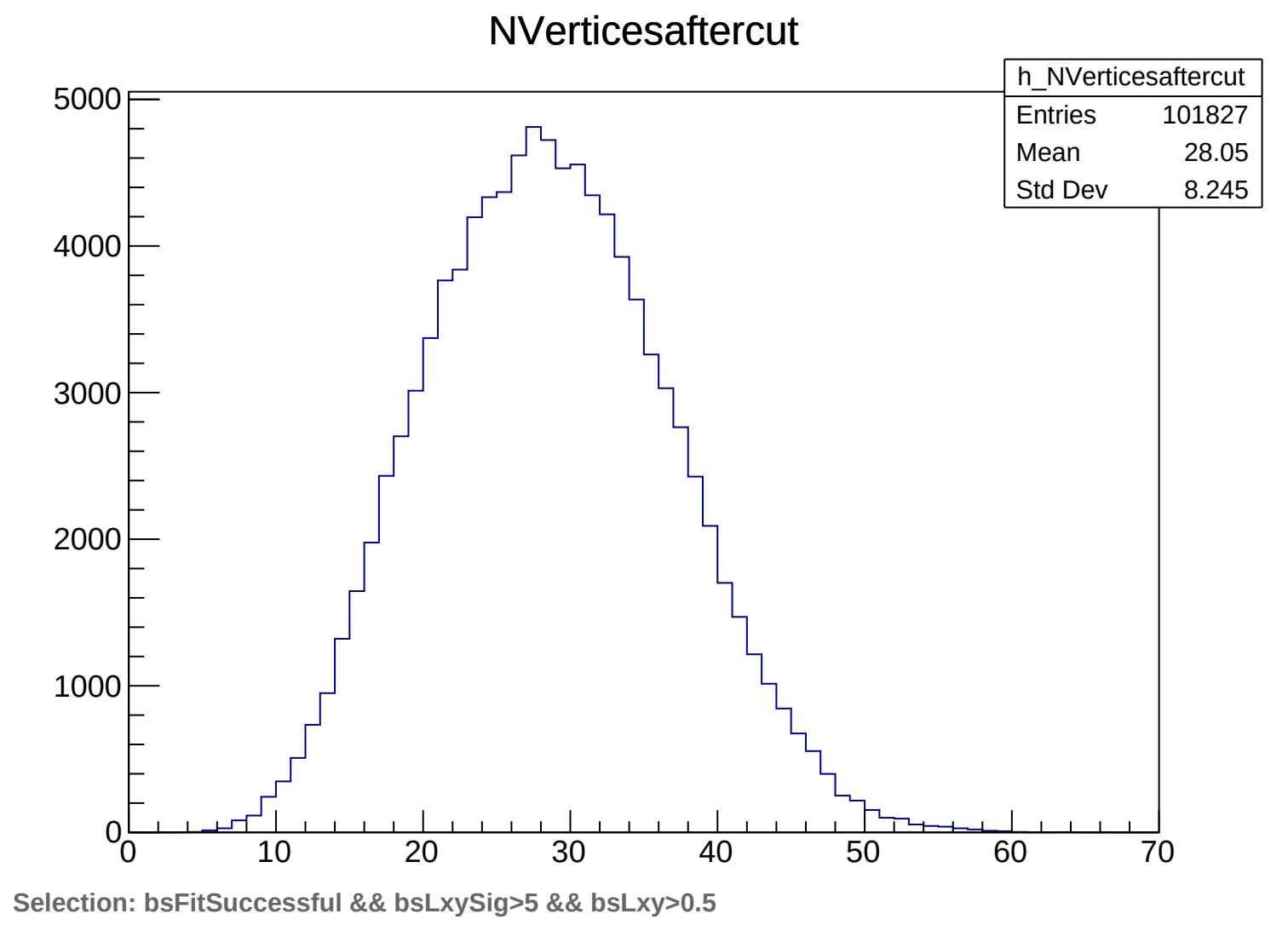

genBDecayVtxY
h_genBDecayVtxY
Entries 101827
Mean −0.02136
Std Dev 0.6964
16000
14000
12000
10000
8000
6000
4000
2000
0
−10
−5
0
5
10
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

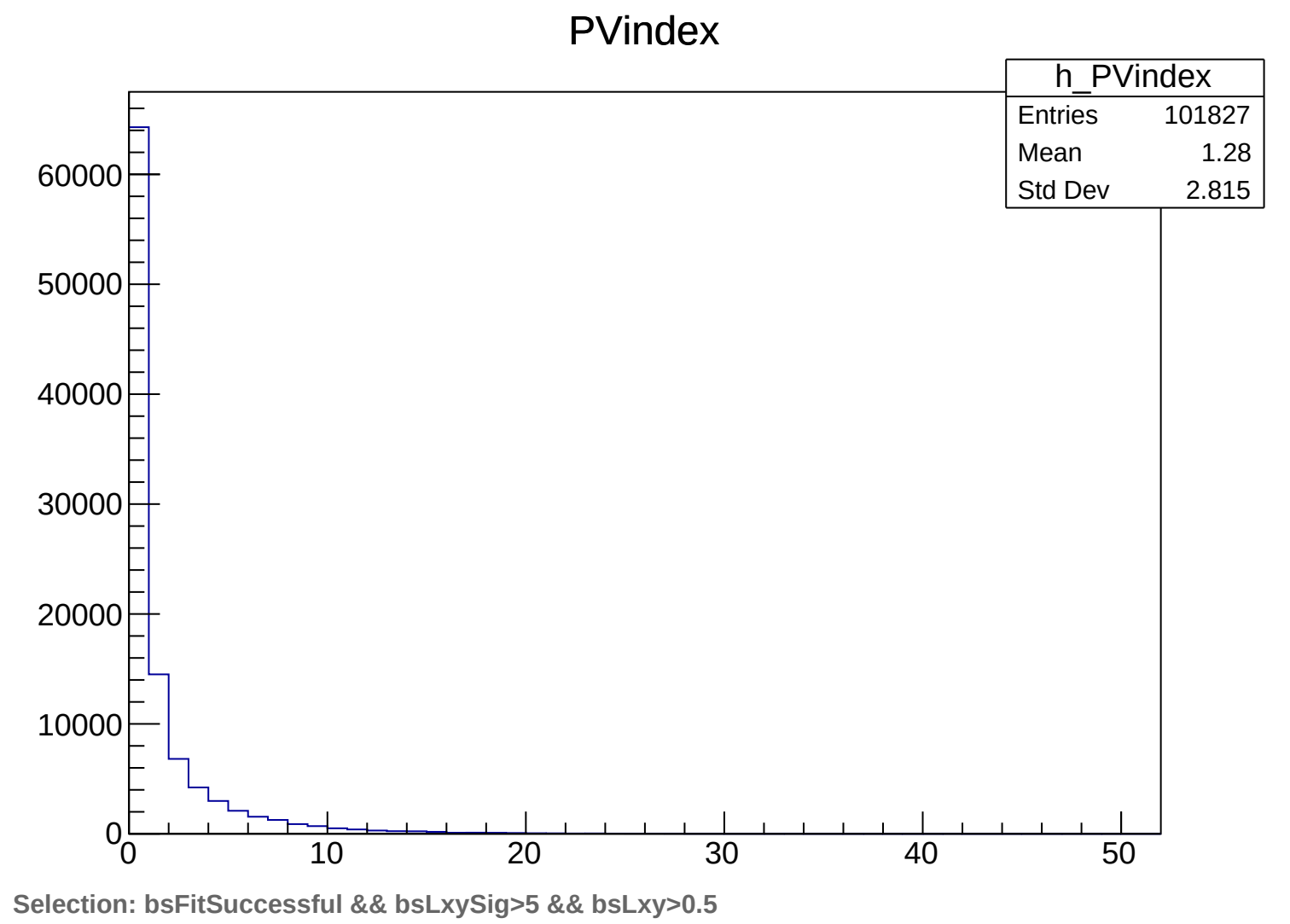

genBDecayVtxZ
h_genBDecayVtxZ
Entries 101827
Mean 1.267
Std Dev 4.209
7000
6000
5000
4000
3000
2000
1000
0
−30
−20
−10
0
10
20
30
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

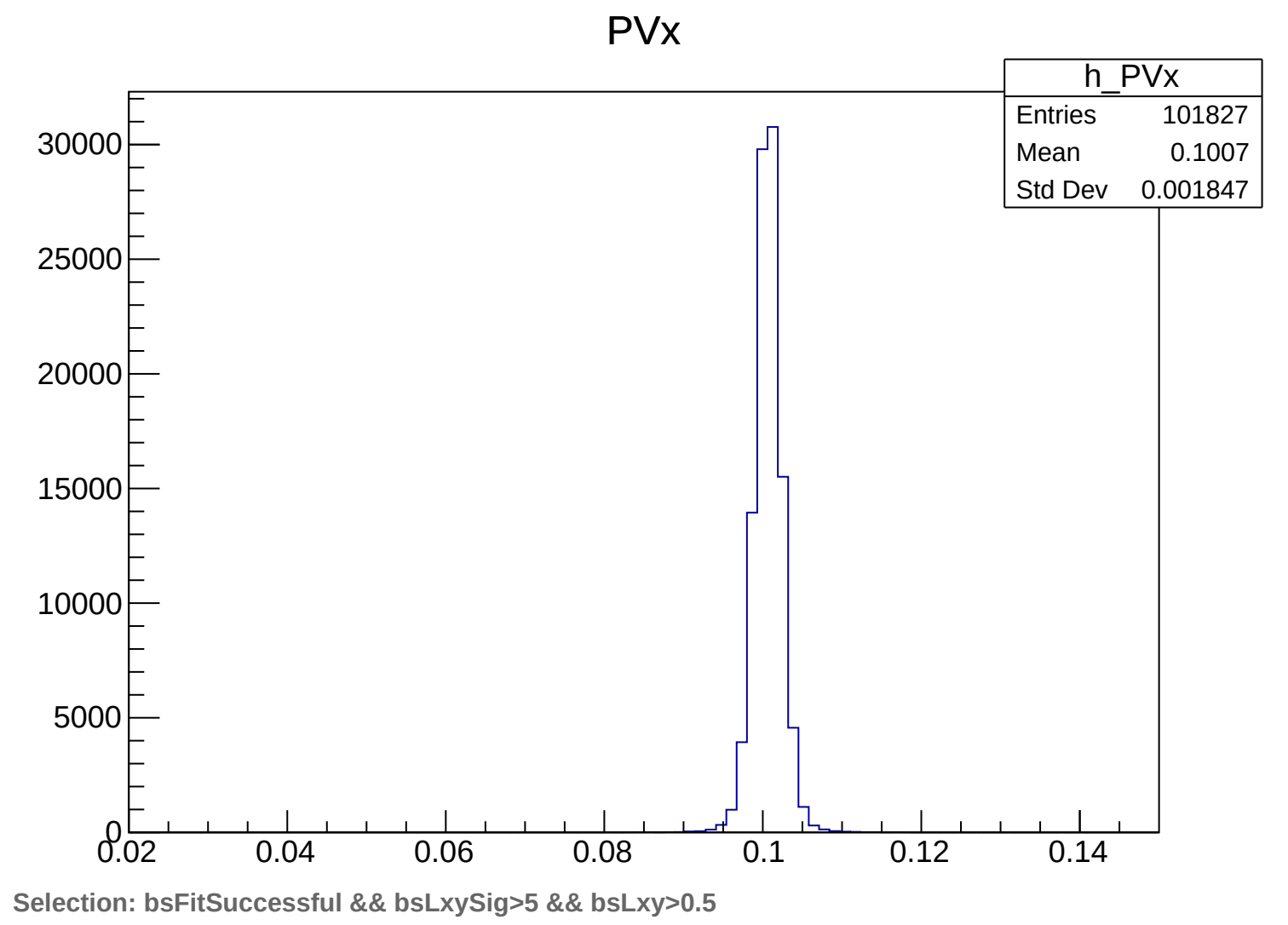

genLxy
h_genLxy
Entries 101827
Mean 0.8733
Std Dev 0.4602
30000
25000
20000
15000
10000
5000
0
0
2
4
6
8
10
12
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

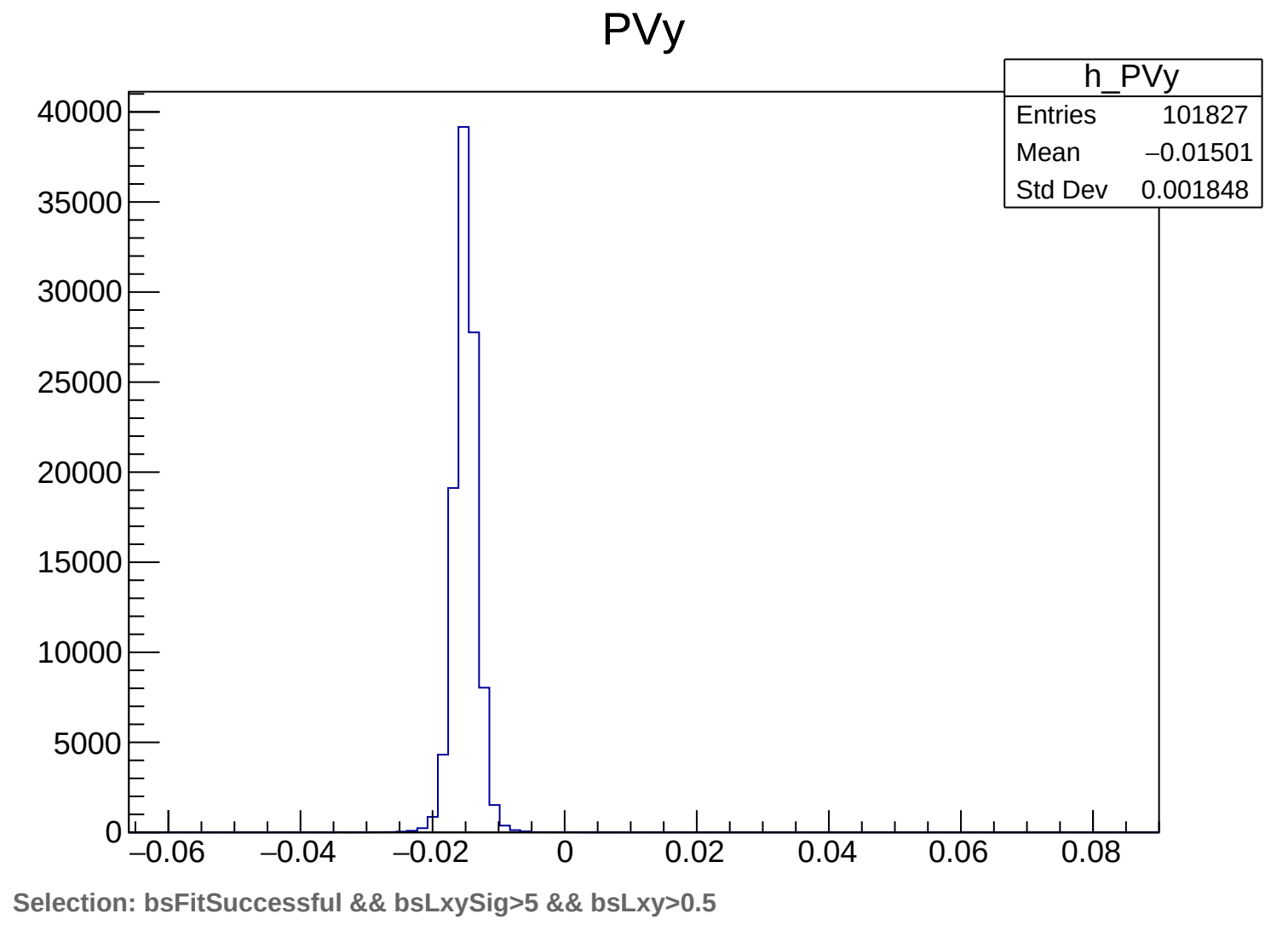

genCt
h_genCt
Entries 101827
Mean 0.1236
Std Dev 0.05542
6000
5000
4000
3000
2000
1000
0
0
0.1
0.2
0.3
0.4
0.5
0.6
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

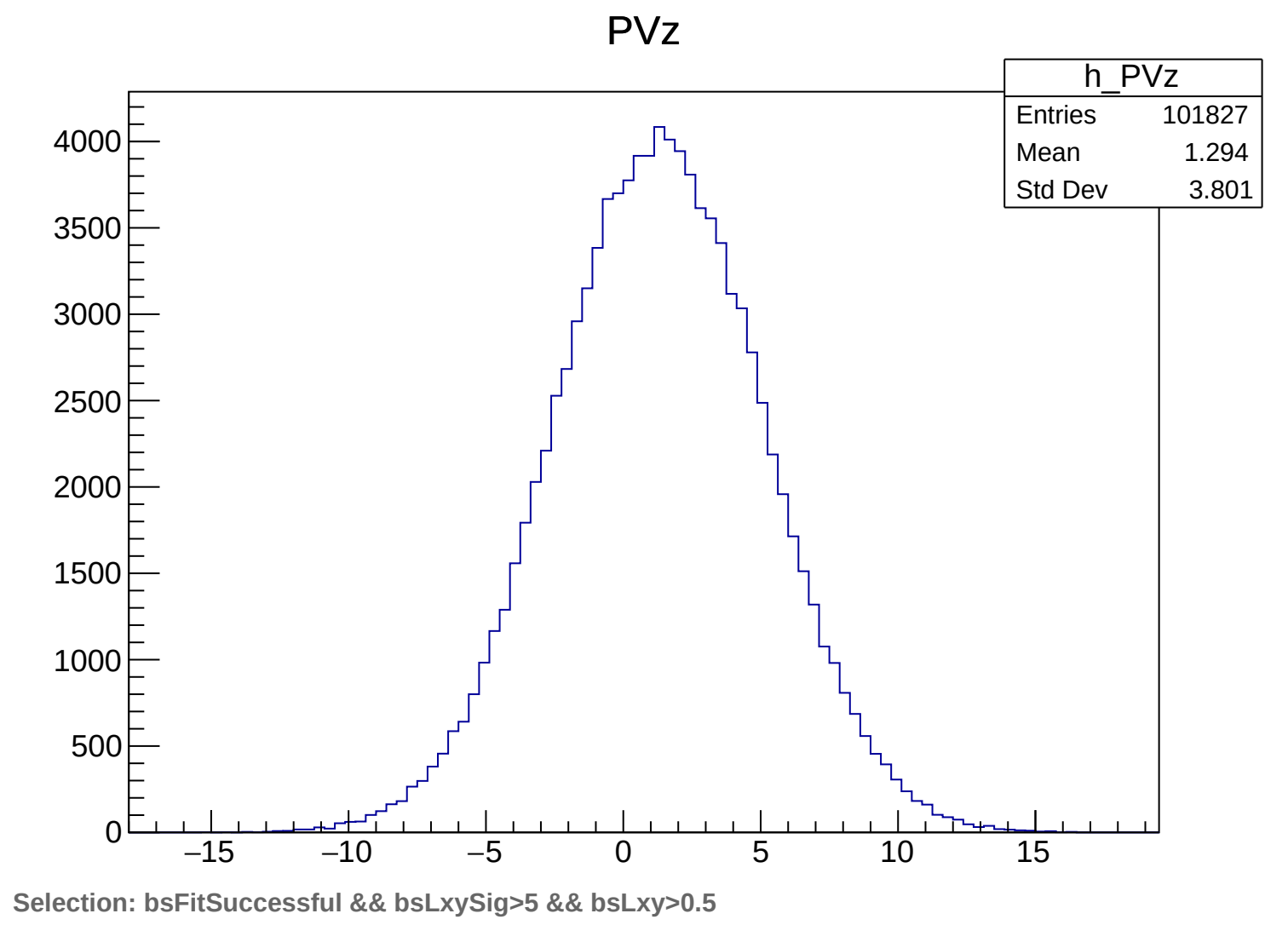

NVerticesbeforecut
h_NVerticesbeforecut
Entries 101827
Mean 30.44
Std Dev 8.74
4500
4000
3500
3000
2500
2000
1500
1000
500
0
0
10
20
30
40
50
60
70
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


NVerticesaftercut
h_NVerticesaftercut
Entries 101827
Mean 28.05
Std Dev 8.245
5000
4000
3000
2000
1000
0
0
10
20
30
40
50
60
70
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

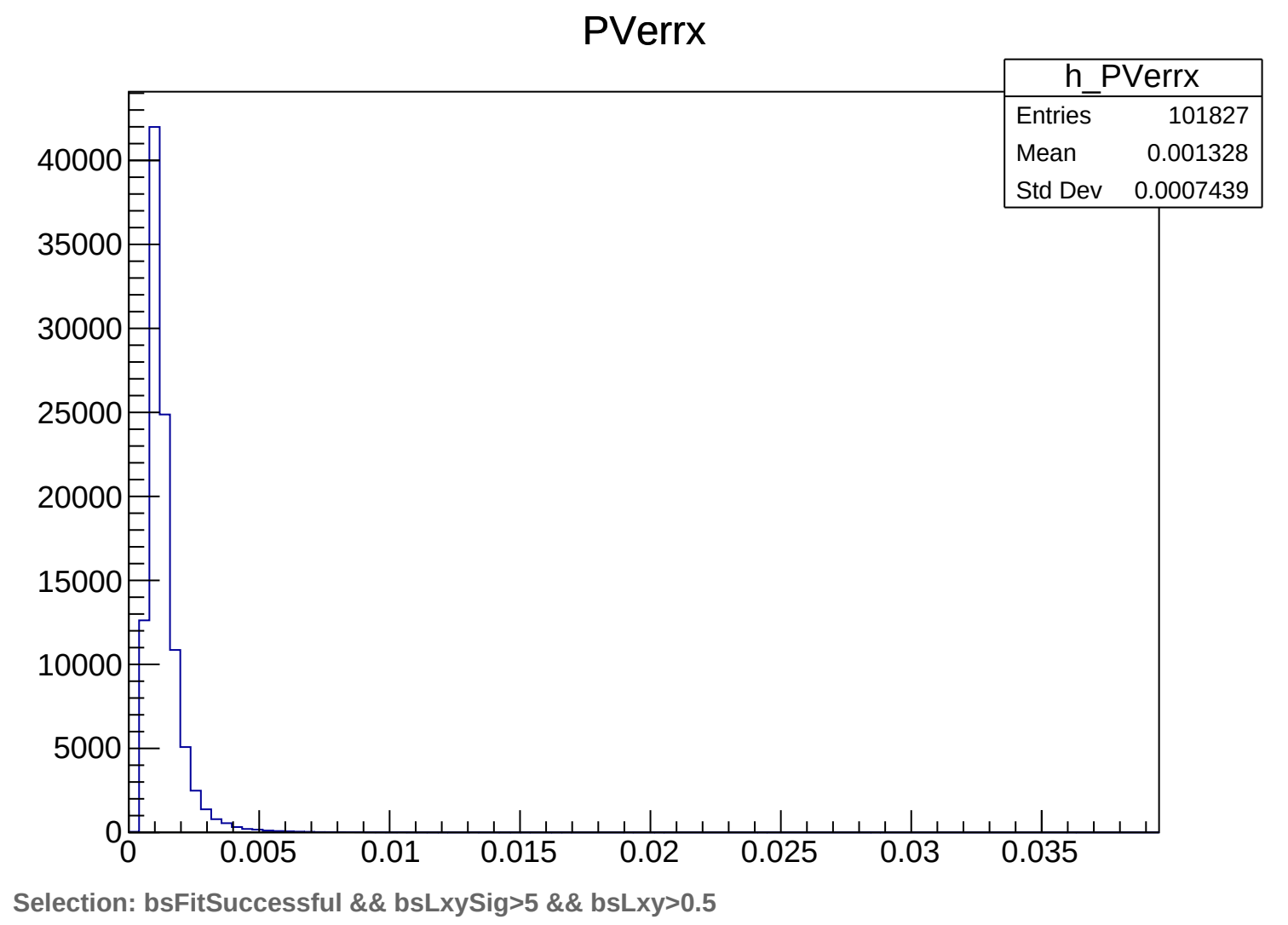

PVindex
h_PVindex
Entries 101827
Mean 1.28
Std Dev 2.815
60000
50000
40000
30000
20000
10000
0
0
10
20
30
40
50
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

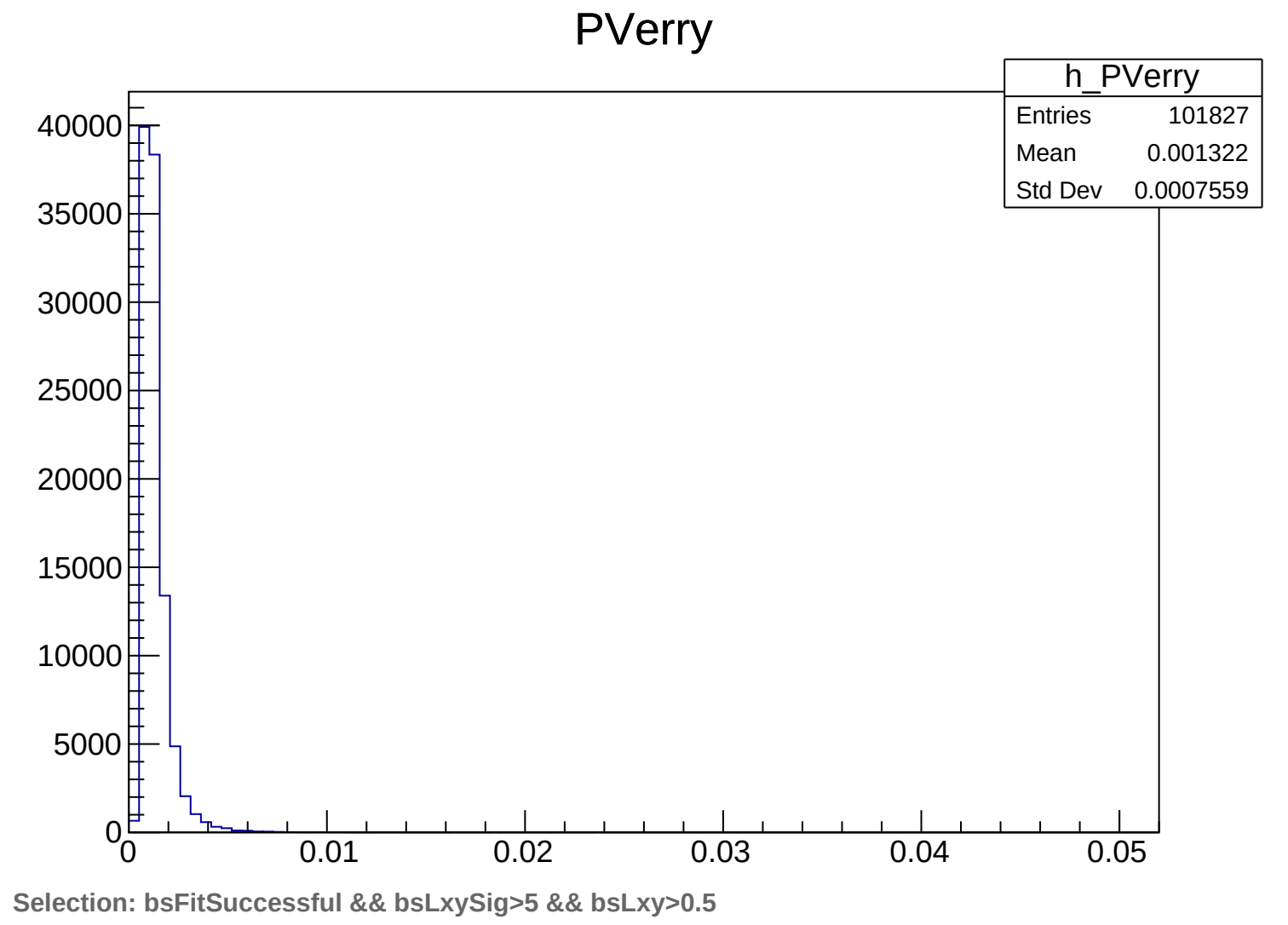

PVx
h_PVx
Entries 101827
Mean 0.1007
Std Dev 0.001847
30000
25000
20000
15000
10000
5000
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

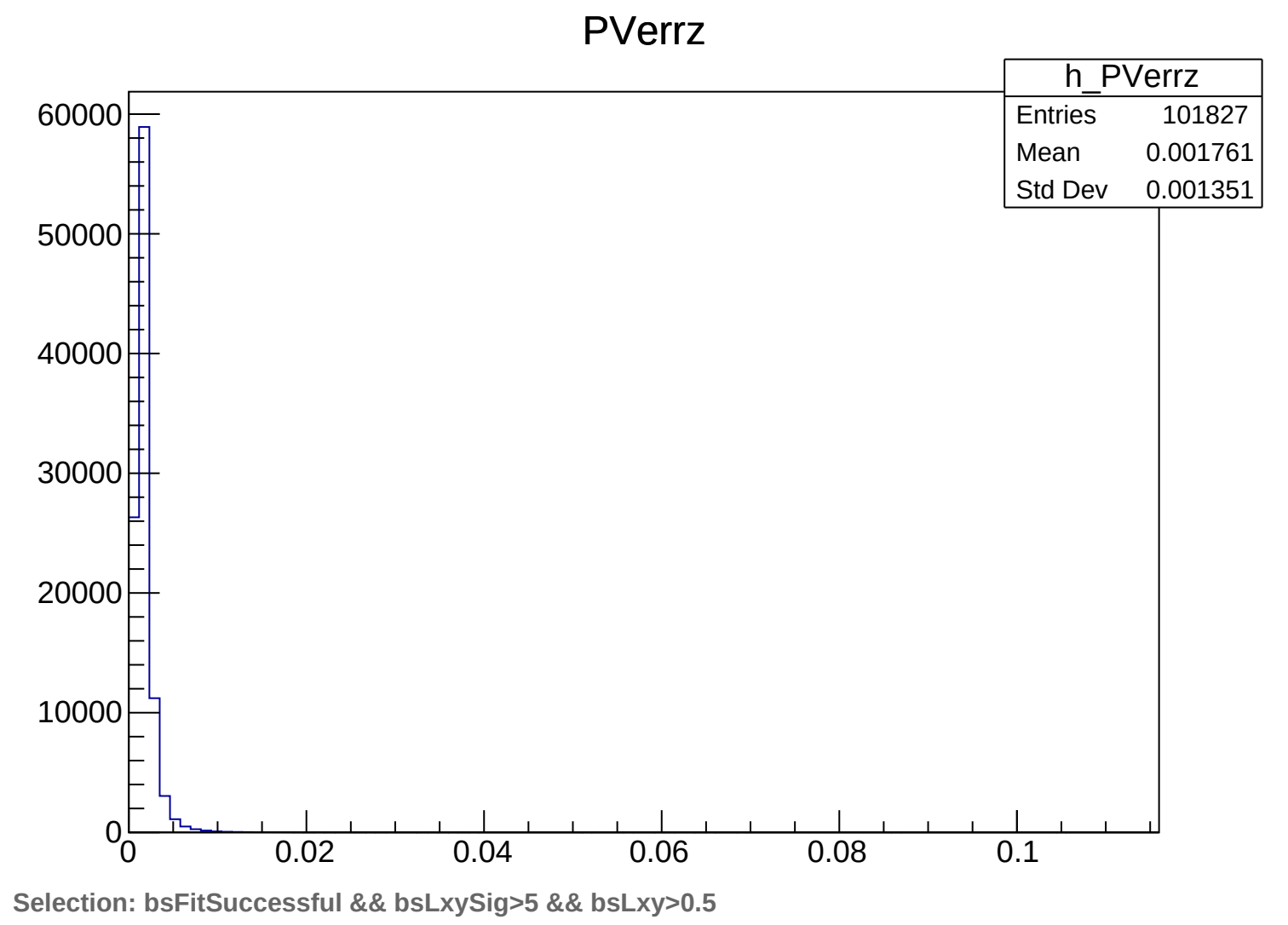

PVy
h_PVy
Entries 101827
Mean −0.01501
Std Dev 0.001848
40000
35000
30000
25000
20000
15000
10000
5000
0
−0.06
−0.04
−0.02
0
0.02
0.04
0.06
0.08
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

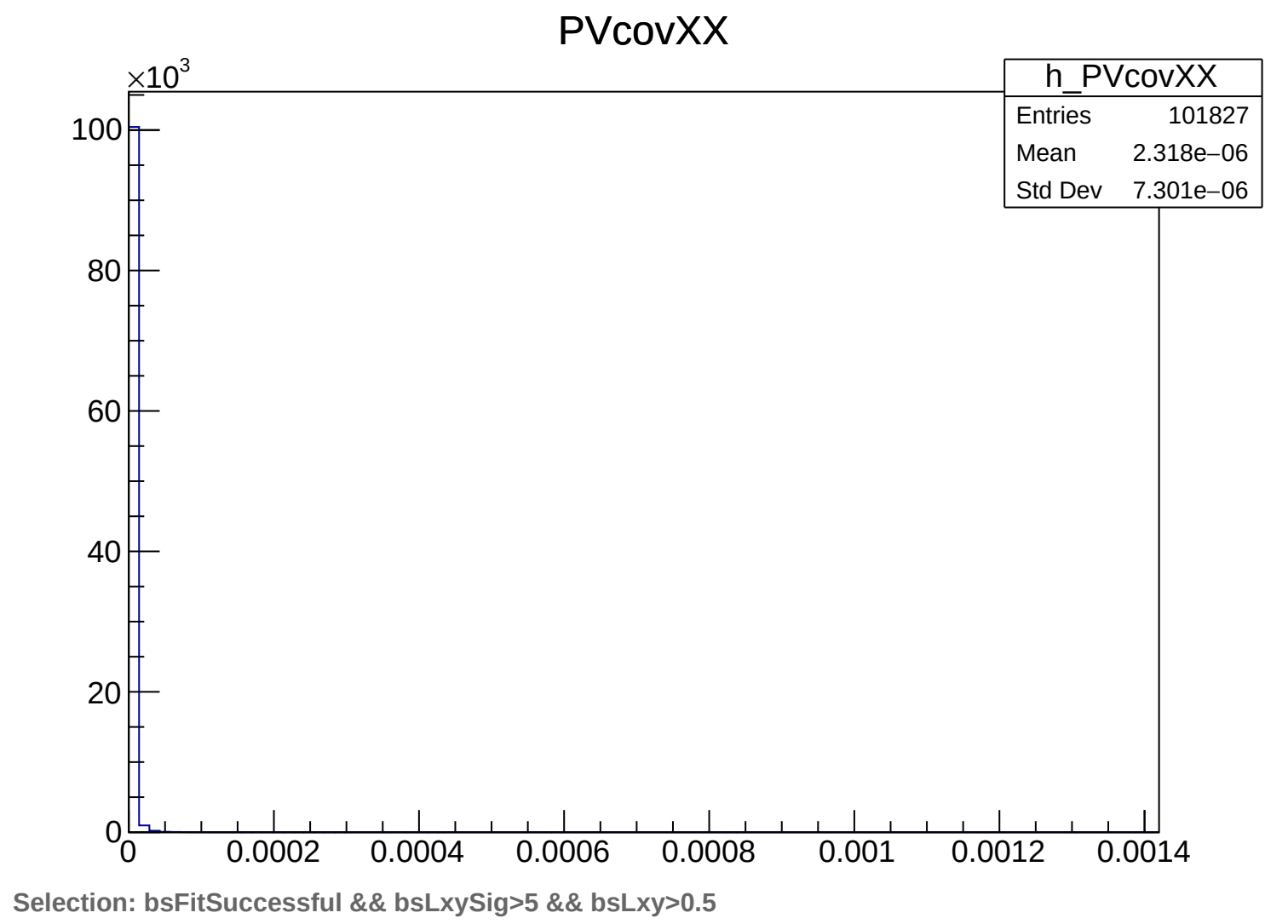


Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

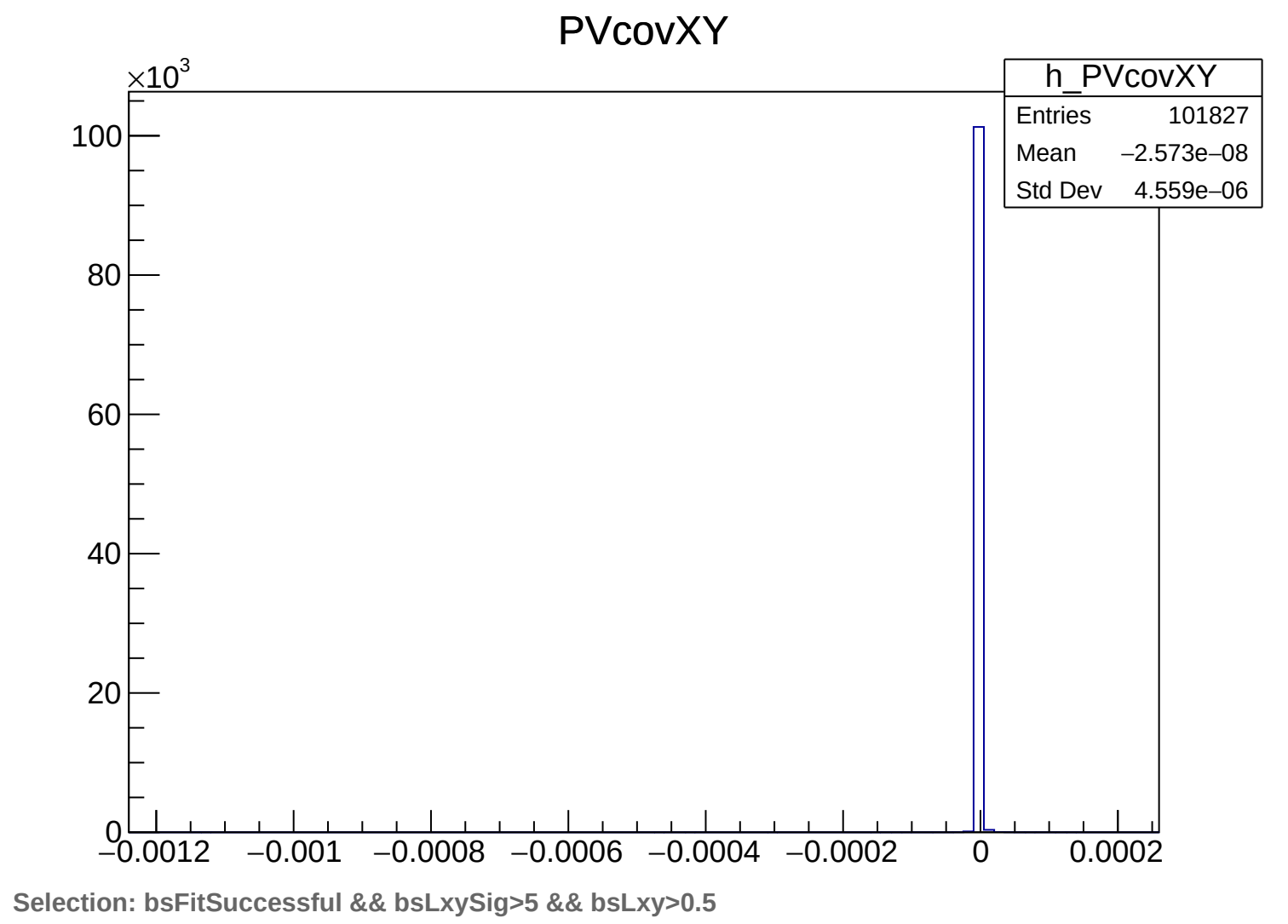

PVerrx
h_PVerrx
Entries 101827
Mean 0.001328
Std Dev 0.0007439
40000
35000
30000
25000
20000
15000
10000
5000
0
0
0.005
0.01
0.015
0.02
0.025
0.03
0.035
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

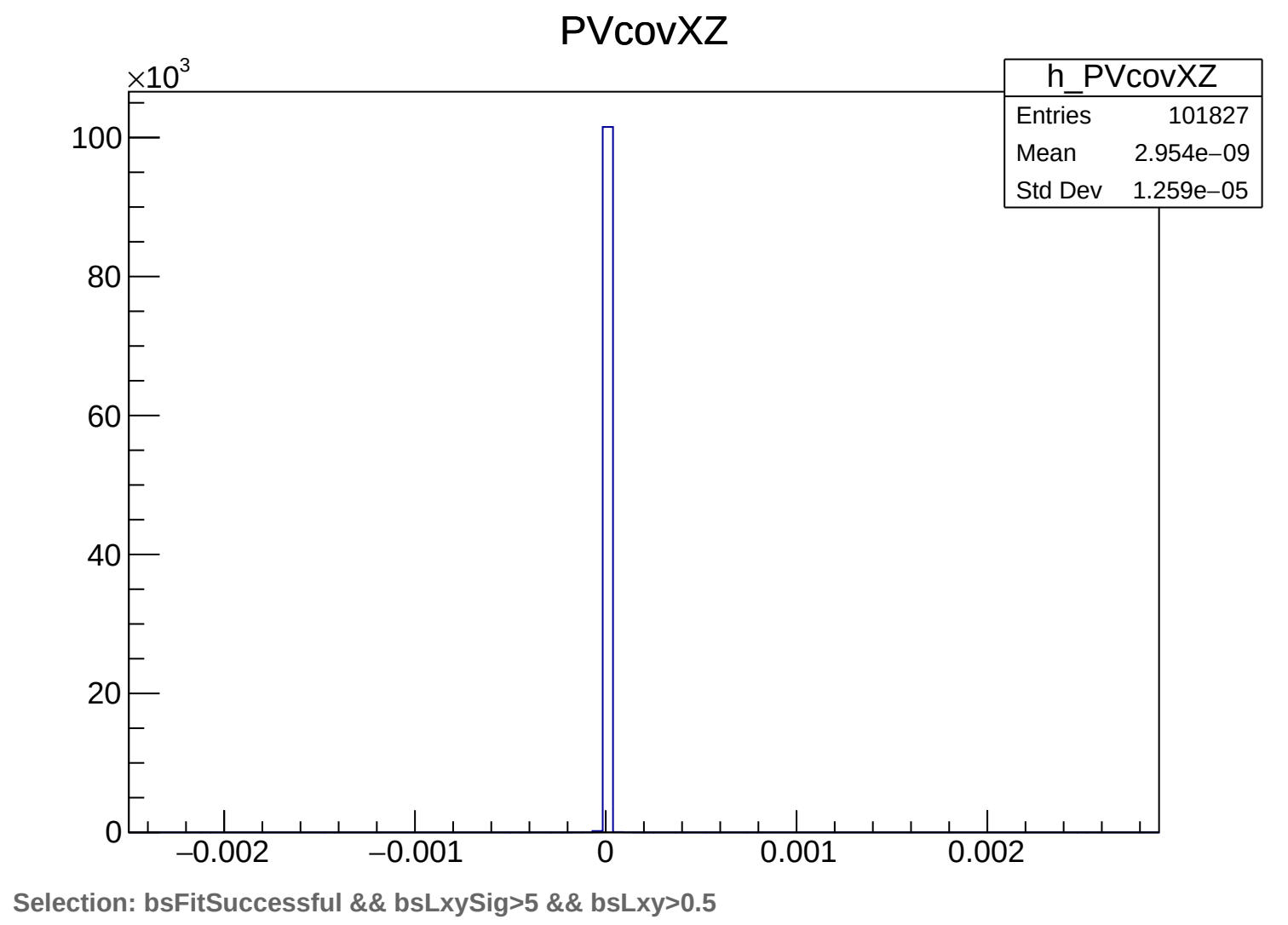

PVerry
h_PVerry
Entries 101827
Mean 0.001322
Std Dev 0.0007559
40000
35000
30000
25000
20000
15000
10000
5000
0
0
0.01
0.02
0.03
0.04
0.05
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


PVerrz
h_PVerrz
Entries 101827
Mean 0.001761
Std Dev 0.001351
60000
50000
40000
30000
20000
10000
0
0
0.02
0.04
0.06
0.08
0.1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

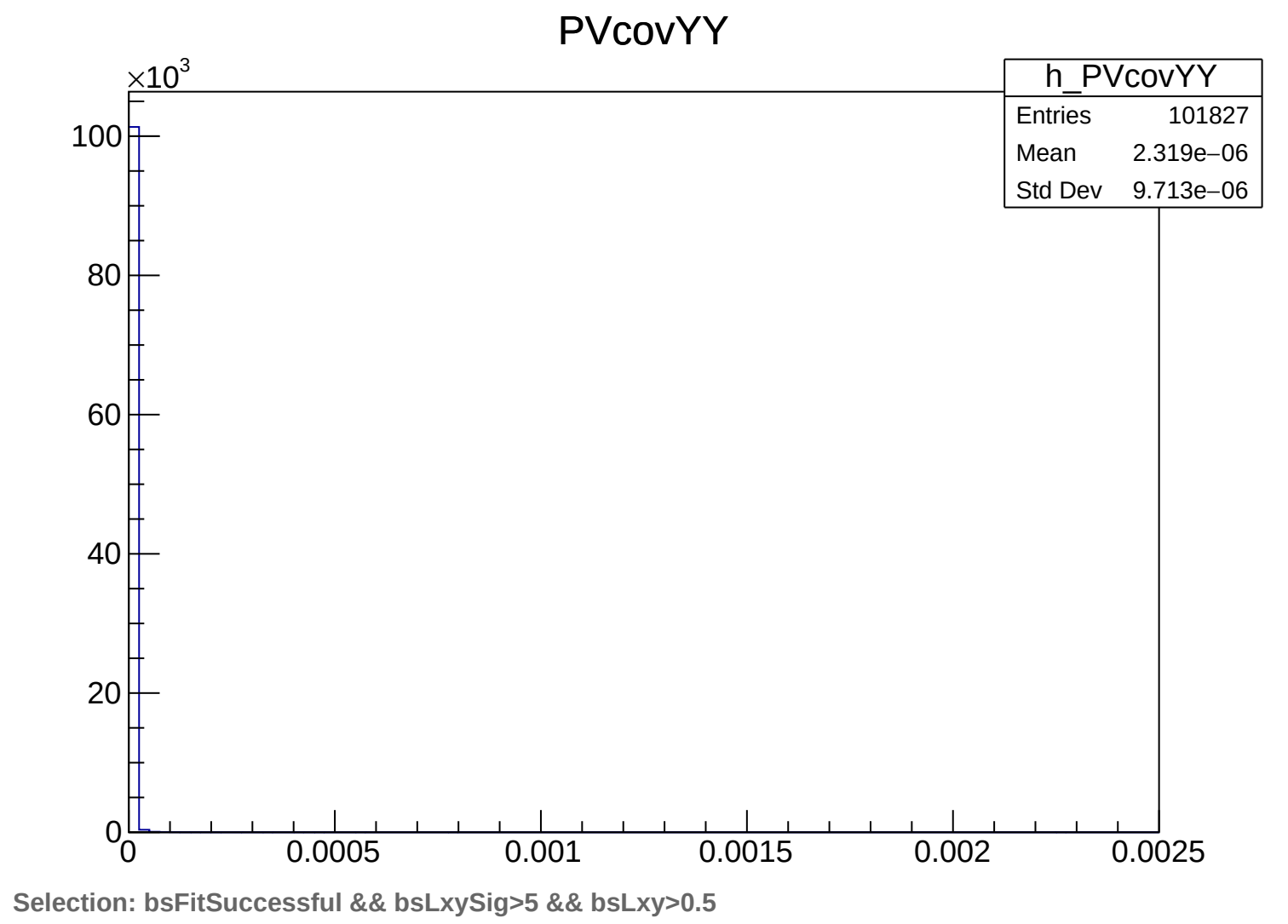


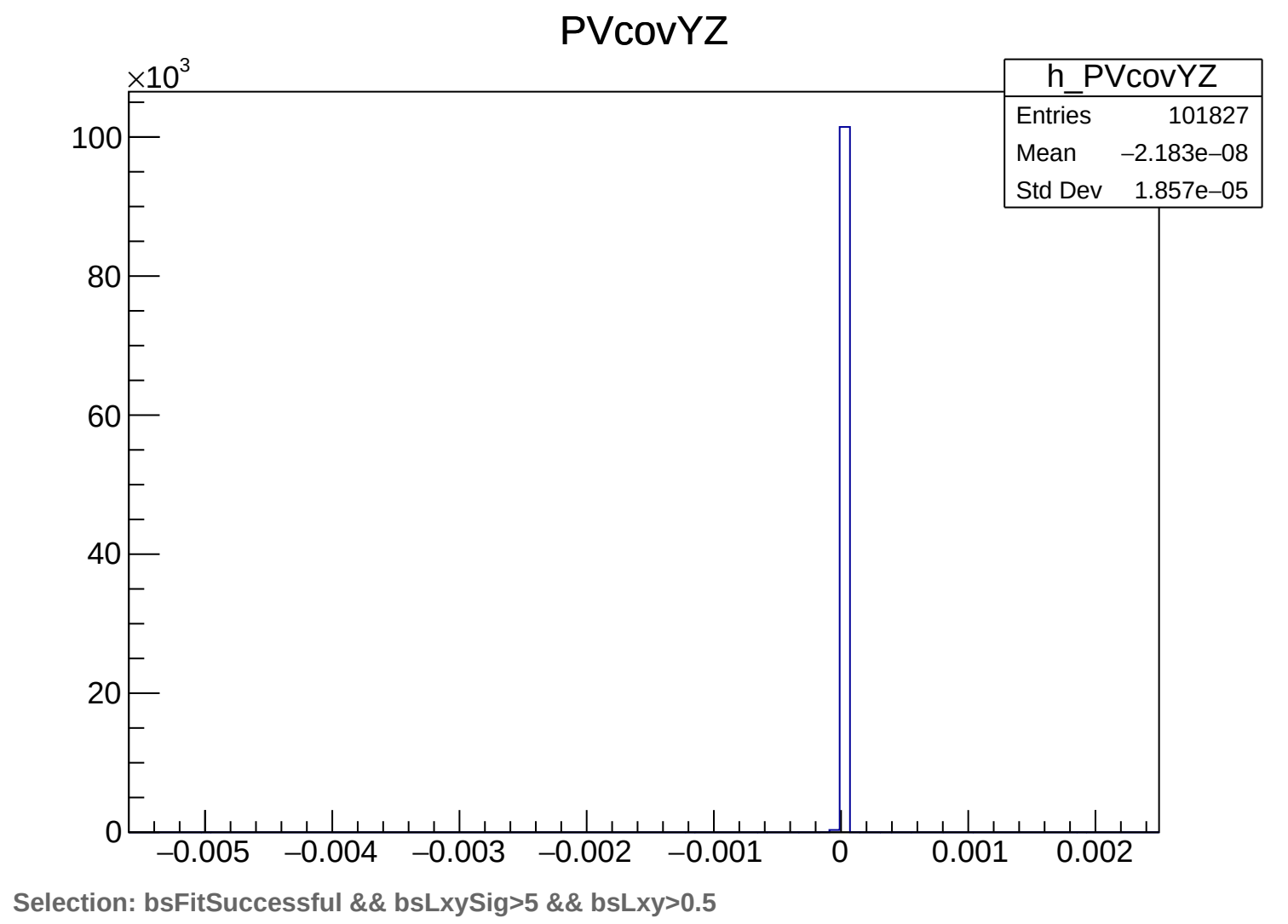
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


PVcovXY
×10^3
h_PVcovXY
Entries 101827
Mean −2.573e−08
Std Dev 4.559e−06
100
80
60
40
20
0
−0.0012
−0.001
−0.0008
−0.0006
−0.0004
−0.0002
0
0.0002
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

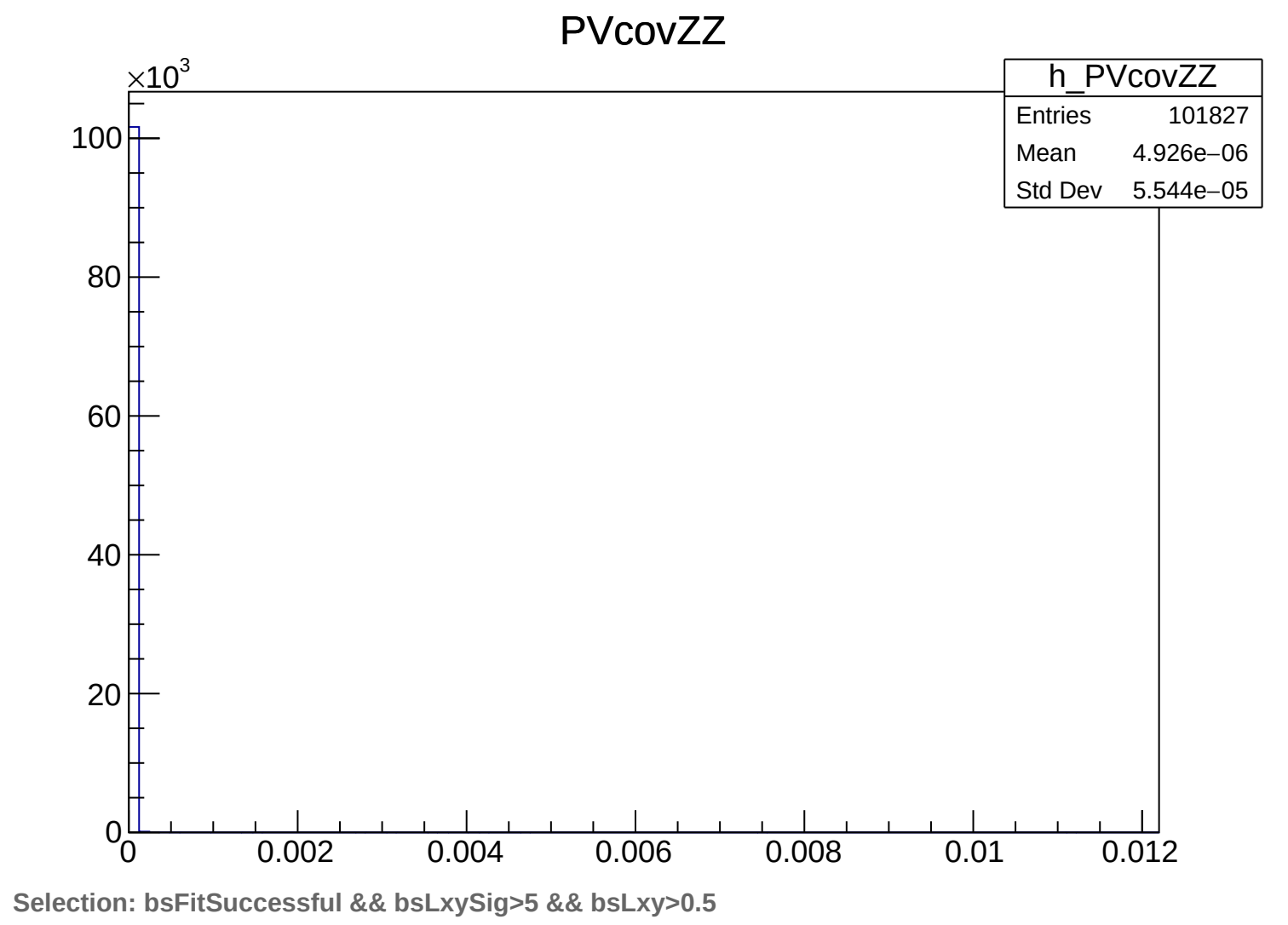


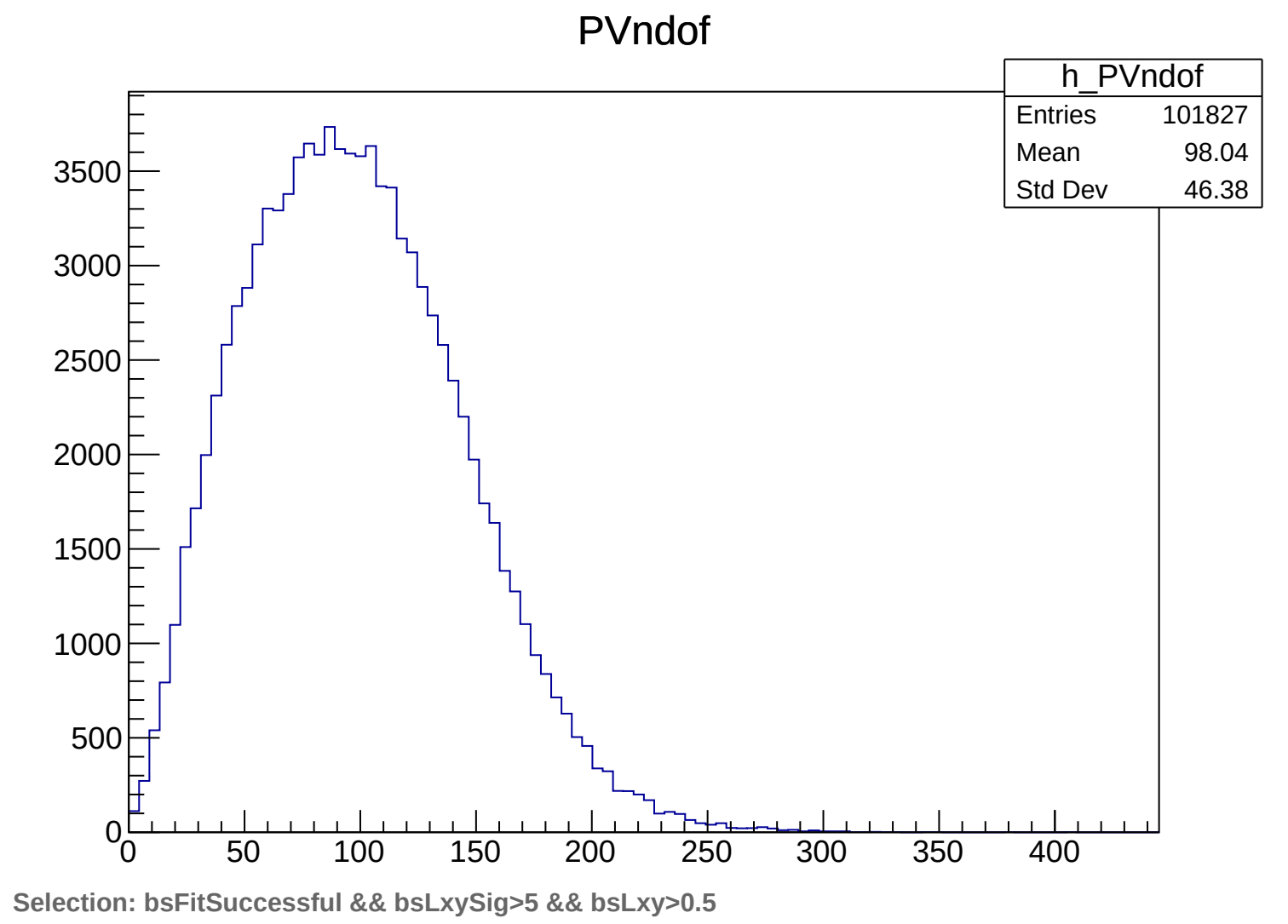
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

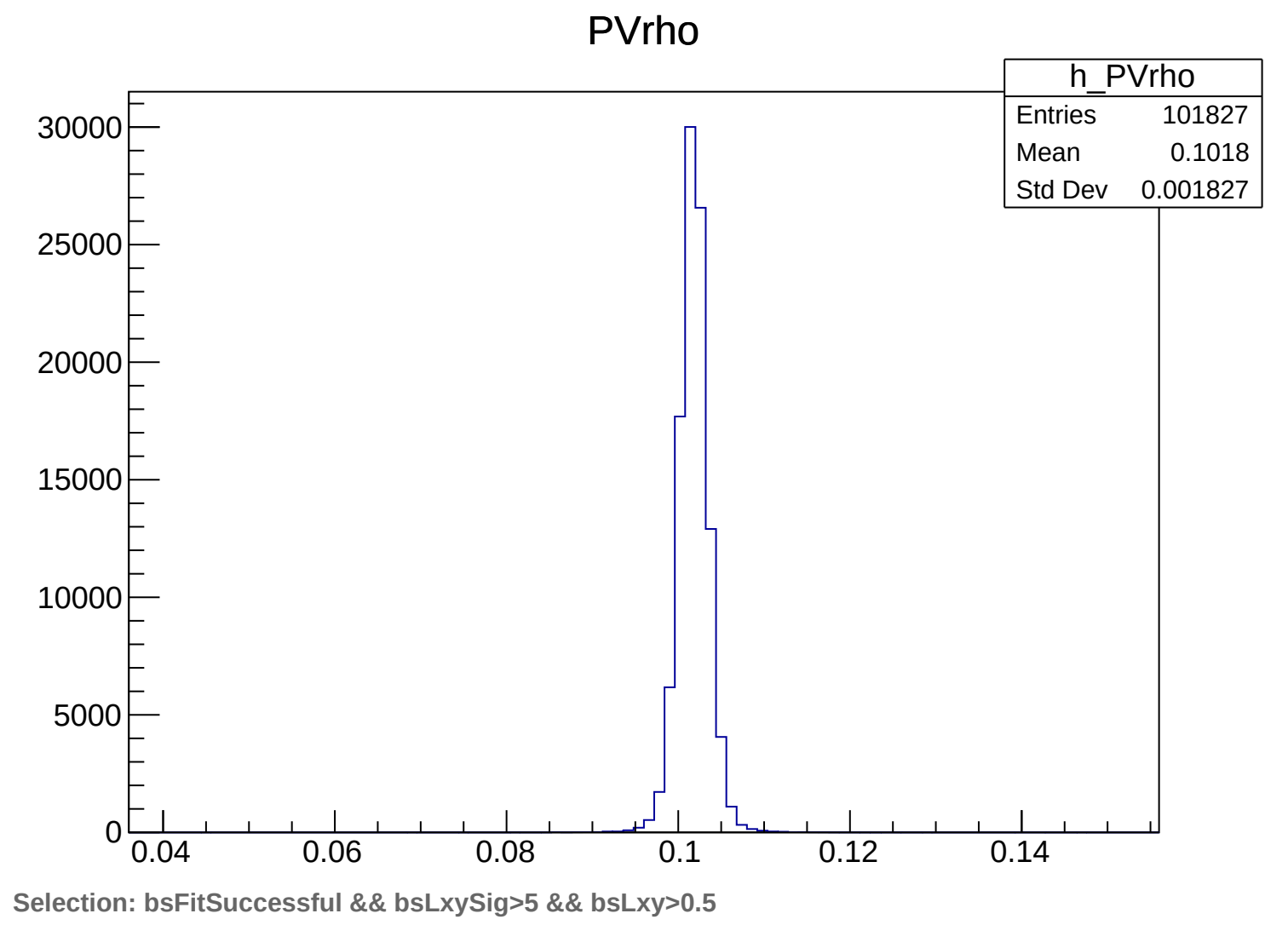

PVcovYY
×10^3
h_PVcovYY
Entries 101827
Mean 2.319e−06
Std Dev 9.713e−06
100
80
60
40
20
0
0
0.0005
0.001
0.0015
0.002
0.0025
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

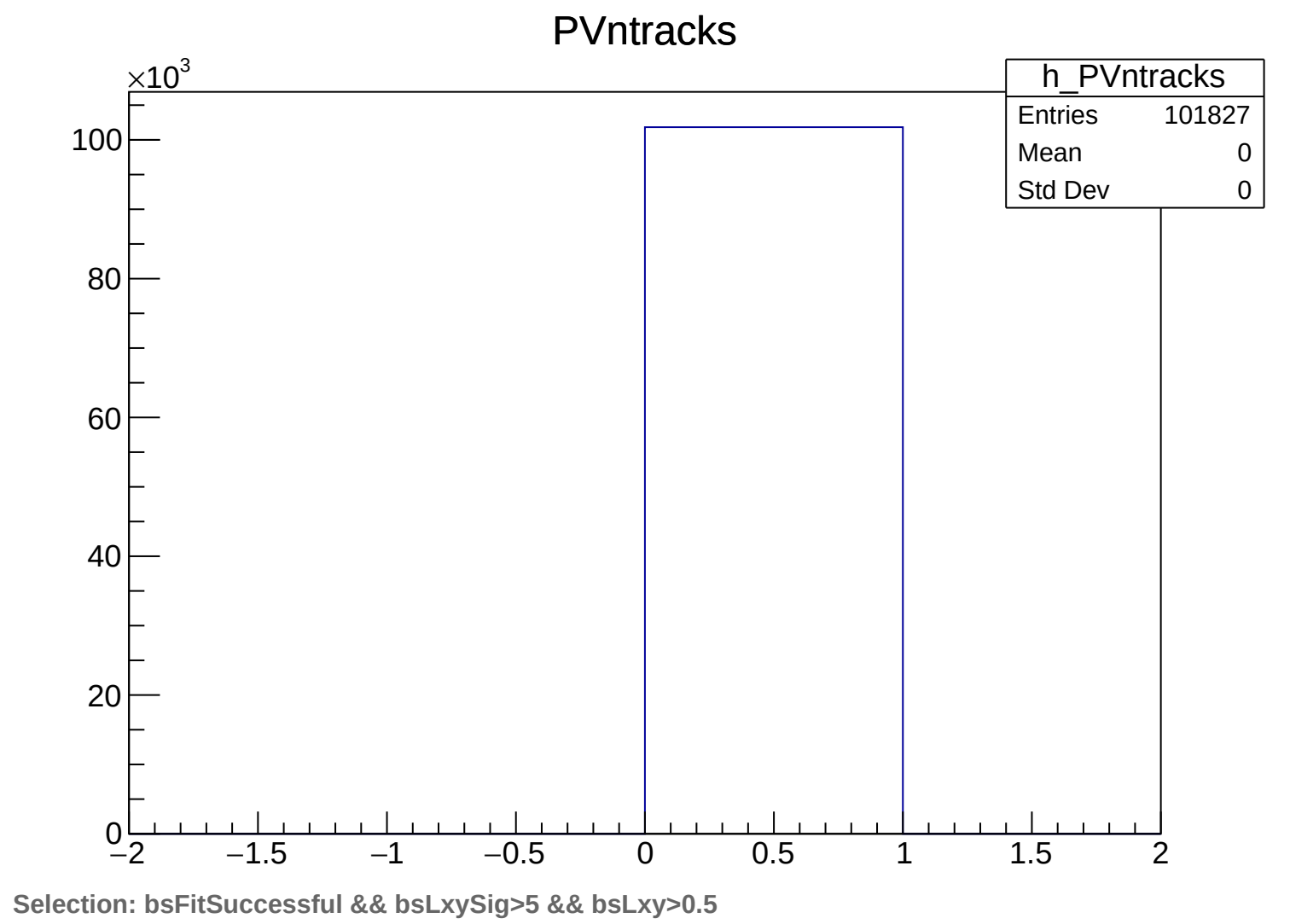


Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

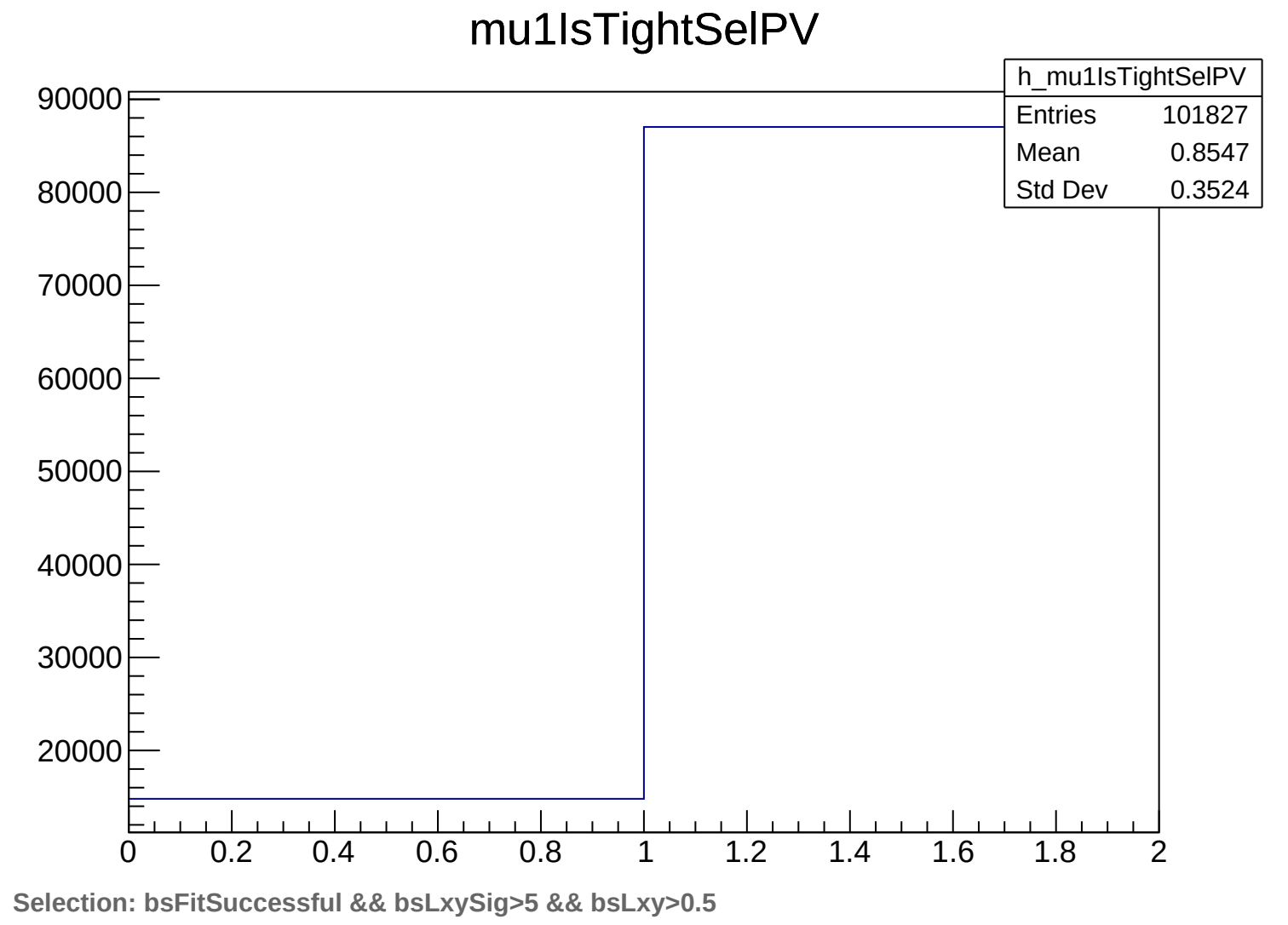

PVcovZZ
×10^3
h_PVcovZZ
Entries 101827
Mean 4.926e−06
Std Dev 5.544e−05
100
80
60
40
20
0
0
0.002
0.004
0.006
0.008
0.01
0.012
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

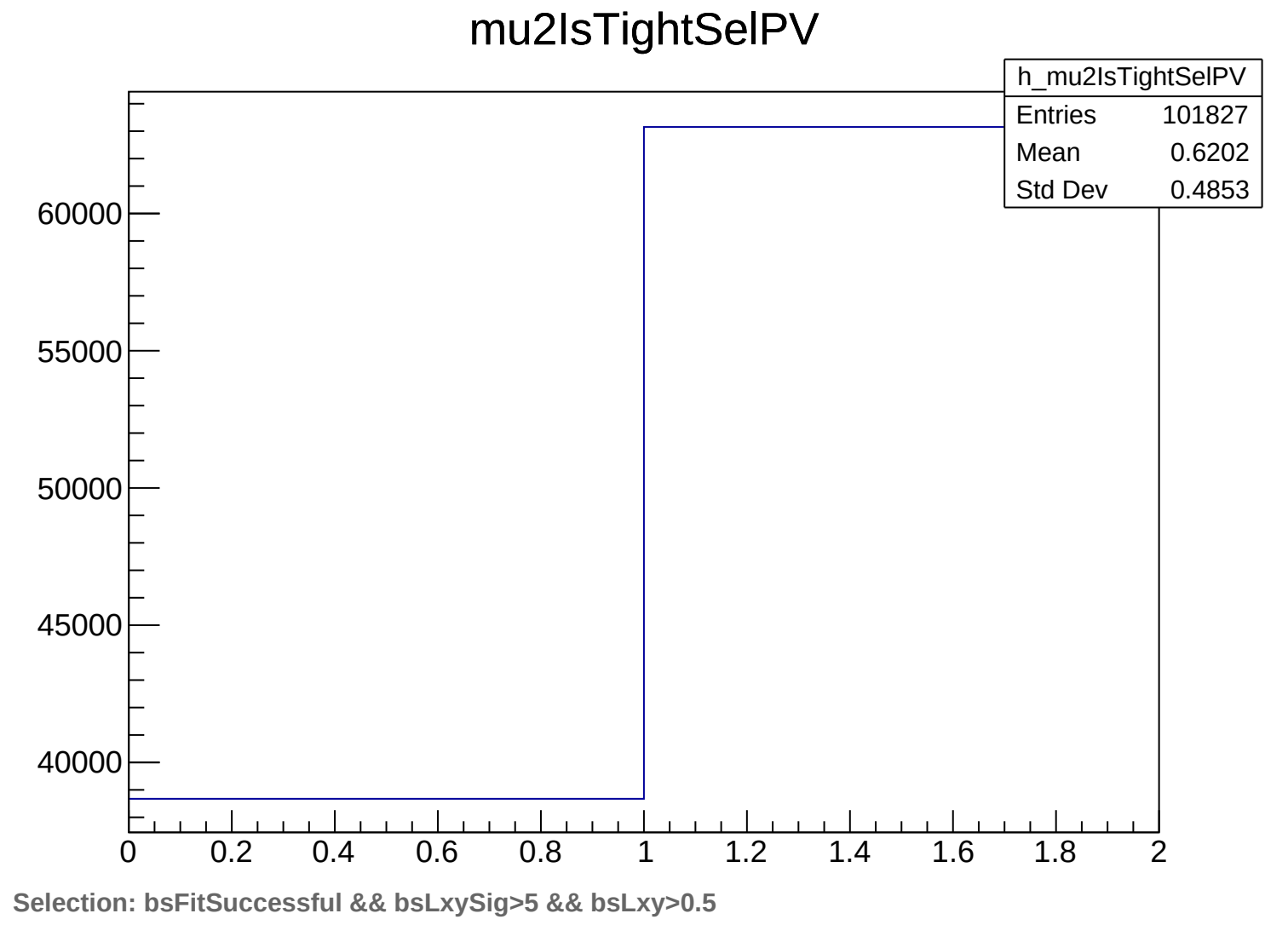

PVndof
h_PVndof
Entries 101827
Mean 98.04
Std Dev 46.38
3500
3000
2500
2000
1500
1000
500
0
0
50
100
150
200
250
300
350
400
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

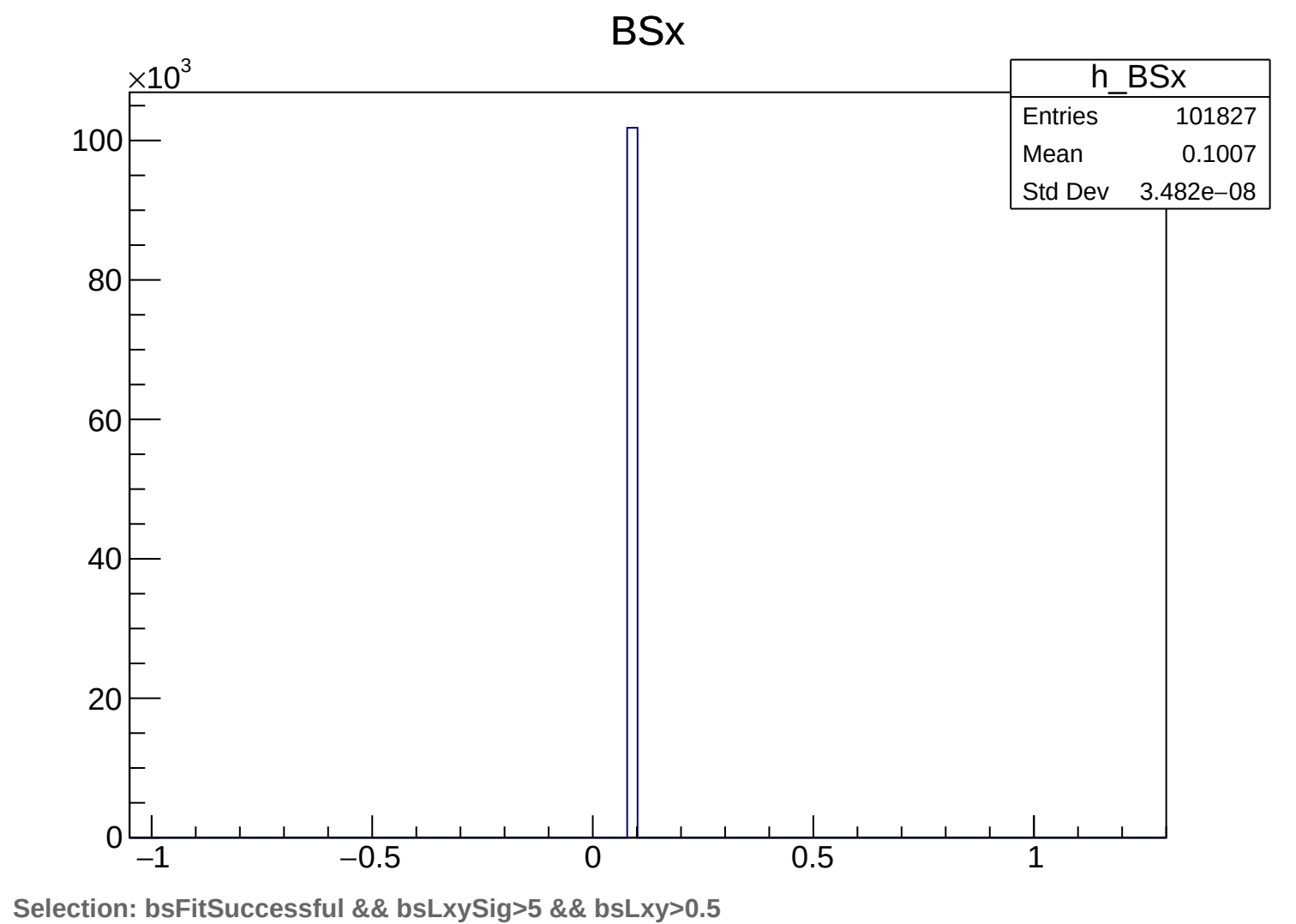

PVrho
h_PVrho
Entries 101827
Mean 0.1018
Std Dev 0.001827
30000
25000
20000
15000
10000
5000
0
0.04
0.06
0.08
0.1
0.12
0.14
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5



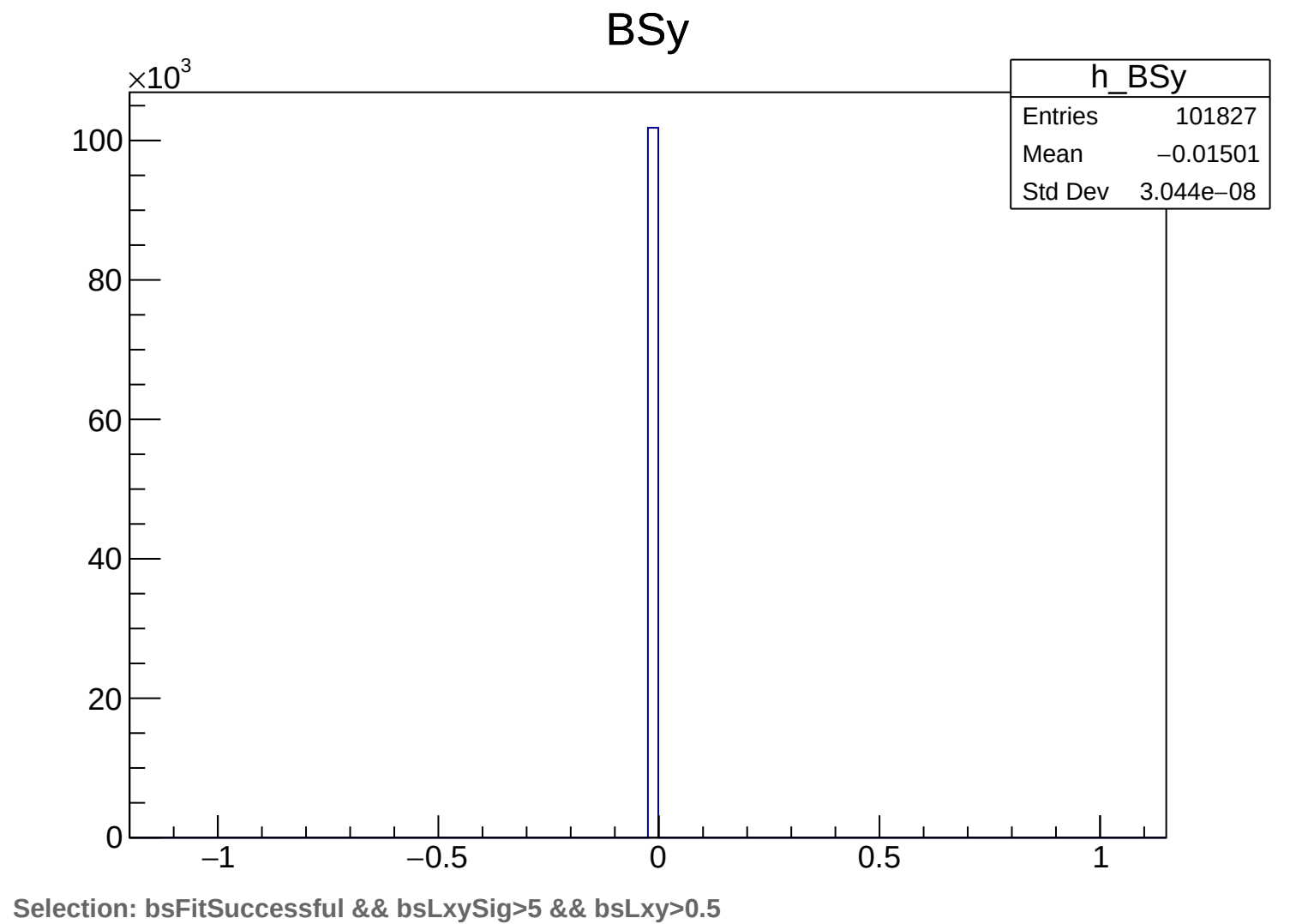
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

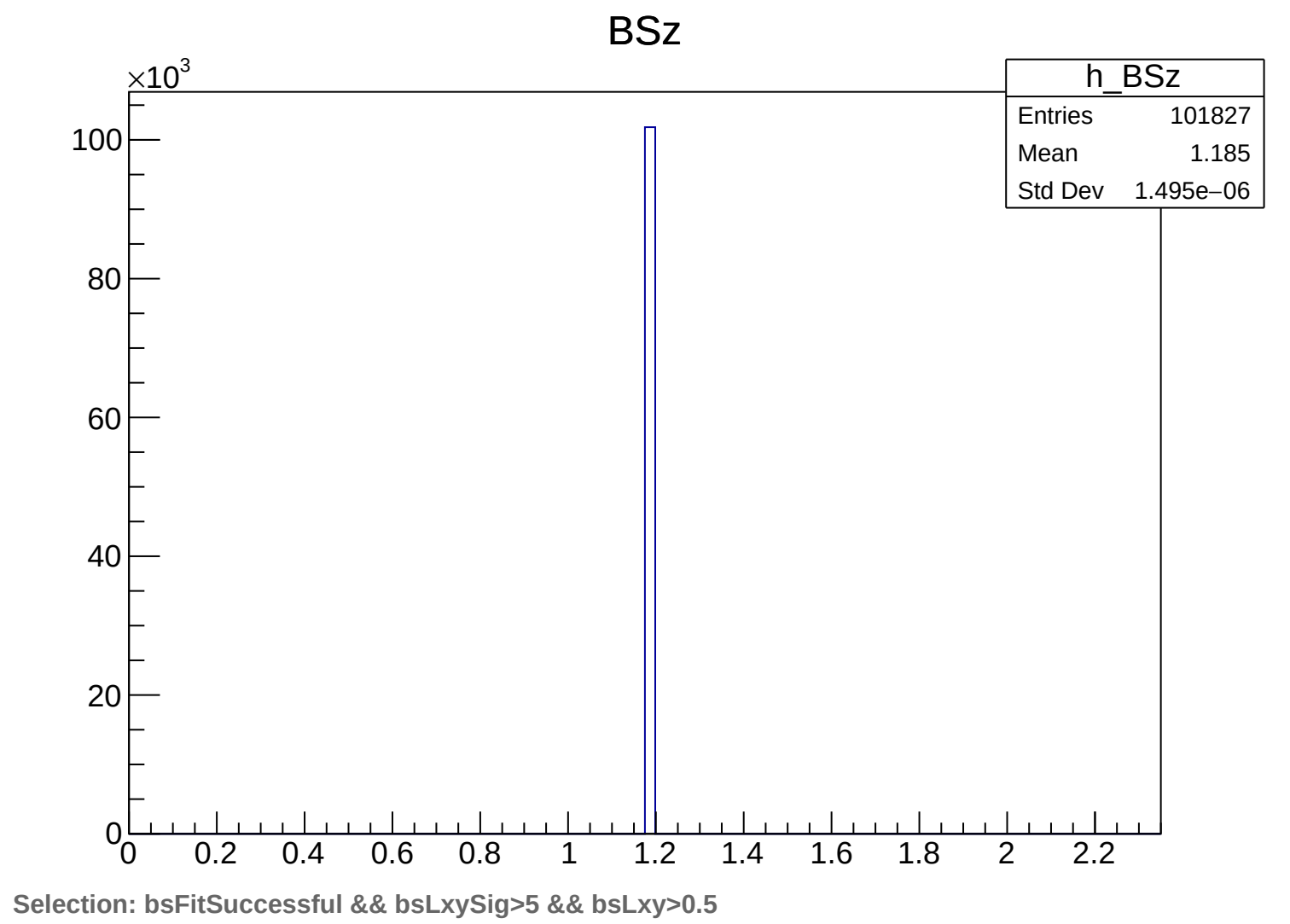

mu1IsTightSelPV
h_mu1IsTightSelPV
Entries 101827
Mean 0.8547
Std Dev 0.3524
90000
80000
70000
60000
50000
40000
30000
20000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

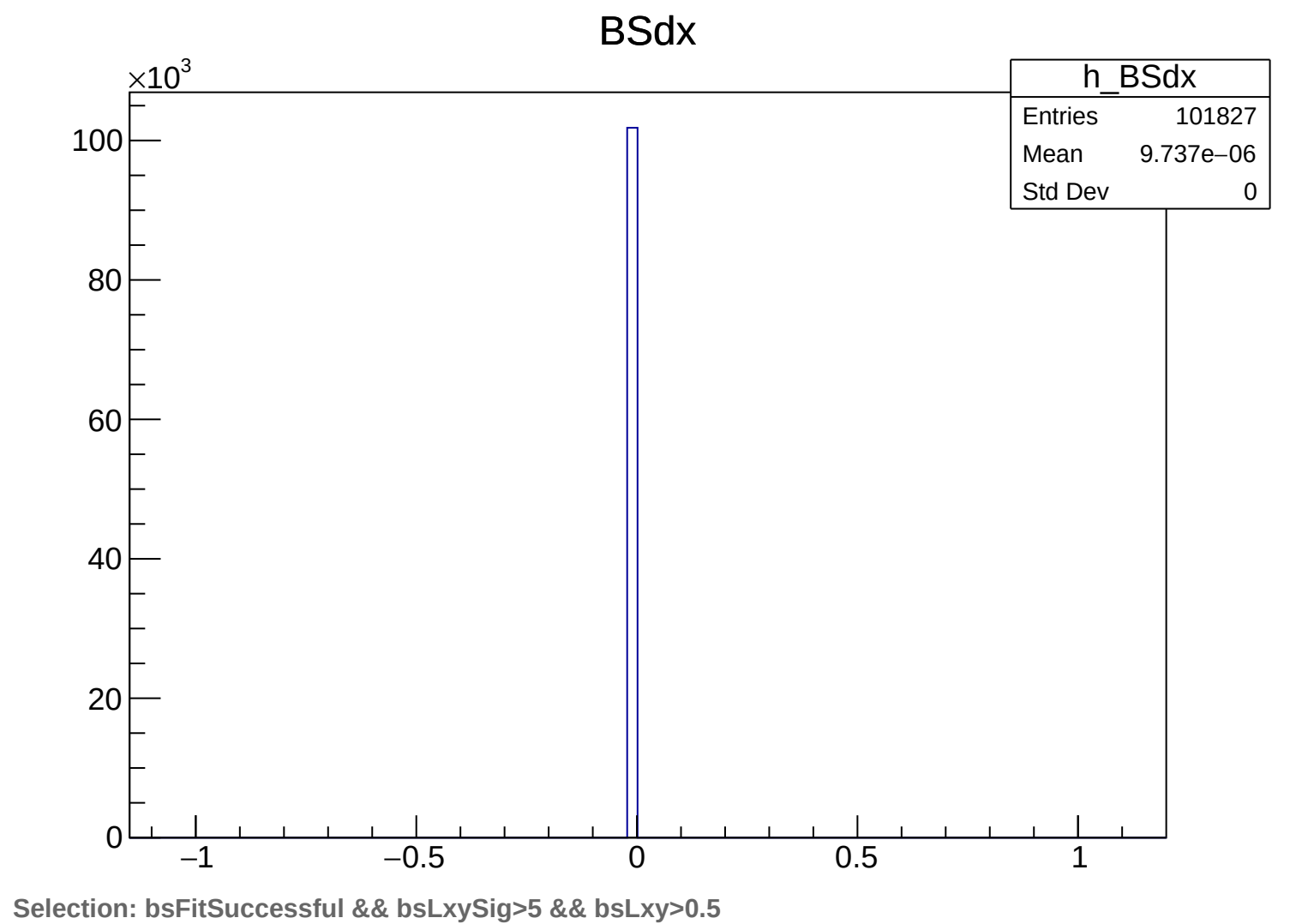

mu2IsTightSelPV
h_mu2IsTightSelPV
Entries 101827
Mean 0.6202
Std Dev 0.4853
60000
55000
50000
45000
40000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5



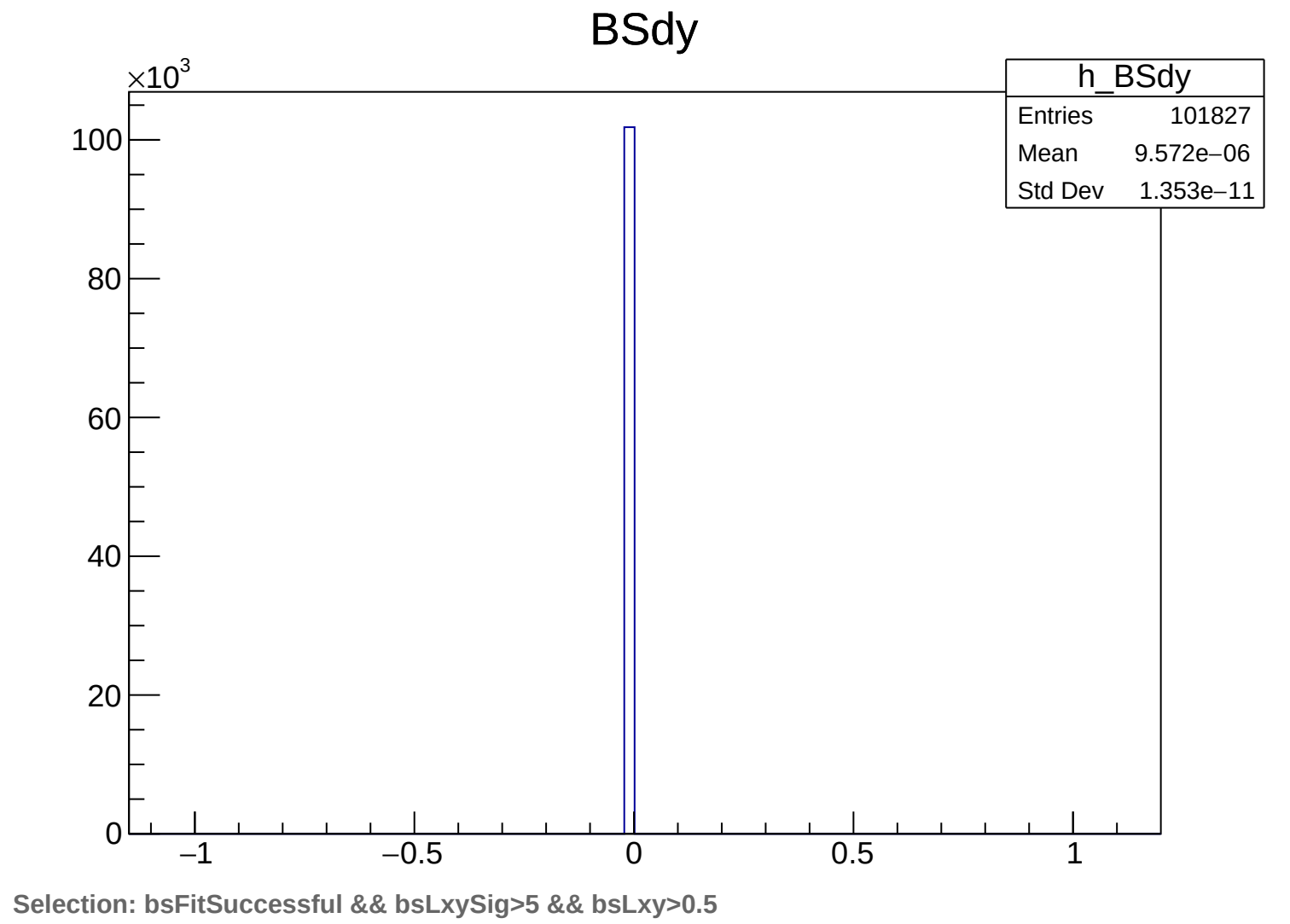
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


BSy
h_BSy
Entries 101827
Mean −0.01501
Std Dev 3.044e−08
×10^3
100
80
60
40
20
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

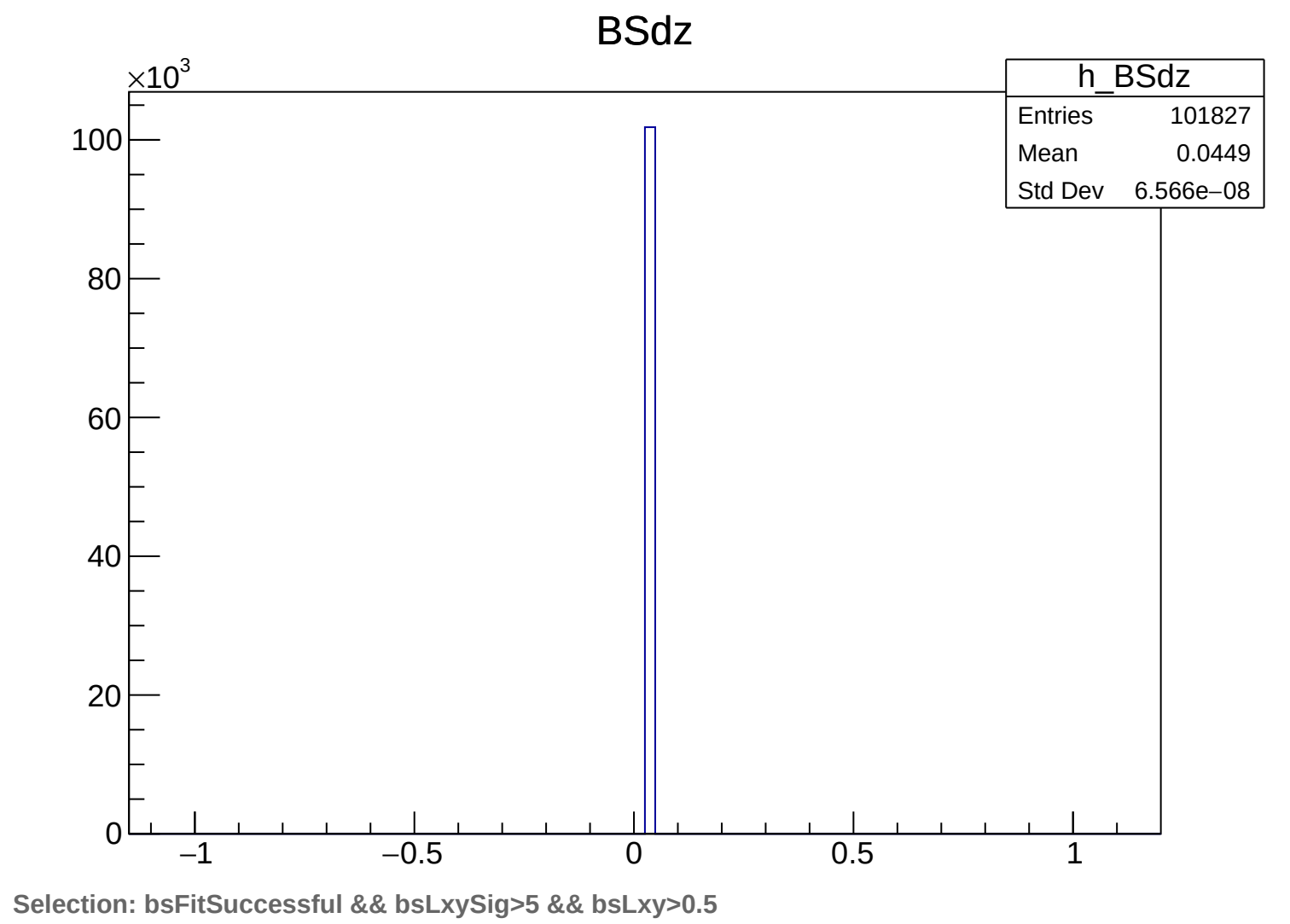

BSz
h_BSz
Entries 101827
Mean 1.185
Std Dev 1.495e−06
×10^3
100
80
60
40
20
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
2.2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5



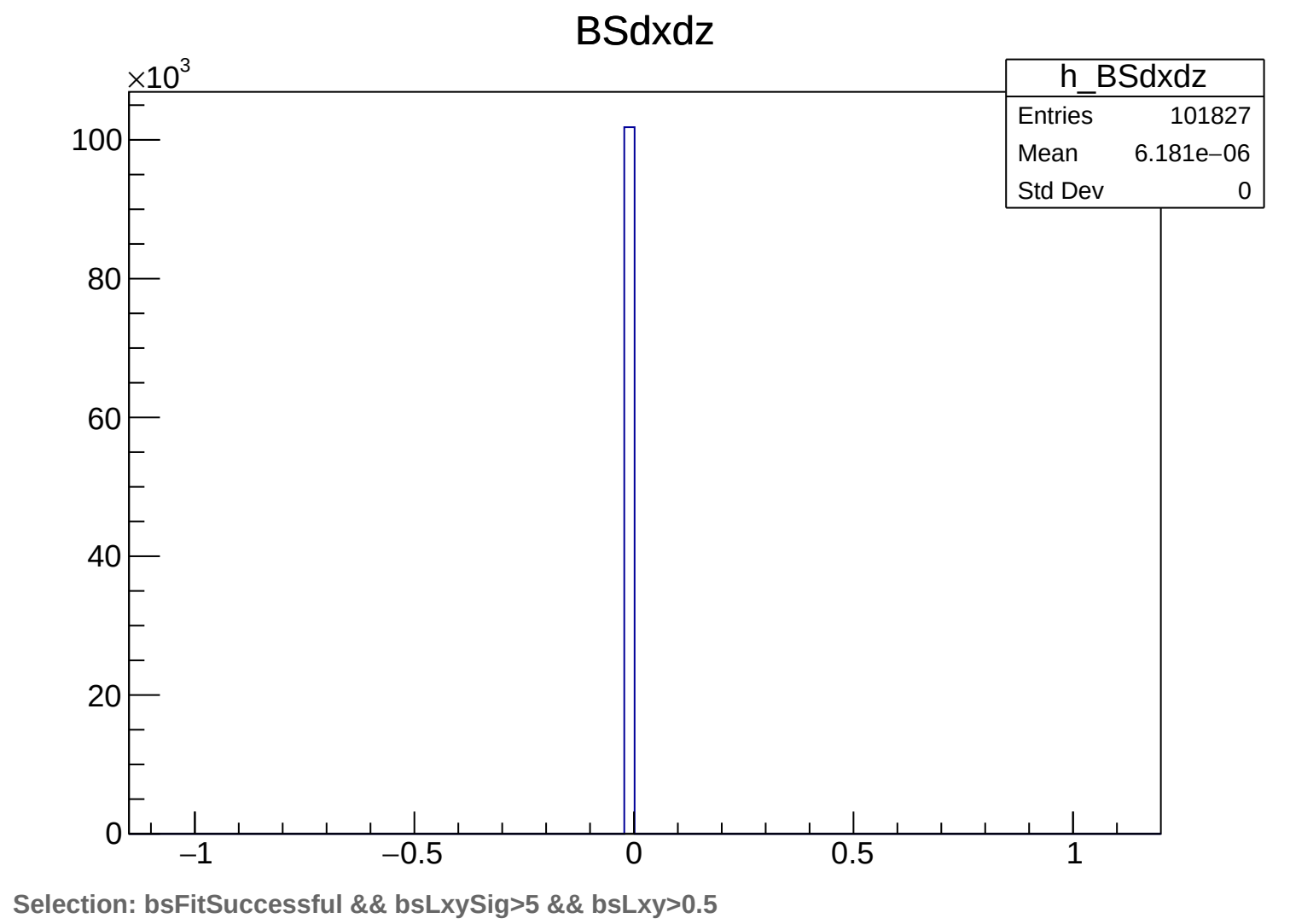
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

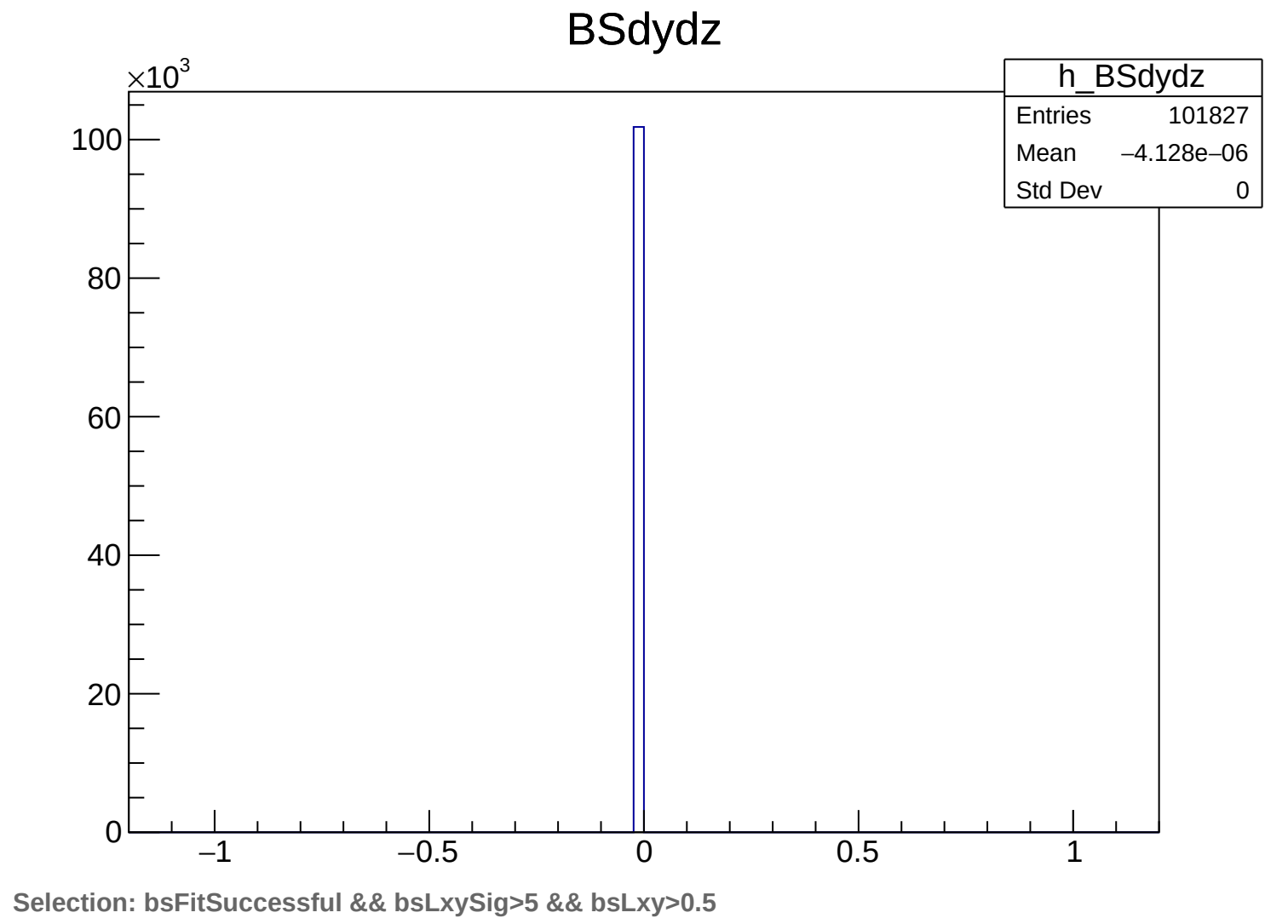


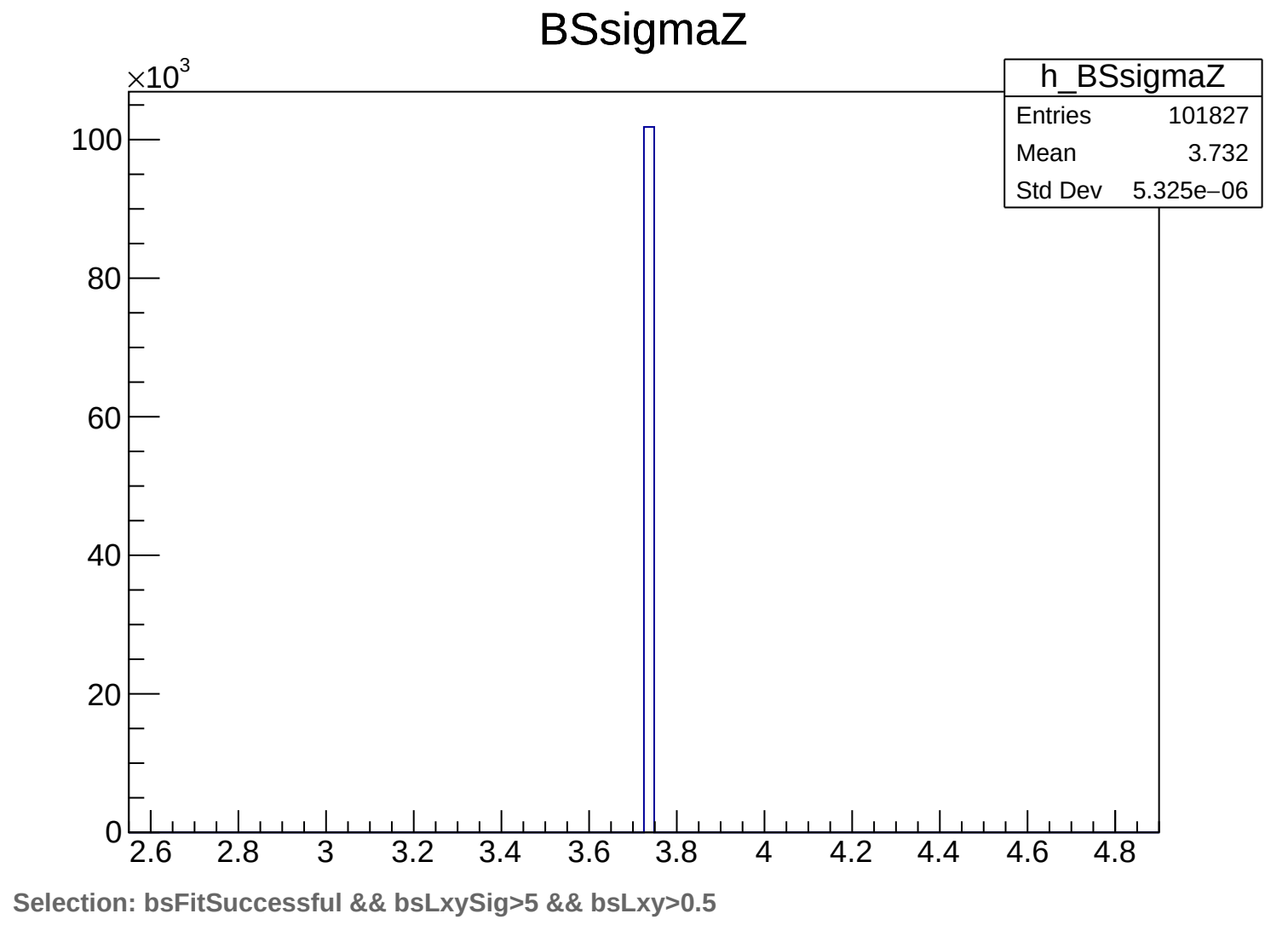
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

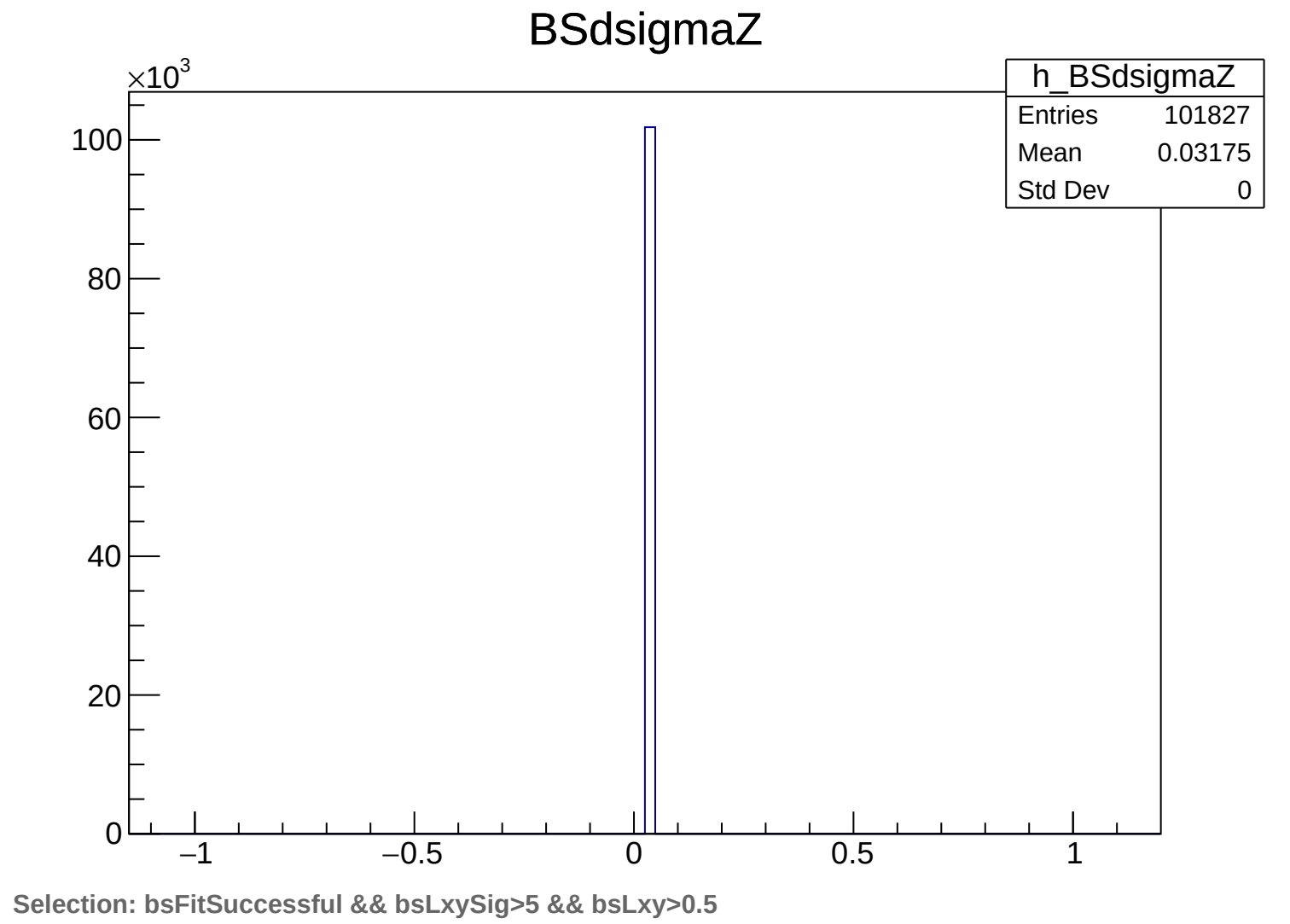


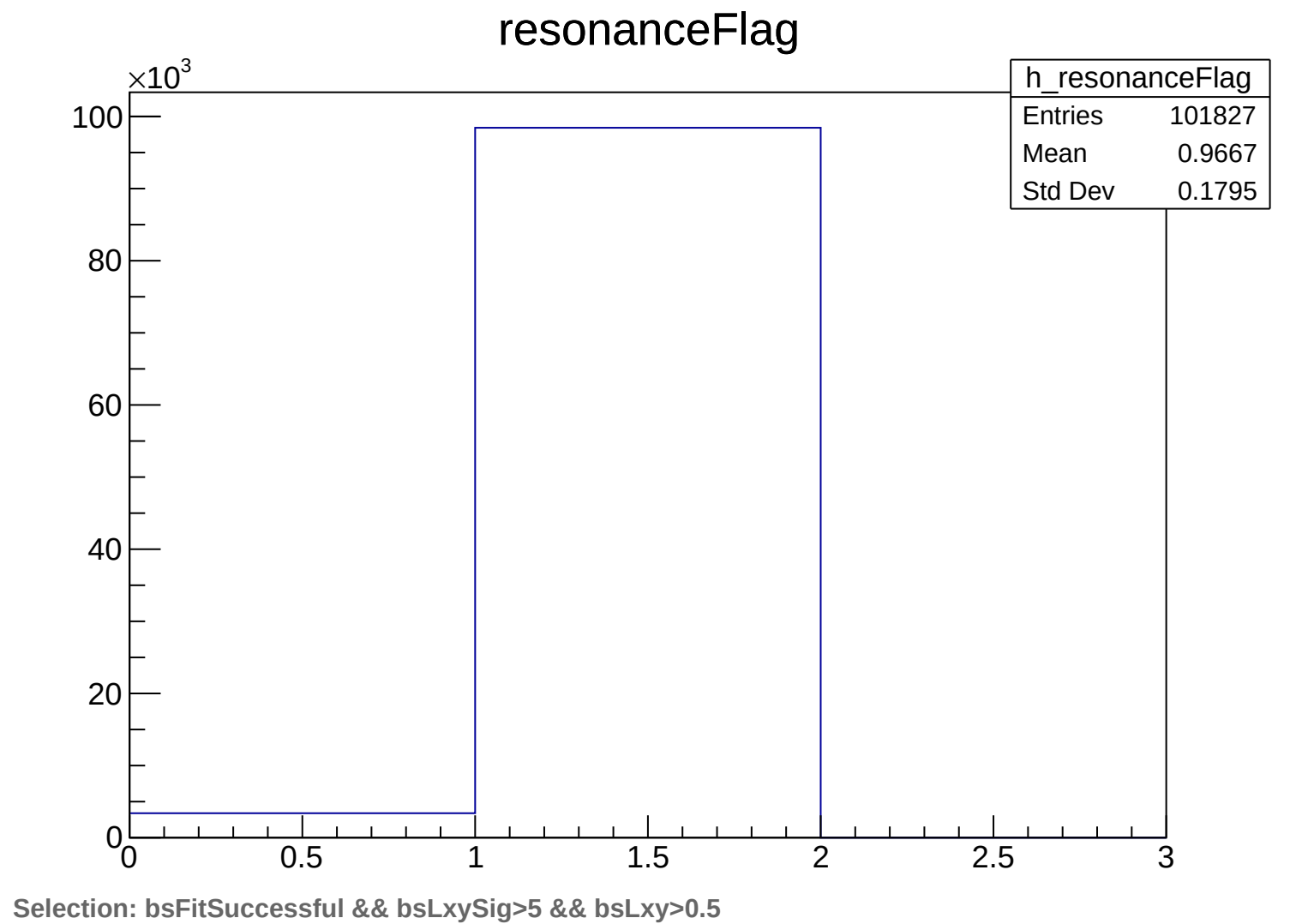
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


BSdxdz
h_BSdxdz
Entries 101827
Mean 6.181e−06
Std Dev 0
×10^3
100
80
60
40
20
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

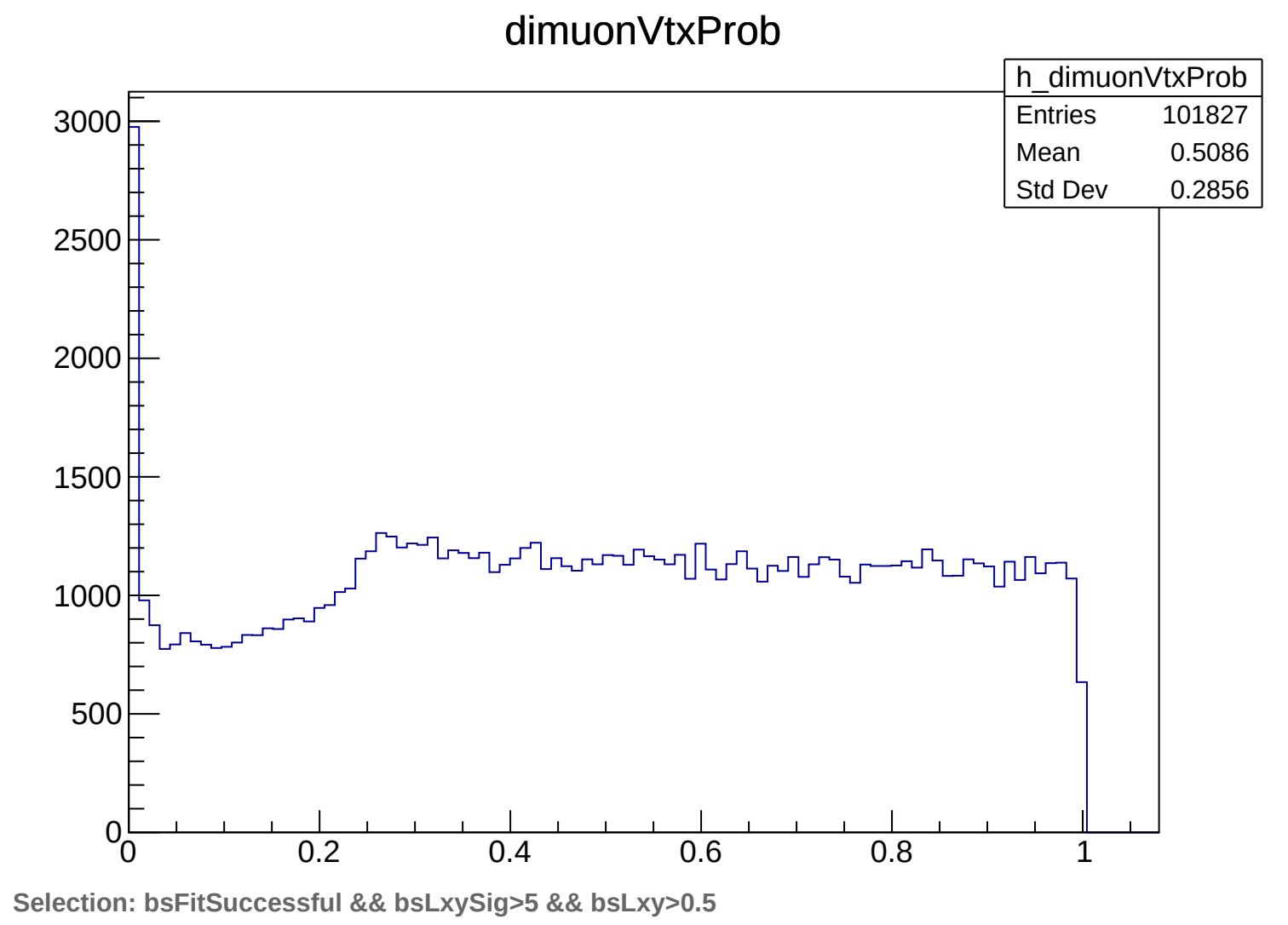

BSdydz
h_BSdydz
Entries 101827
Mean –4.128e–06
Std Dev 0
×10^3
100
80
60
40
20
0
–1
–0.5
0
0.5
1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

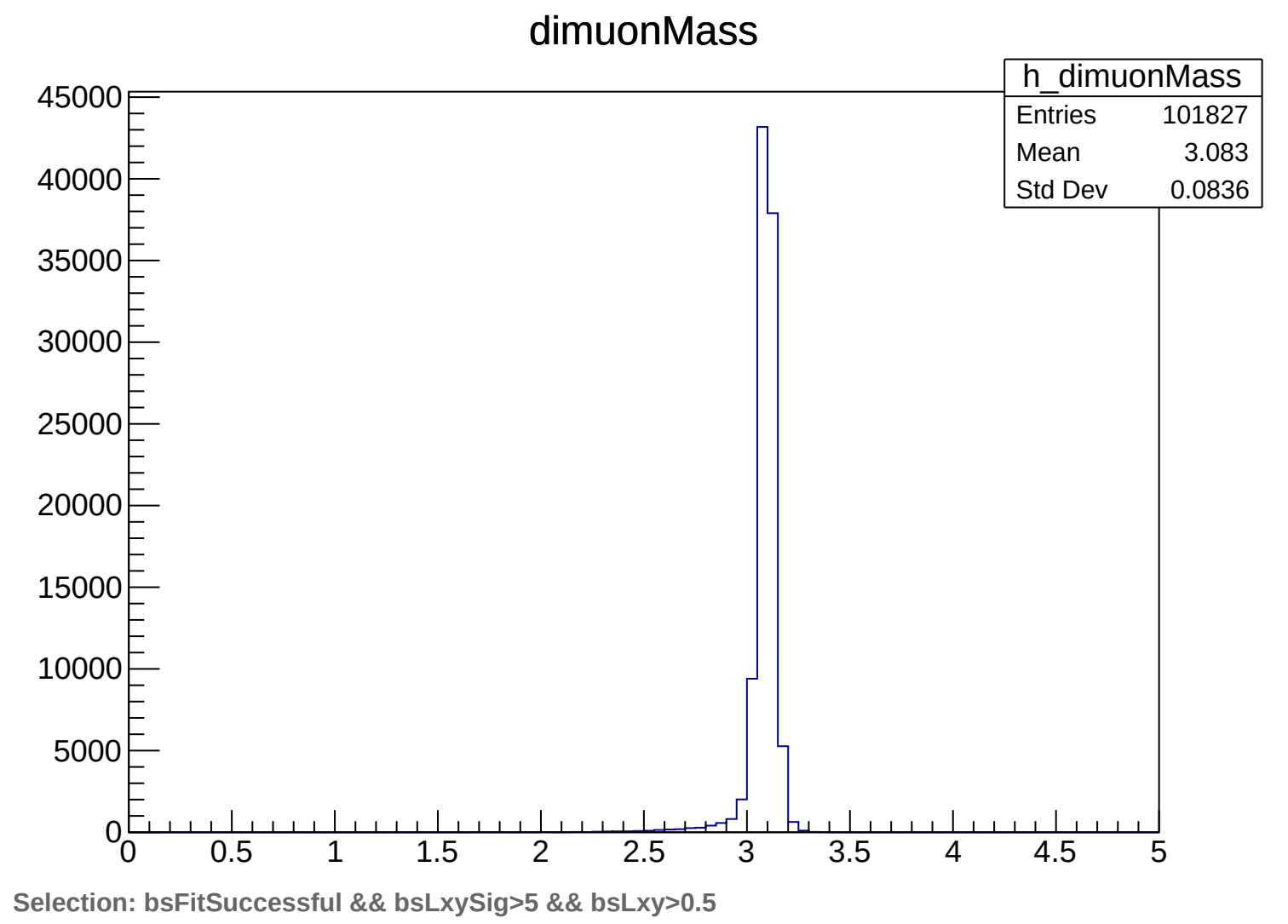

BSsigmaZ
×10^3
h_BSsigmaZ
Entries 101827
Mean 3.732
Std Dev 5.325e−06
100
80
60
40
20
0
2.6
2.8
3
3.2
3.4
3.6
3.8
4
4.2
4.4
4.6
4.8
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

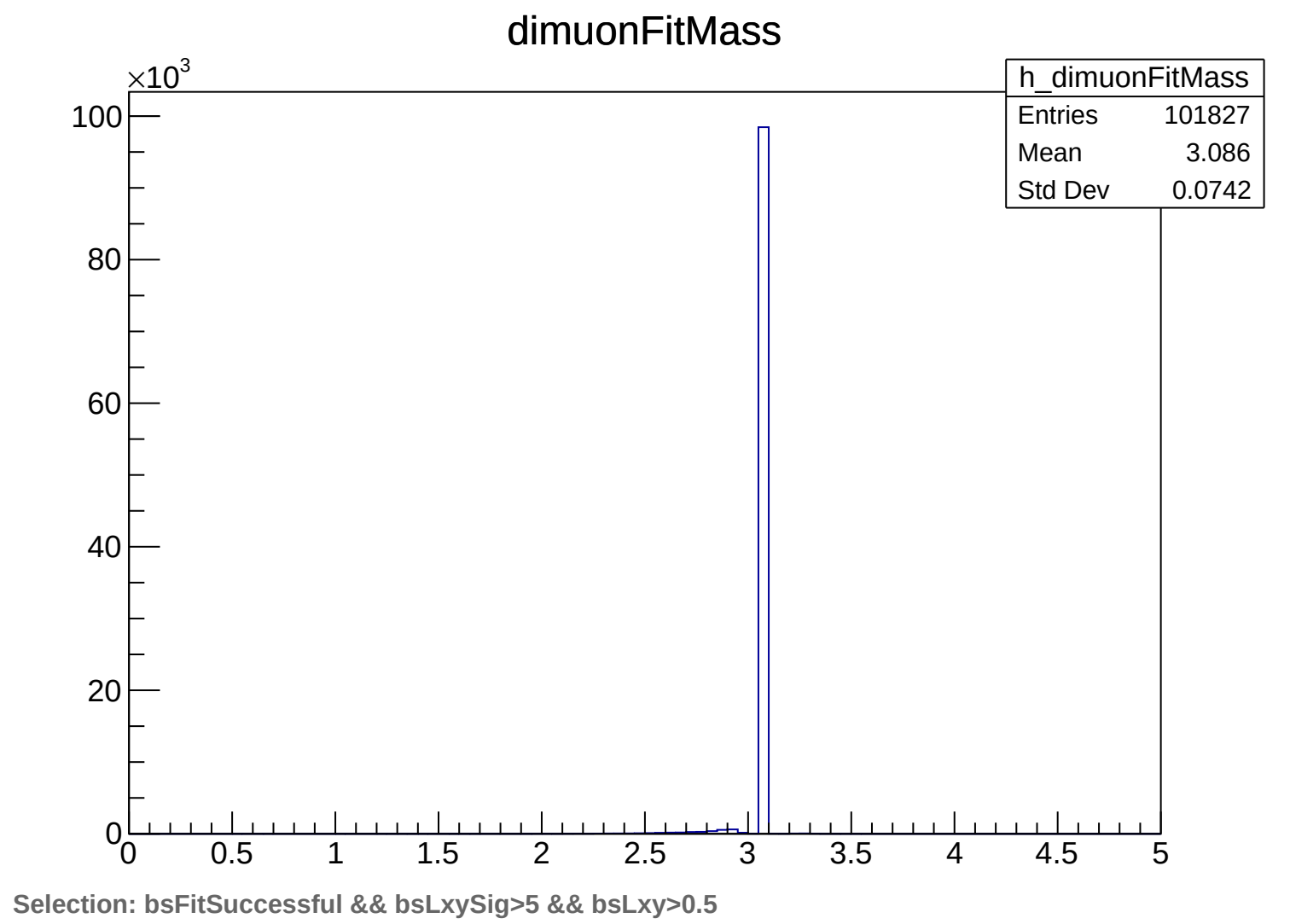

BSdsigmaZ
h_BSdsigmaZ
Entries 101827
Mean 0.03175
Std Dev 0
×10^3
100
80
60
40
20
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

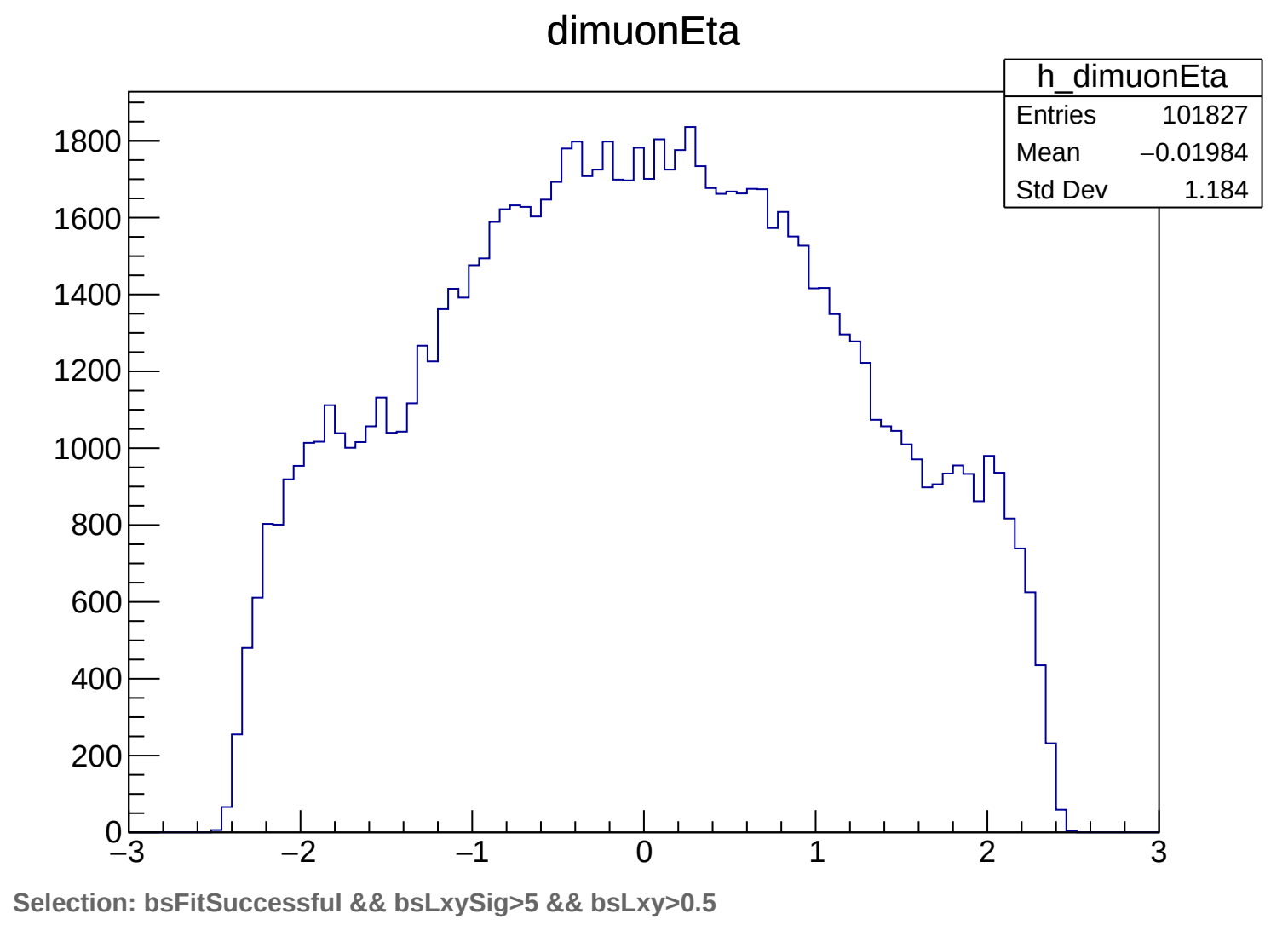


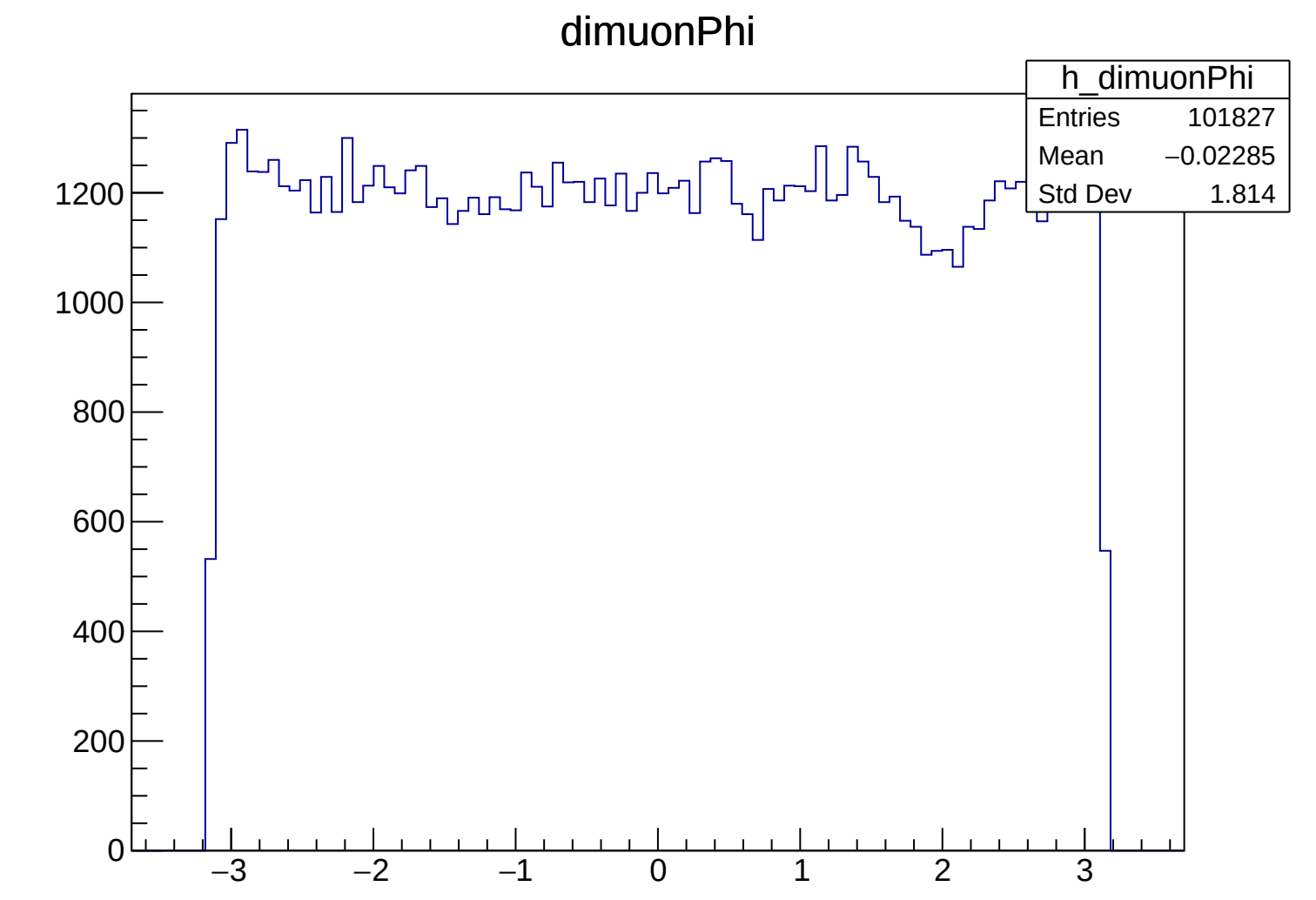
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


dimuonVtxProb
h_dimuonVtxProb
Entries 101827
Mean 0.5086
Std Dev 0.2856
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

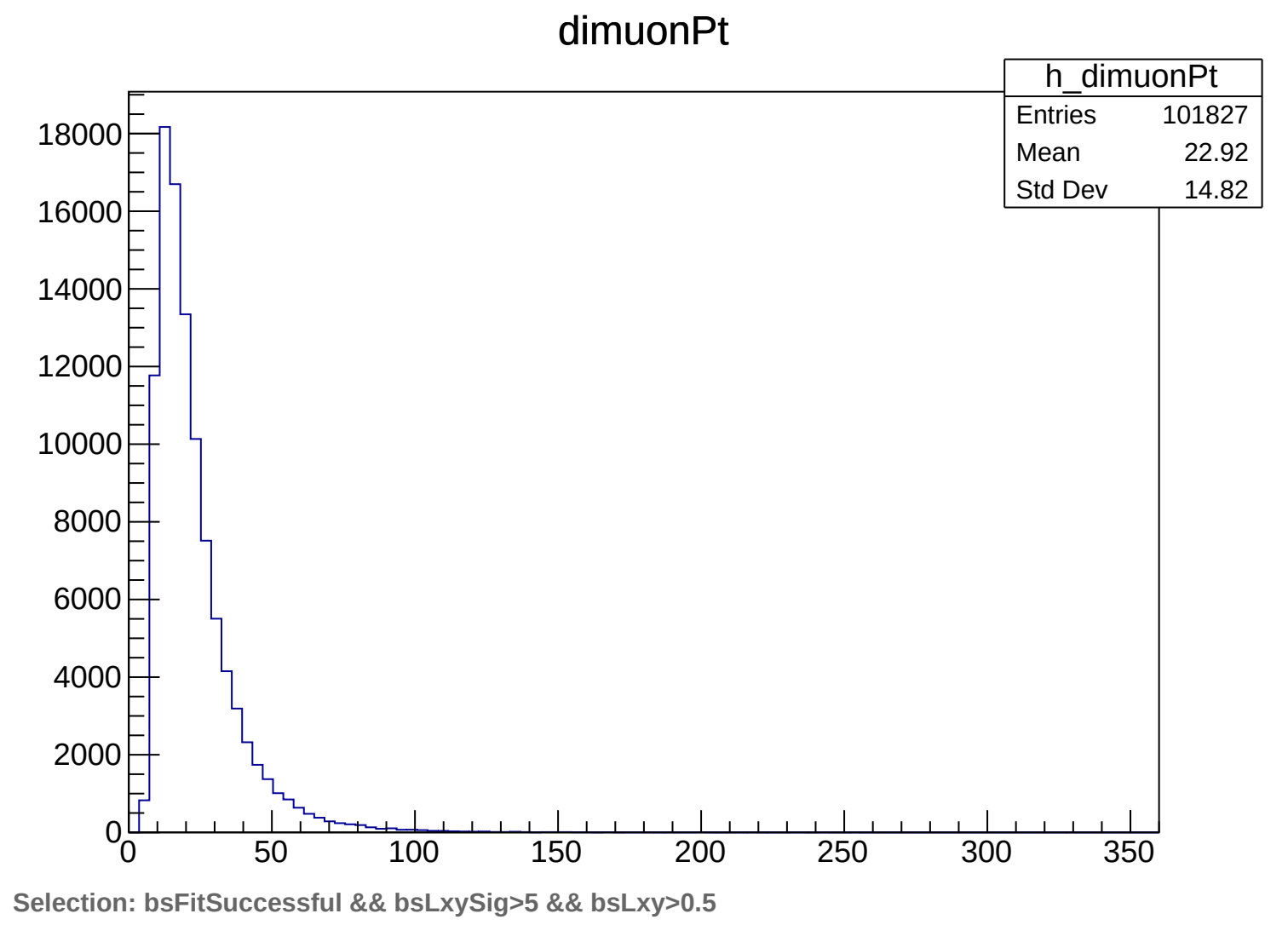

dimuonMass
h_dimuonMass
Entries 101827
Mean 3.083
Std Dev 0.0836
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

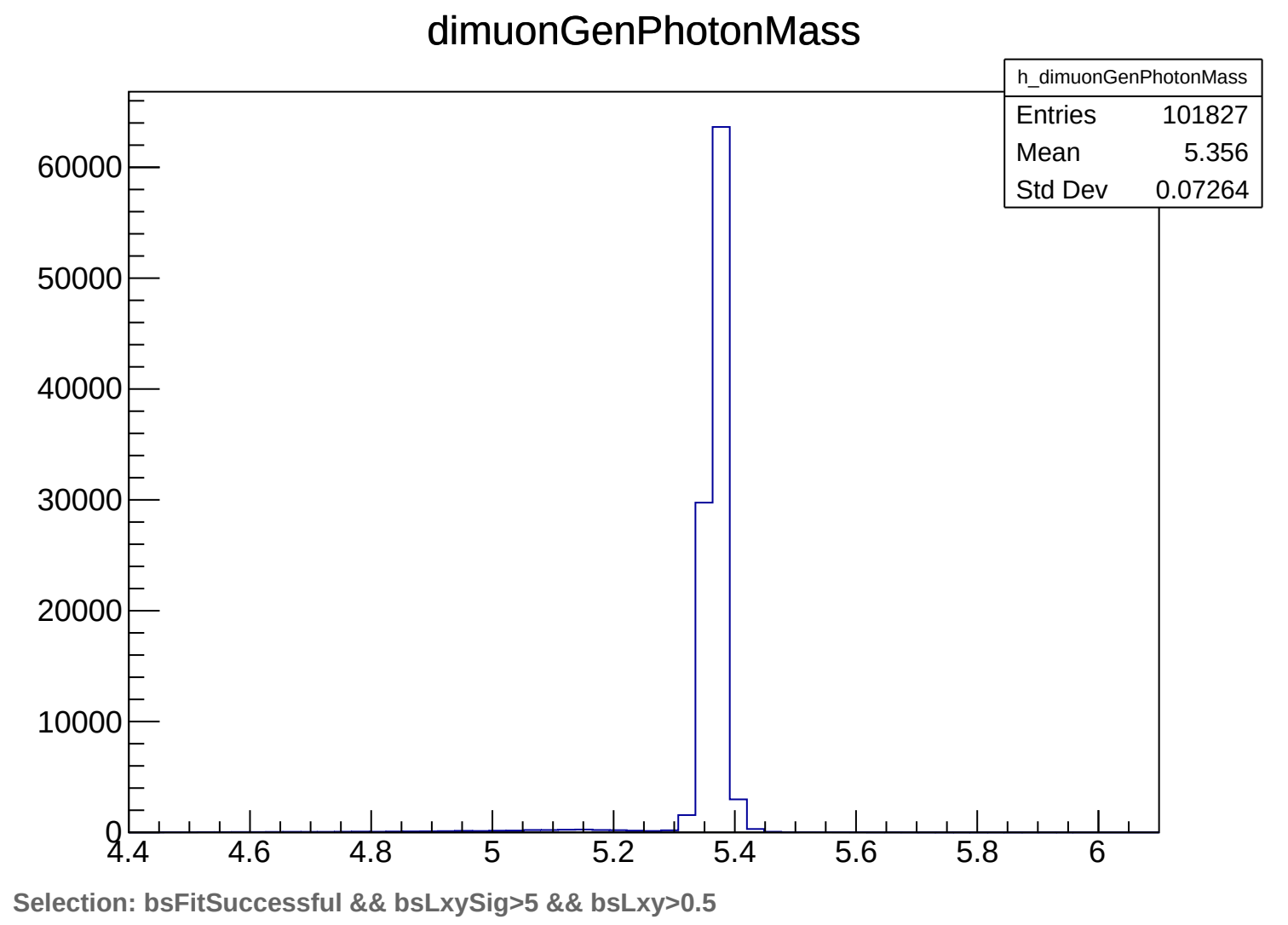

dimuonFitMass
×10^3
h_dimuonFitMass
Entries 101827
Mean 3.086
Std Dev 0.0742
100
80
60
40
20
0
0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

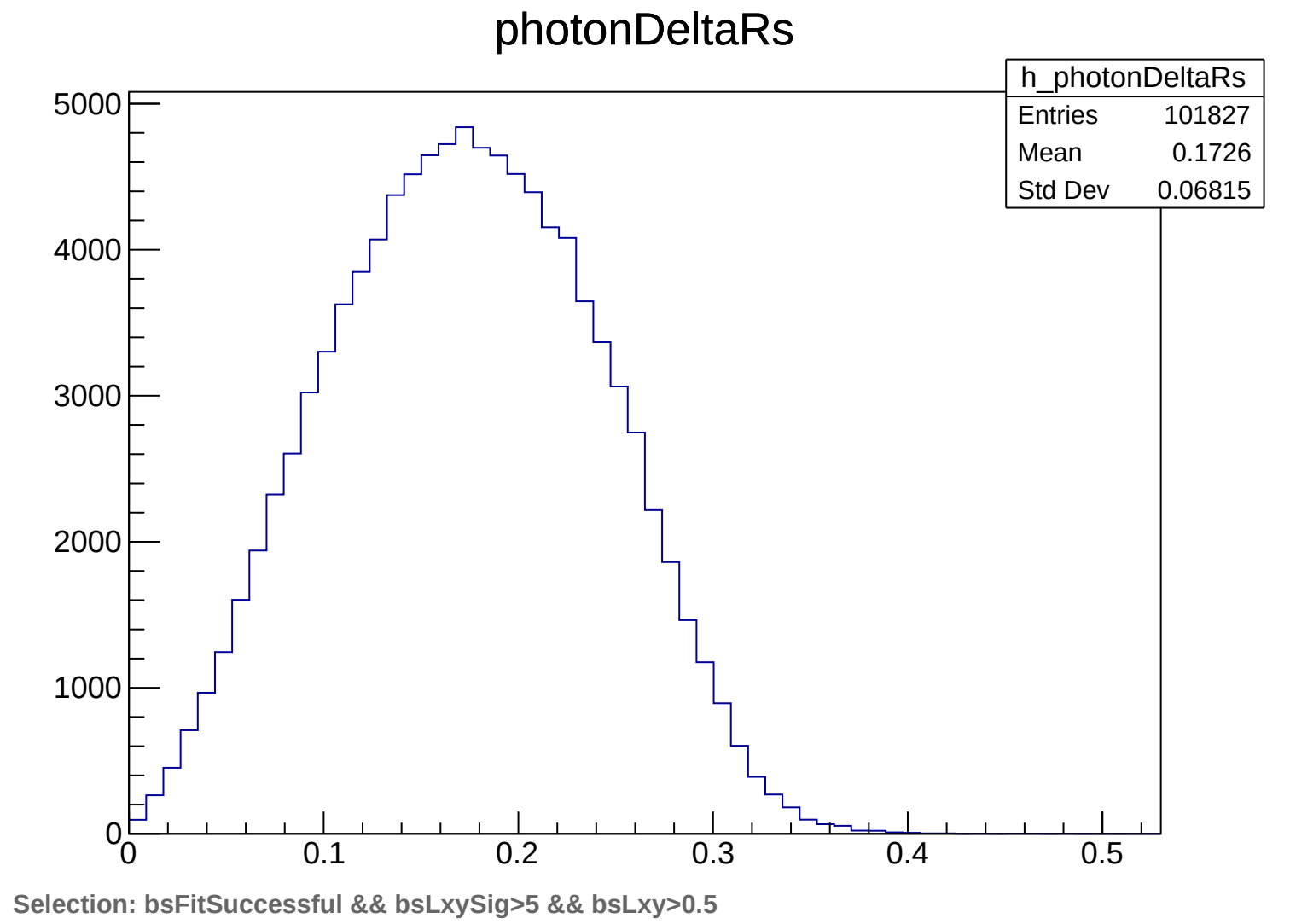

dimuonEta
h_dimuonEta
Entries 101827
Mean −0.01984
Std Dev 1.184
1800
1600
1400
1200
1000
800
600
400
200
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

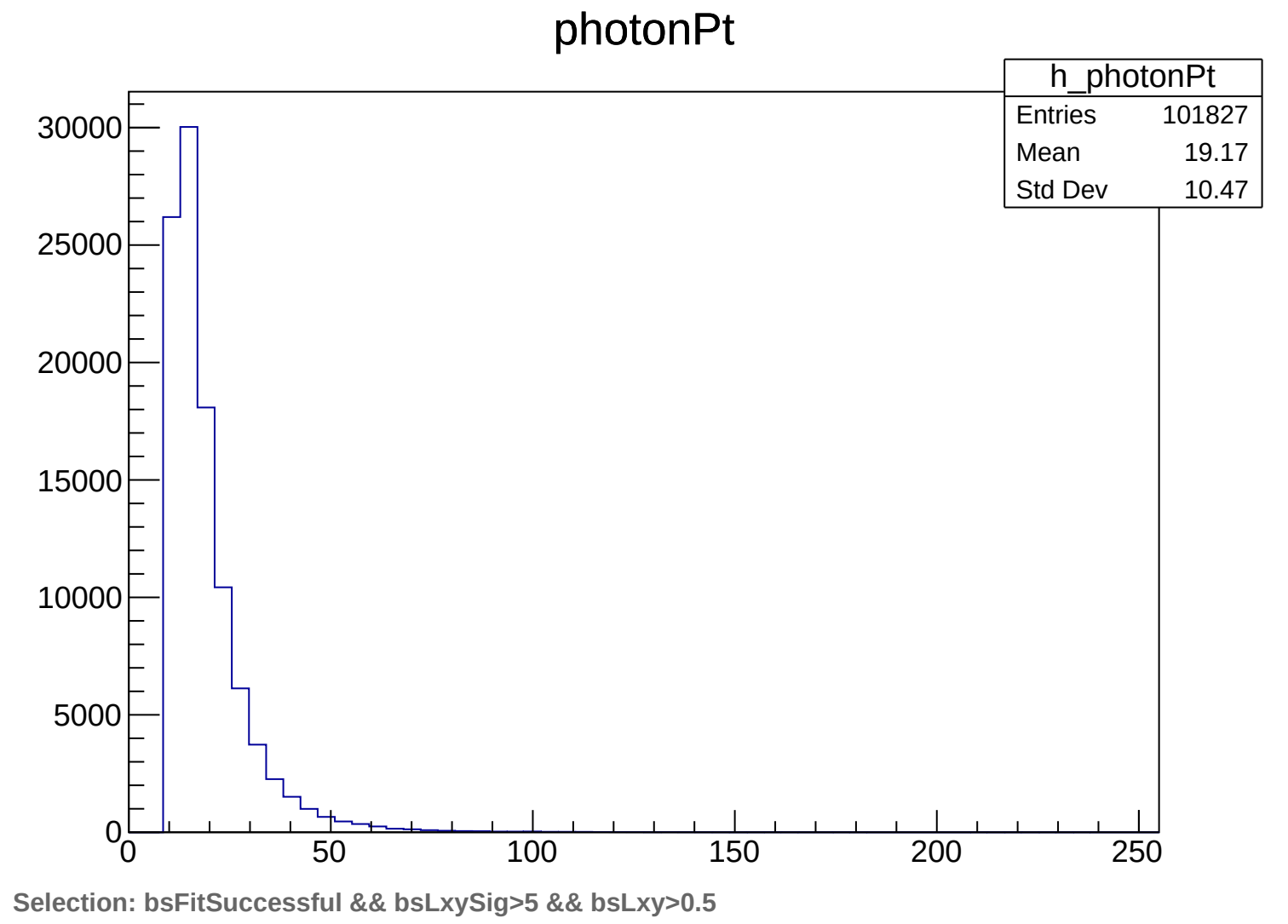


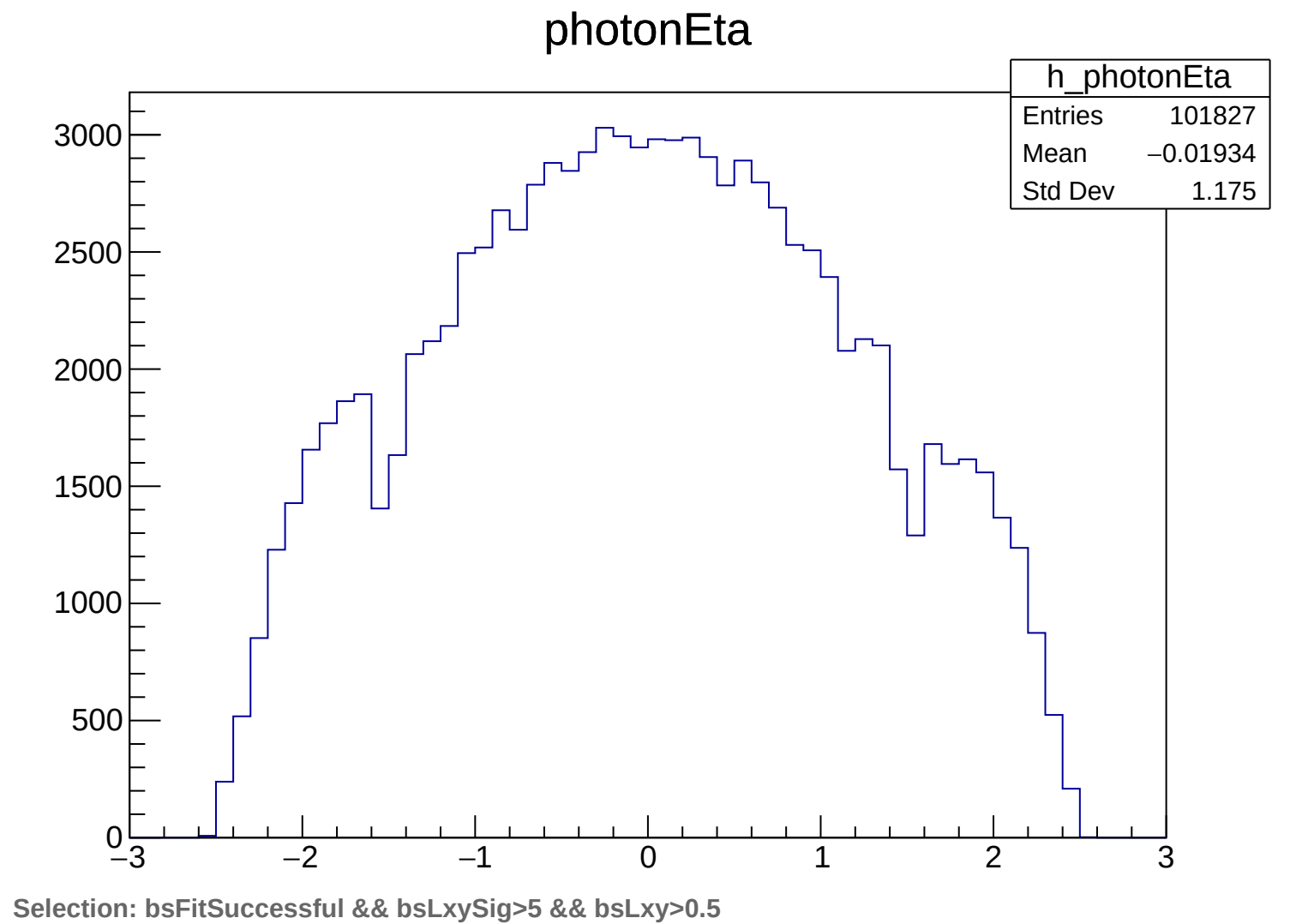
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


dimuonPt
h_dimuonPt
Entries 101827
Mean 22.92
Std Dev 14.82
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
50
100
150
200
250
300
350
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

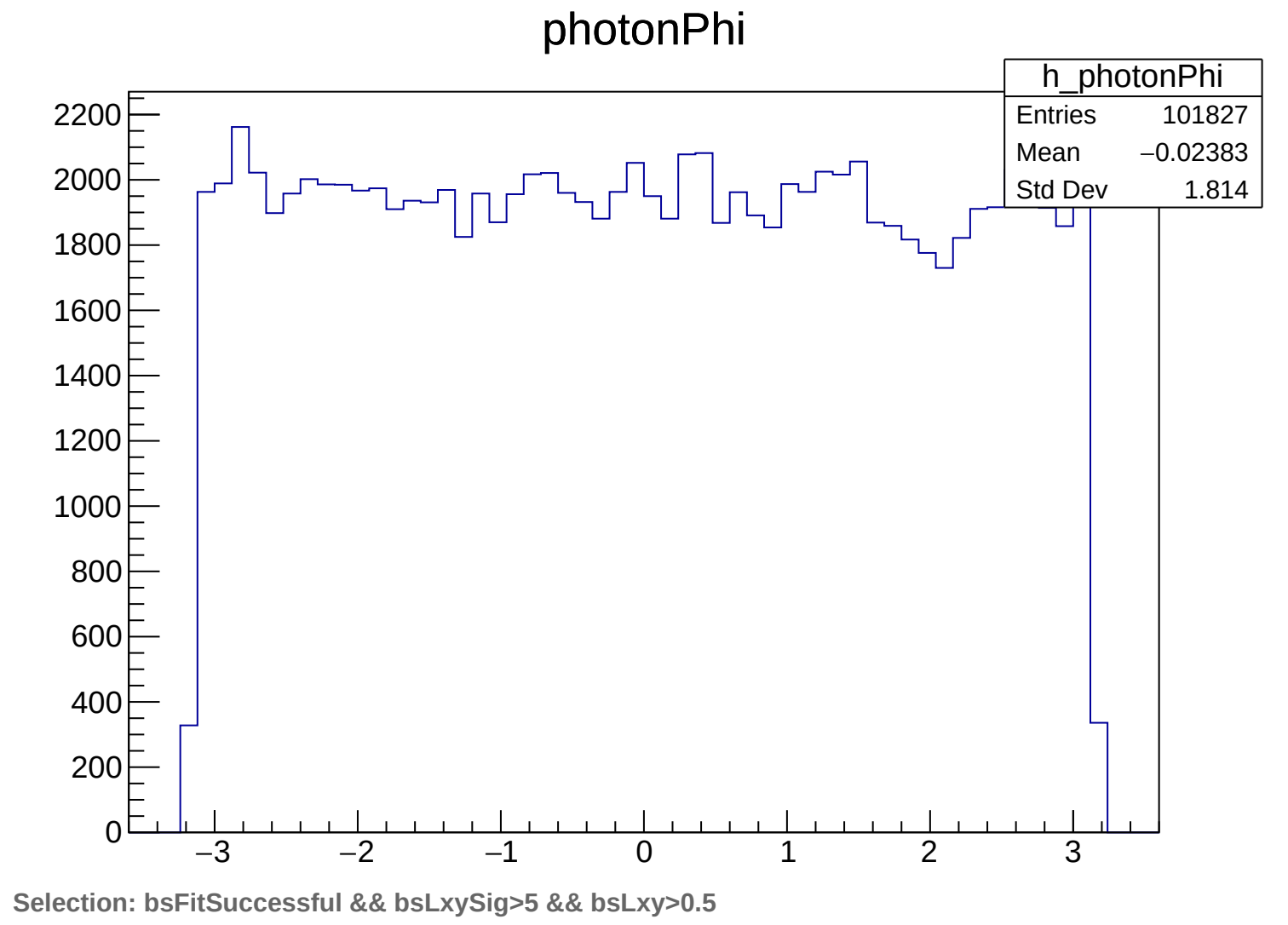

dimuonGenPhotonMass
h_dimuonGenPhotonMass
Entries 101827
Mean 5.356
Std Dev 0.07264
60000
50000
40000
30000
20000
10000
0
4.4
4.6
4.8
5
5.2
5.4
5.6
5.8
6
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


photonDeltaRs
h_photonDeltaRs
Entries 101827
Mean 0.1726
Std Dev 0.06815
5000
4000
3000
2000
1000
0
0
0.1
0.2
0.3
0.4
0.5
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

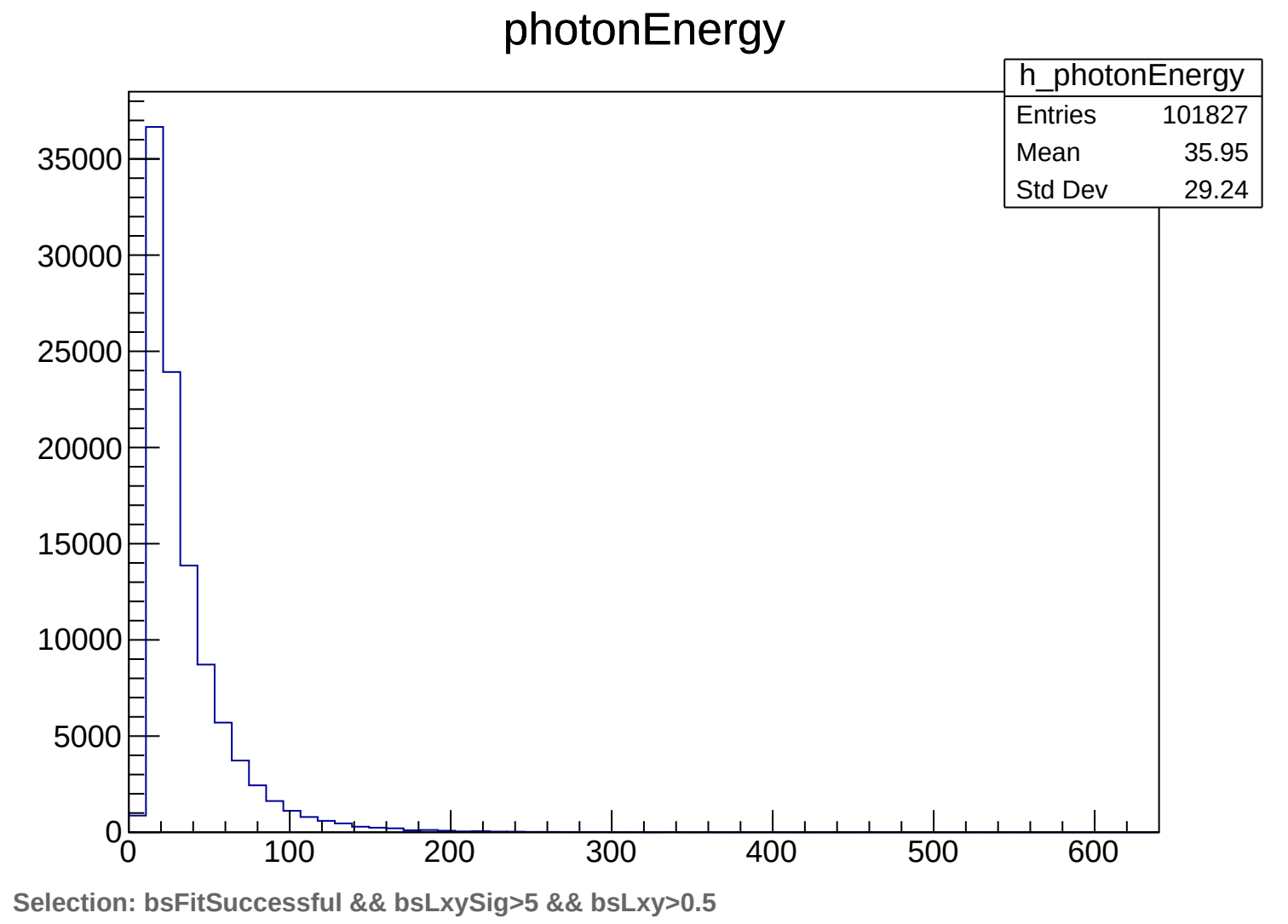

photonPt
h_photonPt
Entries 101827
Mean 19.17
Std Dev 10.47
30000
25000
20000
15000
10000
5000
0
0
50
100
150
200
250
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

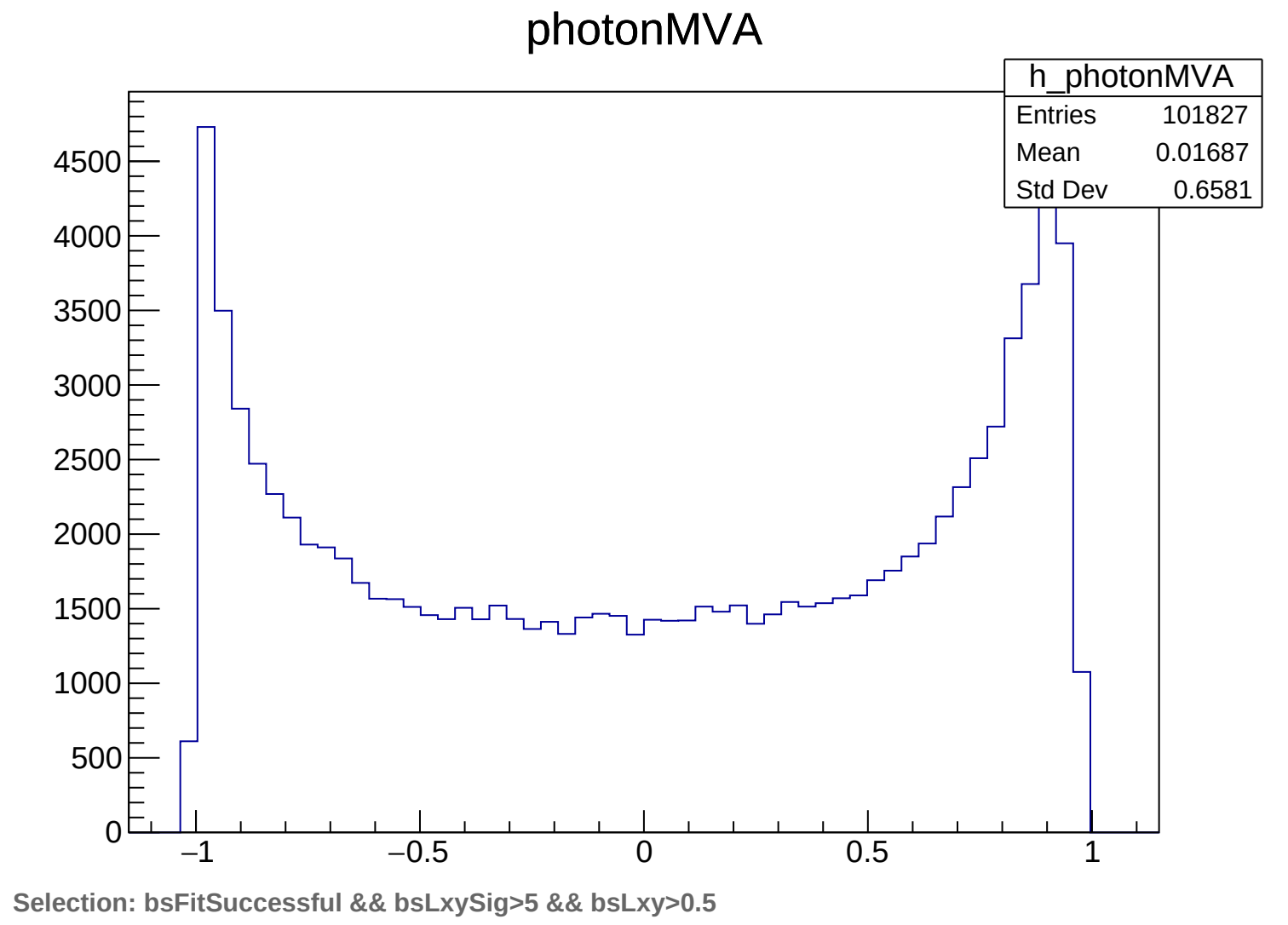


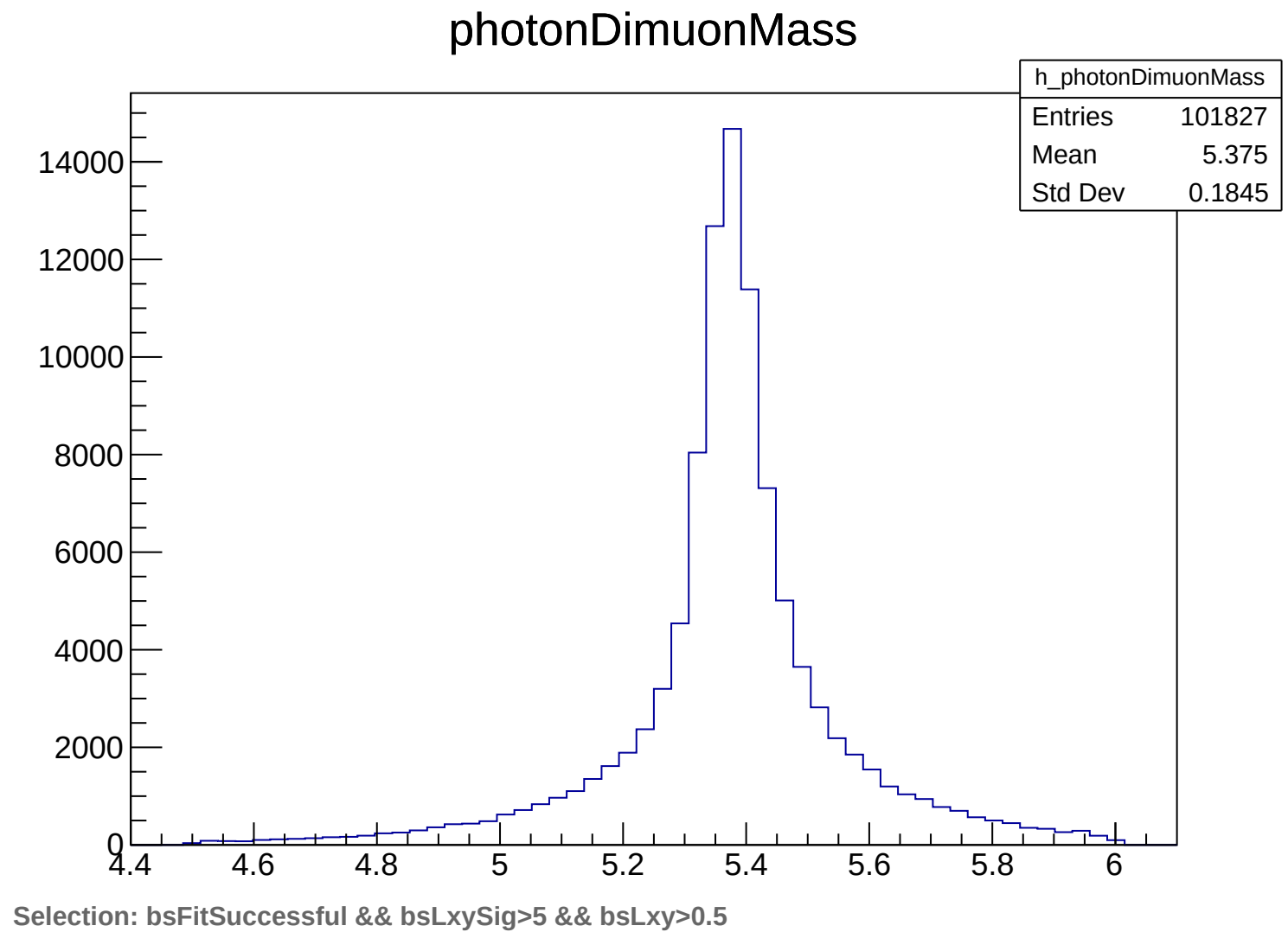
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

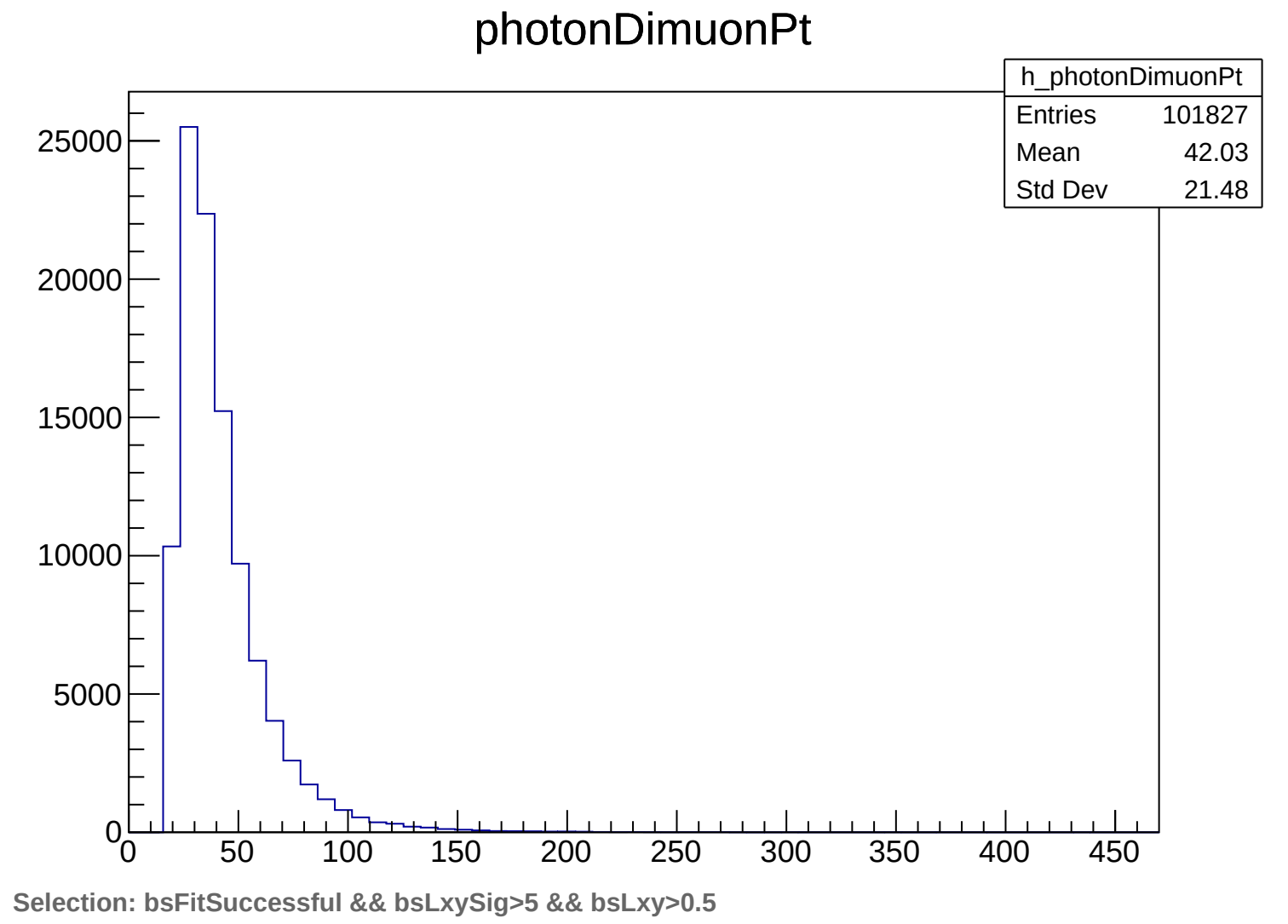


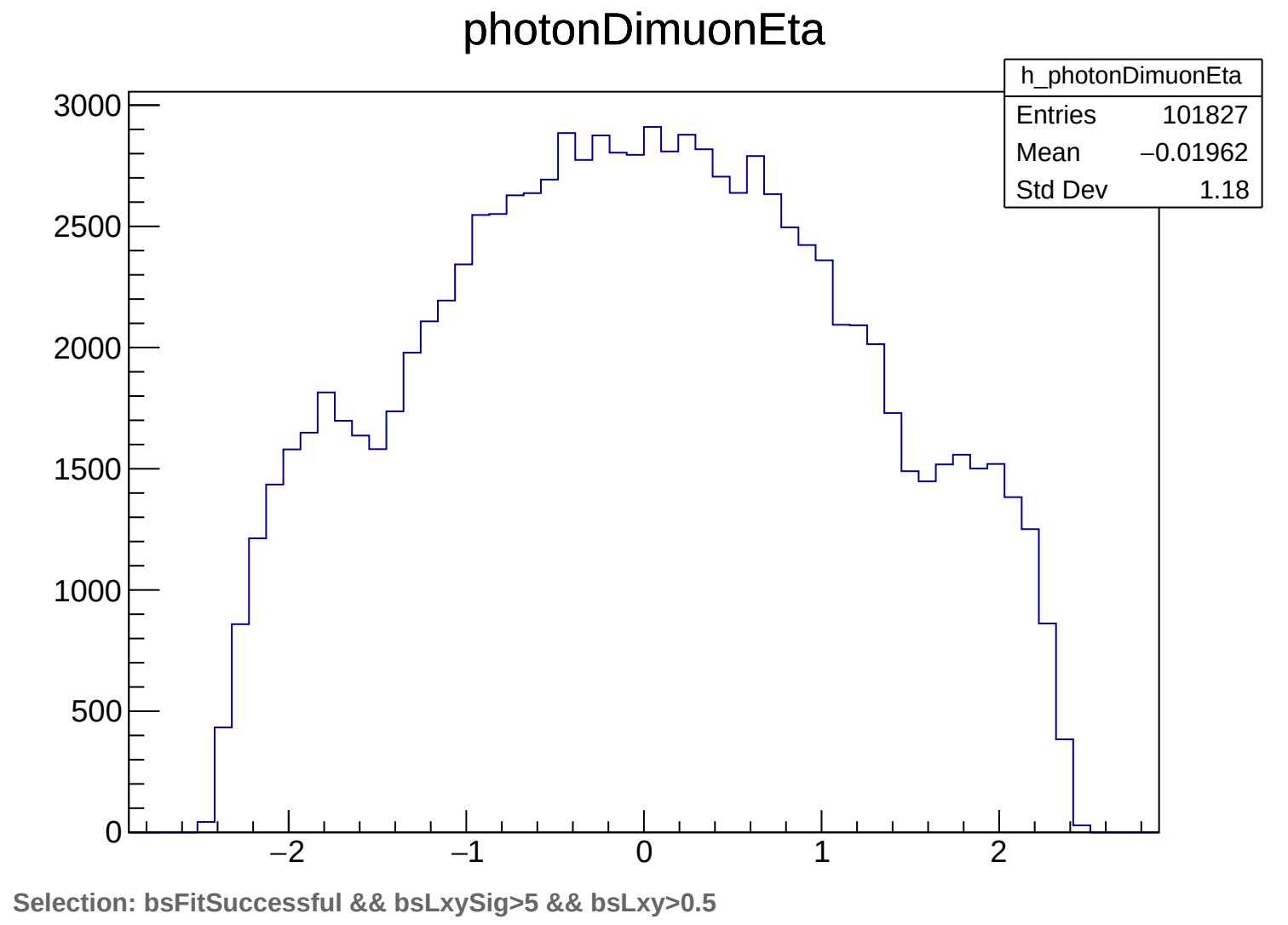
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


photonEnergy
h_photonEnergy
Entries 101827
Mean 35.95
Std Dev 29.24
35000
30000
25000
20000
15000
10000
5000
0
0
100
200
300
400
500
600
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

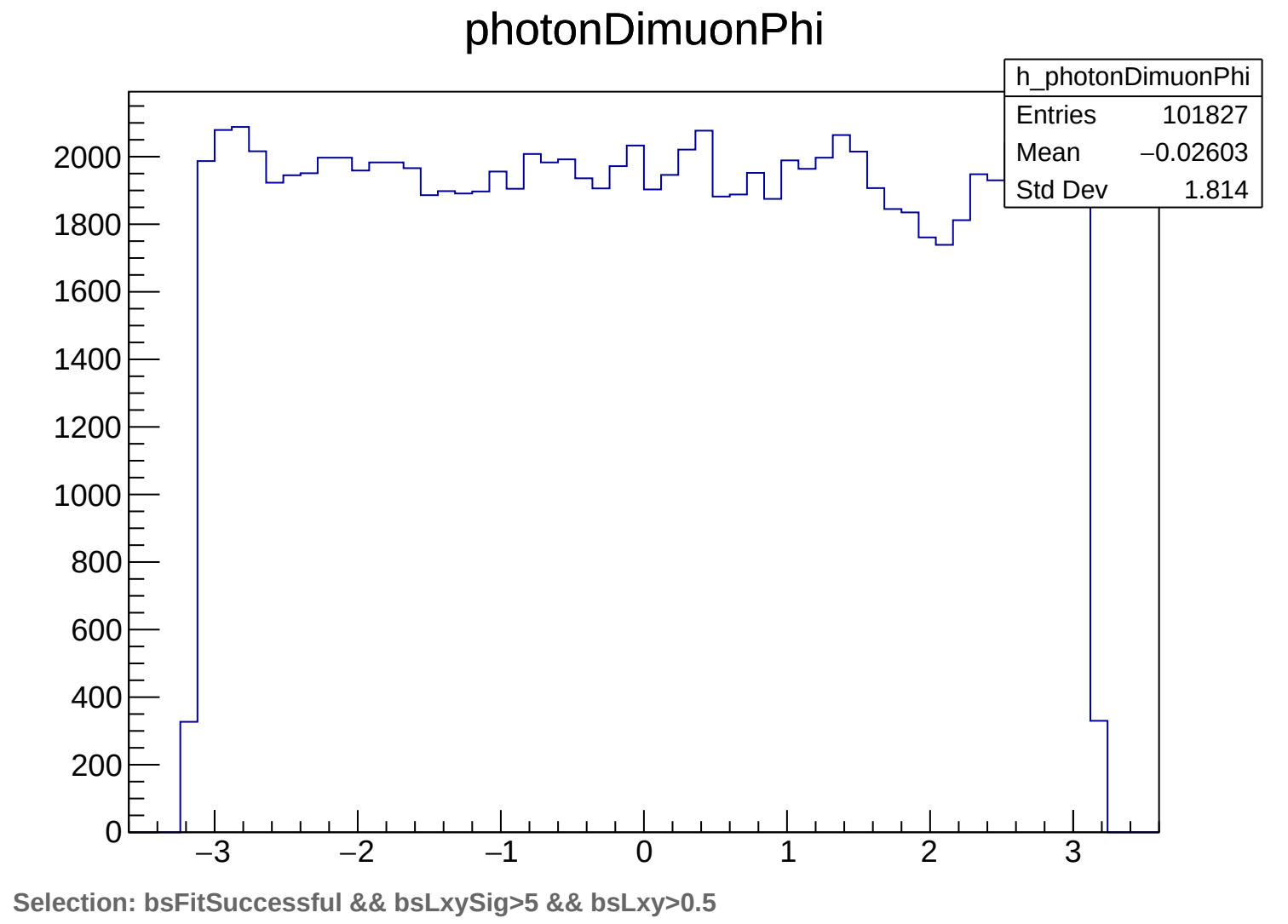

photonMVA
h_photonMVA
Entries 101827
Mean 0.01687
Std Dev 0.6581
4500
4000
3500
3000
2500
2000
1500
1000
500
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

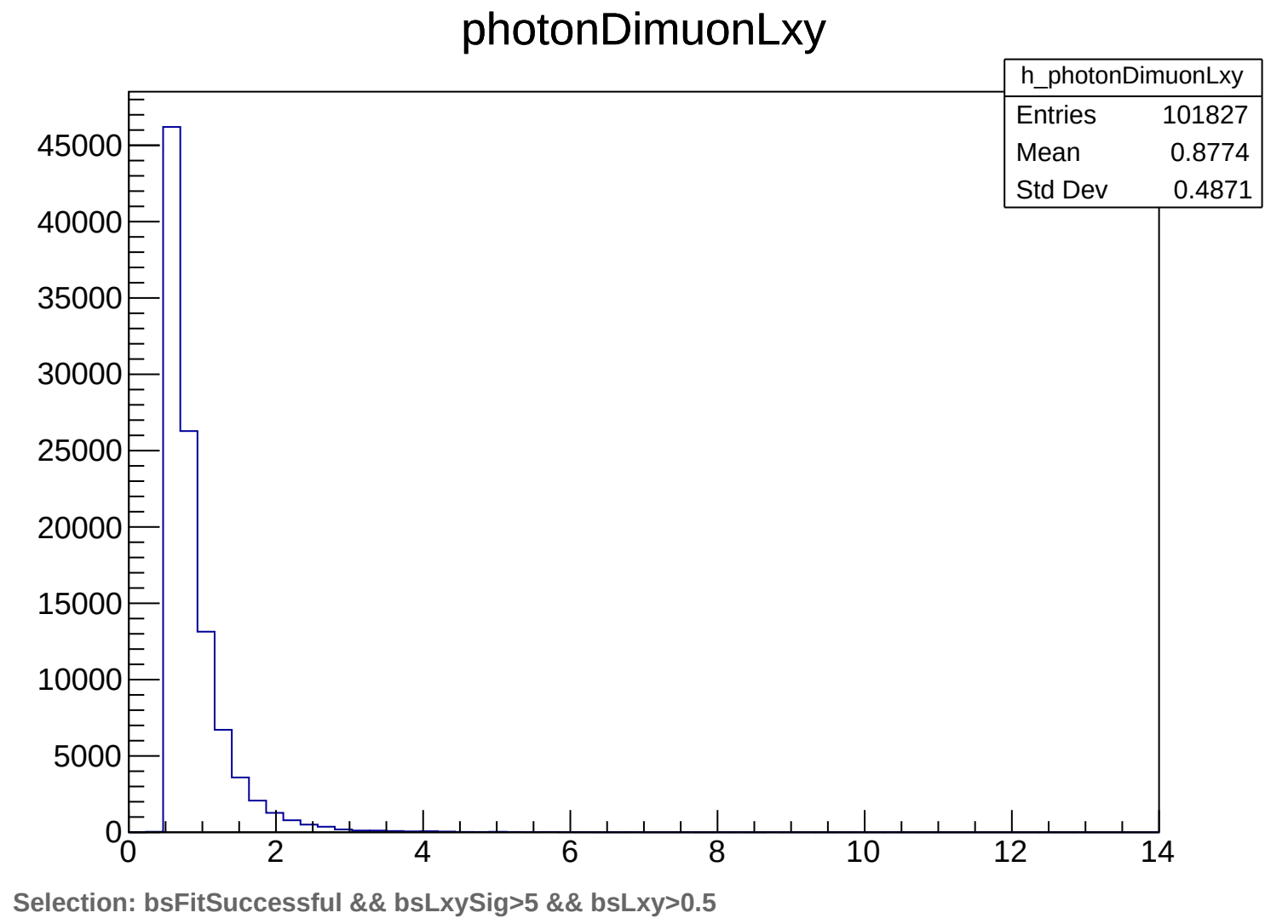

photonDimuonMass
h_photonDimuonMass
Entries 101827
Mean 5.375
Std Dev 0.1845
14000
12000
10000
8000
6000
4000
2000
0
4.4
4.6
4.8
5
5.2
5.4
5.6
5.8
6
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

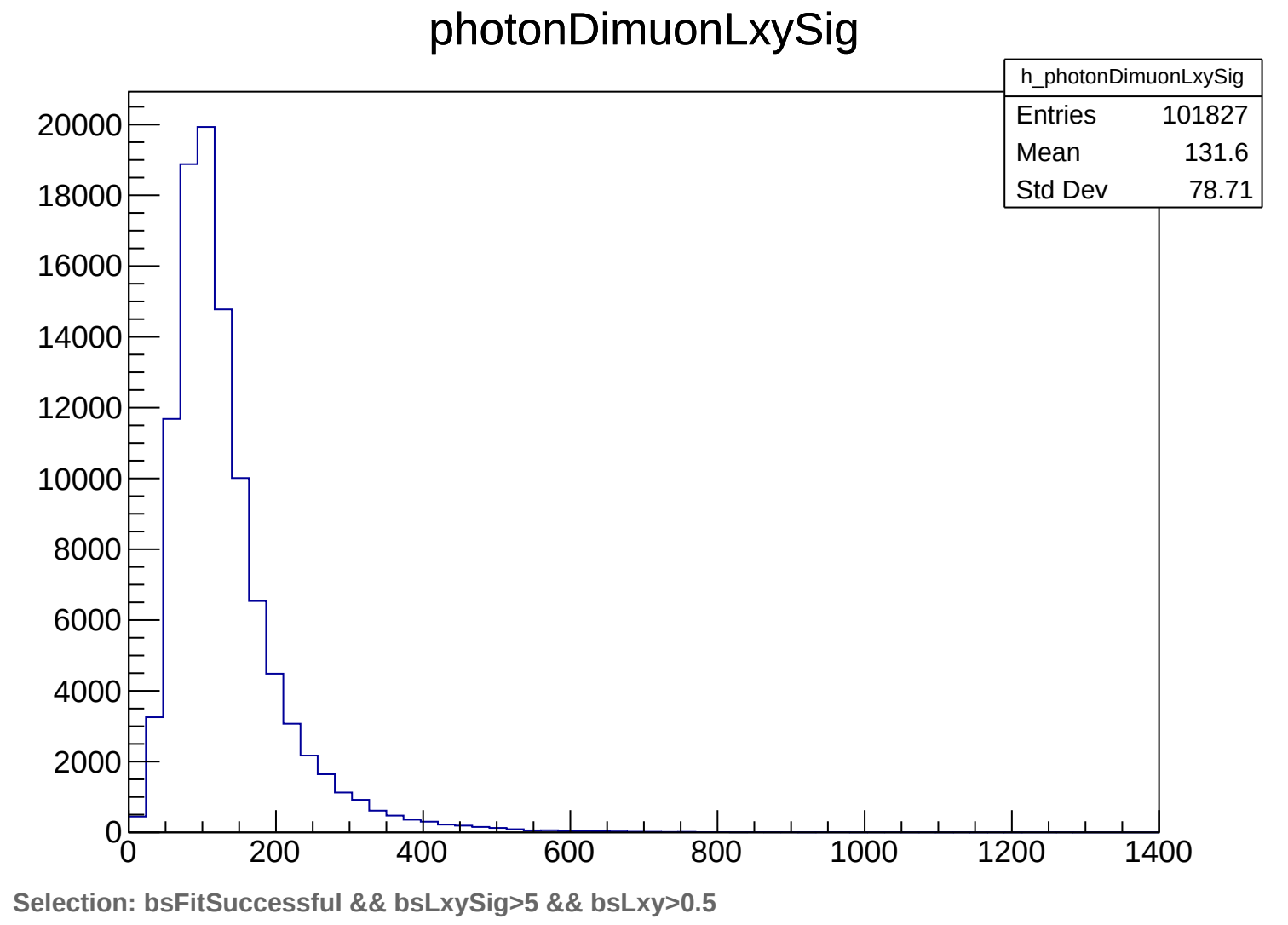

photonDimuonPt
h_photonDimuonPt
Entries 101827
Mean 42.03
Std Dev 21.48
25000
20000
15000
10000
5000
0
0
50
100
150
200
250
300
350
400
450
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

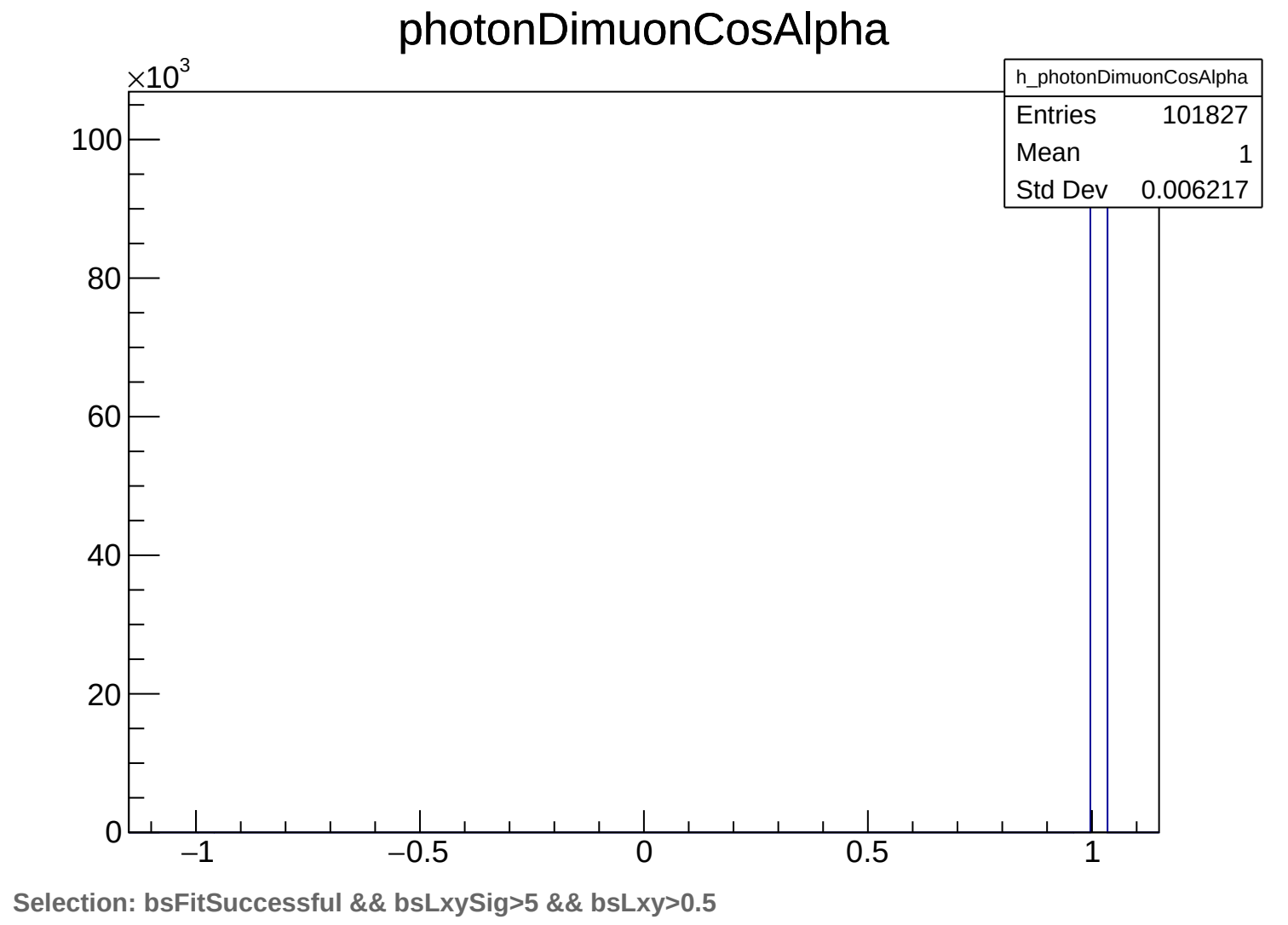


Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

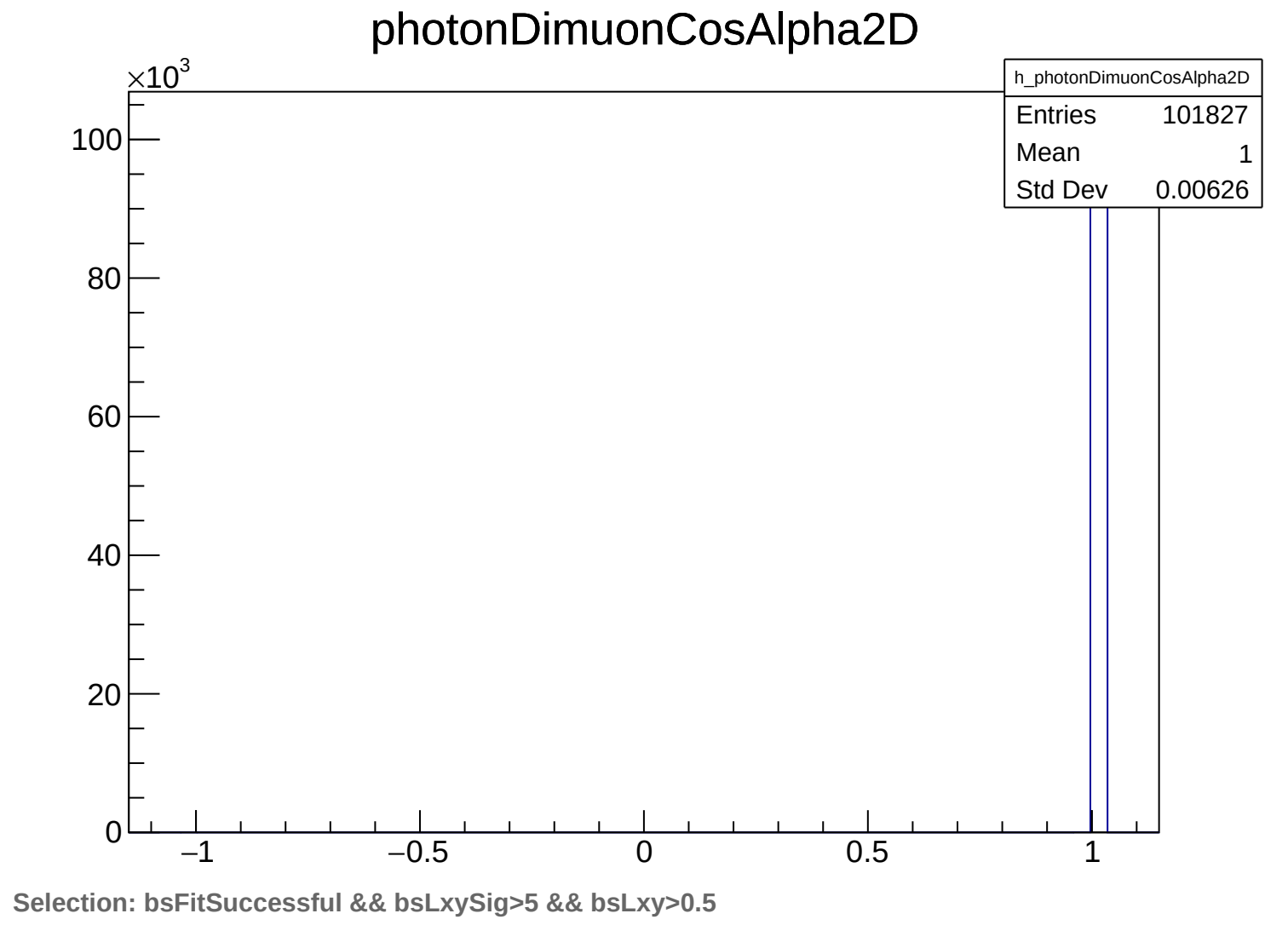

photonDimuonPhi
h_photonDimuonPhi
Entries 101827
Mean −0.02603
Std Dev 1.814
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


photonDimuonLxy
h_photonDimuonLxy
Entries 101827
Mean 0.8774
Std Dev 0.4871
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
0
2
4
6
8
10
12
14
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

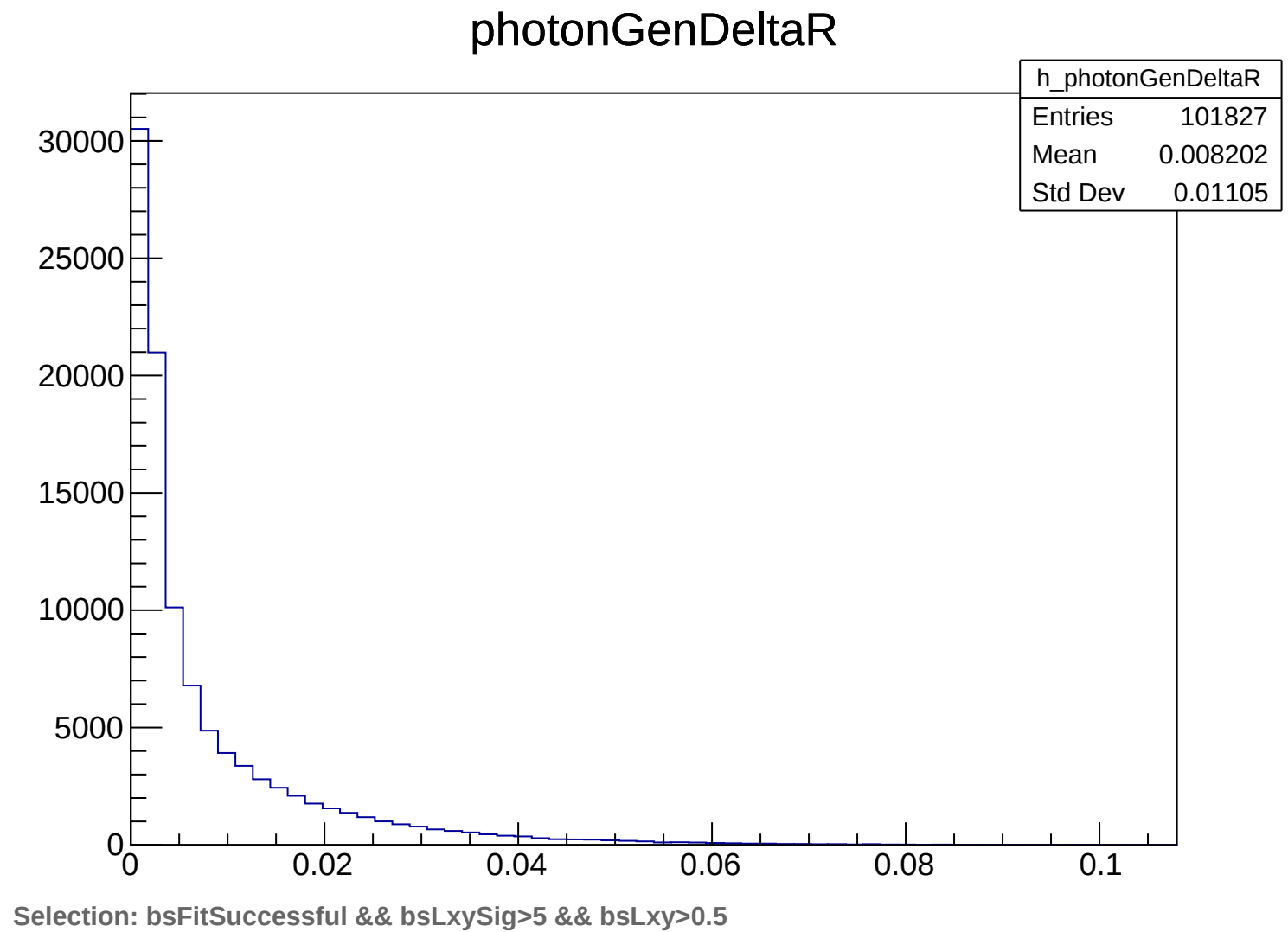

photonDimuonLxySig
h_photonDimuonLxySig
Entries 101827
Mean 131.6
Std Dev 78.71
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
200
400
600
800
1000
1200
1400
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5



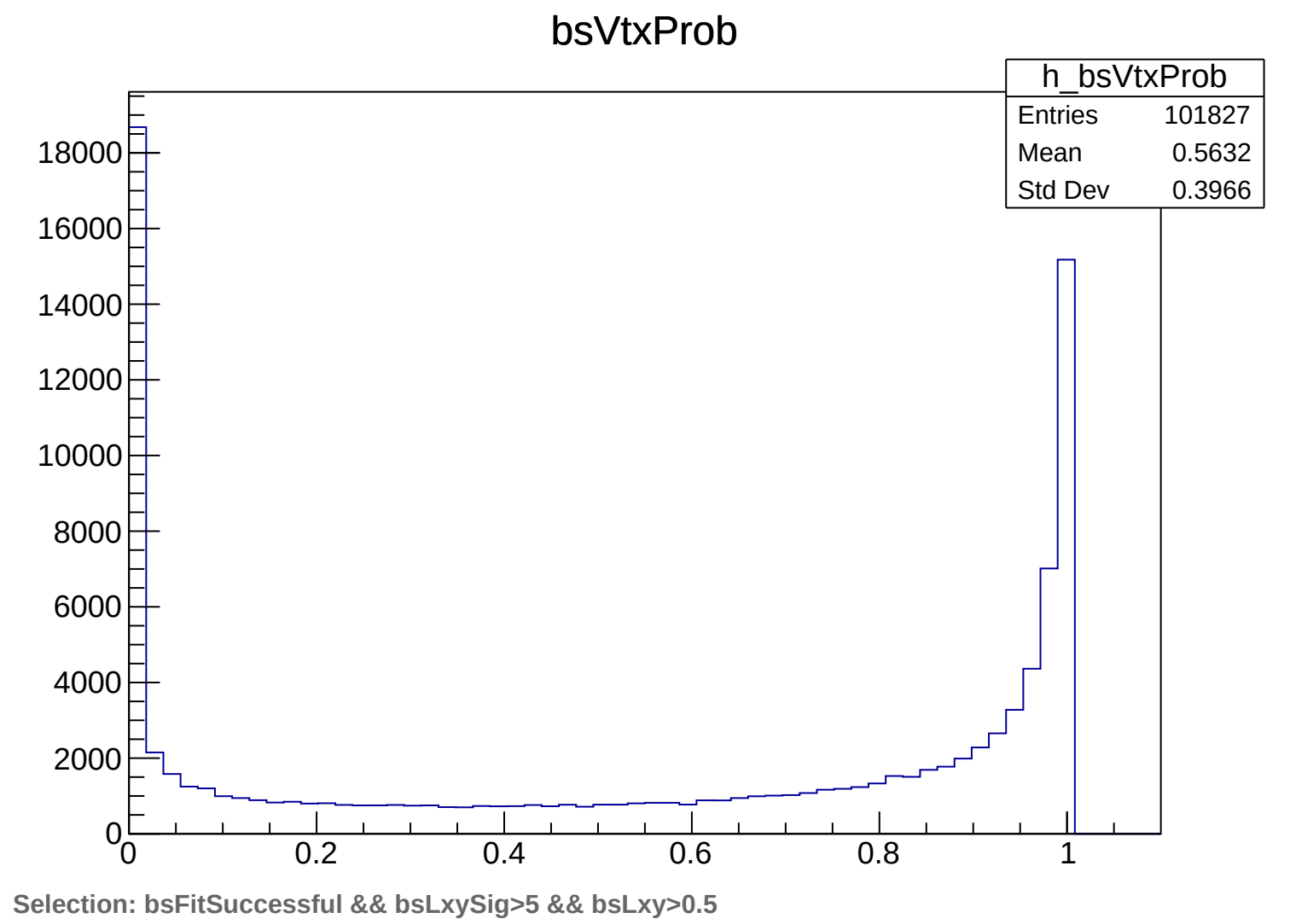
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

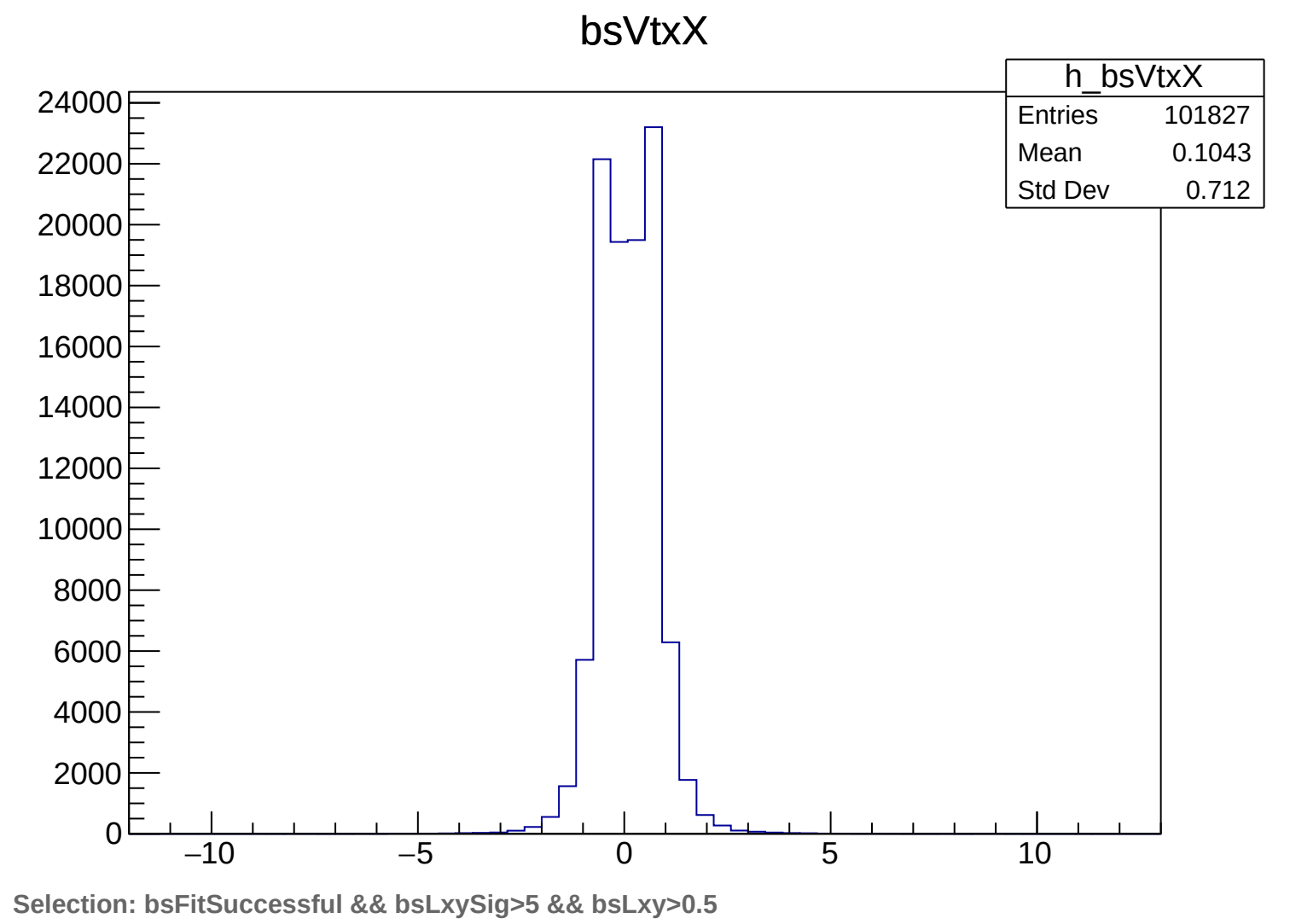


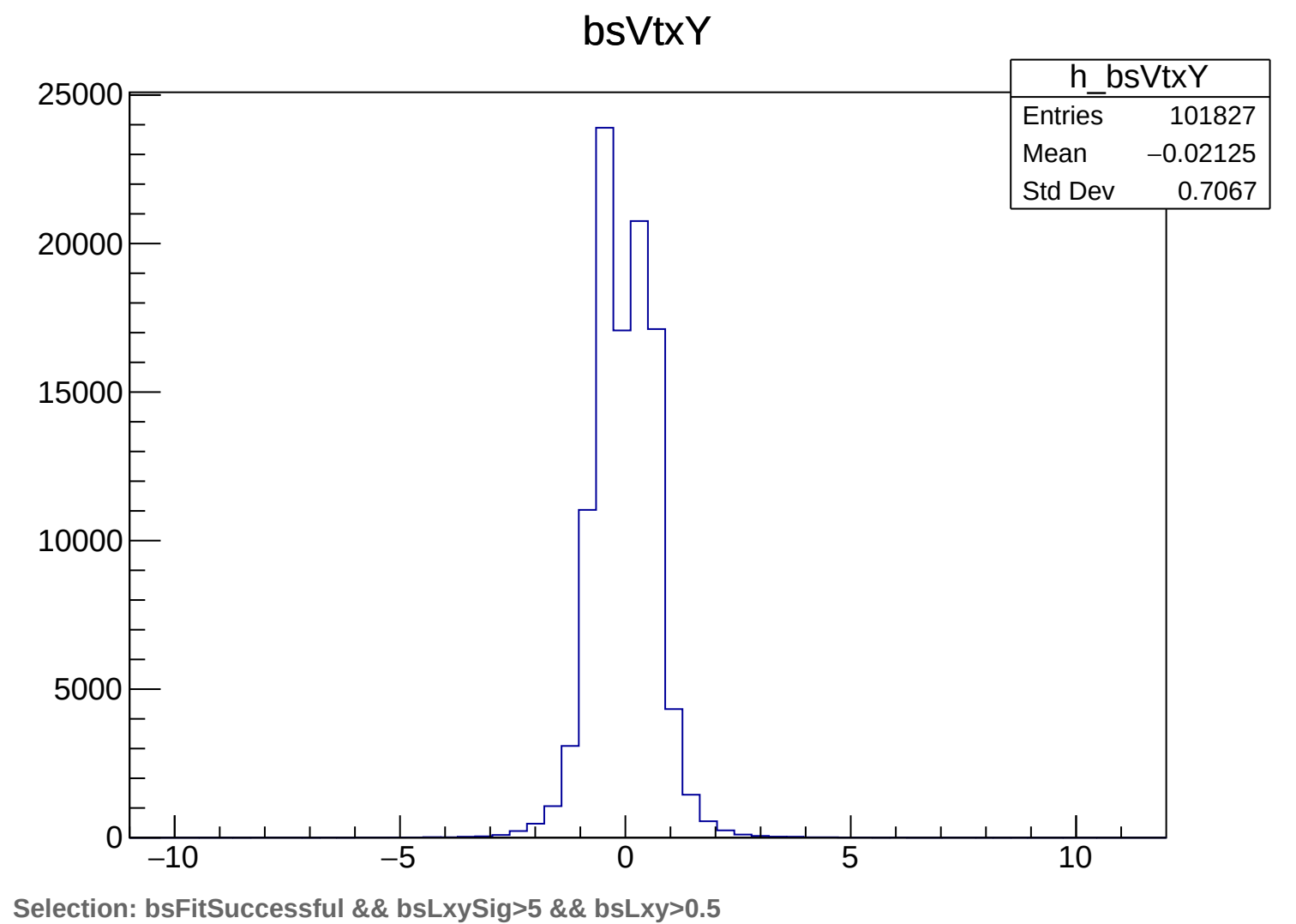
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

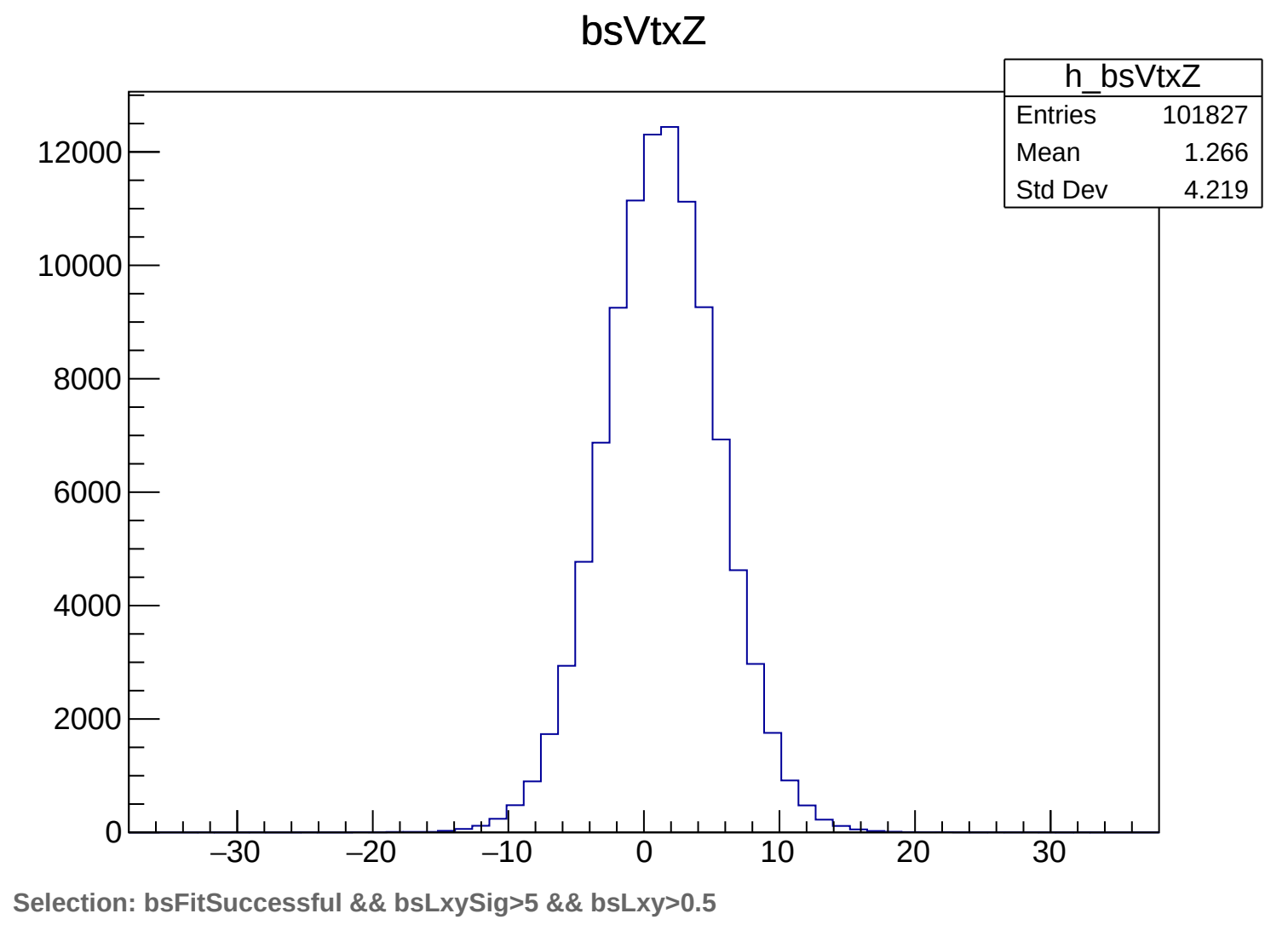


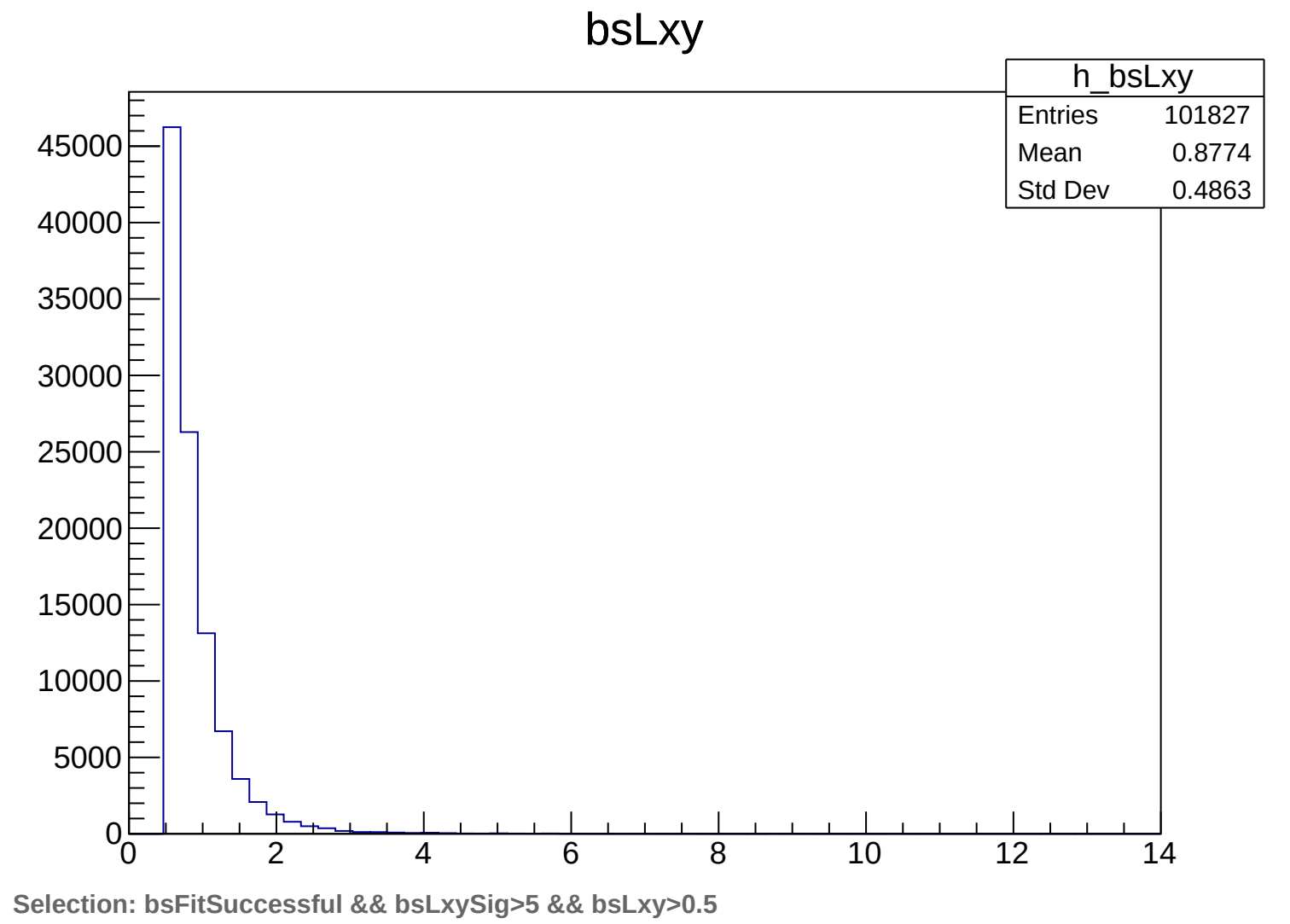
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

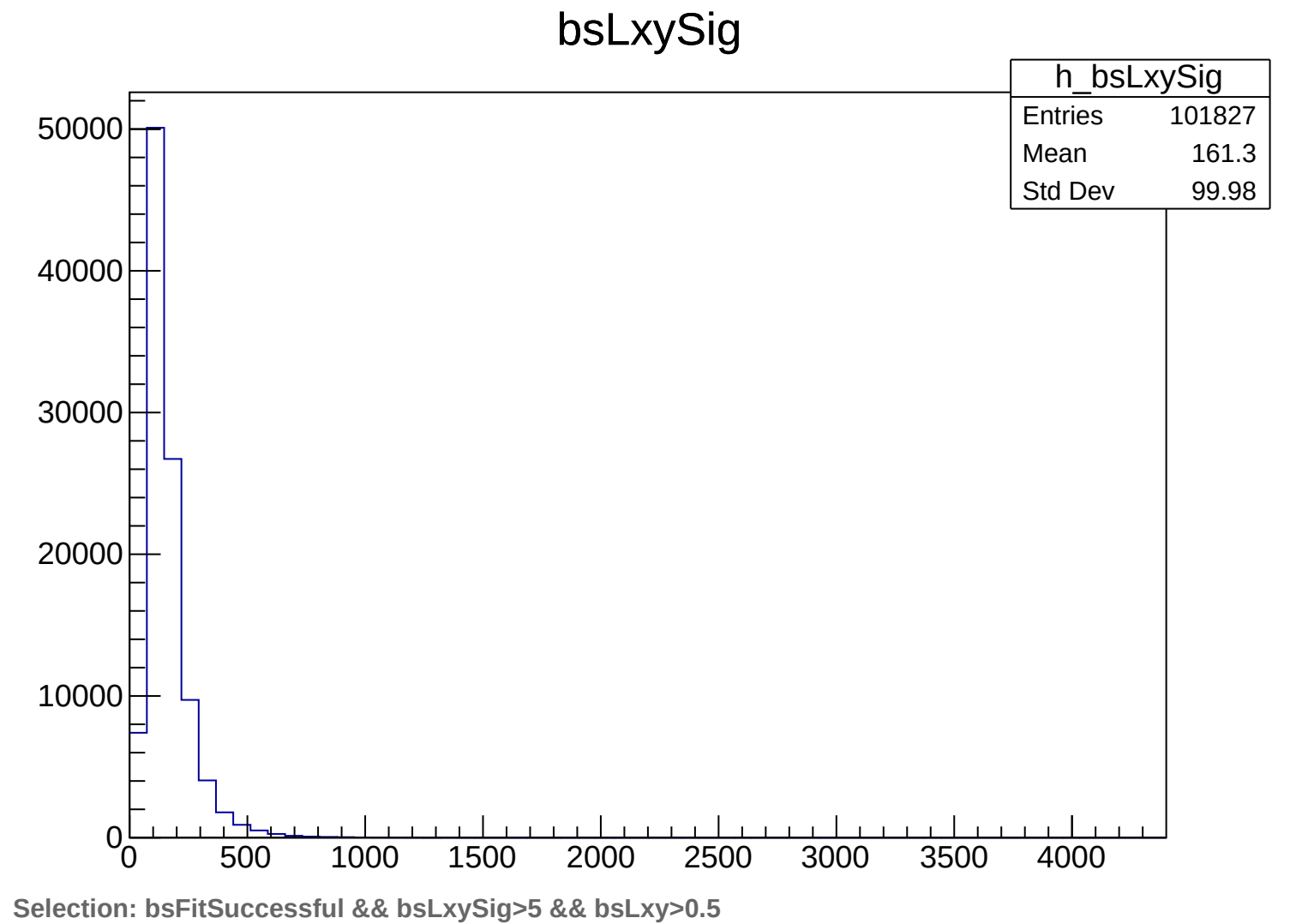

bsVtxProb
h_bsVtxProb
Entries 101827
Mean 0.5632
Std Dev 0.3966
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


bsVtxX
h_bsVtxX
Entries 101827
Mean 0.1043
Std Dev 0.712
24000
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
−10
−5
0
5
10
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

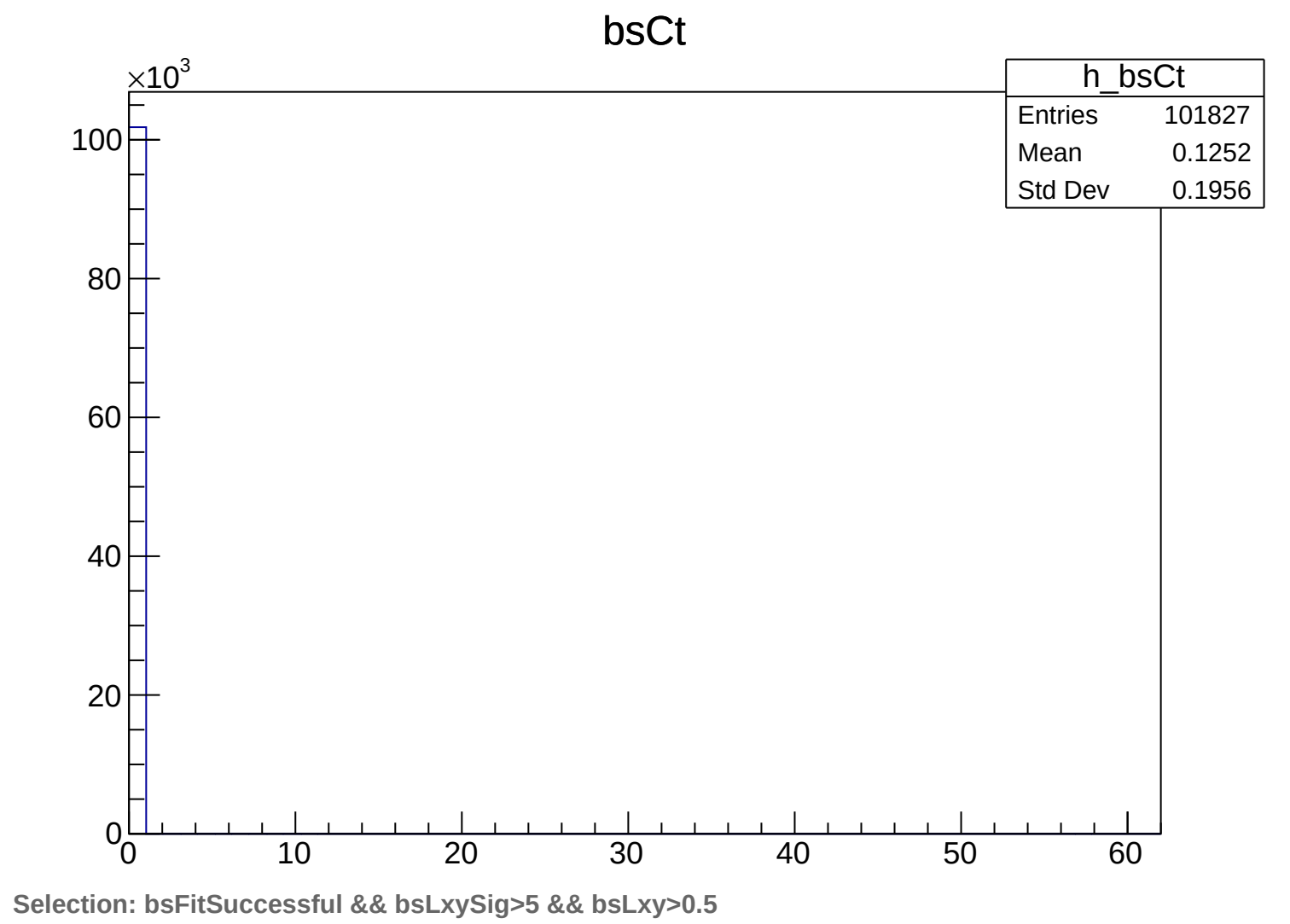

bsVtxY
h_bsVtxY
Entries 101827
Mean −0.02125
Std Dev 0.7067
25000
20000
15000
10000
5000
0
−10
−5
0
5
10
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


bsVtxZ
h_bsVtxZ
Entries 101827
Mean 1.266
Std Dev 4.219
12000
10000
8000
6000
4000
2000
0
–30
–20
–10
0
10
20
30
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

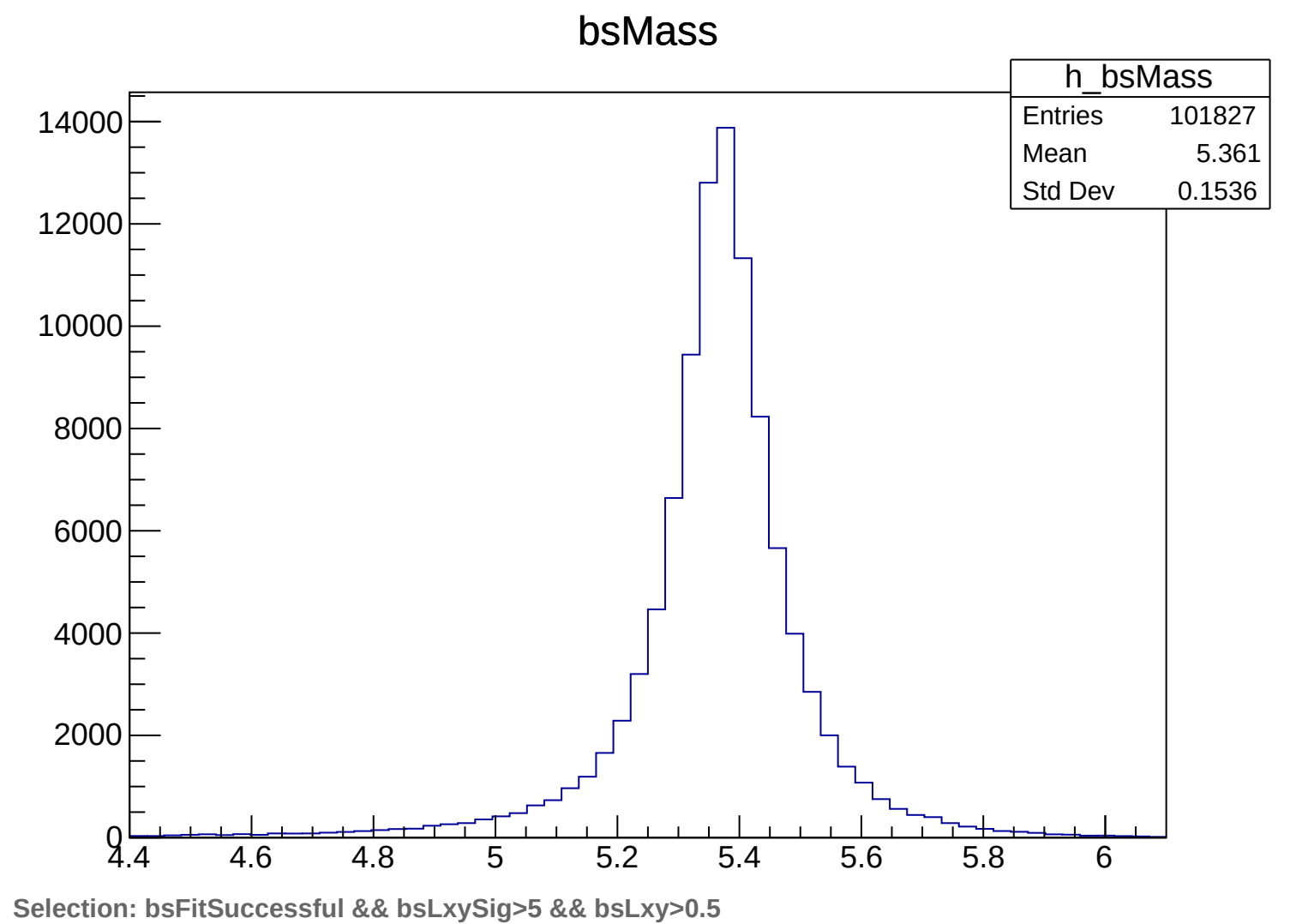

bsLxy
h_bsLxy
Entries 101827
Mean 0.8774
Std Dev 0.4863
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
0
2
4
6
8
10
12
14
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

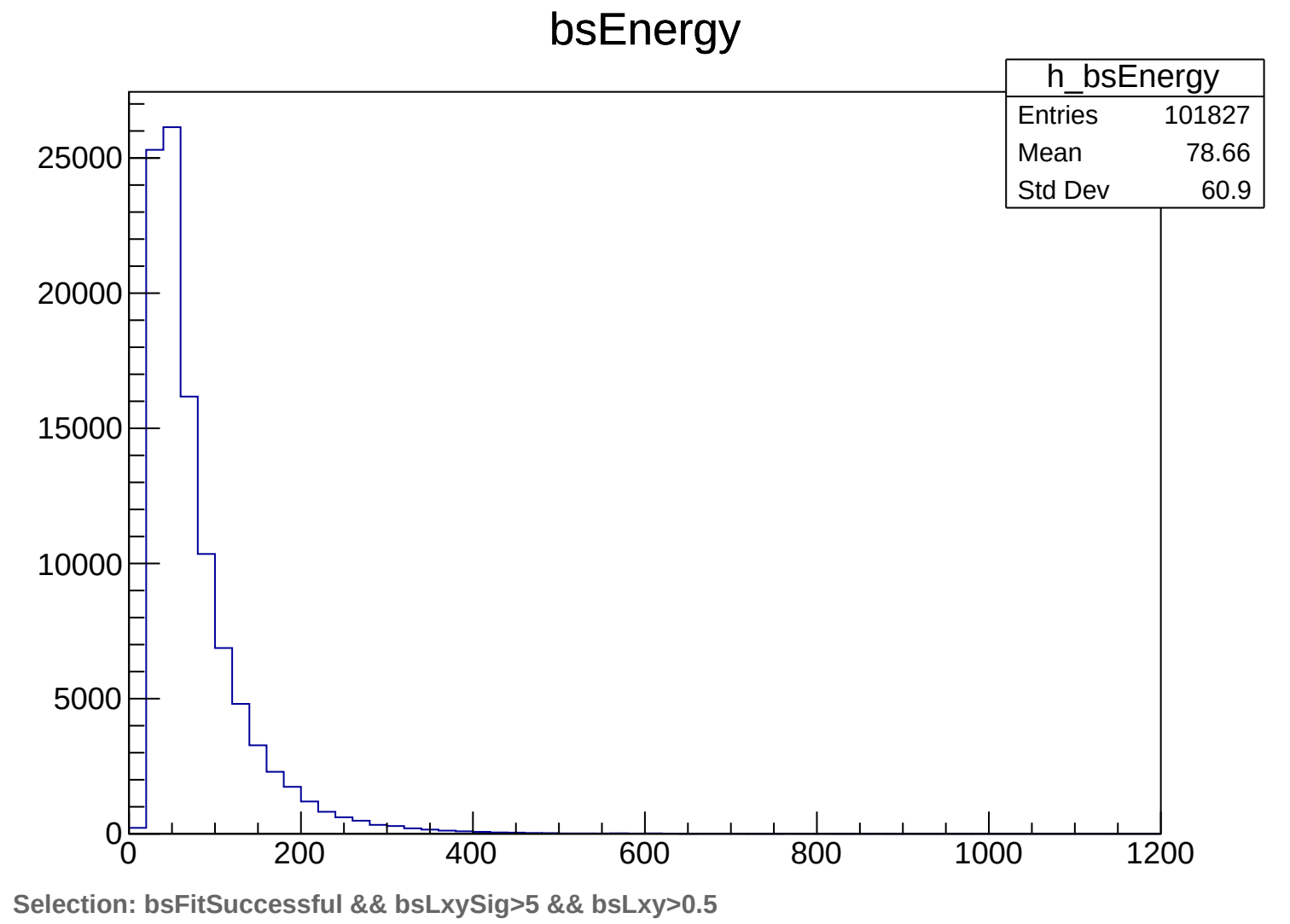


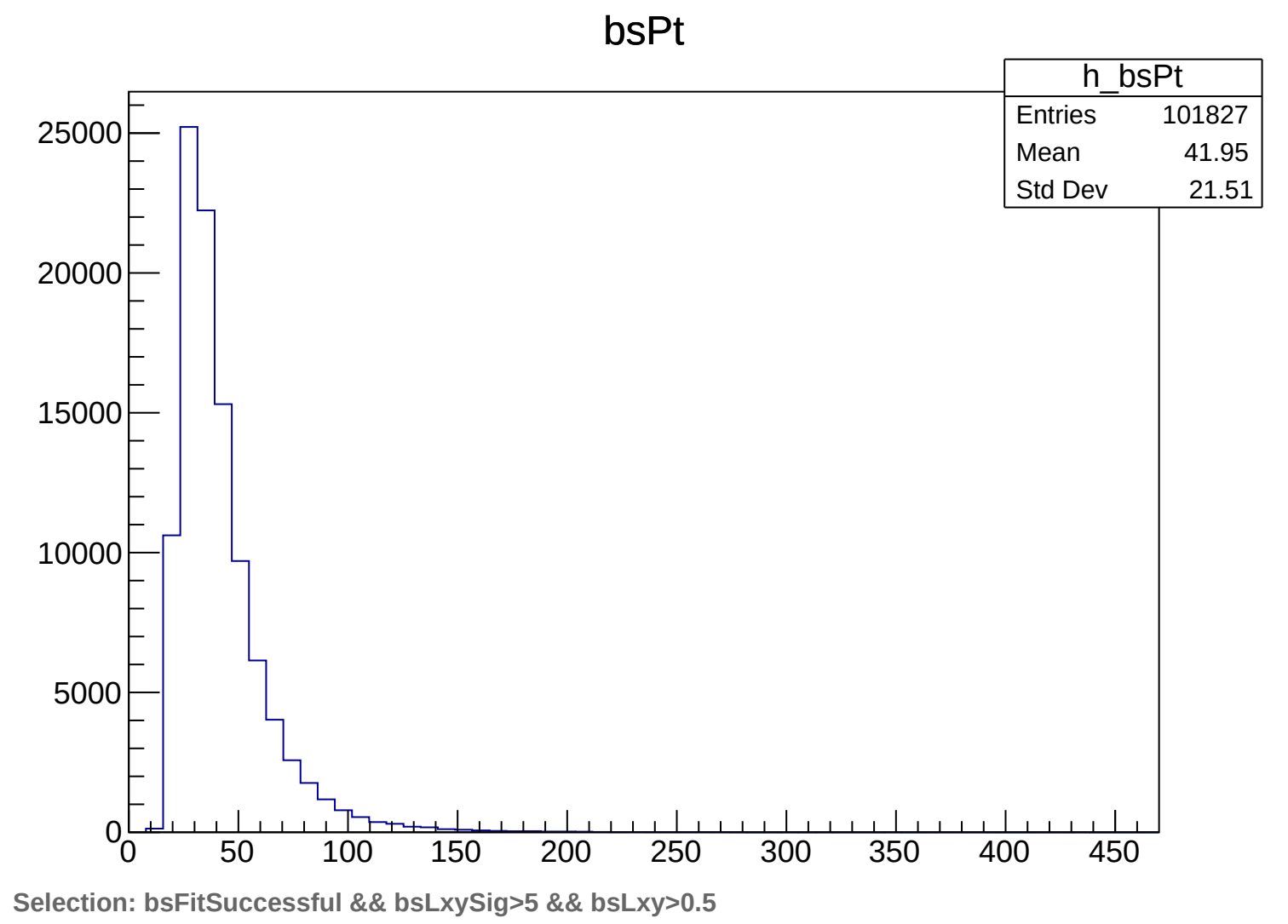
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

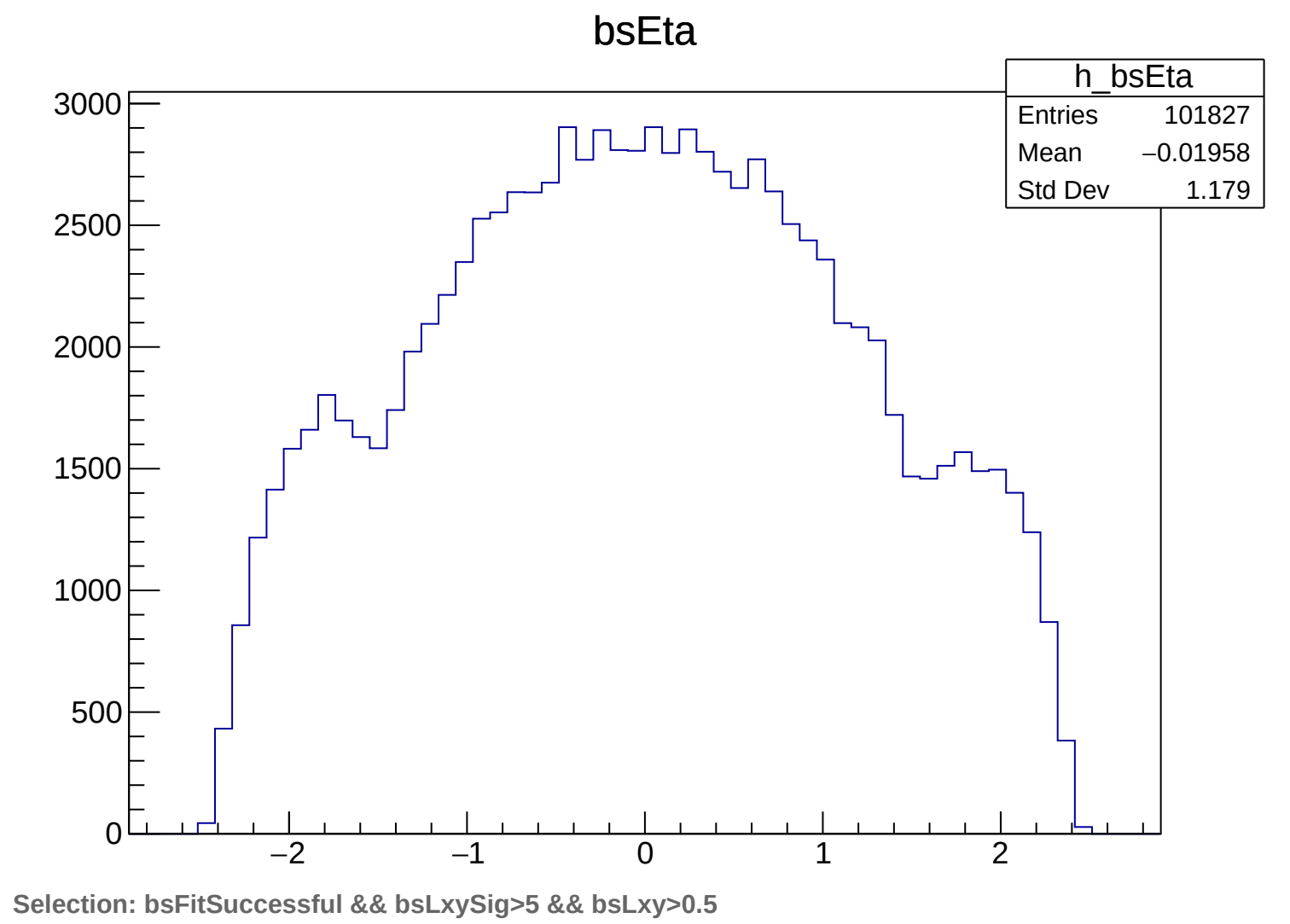


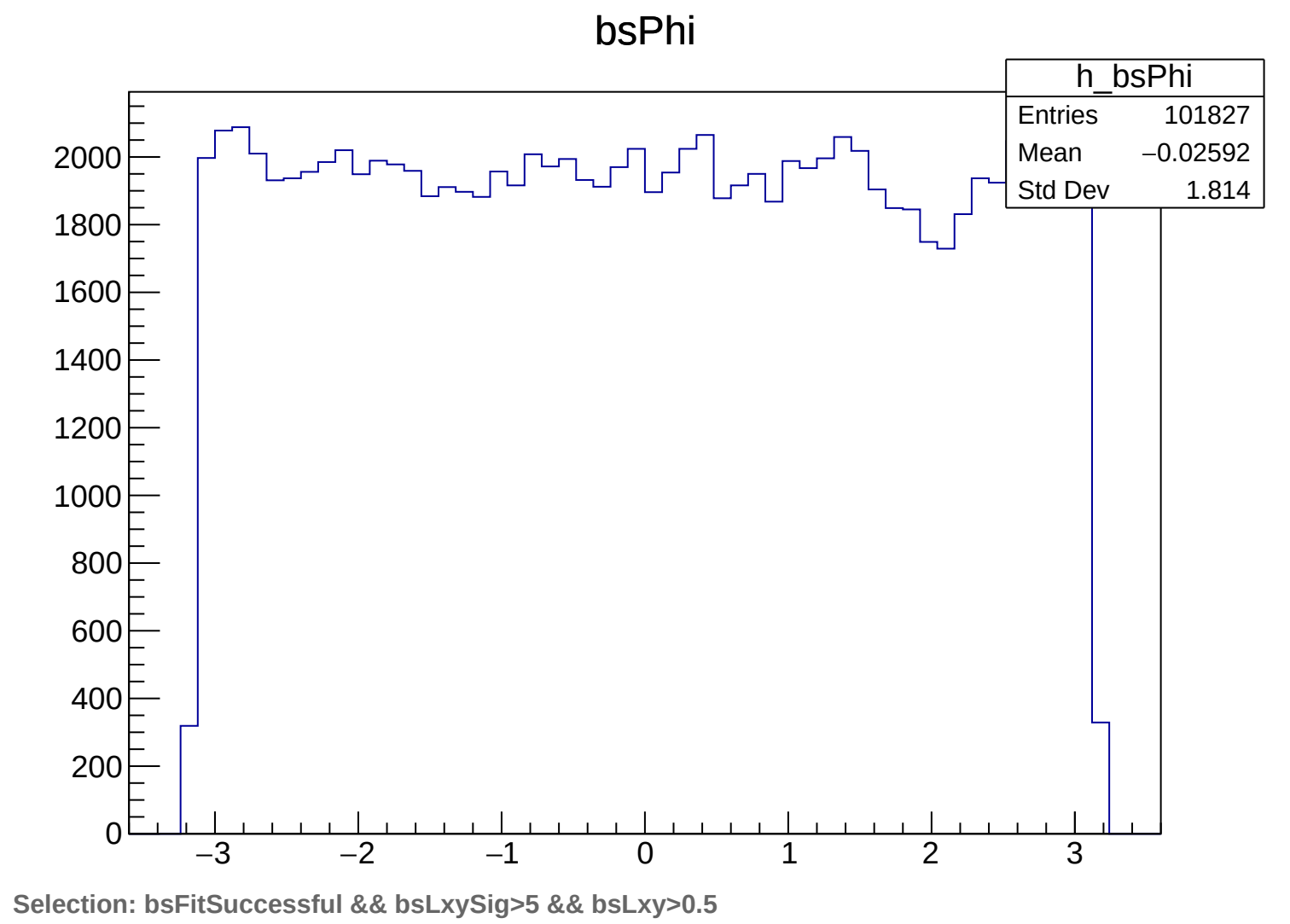
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


bsMass
h_bsMass
Entries 101827
Mean 5.361
Std Dev 0.1536
14000
12000
10000
8000
6000
4000
2000
0
4.4
4.6
4.8
5
5.2
5.4
5.6
5.8
6
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5

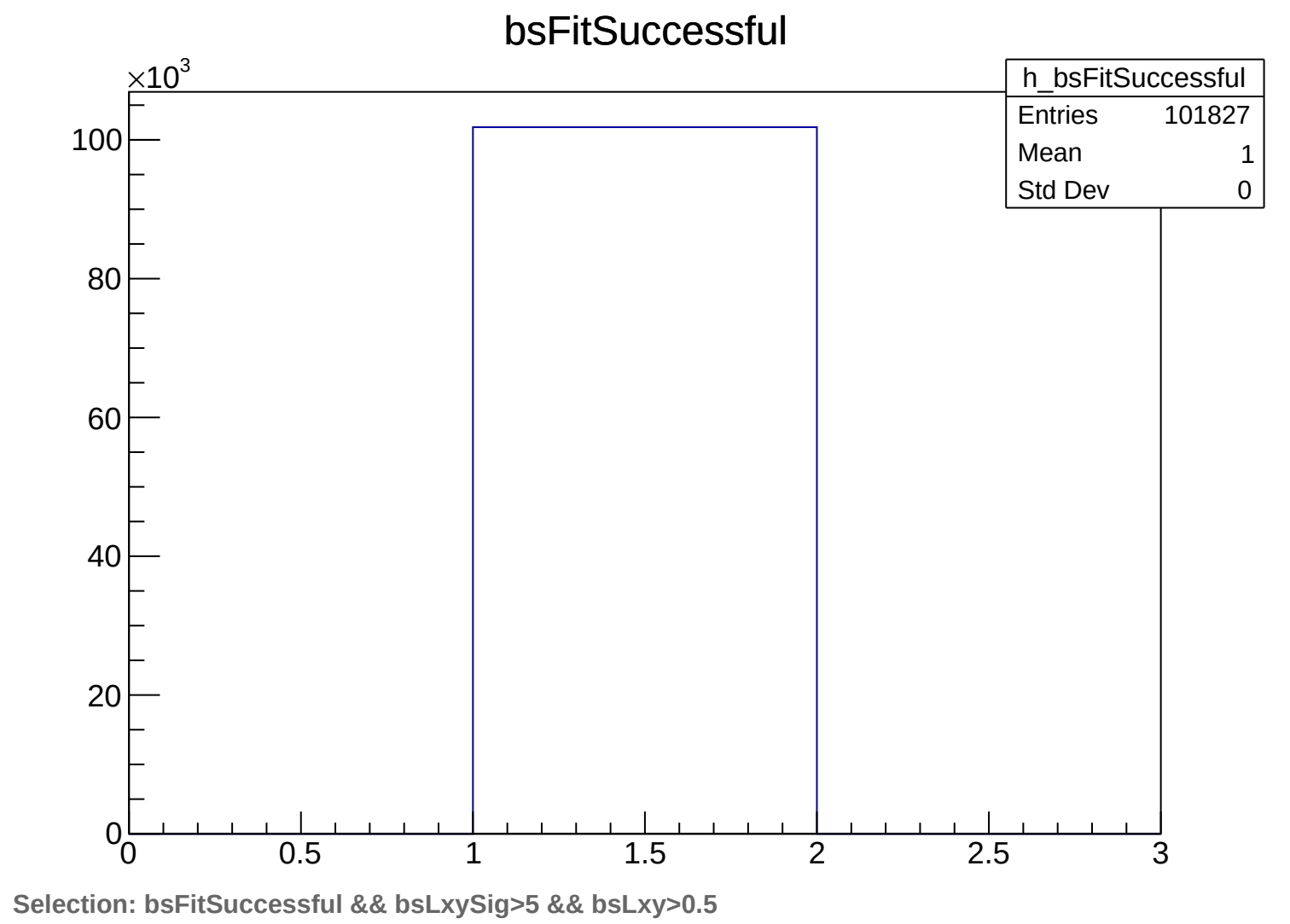

bsEnergy
h_bsEnergy
Entries 101827
Mean 78.66
Std Dev 60.9
25000
20000
15000
10000
5000
0
0
200
400
600
800
1000
1200
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


bsPt
h_bsPt
Entries 101827
Mean 41.95
Std Dev 21.51
25000
20000
15000
10000
5000
0
0
50
100
150
200
250
300
350
400
450
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5


bsEta
h_bsEta
Entries 101827
Mean −0.01958
Std Dev 1.179
3000
2500
2000
1500
1000
500
0
−2
−1
0
1
2
Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5



Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5



Selection: bsFitSuccessful && bsLxySig>5 && bsLxy>0.5